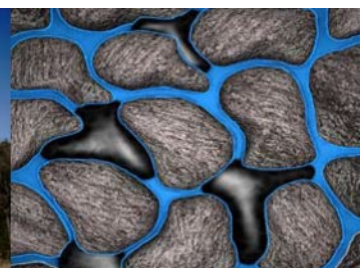


Engineering Resource Material



Baroid Fluids Handbook

Version 10-2012

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1. Rheology

Rheology is the study and science of flowing matter.

In the oilfield this science is typically focused on liquids or particulate suspensions. Examples include liquids such as brine completion fluids; suspensions, such as barite weighted drilling fluids and cements that react with time, temperature and chemistry. These fluid types represent the wide range of rheologically complex and diverse materials that are encountered daily.

Each of these has its own rheological complexity that must be understood to maximize drilling success and minimize non-productive time (NPT). The field engineer must understand how these can impact and drive field operations.

Knowledge of certain rheological terms, principles, and commonly used rheological models is necessary to gain a fundamental understanding of rheology and its impact on field operations.

Basic knowledge of the common language and terms used to discuss rheology is a key component to understanding how and why rheology is important. Basic rheological equations are addressed in Halliburton software packages such as DFG, WellPlan and ICem, which perform calculations and hydraulics predictions. Public domain equations and methods are readily available in the publication API Recommended Practices 13D available at www.api.org.

The key objectives to learning about rheology are as follows:

- Understand the language of rheology
- Understand the physical meaning of the language terms
- Understand why detailed software inputs are sometimes needed.
- Understand some principles of hydraulics and wellbore pressure management

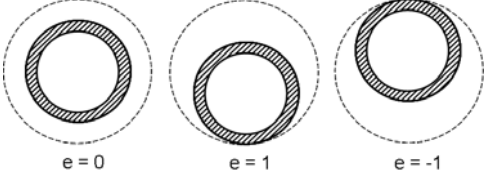
1.1. Rheology and Hydraulics Terminology

The terms and definitions in the following table are fundamental to the discussion of rheology and hydraulics in drilling operations. Some of these terms are common to programs like DFG hydraulics modeling software.

Table 1 Rheology and Hydraulics Terminology

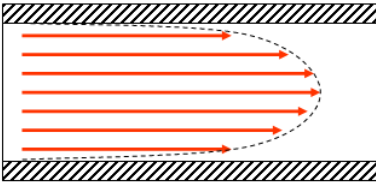
Term	Symbol (s)	Unit(s)	Definition
Annular Velocity	Av	ft/min ft/s m/m m/s	The average velocity of a fluid as it moves through an annular section in the wellbore. Increasing the pump rate increases the annular velocity. Increasing the pump rate tends to improve cuttings transport but also increases down hole pressure or ECD.
Annular Volume	V _a	bbl s ft ³ ft ³ gal	Volume of the wellbore annulus.
Average Specific Gravity	ASG	-	The relative density of all the solids that make up the drilling fluid. For Barite weighted fluid systems the typical ASG is 3.8 to 4.1. Typical drilling fluids solids consist primarily of Barite, Barite impurities, drill solids, LCM and other solid products. Pure barite has an SG of about 4.5. Thus, standard 4.2 API Barite is about 15% impurities. 4.1 Barite is about 21% impurities. DFG requires an ASG input in its density modeling algorithms since it is based on conservation of mass and equation of state methods.
Base oil	-	-	Each oil has a unique equation of state to describe its density as a function of temperature and pressure. DFG algorithms used to calculate the downhole pressure (ESD or EMW) exerted by the drilling fluid uses these to do accurate simulations of fluid density downhole. It is not sufficient to model downhole pressure without these equations of state. Sometimes engineers will be required to model competitor fluids though DFG does not have models for those specific fluids. In these cases, fluid engineers should try to match the type of fluid as much as possible. For example, if a competitor fluid is a mineral oil then use one of the Baroid mineral oils, etc.
Barite sag			Barite sag can have a large impact on field operations and wellbore hydraulics. There have been many methods used to test drilling fluids for barite sag in the lab. None of these methods are 100% reliable to predict barite sag in the field. Furthermore, testing lab formulated fluids is always questionable when comparing sag treatments and performance. Every effort should be made to perform sag testing on field submitted samples. As a rule of thumb drilling fluids should have a minimum tau ₀ of 4.0 lb/100ft to minimize sag occurrence.
Bingham Model	-	-	An old hydraulics model for calculating wellbore pressures. This model tends to over predict drilling fluid hydraulics especially for shear thinning fluids. Baroid does not recommend using the Bingham model for hydraulics calculations. One interesting and remaining use of this model is the PV and YP numbers fluid engineers use to discuss and compare fluids. This is a practice that was good before computers, but is not the best today. The Bingham YP does not capture the lower shear rate rheology and fluid performance adequately.
Bed Height	-	in cm	DFG uses a bed height algorithm for the sliding algorithms. Cuttings transport with no pipe rotation is difficult and is practical only in a narrow annulus.

Term	Symbol (s)	Unit(s)	Definition
Bottom Hole Temperature	BHT	F C	Temperature at the bottom of the wellbore. DFG used BHT in modeling the dynamic temperature profile for the thermal and compressibility calculation of EMW and also for predicting downhole rheology.
Compressibility coefficient	b	-	Compressibility is a measure of the relative volume change in a fluid (or solid) in response to a pressure change. Its basic form is: $\beta = \frac{1}{V} \frac{\partial V}{\partial p} \text{ or } \Delta V = \beta V (\Delta P)$ Where: V = Volume P = Pressure beta = Compressibility coefficient DFG will calculate the compressibility coefficient for whole fluids considering the composition, OWR, density, salinity and ASG.
Consistency index	K	(eq) cP Pa sec ⁿ lb/100 ft ² sec ⁿ	A term used to determine the "viscosity" effects of a flowing fluid used in the power law and Herschel-Bulkley models. It is similar in concept to the PV in the Bingham model. Its units are not viscosity units in the true sense. Viscous effects attributed to a fluid's yield stress are not part of the consistency index as this parameter describes dynamic flow only. DFG calculates this parameter for both the Herschel-Bulkley and Power Law models.
Critical Velocity	-	-	Flow velocity at which the flow changes from laminar to turbulent. In DFG critical flow rate is used to describe this transition point.
Critical Flow Rate	Q _c	gpm bbl/min m ³ /m	Flow rate at which the flow changes from laminar to turbulent.
DrillAhead Hydraulics	DAH	-	DrillAhead hydraulics is the Baroid hydraulics simulation. In this simulation all the fluid downhole rheological properties, drill pipe rotation and operational methods such as pump and rotate and sliding are used in drilling simulation to monitor cuttings transport in the wellbore. DFG provides excellent accuracy for wellbore pressure management.

Term	Symbol (s)	Unit(s)	Definition
Eccentricity	e	-	This dimensionless term refers to the position of a pipe inside another pipe or hole. In the oil field it usually refers to the position of the drillpipe in an annulus. When the drillpipe lies directly in the middle of the annulus, the drillpipe position is concentric and the eccentricity factor is 0.  As the drillpipe moves to one side of the annulus, the drillpipe becomes increasingly eccentric. If the sides of the drillpipe comes in contact with the wall of the annulus, the drillpipe is fully eccentric and the eccentricity factor is 1.0. In high-angle or horizontal wells, the drillpipe usually lies on the low side of the hole and its eccentricity factor is 1. If the drillpipe lies on the upper side of the hole, its eccentricity factor is -1. Drillpipe eccentricity can affect pressure drops in the annulus by reducing the frictional forces of fluid flow. A fully concentric drillpipe in an annulus has higher pressure drops than an eccentric one. In some disciplines like cementing eccentricity is called standoff. If Standoff = 1 then $e = 0$ and if standoff = 0 then $e = 1$. It is very important in cementing operations to make the casing as close to concentric as practical to minimize cement channeling to the widest gap and not fully filling the narrow gap.

Term	Symbol (s)	Unit(s)	Definition
Equivalent circulating density	ECD	lb/gal SG Kg/m ³	Equivalent Circulating Density, ECD is the pressure exerted downhole by the fluid(s), choke pressure, transported cuttings and the hydraulics losses in the annulus. It can be calculated at any vertical depth. It is a pressure expressed in terms of a fluid density that is required to get an equivalent pressure at a given depth. $ECD = \frac{\text{Hydrostatic Pressure} + \text{Hydraulic Losses}}{0.052 \times \text{True Vertical Depth, ft}}$ $ECD, \frac{\text{lb}}{\text{gal}} = \frac{\text{SG} \times 8.33 \frac{\text{lb}}{\text{gal}}}{0.052 \times \text{True Vertical Depth, ft}} + \frac{\text{Total Hydraulic Losses, psi}}{0.052 \times \text{True Vertical Depth, ft}}$ In TERM ONE of the above equations, hydrostatic pressure, DFG considers the following. • Fluid density at a reference temperature • OWR • Salinity of the water phase • Oil type • ASG of the solids • Compressibility • Thermal expansion • Thermal gradient, static or pumped • Heat transfer if pumped • TVD • Pit suction temp • Choke pressure In TERM TWO, Total Hydraulics losses, DFG considers the following: • All the above • Downhole rheology- (Herschel- Bulkley modeling) • Cuttings Diameter and SG • Pump rate • Booster pump rate • Cuttings transport • Operational procedures- Pump and rotate, rotary drilling sliding, % sliding and % rotary and connection times • Wellbore and tubular geometry, including tool joints • Drillstring RPM DFG is used to accurately calculate ECD. It has unparalleled accuracy in blind testing when compared to other programs used in the industry. Simulation differences in ECD are typically less than 0.1 lb/gal when compared to PWD. The key difference in DFG and other hydraulics/cuttings transport simulators is DFG simulates the transport of discrete cutting elements. It does not use correlations of poor, average or good etc. cuttings transport or provide any method to calibrate to PWD. Typically, DFG matches PWD data very well. When DFG is over or under predicting, some possible reasons are: • Hole erosion • PWD error in communication and calibration • Data input errors • Drillstring and hole geometry • Cutting size

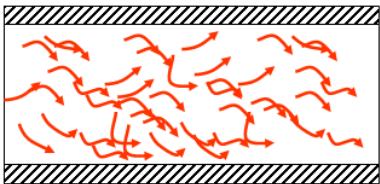
Term	Symbol (s)	Unit(s)	Definition
Equivalent mud weight Equivalent static density	EMW ESD	lb/gal SG Kg/m ³	Equivalent Mud Weight (ESD) is the pressure exerted by a static fluid column at bottom hole(or any TVD) expressed in terms of a fluid density that is required to get an equivalent pressure at a given vertical depth. The simple equation for EMW in lb/gal is: $EMW = 0.052 * Density * TVD$ Where: Density = Fluid density, lb/gal TVD = true Vertical depth, ft This equation is very simple and only useful in the simplest terms. It is not sufficient to take surface mud weight and simply put it into this equation and expect to get accurate downhole static pressures. The problem is fluid expansion due to formation temperature; and fluid compressibility due to pressure will change the density of the fluid downhole and thus the pressure it exerts. In DFG the fluid densities are modeled based on fluid composition, OWR, salt content, oil type, ASG, heat transfer from drilling and the formation thermal gradient. If this attention to detail is not considered, then calculation errors up to 0.5 lb/gal or more are possible. While drilling or circulation the formation gradient will change as well. After many hours the near wellbore formation temperatures will dramatically change. These changes will impact EMW and ECD. DFG can model the changes in downhole fluid temperature due to circulation and drilling. In some cases it can take several days for the near wellbore formation gradient to return to its natural state when circulation stops. Models are available in WellCat to simulate this behavior.
Friction factor	f	-	A dimensionless number used in fluid flow calculations. Refer to API bulletin for methods to use.
Effective viscosity	-	cP Pa sec	The viscosity used to describe fluid flowing through a particular geometry; as hole geometries change, so does the effective viscosity (viscosity = shear stress/shear rate). This is automatically taken care of in the various software models used in DFG and other hydraulics software packages.
Flow index	n	none	The numerical relation between a fluid's shear stress and shear rate on a log/log plot. This value describes a fluid's degree of shear-thinning behavior.
Flow regime	-	-	There are three types of flow regimes commonly dealt with in drilling. These are laminar, turbulent and transition. See definitions of each.
Fracture gradient	FG	-	The pressure that the formation can withstand without losing fluid or fracturing, usually expressed in EMW units at a given depth. Often FG is expressed in term of psi/ft. Knowing the TVD and the FG one can express the FG in EMW or pressure terms for a specific depth.
Gel strength	None	lb/100ft ² Pa	Time-dependent measurements of a fluid's shear stress under static conditions. Gel strengths are commonly measured after 10-second, 10- minute, and 30-minute intervals, but they can be measured for any desired length of time. It is important to manage the peak gel strength of all fluids used downhole. The gel structures can cause spikes in wellbore pressure during pumps-on and when tripping.

Term	Symbol (s)	Unit(s)	Definition
Hole Cleaning	-	-	The hole-cleaning model in DFG is a discrete element model found in the DAH section. Some generalizations of hole cleaning are below. To improve hole cleaning: • Increase pump rate • Increase fluid density • Increase pipe rotation speed • Manage cutting size through bit selection and weight on bit. As cutting size increases, transporting difficulty also increases. • Increase fluid viscosity, especially the low end rheology • Reduce ROP • Pump sweeps (higher density is preferred if wellbore pressures permit) Each of these has limitations and proper fluid design and engineering practices must always be used when considering all of these methods.
Hydraulic Horse Power	-	HHP	The horsepower consumed by pressure losses in the bit nozzles.
HSI	-	HHP/in ²	Hydraulic horsepower per square inch. This parameter is used in the DFG optimization as a lower boundary for “window of opportunity” in the DAH optimization.
Jet Impact Force	-	lb N	Force impacting the formation from the fluid flowing through the nozzles
Laminar Flow	-	-	Typical in annular sections and surface equipment  Laminar flow occurs at low-to-moderate shear rates when layers of fluid move past each other in an orderly fashion. In this example the parallel arrows are the streamlines. This motion is parallel to the walls of the channel through which the fluid is moving. Friction between the fluid and the channel walls is lowest for this type of flow. Rheological model parameters are important in calculating frictional pressure losses for muds in laminar flow. In simple terms, as model parameters such as K and PV increase so does the frictional pressure,
LCM – viscosity			Tool in the DFG WellSet program to predict the increase in viscosity of any fluid with the addition of LCM materials such as BARACARB, STEELSEAL, SWEEPWATE and BAROFIBER in any combination and concentration. The base fluid properties and FANN 35 rheology inputs are used with the LCM product additions to predict treated fluid rheological parameters. This tool is very helpful when combined with the sweep simulation in DFG to predict ECD with respect to pumped volume.
Local Mud Weight	LMW	lb/gal SG Kg/m ³	Actual fluid density changed from surface density by the temperature and pressure at some specific depth in the wellbore.

Term	Symbol (s)	Unit(s)	Definition
Model	-	-	A mathematical representation of the shear stress versus shear rate responses of a fluid. Typical models used in the field are the Power Law, Bingham and the Herschel-Bulkley. The Herschel-Bulkley is the preferred model for drilling fluid hydraulics. In DFG only the Herschel-Bulkley model is used for cuttings transport simulations because it is proven the best model for drilling fluids.
Newtonian	-	-	Newtonian fluids are materials like diesel fuel, water and glycerin. These fluids have a constant viscosity at a given temperature and pressure. In numerical terms; Shear stress = Viscosity * Shear rate
non-Newtonian	-	-	Non-Newtonian fluids are fluids like cross-linked gels, cements and most drilling fluids. If a fluid has gel strength, shear rate dependency, time dependency or yield stress then it is non-Newtonian.
Plastic viscosity	PV	cP Pa sec	PV is the viscosity term in the Bingham model. PV is calculated using shear stresses measured at 600 rpm and 300 rpm on the FANN 35 viscometer.
Pore pressure	PP	lb/gal SG Kg/m ³	The pressure of the formation fluids usually expressed in EMW
Pressure drop	-	psi Pa	Frictional forces develop when fluids flow through a pipe or an annulus. As a result, fluid energy dissipates. These frictional forces are referred to as pressure drops, and are usually referred to as a pressure per unit length. The longer a pipe or annulus, the greater is the pressure drop. Some factors that can affect the magnitude of pressure drop include: • Length • Flow rate (flow regime type laminar or turbulent) • Fluid rheological properties • Fluid density • Pipe eccentricity • Pipe/annulus geometry • Pipe roughness
Reference Temperature	-	F C	The temperature of the mud when the density is measured.
Reynolds Number	Re NRe	-	A dimensionless term that relates the inertial forces in a flowing fluid to the viscous forces. It is commonly used to determine whether a flowing fluid will be in laminar or turbulent flow. Generally for pipe flow, a Reynolds number greater than 2,100 will mark the onset of transitional to turbulent flow, but this is not always so because of many reasons. These include fluid elasticity and shear thinning or shear thickening of the fluid.
Rheogram	-	-	Graph of Shear Stress vs. Shear Rate
Rheology	-	-	The study and science of flowing matter.
Running Speed	-	ft/min ft/s m/min m/s	Velocity or speed of the pipe moving into or out of the wellbore.

Term	Symbol (s)	Unit(s)	Definition
Shear Rate	$\dot{\gamma}$ SR	sec-1 1/s	This term is appropriate for laminar flow only. In a flowing fluid, numerically it is the change in fluid velocity from one streamline to another divided by the distance between them.
Shear Stress	σ SS	lb/100ft ² Pa	Definition 1 The force per unit area required to shear a fluid at a given shear rate Definition 2 Shear stress is measured on oil field viscometers by the deflection of the FANN 35 meter's dial at a given shear speed. The rheometer dial reading is sometimes incorrectly referred to as shear stress. The reason it is incorrect is the dial reading is dependent on the torsion spring the rheometer is equipped with and requires a numerical factor to be converted to shear stress units. For example, on the standard R1, B1 rotor and bob configuration and a standard spring of a fann 35, the factor is 1.065 to convert the Dial reading to shear stress with units of lb/100ft² or 5.11 to convert to dynes/cm². DFG has rotor and bob configurations built into its engineering tool. Additionally, it will use any of the spring factors available from fann Instruments to calculate Herschel-Bulkley, Bingham or Power Law model parameters.
Shear Speed	rpm	rpm	The rotational speed for standard oilfield viscometers like the FANN 35 at which a dial reading is observed. The shear speed is not the same as shear rate though it is commonly misused that way. For example, 300 rpm on a FANN 35 is not a shear rate of 3001/s. Also, the dial reading on a standard FANN 35 viscometer is not a true shear stress since it must be converted to units like lb/100ft² or Pa. On a standard R1, B1 Fann 35 rheometer from Fann Instruments the Newtonian shear rates corresponding to the standard RPMs below are: 600 rpm = 1022 1/s 300 rpm= 511 1/s 200 rpm = 341 1/s 100 rpm =170 1/s 6 rpm = 10.2 1/s 3 rpm = 5.1 1/s For non-standard Fann 35 rpm values, multiply the rpm by 1.703 to obtain the 1/s value. It is important to know that these shear rates calculated for Newtonian fluids will not be the same for non-Newtonian fluids even though the instruments rpm and configuration are identical. The reason being that the shear thinning (or thickening) nature of these fluids changes the average shear rate calculated in the rheometer gap. DFG takes this into account and corrects for non-Newtonian effects in the viscometer when using the Herschel-Bulkley model. The API methods for calculating n, K and tau0 do not make this correction.
Shear thinning			Most drilling fluids are shear thinning. This means that the effective viscosity is lower at higher shear rates. In the Herschel-Bulkley and power law model the parameter, n, models the degree of shear thinning. If n=1 then the fluid is not shear thinning. As n becomes smaller, the fluid is more shear thinning. A typical drilling fluids range for n is 0.6 to 1 for either model.

Term	Symbol (s)	Unit(s)	Definition
Slip Velocity	Vs	ft/min ft/s m/min m/s	Slip velocity is relevant for vertical holes only when referring to cuttings transport. It is characterized as the difference in the average annular velocity of the fluid and the cuttings that are being transported by the fluid. It can be represented by the following equation. $V_{slip} = V_{fluid} - V_{cuttings}$ The Chien and Moore methods are typically used to calculate slip velocity. DFG will calculate both of these. However, DFG corrects some errors in the published assumptions of these models.
Streamline	-	-	The pathline a fluid volume element will move with respect to time.
Surge	-	lb/gal SG Kg/m ³	Frictional pressure exerted on the wellbore due to the drill string and BHA being run into the hole. DFG will correct for acceleration and deceleration of the drill string when performing these calculations. DFG will calculate ECD for surge and swab pressures at any point in the wellbore as well as bottom hole. DFG will also provide calculations of ECD at the bit.
Swab	-	lb/gal SG Kg/m ³	Friction pressure that causes the wellbore pressure to be lower when the BHA and drill string is removed from the hole.
TFA	-	in ² cm ²	Total Flow Area of the drill bit nozzles. This is simply the sum of the cross-section area for each nozzle.
Thermal Expansion Coefficient Or Coefficient of thermal expansion	α_v	-	Thermal expansion coefficient is a parameter represents the relative volume change in a fluid (or solid) in response to a temperature change. Its basic form is: $\alpha_v = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right) \text{ or } \Delta V = \alpha V \Delta T$ Where: V = Volume T = Temperature DFG will calculate the thermal expansion coefficient for whole muds. In this simulation mud composition is used.
Transitional flow	-	-	Typical in the drill pipe, collars and downhole tools while circulating and drilling Transitional flow occurs when the flow shifts from laminar flow to turbulent flow or vice versa. The critical velocity of a fluid is the particular velocity at which the flow changes from laminar to turbulent or vice versa.

Term	Symbol (s)	Unit(s)	Definition
Turbulent Flow	-	-	Typical in the drill pipe, collars and downhole tools while circulating and drilling  Turbulent flow occurs at high shear rates (high pump rates) where the fluid moves in a chaotic fashion. Turbulent flow is characterized by random loops and current eddies. Friction between the fluid and the channel walls is highest for this type of flow. Fluid rheological parameters are not significant in calculating frictional pressure losses for fluids in turbulent flow. Generally, turbulent flow is avoided in annular open-hole sections to minimize erosion of the formation. Erosion of the formation can cause a number of problems such as: • Lowering cuttings transport efficiency because of lower annular velocities • Causing hole stability problems • Drilling fluid viscosity increase because of the drill solids added to the fluid
Viscosity		cP Pa sec	In everyday terms viscosity is the thickness of a material or resistance to flow. Common units of measure are centipoise, cP and Pascal seconds. Fluid viscosity can be measured over a wide range of shear rates. In the Herschel-Bulkley and power law models the parameter, K, is analogous to viscosity and in the Bingham model, PV.
True Vertical Depth	TVD	ft m	Vertical depth of some point in the wellbore. 
Yield point	YP	lb/100 ft ² Pa	The force required to initiate flow; the calculated value of the fluid's shear stress when the rheogram is extrapolated to the y-axis at shear rate =0 sec-1. Typically YP is one parameter of the Bingham model and it is usually calculated from the 600 rpm and 300 rpm dial readings. Current API guidelines require the calculation of YP and PV using the following equations: PV = 600 dial – 300 dial YP = 300 dial – PV, or YP = (2 x 300 dial) – 600 dial
Yield stress	Tau0	lb/100ft ² Pa	The force required to initiate flow; the calculated value of the fluid's shear stress when the rheogram is extrapolated to the y-axis at shear rate = 0 sec-1. Yield stress is a time-independent measurement and is usually denoted in the Heshchel-Bulkley (yield-power law [YPL]) model as Tau0 and in the Bingham model as YP. It can also be considered gel strength before any time dependent changes in properties are observed

1.2. Rheological Models

Rheological models help predict fluid behavior across a wide range of shear rates. Most drilling fluids are non-Newtonian, pseudoplastic fluids. The most important rheological models that pertain to drilling fluids are as follows:

- Bingham model
- Power law model
- Herschel-Bulkley (yield-power law [YPL]) model

Figure 1 depicts typical rheological profiles for Newtonian, Bingham-plastic fluids, power law fluids, and Newtonian fluids.

The Herschel-Bulkley (yield-power law [YPL]) model is the most accurate model for predicting the rheological behavior of common drilling fluids.

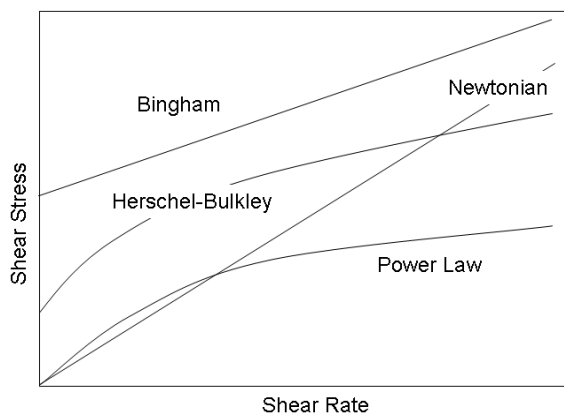


Figure 1 Typical Rheological Profiles for Newtonian, Bingham-Plastic Fluids, Power Law Fluids, and Newtonian Fluids

Bingham Model

The Bingham model describes laminar flow using the following equation:

$$SS = YP + (PV \times SR)$$

Where:

SS = the measured shear stress, lb/100 ft²

YP = the yield point, lb/100 ft²

PV = the plastic viscosity, cP

SR = is the shear rate, sec⁻¹

Because the model assumes true plastic behavior, the flow index of a fluid fitting this model must have $n = 1$.

Unfortunately, this does not often occur and the model usually over-predicts yield stresses (shear stresses at zero shear rate) by 40 to 90 percent. A quick and easy method to calculate more realistic yield stresses is to assume the

fluid exhibits true plastic behavior in the low shear-rate range only. A low shear- rate yield point (LSR YP) can be calculated using the following equation:

$$\text{LSR YP} = (2 \times 3 \text{ rpm dial}) - 6 \text{ rpm dial}$$

This calculation produces a yield-stress value close to that produced by other, more complex models and can be used when the required computer algorithm is not available.

Power Law Model

The power law model describes fluid rheological behavior using the following equation:

$$SS = K \times SR^n$$

This model describes the rheological behavior of polymer-based drilling fluids that do not exhibit yield stress (i.e., viscosified clear brines). Some fluids viscosified with biopolymers can also be described by power-law behavior.

The Power Law Model can produce widely differing values of n and K . The results depend on the shear-stress/shear-rate data pairs used in the calculations.

Herschel-Bulkley Model

(Yield-Power Law [YPL])

Because most drilling fluids exhibit yield stress, the Herschel-Bulkley (yield-power law [YPL]) model describes the rheological behavior of drilling muds more accurately than the Bingham and Power law models.

The YPL model uses the following equation to describe fluid behavior:

$$SS = \tau_0 + (K \times SR^n)$$

Where:

SS = the measured shear stress, lb/100 ft²

τ_0 = the Herschel-Bulkley yield point, lb/100 ft²

K = Consistency index, lb/100ft²

SR = is the shear rate, sec⁻¹

n = Flow index, no units

K and n values in the YPL model are calculated differently than their counterparts in the power law model. The YPL model reduces to the Bingham model when $n = 1$ and it reduces to the power law model when $\tau_0 = 0$. An obvious advantage the YPL model has over the power law model is that, from a set of data input, only one value for n and K are calculated.

The YPL model requires:

- A computer algorithm such as DFG etc. to obtain solutions.
- A minimum of three shear-stress/shear-rate measurements are required for solution.
- Model accuracy is improved with additional data input.

Deriving Dial Readings

The 600 and 300 rpm readings are back-calculated from the plastic viscosity and yield-point values as shown:

- 300 = Plastic viscosity + yield point
- 600 = Yield point + 300
- 3 = 10-second gel (using a hand-crank viscometer)
- 3 = 3 (using a FANN 6-speed viscometer)

Field Tests

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1. Field Tests

1.1. Overview

The procedures for field testing of water-based drilling fluids, oil- or synthetic-based drilling fluids, and completion / workover fluids are listed below.

All field labs follow API procedures as defined in the API Recommended Practices when running lab tests or calibrating lab equipment. Baroid global laboratory work methods are followed in cases where API Recommended Practices do not address a specific procedure or method.

Copyright infringement issues do not allow Baroid to distribute the API documents directly to employees. Each Baroid employee has access to download the API Recommended Practices and print copies for his / her own use. Employees must not print copies of the Recommended Practices for others to use. Employees can print additional copies if needed for their own personal use.

API Standards

The API Recommended Practices are located in the Baroid HMS Lab Document site in Halworld. The pathway to them is as follows:

HALWORLD > PSLs > Global HMS > Support Functions > Lab > Lab Documents > Work Methods

[WM-GL-HAL-BAR-LAB-710](#)

Section 2.2 in the above link provides instructions for accessing API standards electronically. Please follow those instructions to download and print your copy of the Recommended Practices below. You may print as many as needed for your own personal use, but do not distribute to anyone else. Others must download and print their own copies.

- **API Recommended Practices 13B-1** – Recommended Practice for Field Testing Water-based Drilling Fluids
- **API Recommended Practices 13B-2** – Recommended Practice of Field Testing of Oil-based Drilling Fluids
- **API Recommended Practices 13J** – Testing of Heavy Brines

Baroid Global Laboratory Work Methods

The tests and the Halliburton Management System (HMS) reference codes are shown below. The pathway to them is as follows:

HALWORLD > PSLs > Global HMS > Support Functions > Lab > Lab Documents > Lab Test Procedures

Brookfield Viscometer Gel Strengths	WM-GL-HAL-BAR-LAB-TES-001
Chillers Accompanying FANN 35 / 75	WM-GL-HAL-BAR-LAB-TES-002
Compatibility Analysis of Completion Brine and Crude Oil	WM-GL-HAL-BAR-LAB-TES-003
Capillary Suction Time	WM-GL-HAL-BAR-LAB-TES-004
LE SUPERMUL Content in Mud	WM-GL-HAL-BAR-LAB-TES-006
Polyacrylamide Additive Content Using HPK	WM-GL-HAL-BAR-LAB-TES-007
FANN 50 Viscometer	WM-GL-HAL-BAR-LAB-TES-008
FANN 75 Viscometer	WM-GL-HAL-BAR-LAB-TES-009
FANN 90 Viscometer	WM-GL-HAL-BAR-LAB-TES-010
HTHP Corrosion Test	WM-GL-HAL-BAR-LAB-TES-012

1.2. Water-Based Drilling Fluid Test Procedures

API Recommended Practices 13B-1

Drilling Fluid Density (Mud Balance)	Methylene Blue Capacity	Potassium (Concentration > 5000 mg/l)
Alternate Drilling Fluid Density (Pressurized Mud Balance)	pH	Potassium (Concentration < 5000 mg/l)
Viscosity and Gel Strength	Alkalinity and Lime Content	Shear Strength Using Shearometer Tube
Marsh Funnel Viscosity	Chloride Ion Content	Removal of Air or Gas from Fluid Prior to Testing
Direct-Indicating Viscometer Rheological Properties	Total Hardness as Calcium	Drill Pipe Corrosion Ring Coupon
Low Temperature / Low Pressure Filtration	Calcium	HPHT Filtration Using a Permeability Plugging Apparatus with End Caps with Set Screws
High Temperature / High Pressure Filtration	Magnesium	HPHT Filtration Using a Permeability Plugging Apparatus with Threaded End Caps
Water, Oil, and Solids Contents (Retort)	Calcium Sulfate	Resistivity
Sand Content	Sulfide (Garrett Gas Train)	Carbonate (Garrett Gas Train)

Baroid Tests (Global Laboratory Work Methods)

The tests and the Halliburton Management System (HMS) reference codes are shown below. The pathway to them is as follows:

HALWORLD > PSLs > Global HMS > Support Functions > Lab > Lab Documents > Lab Test Procedures

ALDACIDE G Content	WM-GL-HAL-BAR-LAB-TES-018
Bacterial Presence in Aqueous Drilling Fluids	WM-GL-HAL-BAR-LAB-TES-024
BARACOR 95 Content	WM-GL-HAL-BAR-LAB-TES-031
OXYGON Content	WM-GL-HAL-BAR-LAB-TES-027
Polyacrylamide Content Using HPK	WM-GL-HAL-BAR-LAB-TES-007
Polyglycol Content Using Refractometer	WM-GL-HAL-BAR-LAB-TES-029
Silicate Content	WM-GL-HAL-BAR-LAB-TES-030

1.3. Oil- or Synthetic-Based Drilling Fluid Test Procedures

API Recommended Practices 13B-2

Drilling Fluid Density (Mud Balance)	Base Alkalinity Demand	Water, Oil, and Solids Contents (Retort)
Alternate Drilling Fluid Density (Pressurized Mud Balance)	pH	Lime, Salinity and Solids Calculations
Viscosity and Gel Strength	Alkalinity and Lime Content	Shear Strength Using Shearometer Tube
Marsh Funnel Viscosity	Chloride Ion Content	Electrical Stability
Direct-Indicating Viscometer Rheological Properties	Whole Drilling Fluid Alkalinity	Sulfide (Garrett Gas Train)
High Temperature / High Pressure Filtration (up to 350°F)	Whole Drilling Fluid Chloride	Aqueous Phase Activity using an Electro-hygrometer
High Temperature / High Pressure Filtration (350°F to 450°F)	Whole Drilling Fluid Calcium	HPHT Filtration Using a Permeability Plugging Apparatus with End Caps with Set Screws
Oil and Water Content of Cuttings	Aniline Point	HPHT Filtration Using a Permeability Plugging Apparatus with Threaded End Caps
Cuttings Activity (Chenevert Method)		

Baroid Tests (Global Laboratory Work Methods)

The tests and the Halliburton Management System (HMS) reference codes are shown below. The pathway to them is as follows:

HALWORLD > PSLs > Global HMS > Support Functions > Lab > Lab Documents > Lab Test Procedures

LE SUPERMUL Content by HPK	WM-GL-HAL-BAR-LAB-TES-006
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1.4. Completion / Workover Fluid Test Procedures

API Recommended Practices 13J

Density by Hydrometer	Crystallization Temperature	Brine Clarity
Iron Contamination	pH	Solids Evaluation by Gravimetric Procedures

Baroid Tests (Global Laboratory Work Methods)

The tests and the Halliburton Management System (HMS) reference codes are shown below. The pathway to them is as follows:

HALWORLD > PSLs > Global HMS > Support Functions > Lab > Lab Documents > Lab Test Procedures

Crystallization Point of Brines	WM-GL-HAL-BAR-LAB-TES-016
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Specialized Testing

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1. Overview

The Baroid global laboratories have the equipment and staff needed to conduct all standard and virtually all non-standard testing on drilling, drill-in, and completion fluids. The Baroid regional laboratories are capable of conducting all standard tests and many specialized tests. Baroid country and area labs are equipped to support the technologies being implemented in their country or area.

All test equipment that performs measurements is subject to programmed calibration and maintenance in accordance with documented procedures.

Technical service laboratory project requests are entered and the project status is tracked. Lab test data is logged and lab reports are generated using the “Viking” global database.

1.1. Technical Services: Drilling Fluids

The availability of equipment is clearly important. Lab personnel are trained to correctly utilize this equipment and properly interpret the data output, allowing Baroid to deliver tangible fluid performance improvements to our customers. The Baroid labs have a well-established track record for producing customized, state-of-the-art fluid solutions, solving complex drilling problems, and helping operators reduce costs.

Specialized Equipment for Drilling Fluid Testing

FANN® 50 High Temperature Viscometer	Used to evaluate rheological properties up to 500°F (260°C) and 700 psig to determine the temperature stability of a drilling fluid. When the viscosity of the drilling fluid increases or decreases after heating and cooling cycles, the test results can indicate temperature instability.
FANN 70 and 75 High Pressure High Temperature Viscometers	Both instruments are concentric cylinder viscometers capable of providing standard oilfield rheology data on fluids subjected to 20,000 psig and 500°F (260°C). The FANN 75 viscometer can also be used sub ambient (to 41°F/ 5°C) to simulate low fluid temperatures encountered within deepwater risers. FANN 70 and FANN 75 rheometers are used extensively during the planning and drilling of HPHT wells to measure rheology under field conditions. Measurement of fluid rheology under downhole conditions is critical to management of equivalent circulating density (ECD) and must always be considered in conjunction with any measured change in sag performance.
FANN® i77 HPHT Viscometer	Operates at temperatures up to 600°F (315°C) and pressures up to 30,000 psig to allow rheological property measurements on fluids designed for extremely hot, deep wells. The instrument has an embedded electronics control module, data acquisition and control software, and pressure, temperature, and speed controllers.
Permeability Plugging Apparatus (PPA)	Permits fluid loss measurement using ceramic discs available in a variety of permeabilities (5 micron to 190 micron) to simulate reservoir pore throat diameters. Filter cake is built on the underside of the ceramic disc. This orientation eliminates the effects of settlement during formation of the filter cake. Overbalances to 2500 psig can be reproduced and the cell can be heated to 500°F (260°C). PPA is used extensively for optimization of pore throat bridging formulations using BARACARB® bridging agent (sized marble). The continued ability of field muds to provide suitable bridging is typically evaluated using a combination of PPA testing and particle size analysis.
FANN 90 Dynamic Filtration	FANN 90 dynamic filtration testing builds on the capabilities of the PPA in that it utilizes ceramic cores available in a range of different permeabilities. The FANN 90 dynamic filtration test differs from PPA in three important respects: • Filter cake is built on the inner surface of a vertically oriented, cylindrical ceramic core to more accurately replicate the wellbore. • A motor-driven rod inserted through the center of the core simulates the action of drill

	string rotation and annular fluid velocity on filter cake deposition and attrition. Filtrate volume can be measured versus time. Simulates filtration properties downhole and implements unique Baroid filtration models to determine cake deposition index (CDI) and dynamic filtration rate to provide solutions preventing differentially stuck pipe.
Dynamic High Angle Sag Testing (DHATM) Device	The Baroid dynamic sag testing device operates under variable temperature and pressure. Conventional static sag testing does not yield clear answers since sag is highly impacted by temperature, pressure, and low shear rate. The device requires a 40 ml mud sample and is used for pilot testing improvements to correct dynamic sag problems.
Shale Recovery and Shale Erosion Tests	Both tests are very similar but differ only in the amount and sizes of shale particles used. A known weight of dry sized shale is hot rolled in the test fluid (mud or brine) for 16 hours. The shale/brine mixture is then passed through the sieve used to size the original particles. The shale retained on the sieve is washed, dried, and weighed. This recovered weight is expressed as a percentage of the original weight. The greater the inhibition qualities of the mud or brine, the higher the shale return weight will be.
Linear Swell Meter (LSM)	Measures dimensional changes of constrained shale pellets exposed to candidate fluids. Measurement is effected by means of a Linear Displacement Transducer probe maintained in contact with the upper face of the shale pellet. Testing may be conducted at ambient or elevated temperature. Results are recorded as plots of swelling or contraction versus time. The LSM test provides a graphical comparison of up to four inhibitive fluids simultaneously. However, differences between fluids are less apparent than would be expected from shale recovery tests. Ideally comparisons using this technique would involve sections of representative shale cut such that bedding planes lie perpendicular to the direction of measurement. Rarely is it possible to obtain such samples and hence most linear swell testing is conducted using compressed shale pellets formed from powdered shale.
Capillary Suction Time (CST)	The CST instrument measures the water retained by shale/brine slurries. Water retained by the shale will result in shale swelling and loss of mechanical properties. Water retention is measured as the time taken for 'free' water from the slurry to travel radially between two electrodes on thick, porous filter paper. CST testing is used principally to validate increases in brine salinity and cation selection.
Slake Durability	Samples of sized test shale are placed into mesh-covered cylindrical cages. The cages are then rotated at a constant 20 rpm while immersed in the drilling fluid. Tests are typically run for four hours at room temperature. However, longer runs at elevated temperature can be conducted where appropriate. The weight of shale recovered as a percentage of the original weight enables the inhibitive qualities of the drilling fluid to be compared. Results obtained using the slake durability test generally follow the same trends as those obtained from shale recovery testing. However, shale samples that are particularly susceptible to mechanical damage will give lower recoveries in this test than those in shale recovery tests. Hence, data from both test methods provide an insight on the effects of candidate fluids on shale hydration/dispersion and attrition.
Filter Cake Removal Pressure Apparatus	This device is essentially a flow loop incorporating a pump, pressure transducer, double-ended cell and valve arrangements. The valves permit control of flow in either direction through the double-ended cell. The cell can accommodate a variety of 'filter' media including gravel pack screens and ceramic discs of the type used in the PPA test described above. Equipment is used to optimize fluids for gravel packing and minimize filter cake 'pop-off' pressures.
FANN Lubricity Meter	Measures reduction in metal-to-metal friction. A constant force is applied to a contoured metal test block. The applied force presses the test block against a rotating metal ring. Both metal components are immersed in the test fluid. The motor torque required to maintain rotation of the test ring is measured and used in conjunction with the metal-to-metal contact area to calculate a "lubricity coefficient". Water-based mud lubricants are evaluated by measurement of lubricity coefficient reduction following addition of the lubricant to the drilling fluid.

	The relative performance of lubricants is often dependent upon the fluid type with factors such as solids loading and pH having a marked effect on the performance of certain lubricants. Lubricity coefficient is typically found to reduce with increasing lubricant concentration. However, it is usual to find one concentration at above which the addition of further lubricant is no longer cost effective. Hence, use of the lubricity meter can help determine the optimal lubricant and lubricant concentration for a particular application.
Particle Size Analyzer	Malvern, Microtrac or Coulter laser diffraction particle size analyzers are used in a majority of the laboratory locations. The analyzers measure the distribution of the sizes of particles in a fluid or powder. The results are presented in a table and graph. The table lists the amount of particles classified by size (microns). The graph shows the concentration in percent by volume of solids in a particular range. A useful value determined by the instruments is the D50, which is the median size of the particles in the sample.

1.2. Technical Services: Completion Fluids

Completion fluids services encompass a wide range of testing capabilities, including but not limited to formation damage assessment resulting from exposure to completion and drill-in fluids, shale inhibition properties, determining effects of filter cake breakers, and drill pipe, casing, and tubing corrosion prevention. Proven processes are established for brine evaluation, treatment recommendations for fluid reconditioning, and preventing possible permeability impairment by contaminants.

Specialized Equipment for Completion Fluids Testing

Variable Pressure Chrystallometer	Provides the ability to test crystallization points of brines under elevated pressures for deepwater applications.
Automated Return Permeameter (ARP)	Core samples are tested with the ARP to identify the least damaging drilling/completion fluid for a particular operation. The samples can be tested at elevated temperatures and pressures and under dynamic and actual downhole conditions. The ARP provides the ability to program remedial steps, such as cleanup with acids and oxidizers.
Manual Return Permeameter	Similar in use to the ARP; however test fluid exposure is static rather than dynamic. The operation of the permeameter is mostly manual rather than automatic. Very useful for determining damage with solids-free fluids such as displacement pills or completion fluids.
Screen Tester	Used to evaluate sealing capabilities of inside screen pills on screen coupons, with or without gravel packs, under variable temperatures and differential pressures. Also used as a screen flow-through device to ensure the fluid will pass through the production screen without plugging or hindering flow, such as when running screens in mud before displacement to a completion fluid or gravel pack fluid.

1.3. Technical Support: Analytical

The Baroid Analytical Laboratories in Houston and Pune, India offer a broad spectrum of chemical and material characterization capabilities. Analyses ranging from bulk properties down to ultra-trace elemental quantification can be performed in-house via various types of instrumental and wet chemical techniques.

The analytical group provides direct support to every technically-oriented function within Baroid and provides the data required to help customize fluid formulations or to identify non-compliant materials.

In addition to the capabilities shown in the table below, the Baroid Analytical Laboratory can perform many classical gravimetric and volumetric “wet” chemical analyses utilizing modern automated instrumentation, Kjeldahl Nitrogen analysis, and aniline point determination.

Analytical Instrumentation

		Houston	Pune
X-Ray Diffractometer	Determines the mineral composition of cores, cuttings, and ores, identification of scales, corrosion by-products, and detection impurities in products.	✓	✓
X-Ray Fluorescence Spectrometer	Used to determine the elemental components in barites, clays, brines, non-crystalline materials, scales, and corrosion by-products and ore assays.	✓	
Scanning Electron Microscope (SEM) with Energy Dispersive Spectroscopy	Provides measurement of pore sizes in cores, identifies the location of clays in cores, determines causes of metal failure, characterizes particle sizes and shapes, and identifies corrosion.	✓	
Gas Chromatography with Mass Selective Detector	Used for the determination of crude oil contamination in synthetic muds, base oil “fingerprinting” identification of volatile organic components of products and drilling fluids.	✓	
Pyrolysis/Gas Chromatography with Mass Selective Detector	Used for the molecular characterization of non-volatile organic materials such as fluid-loss polymers, and surface-adsorbed coating agents.	✓	
Infrared Spectrometer	This test is performed to identify polymer, surfactant, and emulsifier content, and to determine sludge composition.	✓	✓
Infrared Microscope	Used to identify organic coatings on solid surfaces and evaluation of corrosion inhibitor coverage, surfactant coatings, and polymer homogeneity.	✓	
Inductively Coupled Argon Plasma Spectrometer	Used to determine the presence of heavy metals in barites, clays, and soil samples, and to identify trace elements in brines, effluents, mud filtrates, acid leachates, and production discharges.	✓	✓
Ion Chromatograph	Used to determine cations and anions in brines, makeup waters, effluents, and discharges, and ion composition in water leachates from solid products, soils, and ores.	✓	✓
Laser Diffraction Particle Size Analyzer	Provides grind size analysis of barite, limestone, and hematite, and determination of particle size distribution in drilling fluids and brines.	✓	✓
High Performance Liquid and Gel Permeation Chromatograph	This specialized chromatograph identifies and quantifies nonvolatile organic components such as surfactants, emulsifiers, rheology modifiers, and filtration control agents.	✓	
Optical Microscope	Helps determine the size and shape of sands and ground products and helps with micro-fracture identification.	✓	✓
Digital Imaging Microscope	Provides 3-dimensional surface mapping and measurement to enhance characterization of small particles and materials failure-analysis.	✓	
Thermo-Gravimetric Analyzer	Determines sample weight loss with increase in temperature, moisture content on small sample volumes, and distillation ranges of base oils.	✓	
Differential Scanning Calorimeter	Helps determine exothermic and endothermic reactions of samples with increase in temperature and characterization of polymers and clays.	✓	✓
Flash and Fire Point Tester	Used to determine the flash point of base oils, diesel oils, crude oils, oil-base mud, products, and solvents.	✓	✓
Mercury Analyzer	Determines the mercury content in weight materials, clays, reserve pit water, and waste water.	✓	
High-Resolution Densitometer	Used to measure the density of brines, base oils, and liquid products with a very high level of resolution to evaluate contamination, decomposition, evaporation, or alteration.	✓	

1.4. Technical Support: Bioassay

At Baroid, we strive to conduct business without affecting the environment in which we work. We comply with environmental rules and regulations, and provide our customers with products and services that help them do the same.

The Baroid Bioassay Laboratory Group performs aquatic toxicity tests on water, oil, and synthetic-based drilling fluids, stock base fluids, products, brines, effluent discharges and cuttings from land or offshore. Sheen tests are performed on drilling and completion fluids, brines and some products. Biodegradation tests are performed on existing products and research product for North Sea or other International applications. Specific EPA, ASTM, COE, and OECD test protocols are followed to meet state, federal and International discharge monitoring regulations.

The Baroid Bioassay Laboratory's Quality Assurance Program guarantees that the accuracy and precision of reported results from the lab have been thoroughly monitored and exceed minimum reliability requirements for the appropriate protocols. The Program meets NELAC and ISO9001:2008 certification criteria. The lab is also Good Laboratory Practices (GLP) qualified to submit test results acceptable to the EPA, North Sea and other International regulatory agencies.

Aquatic Organisms Cultured and Tested in the Lab

Mysid Shrimp	Primarily to determine aquatic toxicity of both Water-Based and Synthetic-Based drilling fluids and product components for use in the USA NPDES regulated waters.
Leptocheirus Amphipod	To determine sediment toxicity of base synthetic fluids, synthetic drilling fluids and product components for use in the USA NPDES regulated waters.
Sheepshead Minnow Skeletonema Algae Acartia Copepods	For product/ component toxicity tests as required by North Sea and other International regulatory agencies including the EPA, specifying Good Laboratory Practices (GLP) methods.
Daphnia (Freshwater Crustacean)	To determine freshwater aquatic toxicity of product components, inland drilling fluids and effluent discharge to freshwater areas.
Fathead Minnow	To determine freshwater aquatic toxicity of product components, drilling fluids, land-based cutting and effluent discharge to freshwater areas.

Bioassay, Biodegradation, and Sheen Tests Specified by State, Federal or International Regulations

48hr Rangefinder and Definitive Acute Toxicity Tests	For international regulatory agencies including the EPA, using many of the cultured species.
96hr Rangefinder and Definitive Acute Toxicity Tests	For drilling fluids or component products for International regulatory agencies including the EPA, using many of the cultured species.
96hr Leptocheirus Sediment Toxicity Tests	On synthetic-based drilling fluids for use in the USA NPDES regulated waters.
10day Leptocheirus Sediment Toxicity Tests	On base synthetic fluids for use in the USA NPDES regulated waters
28day OECD 306 and BODIS Seawater Aerobic Biodegradation Tests	On drilling fluids, product and components for North Sea and other international regulatory agencies.
Static Sheen Tests	To indicate the presence of free oil for the use in the USA NPDES regulated waters.

Water-Based Fluids

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1. High-Performance Water-Based Fluids

Overview

For many decades, oil and gas producers have relied on invert emulsion oil- and synthetic-based drilling fluids as key contributors to trouble-free drilling of high-quality wellbores.

Despite the excellent track record demonstrated by invert emulsion fluids, operators continue searching for a water-based system that will perform like an invert fluid with regard to wellbore stability, fast rate of penetration (ROP), high tolerance to contaminants, effective inhibition, and excellent lubricity.

Environmental regulations are increasingly stringent and discharge of oil- contaminated drilling waste has been prohibited in environmentally sensitive areas, thus making a water-based alternative attractive.

By definition, a high performance water-based system emulates the performance of an invert emulsion fluid while eliminating most, if not all, of the risk and cost associated with managing wastes generated while drilling with invert emulsion systems.

Baroid's High-performance Water-based Fluids (HPWBF)

A brief description of each HPWBF is shown below. Detailed discussion is included later in this section.

These are low-solids non-dispersed (LSND) fluids that exhibit the essential performance characteristics for emulating invert emulsion fluids.

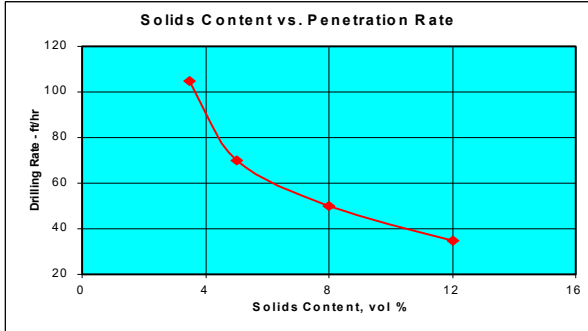
HYDRO-GUARD®	Formulated with brine solutions with a concentration of 10% NaCl or higher. Freshwater formulations have also been successful in applications where the shales are less reactive. However, the specific clay mineralogy should be investigated and the proposed HYDRO- GUARD formulation should be tested in the lab before running the system with fresh water.
PerformaDril®	Formulated with fresh water or monovalent brine made with KCl and/or NaCl in seawater or drill water. Designed to provide maximum shale stabilization in highly reactive clays.
BOREMAX®	Formulated with fresh water. Designed to help maximize ROP and wellbore stability while reducing dilution requirements, disposal costs, and environmental concerns.
SHALEDRIL® Fluids	SHALEDRIL F&B: Formulated with fresh water. Designed to be run with potassium based products. Can be used on land rigs that are capable of fresh water dilution but do not have a tank for whole mud dilution. SHALEDRIL H: Environmentally friendly, fresh-water based fluid designed to combat the extreme temperatures of the Haynesville shale.

Characteristics

A true high-performance fluid fulfills all, not just some, of the four requirements listed below. These four characteristics work together to effect maximum drilling performance.

Table 1 Characteristics of HPWBF

Non-dispersed system	The use of dispersants sets up a “tail-chasing” scenario: drill solids are dispersed by adding chemicals, leading to the generation of ultra-fine solids, leading to an undesirable increase in rheological properties, leading to more additions of chemical dispersants and water. Solids removal efficiency—absolutely critical for achieving a fast ROP—drops drastically as the colloidal-size solids build up in the system. The strong inhibitive, flocculating, and highly shear-thinning nature of the HPWBFs can eliminate the need for dispersants and puts an end to the cycle of solids
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	contamination.										
Low colloidal solids content	Studies show that the lower the colloidal solids content in a water-based system, the faster the ROP.  <table border="1"> <caption>Solids Content vs. Penetration Rate Data</caption> <thead> <tr> <th>Solids Content (vol %)</th> <th>Drilling Rate (ft/hr)</th> </tr> </thead> <tbody> <tr> <td>4</td> <td>100</td> </tr> <tr> <td>6</td> <td>70</td> </tr> <tr> <td>8</td> <td>50</td> </tr> <tr> <td>12</td> <td>35</td> </tr> </tbody> </table> Minimizing colloidal solids helps lower the plastic viscosity of the fluid, contributing to greater horsepower at the bit. However, removing colloidal solids becomes difficult, if not impossible, if these solids are allowed to accumulate and further degrade in the active system. A true HPWBF should chemically flocculate and encapsulate these particles so that the solids control equipment can strip out the particles at the surface. CLAY GRABBER polymeric encapsulator and flocculant, a field-friendly liquid additive based on molecular modeling, helps prevent the dispersion and disintegration that affects drill cuttings. CLAY GRABBER flocculant encapsulates and flocs colloidal solids to help ensure they reach the surface while still large enough to be removed by conventional solids control equipment.	Solids Content (vol %)	Drilling Rate (ft/hr)	4	100	6	70	8	50	12	35
Solids Content (vol %)	Drilling Rate (ft/hr)										
4	100										
6	70										
8	50										
12	35										
Effective inhibition	An HPWBF should inhibit the reactive clay and facilitate removal of drilled solids over the duration of the operation. A WBF that is designed only to inhibit formation clays may not go the extra step of flocculating and encapsulating the ultra-fine solids that cause most of the slow penetration rates associated with WBFs. Baroid designed its HPWBFs to deliver a gauge hole and help form a barrier that protects the shale matrix from water invasion. Our well-established mineralogy reference is based on extensive testing with actual core samples that provide guidance in the design, formulation, and application of each HPWBF. Each system is designed for the expected formations, resulting in minimal hole erosion or washout. Drilling a gauge hole promotes better quality logging data and can help improve the quality of the cement job. In turn, a good cement job contributes to successful leak-off and formation integrity tests.										
Shear-thinning behavior	Shear-thinning behavior is a key factor in drilling performance. An ideal drilling fluid will become thinner with increased shear. Baroid's HPWBFs have zero or very low bentonite content. The HPWBFs become thin at the bit, maximizing hydraulic horsepower, and then thicken in the annulus to provide good hole cleaning and the suspension properties necessary for mud weights up to 17.5 ppg, at temperatures up to 375°F.										

1.1. HYDRO-GUARD® High-Performance Water-Based Fluid

The HYDRO-GUARD system is one of Halliburton's latest inhibitive water-based drilling fluids. Its design was based upon proven field technology, extensive testing and enhanced product engineering. The versatility and clay control exhibited by this high performance system has made it a credible potential replacement for invert emulsion fluids worldwide.

Classification

HYDRO-GUARD water-based, clay free drilling fluid is a non-dispersed, inhibited system, designed to provide maximum shale stabilization in highly reactive clays such as those found in the Gulf of Mexico (GoM). Non-dispersed – Inhibited fluids utilize ions such as chloride (Cl^-), sodium (Na^+) and/or potassium (K^+) in the continuous phase to suppress clay hydration via ionic replacement and by decreasing the activity of the fluid/formation water exchange.

Chemical thinners or dispersants are not used in HYDROGUARD. Instead, polymeric flocculants and encapsulators are employed along with inhibiting amines to prevent clay from dispersing in the system. This helps prevent the breaking up of drill solids into smaller particles and helps improve the efficiency of the solids control equipment (SCE).

Applications

The demand for high performance water-based drilling fluid has greatly increased over the past few years due to the constraints forced by rig capabilities, logistics, well economics and environmental regulations. With increasing frequency the role of water-based fluids has been to directly replace invert emulsion fluids without a sacrifice in the drilling performance. While traditional water-based muds (WBMs) have too often struggled to achieve this, the HYDRO-GUARD system has consistently proven to have a high degree of clay control, wellbore stability, performance based rates of penetration (ROP), low coefficients of friction (CoF) and rheological control over a wide range of temperatures (40 – 300°F). Furthermore, with such achievement in the systems versatility there is added benefit of unrestricted cutting discharge based on most worldwide WBM environmental regulations.

The advantages gained by use of the HYDRO-GUARD system make this an excellent choice for tackling the following drilling challenges:

- Gumbo and reactive shales
- Dispersive clays
- Permeable sands
- High temperature wells
- Directional wells
- Extended reach wells
- Deepwater wells
- Evaporite sequences
- Reservoirs
- Slim holes

The use of a low colloidal polymer inhibitive system is a sound engineering approach to provide borehole stability, high ROPs, minimized formation damage, and lower overall well costs.

Key factors in the design of such a fluid include the following:

- Inhibitive ions
- Polymeric inhibitors
- Polymeric encapsulators
- Maintenance of low solids content
- Low pH to discourage clay dispersion

The HYDRO-GUARD system is built upon the foundation of such a fluid. With the addition of three essential products, this system then becomes the highest performing water-based system in the industry.

Fluid Phase

Seawater

Drill water brine

KCl or NaCl

Components

Product

Viscosifier

BARAZAN® D PLUS

Filtration control

N-DRIL™ HT or PAC™-L

Alkalinity

KOH or NaOH

Clay control

CLAY GRABBER®
 CLAY SYNC™ *
 CLAYSEAL® PLUS
 BORE-HIB®

* N-DRIL HT filtration control agent can be replaced or supplemented with FILTER-CHEK™ filtration control agent

* CLAY SYNC shale stabilizer can be replaced or supplemented with CLAY FIRM® shale stabilizer (CLAY SYNC polymer in a carrier liquid)

Table 2 Specialized inhibition additives

Product	Order of Attachment	Characteristic	Benefit
CLAY-SYNC™	First	Adheres to clay via hydrogen bonding. Encapsulates and inhibits. Illite specific.	Prevents hydration and dispersion. Seals micro-fractures/fissures in shale.
CLAYSEAL® PLUS	Second	Ionic inhibition. All-purpose can inhibit smectite, illite and mixed-lattice clays.	Amine locks on to sites on clay. Hydroxyl ions attach to exposed oxygen sites. Prevents hydration. Seals micro-fractured/fissures in shale.
CLAY GRABBER®	Last	Adheres to clay via hydrogen bonding. Encapsulates and flocculates. Smectite specific.	Provide dispersion. Flocculates colloids. Wraps around clay and prevents breakdown.

Formulation and Preparation

The chart below is only meant to be used as a guideline. The true formulation should be designed in conjunction with a Baroid technical services representative to tailor the fluid to suit specific well requirements. Formulations can be designed using various concentrations of salt, polymers and at different mud weights. BARACARB bridging agent can also be included, if required, to help negotiate drilling through high permeability sands.

Table 3 Basic HYDRO-GUARD system formulation (actual formulations may vary due to well specifics)

Product	Description	Primary Function	Concentration
Brine	Monovalent brine made with KCL and/or NaCl in seawater or drillwater (treat out Ca^{2+} ion with soda ash if seawater is used)	Continuous (fluid) phase	To make 1 bbl
BARAZAN® D PLUS	Xanthan gum	Viscosifier	1.0 – 3.0 ppb
N-DRIL™ HT PLUS	Modified starch	Filtration control	2.0 – 4.0 ppb
PAC™-L	Polyanionic cellulose This can be used as a supplement to N-DRIL HT filtration control agent	Filtration control	1.0 – 3.0 ppb
CLAY GRABBER®	High molecular weight non-ionic proprietary polymer (35-40% active in carrier fluid)	Flocculant/Encapsulator (smectite specific)	0.5 – 1.0 ppb
CLAY SYNC™	Low molecular weight non-ionic proprietary polymer (dry powder)	Clay Inhibitor (illite specific)	2.0 – 5.0 ppb
CLAY FIRM®	Low molecular weight non-ionic proprietary polymer (35-40% active in carrier fluid) This can be used instead of, or as a supplement to, CLAY SYNC shale stabilizer	Clay Inhibitor (illite specific)	2.0 – 5.0 ppb
CLAYSEAL® PLUS	Amphoteric amine	Clay Inhibitor (smectite, illite and mixed lattice clays)	4.0 – 8.0 ppb
BORE-HIB®	Potassium silicate/glycol blend	Shale stabilizer	1.0 – 4.0 % by vol
GEM™	Polyalkylene glycol	Shale stabilizer	2.0 – 5.0 % by vol
KOH / NaOH	Potassium or sodium hydroxide	Alkalinity source	0.25 – 0.50 ppb
BARACARB®	Sized CaCO_3 (pure ground marble)	Bridging agent	If/As Req.
BAROID®	Barite	Weight agent	If/As Req.

Salt Concentration

Supplementary ionic inhibition is provided by building the system in monovalent brine (potassium or sodium chloride). Any concentration of these salts can be used. However, field results indicate that for high-gumbo formations at least 8%wt KCl or NaCl brine is required for optimum performance and minimized system dilution. The level of ionic inhibition and salinity provided helps the non-ionic PAs work much more efficiently. Furthermore, as a general practice, drilling gumbo with less than 8%wt KCl or NaCl in a low solid, non-dispersed (LSND) system is not recommended.

NOTE: To pass toxicity tests in the GoM KCl content cannot exceed 2%. Therefore, NaCl is used instead.

Mixing Procedures

The order of mixing for the system is as follows, with the product concentration being as per well requirements:

1. Treat out hardness as required getting total hardness below 200 mg/l.
2. Adjust water to required salinity using KCl or NaCl.
3. Mix BARAZAN D PLUS viscosifier/suspension agent to ensure that Tau zero values are 75% of the open hole size in inches, or around 8+ lbs/100ft 2. Or, engineer the 6 -rpm to be around 1.1 – 1.5 times the open hole diameter in inches.
4. Add alkalinity control agents (NaOH or KOH) to attain the desired pH.
5. Mix N-DRIL HT additive for desired filtration control.
6. Mix CLAYSEAL PLUS shale stabilizer.

7. Mix CLAY GRABBER flocculant and CLAY SYNC or CLAY FIRM shale stabilizers.
8. Mix GEM CP or GEM GP shale stabilizers 2% by volume.
9. Mix BORE-HIB shale stabilizer 1% by volume.
10. Adjust weight as desired. Add bridging agents if required.
11. Add defoamer as required. This may be essential in high salt systems as they are often run in deepwater environments.

NOTE 1: Add PAs only after all preceding products have been thoroughly mixed.

NOTE 2: PAs can be mixed separately to produce better dispersal. They can then be circulated through a shearing device to condition and reduce viscosity. Then, they can be bled into the full system. This may prevent viscosity humps caused by direct addition and reduce shaker overflow losses.

NOTE 3: Use liquid PAs on rigs with limited shearing/mixing capability (CLAY GRABBER flocculant and CLAY FIRM shale stabilizer). This will remove the need to shear dry polymer to work past the viscosity hump.

NOTE 4: PAs are encapsulators. If a dry product is added, they will naturally encapsulate and prevent its dispersion. As a result, do not add any dry polymer (e.g. xanthan gum or PAC filtration control agent) to the HYDRO-GUARD system. The polymer will be encapsulated and form fish-eyes in the fluid. Make all additions of such polymer in a fully hydrated form once the CLAY GRABBER flocculant and CLAY SYNC or CLAY FIRM shale stabilizers have been introduced.

NOTE 5: Use coarse shaker screens for the initial circulations of new fluid. This helps prevent overflow losses caused by unsheared polymer.

Displacement of the HYDRO-GUARD System

The requirement for displacement will depend upon the operations and well design. However, it is recommended to attempt to get a clean displacement when using the HYDRO-GUARD system.

This helps reduce cross-contamination that can lead to wasted volume (dumping) or system contamination. Both of these can lead to additional costs in either building more volume or treating the system back.

Spacer

Covering 300-500 feet (91-152 meters) of annulus length, a spacer should be designed to minimize interface between the fluids in the well.

Being water-based, it should be high-vis or weighted/high-vis to create a barrier between the displaced and displacing fluids. Alternatively, BAROLIFT® sweeping agent can be used instead of raising the rheology. This monofilament fiber acts to prevent channeling and reduce fluid interface.

Density

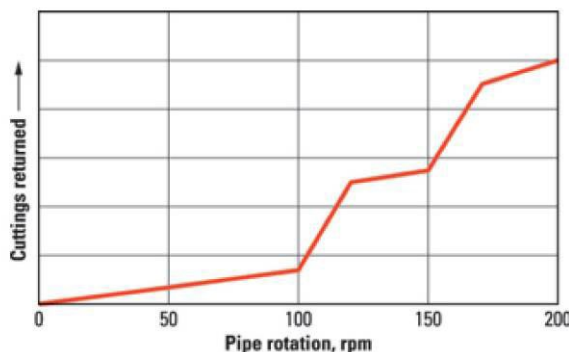
It is important that the fluid going into the well is heavier than that displaced out. If a lighter fluid is being displaced into the well some extra consideration will be required in spacer design (or employ reverse circulation). If not addressed, lighter fluids tend to channel upwards and create large contaminated interfaces between the fluids.

Pump Rate

Displacement should occur at the highest pump rate possible. Higher annular velocity (AV) discourages channeling by pushing the fluids towards turbulent flow.

Pipe Position and Movement

A drillstring centered in the well will create the best conditions for low contamination displacement. If the pipe is positioned to the high or low side of the hole (as occurs in deviated wells), then the fluids will begin to channel in the open side, with old mud being left in the narrow, restricted area around the drill pipe. The best way to prevent this is to rotate the drill string at a high rate (150+ rpm) and reciprocate the drill string over one stand. This movement increases the effective velocity of the fluid around the pipe. This improves displacement as well as hole cleaning efficiency.



Increased fluid efficiency with increased drillstring rotation speed.

Contamination

Cross-contamination of the fluids is common during displacement. If possible, divert any contaminated HYDRO-GUARD fluid to a separate pit. This can be tested and treated as required so it can be returned to the active system. Pretreatments are not advised. The most effective form of treatment is made after the fluid has been successfully displaced. If no such facilities exist to divert fluid interface, consideration will have to be given to dumping excessively contaminated HYDRO-GUARD fluid.

Displacement Indicators

Clean interfaces between water-based fluids can make visual identification of the displacement boundary difficult, even when using calculated pump strokes as a guide. Other guides to recognizing displacement fluid types include the following:

- MW measurement change
- pH measurement change
- Viscosity change
- Material appearing or being eliminated from shaker screens
- LCM or 'marker' added to spacer to indicate return at shakers

A spacer return earlier than calculated pump strokes indicates that channeling could have occurred. The spacer design, pump rate, and pipe movement should be reviewed before pumping the next spacer.

Maintenance and Testing

The system can be maintained similar to a traditional PHPA system. However, given the tougher drilling requirements that the HYDRO-GUARD fluid is used for, staying on top of the system maintenance becomes much more important. However, it requires no specific or additional field testing. It can be tested using all accepted API and Baroid testing methods and procedures.

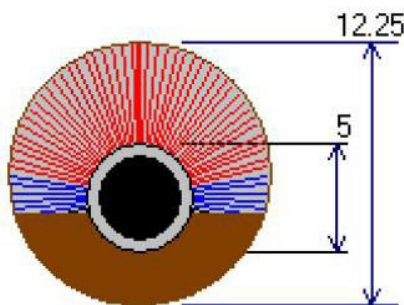
Rheology

The exact requirements for the system rheology cannot be made. They are very much dependent upon pressure and hydraulic constraints of each specific rig and well.

Many variables can affect the rheology of a WBM such as temperature, solids type / size / concentration, polymer content, and presence of contaminants.

General ‘rules of thumb’ are as follows. Adjust the concentration of BARAZAN D PLUS viscosifier/suspension agent to suit accordingly.

1. Keep the PV as low as practically possible. This can reduce friction, plus improve hole cleaning and ROP's. Increasing PV at a constant MW can indicate low gravity solids (LGS) build up in the system.
2. Use YP as primary rheological parameter for vertical to sub-vertical wells (30° inclination or less). The YP is a measure of the fluids carrying capacity.
3. Use 6-rpm and Tau zero as the primary rheological controls for deviated wells (greater than 30° inclination). The 6-rpm is often referred to as the low-end rheology (LER) of the fluid and is coupled with the 3-rpm value. The Tau zero is the measure of the fluids shear stress at almost zero shear rate. These figures give a better indication of mud hole cleaning capabilities in areas of the well where flow (shear rate) is approaching zero, mimicking the conditions around the low side of the drillstring in highly inclined wells where flow is restricted due to cuttings build up and pipe eccentricity.



Cuttings bed position and flow paths in an inclined wellbore during drilling (cuttings bed marked by solid brown shading). This is calculated as part of the Baroid DFG hydraulics modeling software package.

4. Maintain a 6-rpm between 1.1-1.5 times the open hole diameter in inches, OR, a Tau zero value that is at least 75% of the open hole diameter in inches.
5. Maintain a clean system by limiting the solids build up. Monitor the LGS, MBT, and sand content of the system. This will help minimize increases in PV and gel strength to problematic levels.
6. Reduce rheology with use of THERMA-THIN co-polymer thinner or brine dilution. Increase rheology as required with additions of prehydrated BARAZAN D PLUS viscosifier/suspension agent.

Hole cleaning can be supplemented with sweeps when required in tougher drilling conditions.

Reduced pipe rotation, high ROP, excessive sliding, large annuli and limited flow rates all reduce hole cleaning capability. As covered in the “displacement” section, both BAROLIFT sweeping agent and SWEEP-WATE weighting agent can be used in sweeps to improve the lifting capacity.

Additionally, sweeps consisting of N-VIS® HB viscosifier can also be used. This is a dispersed, micro-fibrous cellulose viscosifier that provides excellent suspension and displays optimum rheology at low shear rates. This product has already been used to great success in the Gulf of Mexico with the HYDRO-GUARD system.

Table 4 Suggested sweep selection to aid hole cleaning

Hole Angle	Common Pill Type	Specialist Additives
ZONE I (0 - 40°)	Hi-vis sweeps	BAROLIFT sweeps
ZONE II (40 - 60°)	Weighted Hi-vis sweeps Low-vis/Weighted Hi-vis sweeps	BAROLIFT SWEEP-WATE N-VIS HB
ZONE III (60°+)	Low-vis/Weighted Hi-vis sweeps	BAROLIFT SWEEP- WATE N-VIS HB

Solids Content

Since HYDRO-GUARD fluid is designed as a LSND system, the monitoring of solids build-up is very important. This is especially so when drilling through reactive clays and gumbo formations.

Maintain the following solid levels with regular field testing:

- Low Gravity Solids (LGS) less than 6%vol.
- Sand content less than 1%vol.
- Methylene Blue Test (MBT) less than 10 - ppb eq. clay.

Use solids control equipment (SCE) and dilution as required keeping on top of the required (MBT) levels. Regular dumping of sand traps and SCE tanks will help remove high solids fluid from the system.

Filtration Control

Adjust the concentration of N-DRIL HT additive to achieve the desired fluid loss levels. For higher temperature wells, additional fluid loss agents will be required to achieve the same results.

Use the API Filtrate test as a measure of trends only. HPHT fluid loss tests should be done at estimated bottom hole temperatures and pressures (or standard API test P/T values).

These values will give a better indication of the properties the fluid is displaying downhole. A rule of thumb would be to get a HPHT Fluid Loss value around 2-4 times greater than the API fluid loss. However, this depends greatly upon the P/T values used for the HPHT test. Look to keep the API filtrate below 10 cc/30 mins at all times. However, it is advisable to seek to reduce this level further so it is at 5 cc/30 mins or less. This can provide optimum filtration control across permeable sands and help limit invasion into microfractures in shales (which can lead to shale instability).

Alkalinity

Although it is not the same as traditional PHPA polymer, the non-ionic polyacrylamide that makes up such an important part of the HYDRO-GUARD system does operate under some of the same constraints.

The pH for this system needs to be maintained below 10 at all times, otherwise the polymer can hydrolyze and “burn” away. This is an easy condition to cater to. Avoid direct addition of caustic soda or potash to the fluid. Instead mix in a chemical barrel and slowly bleed into the fluid. This helps allow for better mixing and helps prevent high pH “hot -spots” from depleting the polyacrylamide levels.

Furthermore, pretreatment of the system with sodium bicarbonate (and citric acid if required) can protect the system from cement contamination during operations. These products can help suppress alkalinity and keep the pH below 10.

Ideally, run the HYDRO-GUARD system with a pH around 8.8 – 9.5. This can be maintained with potassium hydroxide (KOH) or sodium hydroxide (NaOH). KOH is usually the preferred source since it can impart additional potassium ions (K +) to aid in clay inhibition.

Hardness

As with traditional PHPA fluids, Ca^{2+} and Mg^{2+} ions in drill water or seawater need to be treated out with addition of soda ash, caustic soda or sodium bicarbonate. The choice of which depends upon the pH sensitivity at the time. Soda ash and caustic soda will raise alkalinity levels, whereas sodium bicarbonate will not. The OH^- ions found in caustic soda/potash are better at removing magnesium.

Furthermore, cement contamination that leads to an increase in Ca^{2+} can be addressed with sodium bicarbonate treatment. Maximum hardness levels should be maintained below 400 mg/l at all times. If the levels rise above this, the divalent ions present will cause a decrease in rheological efficiency due to the screening effects, leading to a hydrodynamic diameter decrease of the polymeric additives and loss in system efficiency.

Salinity

Field results indicate that running a minimum of 80,000 ppm helps the PAs run more efficiently.

Dilution volume decreases considerably when salinity is above this level. Also, drilling performance improves as the polymers become more effective.

Polyacrylamide Levels

Once drilling ahead, the ideal active levels for the two PAs should be held as follows:

- 0.5-1.0 ppb CLAY GRABBER flocculant.
- 2.0-2.5 ppb CLAY SYNC or CLAY FIRM shale stabilizers.

The exact levels required for a well will depend upon the reactivity of the clay being drilled. Additionally, early analysis of shale samples to determine smectite and illite levels present may help redefine the concentration needed. Make sure that regular maintenance additions are made to the active system to allow for polymer depletion.

During drilling, CLAY FIRM shale stabilizer and CLAY GRABBER flocculant should be added to the active system on a regular basis to prevent depletion. This is especially recommended during areas of highly reactive shale. Additions should be made as follows:

- CLAY SYNC shale stabilizer: 10-15 lb per barrel of active shale drilled.
- CLAY GRABBER flocculant: 1-1.5 lb per barrel of active shale drilled.



Inhibited gumbo coming over the scalping unit

If cuttings integrity at the shale shakers begins to degrade or become “sticky”, it is recommended to increase the level of CLAY FIRM or CLAY SYNC shale stabilizers (and CLAYSEAL PLUS shale stabilizer).

If MBT or LGS levels begin to rise, then further addition of CLAY GRABBER flocculant should be made (in conjunction with any required dilution and/or SCE manipulation).

CLAYSEAL PLUS Shale Stabilizer Levels

Once drilling ahead, the ideal level is to maintain around the concentration in the initial formulation (4-8 ppb active).

If cuttings integrity at the shale shakers begins to degrade or become “sticky” it is recommended to increase additions of CLAYSEAL PLUS shale stabilizer (and CLAY FIRM or CLAY SYNC shale stabilizers).

Additions should be made as drilling conditions dictate. Recommended minimum treatment rates are as follows:

- 17 1/2” hole: 10-gal every 50 feet (15 meters) drilled.
- 12 1/4” hole: 10-gal every 66 feet (20 meters) drilled.
- 8 1/2” hole: 10-gal every 82 feet (25 meters) drilled.

However, the exact treatment rates should be based on cuttings integrity and the amount of reactive clay being drilled. More reactive clay will result in more rapid depletion.

Troubleshooting and Guidelines

Table 5 Troubleshooting the HYDRO-GUARD system

Problem	Indication	Cause	Solution
A. Low rheology/gels	Rheology decrease – profile ‘out of spec’ Poor hole cleaning Poor suspension	Excessive dilution/thinner Insufficient product concentration	Increase BARAZAN D PLUS concentration Use PAC-R to increase rheology without increasing gels
B. High rheology/gels	Rheology increase –profile ‘out of spec’ Increase in MBT Increase in LGS Water % decrease coupled with salinity increase High rheology, high gels if flocculated	Increased solids content Drill solids Dehydration / water loss Flocculation Over-treatment	Optimize SCE use Dilution with brine or water Reduce BARAZAN D PLUS concentrations Add THERMA -THIN Increase PA’s and CLAYSEAL PLUS
C. Increase in Fluid Loss	Increased spurt and/or total fluids loss (API and/or HPHT tests)	Excessive dilution Decrease or lack of filtration control agent Increasing temperature downhole Contamination Decrease or lack of bridging	Make additions of filtration control agent such as N-DRIL HT PLUS Address fluid formulation is drilling conditions are becoming tougher (e.g. higher temperature) Add bridging agents such as BARACARB Address source of contamination
D. High hardness	High fluid hardness (above 400 mg/l) Reduced PA concentration in active mud	Drilling cement Contaminated drill water Untreated seawater	1-2 ppb sodium bicarbonate 0.5-1 ppb soda ash 0.5-1 ppb sodium hydroxide
E. High pH	High pH (above 10) Reduced PA concentration in active mud	Drilling cement Over treatment	1-2 ppb sodium bicarbonate 1-2 ppb citric acid

Problem	Indication	Cause	Solution
F. High shaker overflow losses	High shaker losses during drilling Contaminated mud on bottoms up	High flow rate High ROP Sand blinding Shaker screens to fine Unsheared polymer Dehydrated or contaminated mud	Reduce ROP and flow rate Select screens size/type to eliminate blinding Coarse shaker screens for unsheared (new) mud Use liquid PA polymers (CLAY FIRM, CLAY GRABBER) Use THERMA-THIN Check for contamination or dehydration
G. Poor hole cleaning	Increased ECD High MW out Tight trips Pack off Lost circulation Stuck pipe	Low rheology High ROP Sliding / Low rotation Low flow rate Hole instability / washout	Model well conditions using Baroid DFG hydraulics modeling software packages Increase rheology with BARAZAN D PLUS addition Reduce ROP and sliding Increase pipe rotation Increase flow rate and AV's Address hole instability Pump sweeps
H. High solids content	Increased MW Increased LGS Increased MBT or sand content Increased PV and gels	Lack of system inhibition Reduced clay control High ROP SCE inefficiency Holes in shaker screens	Fine-up shaker screens Run all SCE Dilution with brine, water or new mud Add THERMA -THIN Increase concentrations of PA's and CLAYSEAL PLUS
I. Swelling, dispersive and sloughing shale	Increased MBT Tight hole on trips PA and CLAYSEAL PLUS depletion Washout Excessive cuttings	Chemical / mechanical instability of shale Lack of mud inhibition Lack of encapsulator in mud Lack of MW	Increase PAs and CLAYSEAL PLUS concentrations Increase salt concentration Increase MW
J. Bacterial attack	Reduced rheology Increased fluid loss Foul odor in surface pits when not circulating Reduced pH	Unclean sand traps, SCE tanks, possum belly or surface lines/pits Use of unclean drill water Use of old, stagnant mud	Treat drill water, seawater or mud with biocide such as ALDACIDE® G Clean out surface pits/lines with clean water and biocide Dump old mud Raise pH of mud
K. Differential sticking	'Sticky' hole Jarring to free pipe Stuck pipe	Poor filter cake quality Lack of mud/filter cake lubricity Depleted sands Permeable sands High mud overbalance High ECDs Mud losses to formation High solids content	Increase filtration control agent concentration to improve filter cake Add lubricant to the mud Add bridging agent or LCM to the mud Reduce MW/overbalance Lower rheology Reduce flow rate Improve hole cleaning Reduce solids content Good drilling practices Spotting pills

Problem	Indication	Cause	Solution
L. Hydrogen sulfide (H ₂ S)	Drill string corrosion Drop in Pom and pH Rotten egg odor (below ±100-ppm) Gas in mud	Gas influx from formation Thermal degradation of organic mud additives Bacterial degradation of organic mud additives	Raise pH above 11 with lime or caustic soda – this will affect the PA levels! Add H ₂ S scavenger to the mud Add scale or corrosion inhibitor to mud Raise MW Add biocide to mud Run degasser
M. Carbon dioxide (CO ₂)	Drill string corrosion Drop in Pom, Pf and pH Gas in mud Carbonate contamination	Gas influx from formation Thermal degradation of organic mud additives Carbonate contamination from drilling products Sodium carbonate and bicarbonate over-treatments	Raise pH with lime or caustic soda Add CO ₂ scavenger to the mud Add scale or corrosion inhibitor to mud Raise MW Run degasser
N. Foaming	Foam on surface of pits and in flow lines during mixing	Mixing/shearing additives in high salinity fluid	Add defoamer to the mud or base brine This is NOT the same as gas-cut mud, which would require running through a degasser
O. High temperature (> 300°F)	Increased rheology and gels Decreased water content with increased salinity Difficulty maintaining API and HPHT fluid loss levels	Static mud at bottom of high temperature well Build up of mud temperature during drilling Geothermal drilling conditions Product instability at increased temperatures	Dilution with drill water or seawater Formulate/supplement with high temperature tolerant/stable additives Add 20-ppb sodium formate to stabilize fluid
P. Lost Circulation	Loss of circulating mud volume to the formation	Permeable sands Cavernous formation Natural fractures and fault zones Induced fractures	Reduce parameters affecting ECD (flow rate, MW, rheology) Address any poor hole cleaning Select appropriate treatment (squeeze, cement, LCM, etc) See overview below.

Lost Circulation

Lost circulation with the HYDRO-GUARD system should always be prevented. Given the degree of difficulty in drilling many of today's wells, there could be many operational difficulties that result from losses. There are less loss problems with the HYDRO-GUARD system when compared to OBM/SBM fluids. Since there are not the wide variances in rheology and density (caused by different pressure / temperature regimes) the circulating hydraulics are much easier to predict and plan around.

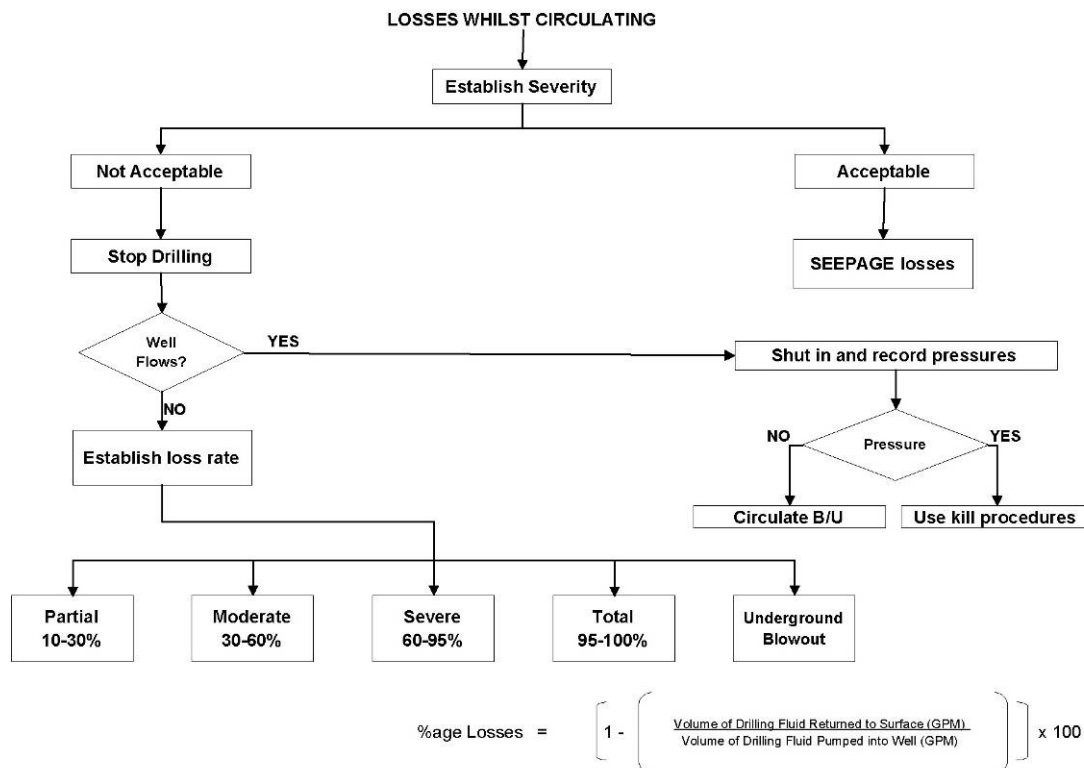
Losses can be classified in four basic types or sources:

1. Highly permeable formations
2. Low or impermeable formations
3. Naturally fractured formations
4. Cavernous formations
5. Induced fractures due to pressure imbalance

The degree of losses varies with the source and may be referred to as a percentage of the circulating volume being lost to the well; or alternatively by a specified volume lost, e.g.:

- Seepage: < 10% or <10 bph
- Partial: 10 – 30% or >10<25 bph
- Moderate: 30 – 60% or >25<50 bph
- Severe: 60 – 95% or >50<100 bph
- Total: 95 – 100% or > 100 bph.

Procedures for gauging the extent of downhole losses can be seen on the chart below.



If losses are caused by permeability, then LCM has a high probability of solving the problem. However, the mud system should not be overloaded with material. This could result in blocked bit nozzles, MWD tools and increased rheology that could exacerbate ECDs and promote further losses. Instead, batch treatments of pills spotted over the loss zone are more advisable.

Sealing natural fractures and caverns is far more complicated than dealing with permeability losses. These loss zones are much larger and the particles size distribution needed to tackle the losses is much greater. Usually, particulate LCM is not successful at curing these problems since the volume that must be plugged is too big. Instead, gunk squeezes and specialty pills are more successful. Baroid's HYDRO-PLUG® lost circulation material (dual acting particulate plus a swelling polymer pill) or EZ-PLUG® lost circulation material (acid soluble palletized pill for the reservoir) can provide a better probability of success than conventional particulate pills. For very high loss rates, chemical sealants such as FUSE-IT® lost circulation material may be required.

The most common mud loss, particularly in deep wells, is due to mechanically induced fractures.

Losses can be slow, moderate or complete, at any depth. Although the mechanics of induced fractures are well documented in most cases, they are treated improperly when it comes to LCM.

The primary cause is usually when the ECD exceeds the formation fracture gradient. In some situations (wellbore “breathing” or “ballooning”), this fluid can be given back once the mud pumps are turned off (and the fractures close). Common causes of induced fractures are as follows:

- Connection techniques (high pipe running speed or bringing pumps on-line too rapidly).
- Running drillstring or casing in hole too fast producing high surge pressures.
- Running drill string in hole with a plugged bit.
- High ECD due to excessive MW and rheology.
- Restricted annulus from swelling clays or balling.
- Insufficient hole cleaning leading to high cuttings concentration in the annulus.
- Excessive ROP, beyond the ability of the drilling and mud parameters to clean the hole.

1.2. PerformaDril Inhibitive Water-Based Fluid

The PerformaDril system is a clay-free, non-dispersed drilling fluid designed to provide maximum shale stabilization in highly reactive clays. Its versatility and effective clay control properties have made it a credible replacement for invert emulsion fluids worldwide.

PerformaDril fluid provides a high degree of clay control, wellbore stability, performance-based rates of penetration (ROP), low coefficients of friction (CoF), and rheological control over a wide range of temperatures (40-400°F). The PerformaDril system can also provide the environmental benefit of unrestricted cuttings discharge based on most worldwide water-based fluid environmental regulations.

Return permeability studies demonstrate that the PerformaDril system is non-damaging. Therefore it is recommended for use in reservoirs containing reactive clays.

PerformaDril Formulation

- Density range 18.5 ppg (2.21 sg)
- Temperature tolerance 250°F (121°C) –temperature stability can be increased with the addition of oxygen scavengers

The chart below should be used as a guideline. The actual formulation used at the rigsite should be designed to meet well-specific requirements. The system can be formulated with various concentrations of salt and polymers and at different mud weights. BARACARB sized calcium carbonate can also be included if required while drilling through high-permeability sands.

Table 6 Basic PerformaDril Formulation

Product	Description	Primary Function	Concentration
Water / Brine	Fresh water or monovalent brine made with KCl and/or NaCl in seawater or drillwater Treat out Ca ²⁺ ion with soda ash if seawater is used.	Continuous (fluid) phase	To make 1 bbl
BARAZAN D	Xanthan biopolymer	Viscosifier / Suspension	0.1-2.0 ppb
DEXTRID E	Modified starch	Filtration control	2.0-6.0 ppb
PAC-R / PAC-L	Polyanionic cellulose May be used as a supplement to DEXTRID E if required.	Filtration control and supplementary viscosity	1.0-3.0 ppb
PerformaTrol	Organic polymer	Shale inhibitor / Flocculant	1.5-5 % by vol
GEM GP	Polyalkylene glycol	Improve shale stability / Lubricity	2-6 % by vol
OXYGON	Granular erythorbate material	Oxygen scavenger	As needed
STARCIDE	Microbiocide solution	Control of sulphate reducing bacteria	0.1-0.5 ppb as needed
CLAYSEAL PLUS	Amphoteric amine	Inhibits clay and shale hydration	4.0-8.0 ppb as needed
Soda Ash	Sodium carbonate	Hardness reducer	0.5-2.0 ppb
KOH/NaOH	Potassium or sodium hydroxide	Alkalinity source	0.1-1.0 ppb

Product	Description	Primary Function	Concentration
BARACARB	Sized CaCO_3 (pure ground marble)	Weighting/Bridging agent	As needed
BAROID	Barite	Weighting agent	As needed

Mixing Procedure

The mixing procedure is as follows, with the product concentration being as per well requirements:

1. Treat out hardness as required maintain total hardness between 200-400 mg/l.
2. Adjust water salinity, if required, using KCl or NaCl.
3. Mix BARAZAN D viscosifier as required
4. Add alkalinity control agents (NaOH or KOH) to maintain the pH between 8.5-10.
5. Mix DEXTRID E/PAC filtration control additives for desired filtration control.
6. Mix CLAYSEAL PLUS shale inhibitor, if required.
7. Mix PerformaTrol shale inhibitor.
8. Mix GEM GP shale stabilizer.
9. Adjust weight as desired. Add bridging agents, if required.
10. Mix STARCIDE microbiocide / OXYGON corrosion inhibitor, as required,
11. Add defoamer as required. This may be essential in salt systems as they are often run in deepwater environments.

Basic Maintenance for the PerformaDril System

The system can be maintained similarly to a traditional salt / polymer system.

However, PerformaDril systems are typically used for more challenging drilling requirements, so maintaining appropriate system properties and product concentrations is critical. Additional field testing is required, as discussed later in this section.

Table 7 PerformaDril Maintenance Recommendations

Fluid Property	Recommended Treatment / Concentration
Rheology	Specifications for the system rheology depend on the pressure and hydraulic constraints of each specific rig and well. Adjust the concentration of BARAZAN D viscosifier to obtain the desired viscosity. Dilute with premixed mud, water or brine if thinning is required. Monitor the low gravity solids (LGS), methylene blue test (MBT) values, and sand content to help minimize increases in PV and gel strength. Guidelines are as follows: PV should be as low as practically possible to reduce friction, plus improve hole cleaning and ROPs. If the PV increases while the mud weight (MW) remains the same, this may indicate a low-gravity solids (LGS) build up in the system. YP is the primary rheological parameter for vertical to sub-vertical wells (30° inclination or less). 6-rpm and Tau zero values can be used as the primary rheological controls for deviated wells (greater than 30° inclination).
Solids Content	PerformaDril fluid is designed as a low-solids non-dispersed (LSND) system Monitoring for solids build-up is critical, especially when drilling through reactive clay and shale formations. Maintain the following solid levels with regular field testing: LGS less than 6% vol. Sand content less than 1% vol. Methylene blue test (MBT) as low as possible and less than 10 ppb equivalent clay.

Specialized Testing / Maintenance Considerations

Shale Dispersion Test / Effect of PerformaTrol Concentration

The purpose of this test is to assess the ability of the fluid to preserve the integrity of drill cuttings as they are transported up the wellbore to surface and removed at the shale shaker. The test involves hot rolling a known weight of sized shale (2-4 mm) for 16 hr, isolating the remaining shale on a 500-micron sieve, and weighing it. The weight of recovered shale is then reported as a percentage of the original weight.

Using even the most reactive shale to conduct this test, the results obtained with PerformaDril fluid approach 100% recovery. However, very good recovery results can be obtained with lower concentrations of PerformaTrol shale inhibitor (tested to simulate the effects of depletion). The recovered cuttings obtained at lower concentration are typically softer and more friable than those obtained using higher levels of the PerformaTrol additive.

PerformaTrol Effect on Bentonite Systems

The PerformaTrol additive is both an effective encapsulator and flocculant. The addition of PerformaTrol liquid to a highly dispersed clay such as pre-hydrated bentonite causes a dramatic increase in rheology and rapid depletion of PerformaTrol liquid.

Breakover from a bentonite system or a less inhibitive system containing dispersed clays should be avoided.

Nitrogen Content as an Indicator of PerformaTrol Concentration

The PerformaTrol liquid is an organic polymer containing nitrogen. Measuring the nitrogen content of a mud sample indicates the content of PerformaTrol liquid in the system. This approach is only possible in fluids that do not contain other sources of nitrogen (e.g., nitrates, amines, etc.)

The test method involves oxidation of nitrogen to nitrate and subsequent reaction of this ion with a phenol to form a colored nitrophenol. The concentration of nitrophenol can be measured using a simple colorimeter.

- The method measures both active PerformaTrol liquid and non-active PerformaTrol liquid. (i.e., that already adsorbed onto the surface of fine solids within the mud).
- There is no single correct value for PerformaTrol additive content for an entire well.
- Test results obtained should be interpreted in conjunction with data on the condition of both the mud and cuttings.

Cuttings Condition as an Indicator of PerformaTrol Concentration

PerformaTrol liquid will be depleted onto the wellbore and drill cuttings. The level of depletion for a given hole section will depend on the nature of the formation being drilled and the surface area of cuttings.

If the condition of the cuttings deteriorates, then the level of active PerformaTrol liquid in the mud is insufficient. Test results may show that the level is as specified. However, this may indicate that either the programmed concentration is incorrect for the formation being drilled or that there is a high proportion of non-active PerformaTrol liquid within the mud.

The presence of significant amounts of non-active PerformaTrol liquid may be indicated by increased MBT values and increased low gravity solids.

The decision to add more PerformaTrol liquid should be based principally on the condition of the cuttings, supported by the test results on the active mud system.

At this stage, it is more cost effective to add further PerformaTrol liquid than to try to cope with the effect of dispersed clay. The rate of PerformaTrol liquid depletion is dependent on the exposed surface area of shale. Inadequate levels of PerformaTrol liquid can result in shale dispersion, an exponential increase in exposed surface area and rapid depletion of PerformaTrol liquid added to try to correct the problem.

Troubleshooting the PerformaDril System

Table 8 PerformaDril Troubleshooting Treatment Guidelines

Problem	Indication	Cause	Solution
Low rheology / gels	Rheology decrease- profile below specification Poor hole cleaning Poor suspension	Excessive dilution Insufficient product concentration	Increase BARAZAN D concentration Use PAC-R to increase rheology without increasing gels
High rheology / gels	Rheology increase- profile above specification Increase in MBT Increase in LGS Water % decrease coupled with salinity increase (in salt system)	Increased solids content Drill solids Dehydration / water loss Flocculation Over-treatment with polymers Insufficient inhibitor	Optimize solids control equipment use Dilution with brine or water Reduce BARAZAN D concentrations Increase inhibitor concentration
Increase in fluid loss	Increased spurt and / or total fluids loss (API and/or HPHT tests)	Excessive dilution Decrease or lack of filtration control agent Increasing temperature down hole Contamination Decrease or lack of bridging	Add filtration control agent such as DEXTRID E, N DRIL HT, or PAC-L Consider adding bridging agents such as BARACARB Address source of contamination
High pH	High pH (above 10)	Drilling cement Overtreatment	Additions of sodium bicarbonate or citric acid as calculated and pilot tested
Swelling, dispersive, and sloughing shale	Increased MBT Tight hole on trips Depletion of inhibitors Washout Excessive cuttings	Chemical/ mechanical instability of shale Lack of mud inhibition Lack of mud density	Increase inhibitor concentrations Increase mud density
Foaming (not gas-cut mud)	Foam on surface of pits and in flow lines during mixing	Mixing / Shearing additives in high salinity fluid	Add defoamer to the mud or base brine Investigate mixing system

1.3. BOREMAX Water-Based Fluid

BOREMAX water-based fluid is a non-dispersed, low-solids high performance freshwater drilling fluid designed to provide maximum penetration rates and wellbore stability while reducing dilution, disposal costs, and environmental concerns associated with diesel and mineral oil invert emulsion systems.

BOREMAX Formulation

- Density range 8.5-17.5 ppg (1.02-2.10 sg)
- Temperature tolerance 350°F (177°C).

Products are listed in order of addition.

Table 9 Basic BOREMAX Formulation

Additive	Description	Function	Typical Concentration, ppb (kg/m ³)
BORE-VIS	Blend of viscosifying agents and extenders	Viscosifier	6.0-10.0 ppb (17.4-29)
CLAY GRABBER	High molecular weight copolymer	Flocculant / Inhibition	0.5-1.5 ppb (1.5-4.4)
POLYAC PLUS	Modified acrylate polymer	Filtration control / Deflocculant	0.5-1.5 ppb (1.5-4.4)
BORE-PLUS	Blend of acrylate copolymers	Filtration control	0.5-1.5 ppb (1.5-4.4)
KOH	Potassium hydroxide	Alkalinity (pH 8.8-9.2)	0-0.5 ppb (0-1.5)
BARO-TROL PLUS	Dispersible asphalt blend	HPHT fluid loss control	2-6 ppb (5.8-17.4)

Mixing Procedure

1. Install water meter(s) to track water additions for accurate product concentrations.
2. Clean pits and fill with freshwater.
3. Build BOREMAX fluid in isolated pits and avoid exposure to cement contamination.
4. Add 8-10 ppb of BORE-VIS viscosifier to build mud if viscosified mud is typically used to drill below surface.
5. Drill cement and discard severely cement-contaminated mud (or high pH water).
6. Before adding CLAY GRABBER flocculant, reduce pH of mud (or water) to less than pH 10.0 using citric acid and calcium to less than 120-180 ppm with bicarbonate of soda.

Basic Maintenance for the BOREMAX System

Fluid Property / Operation	Recommended Treatment / Concentration
Rheology	Maintain 0.5-1.5 ppb CLAY GRABBER flocculant in system while drilling. BARAZAN-D viscosifier (0.1-0.25 ppb) can be used for supplemental viscosity control. THERMA-THIN deflocculant can be used ahead of and in slugs built for tripping to facilitate low stable rheological properties off bottom after trips.
Fluid Loss Control	Add POLYAC PLUS filtration control agent for fluid loss control and monitor product concentration. Above 0.5-1.0 ppb the POLYAC PLUS additive will function as a deflocculant. If the thinning effect is observed, add BORE-PLUS suspension agent for fluid loss control. Supplement BORE-PLUS suspension agent with BARO-TROL PLUS shale stabilizer for API and/or HPHT fluid loss control.

Fluid Property / Operation	Recommended Treatment / Concentration
pH	BORE-VIS viscosifier, CLAY GRABBER flocculant, and water will exhibit a pH 8.7-9.0 without the use of caustic materials. If acid gas is suspected, monitor CO ₂ using Pf / Mf method. Treat with calculated amounts of lime, BARACOR 95 alkalinity control agent, and/or SOURSCAV hydrogen sulfide scavenger if necessary.
Drilling below Surface Casing	If water is typically used to drill below surface casing, build viscosity while drilling using high viscosity sweeps containing 15-20 ppb pre-hydrated BORE-VIS viscosifier. Control MBT at less than 15.0 ppb equivalent.
Drilling Cement	Add 0.5 ppb CLAY GRABBER flocculant after treating out cement contamination. Gradually build concentration to 0.5 ppb (or greater) to prevent screen blinding if insufficient shear was available while mixing in rig pits. Install fine mesh screens on shakers (175-210 mesh).
Lost Circulation Treatment	LCM should be added in sweeps or pills as CLAY GRABBER flocculant will surface coat background LCM, reducing the amount of polymer available for inhibition and increasing LCM lost at shakers.

Specialized Testing / Maintenance Considerations

MBT Testing

The MBT test protocol should be modified to include the following changes for a more definitive end point:

- 20 ml hydrogen peroxide
- 1 ml of 5N H₂SO₄
- Boil for 15 min.

Confirm the CLAY GRABBER flocculant concentration with field samples tested on the HPK apparatus at the nearest regional lab.

Maintenance Guidelines

- If water is typically used to drill below surface, build viscosity while drilling using high viscosity sweeps containing 15-20 ppb pre-hydrated BOREVIS. Control MBT at less than 15.0 ppb equivalent.
- Add 0.5 ppb CLAY GRABBER after treating out cement contamination. Gradually build concentration to 0.5 ppb (or greater) to prevent screen blinding if insufficient shear was available while mixing in rig pits. Install fine mesh screens on shakers (175-210 mesh).
- Add POLYAC PLUS for fluid loss control and monitor product's concentration. Above 0.50-1.0 ppb POLYAC PLUS functions as a deflocculant.
- When POLYAC PLUS starts to function as a thinner, use BORE-PLUS for API fluid loss control and POLYAC PLUS specifically to control viscosity.
- Maintain 0.5-1.5 ppb CLAY GRABBER in system while drilling.
- Barazan-D (0.1-0.25 ppb) can also be used for supplemental viscosity control.
- Therma-Thin can be used ahead of and in slugs built for tripping. This will facilitate low stable rheologies off bottom after trips.
- Supplement BORE-PLUS with BARO-TROL Plus for API and / or HpHt control.
- LCM should be added in sweeps or pills as CLAY GRABBER will surface-coat 'background' LCM. This reduces the amount of polymer available for inhibition and increases LCM lost at shakers.
- BORE-VIS, CLAY GRABBER and water will have a pH 8.7-9.0 without the use of caustic materials. If acid gas is suspected, monitor CO₃ using Pf/Mf method. Treat with BARACOR 95, and/or BARACOR 44 if necessary.
- Add 2 qt of CLAY GRABBER flocculant at the hopper discharge for each ton of barite added.

- Monitor dilution and CLAY GRABBER flocculant additions to ensure that MBT and LGS remain in specified ranges. Rapid addition of product to catch up may cause temporary high viscosity and screen blinding.
- To increase inhibition, consider adding BORE-HIB, or allowing soluble Ca^{++} at 120-180 ppm. Higher calcium content enables flocculation of inert as well as reactive drilled solids.
- If POLYAC PLUS filtration control agent becomes ineffective as a thinner, the two most likely causes are as follows:
 - The dilution rate is inadequate, resulting in a high LGS %.
 - The concentration of CLAY GRABBER flocculant is too low.

Troubleshooting for the BOREMAX System

Contaminant	Indications	Treatments
Carbonates / CO_2	Presence of bicarbonates and carbonates. Increase in rheological and filtration properties Increase in spread between Pf and Mf High and progressive gel strengths	Treat with calculated amounts of lime and/or 0.25-1.4 ppb BARACOR 95.
Cement	Increased: pH Calcium Rheology Loss of polymer at shaker Ammonia odor	Discard severely contaminated fluid, keeping a safe working volume in pits. Reduce pH < 10.0 with citric acid and dilution. Add bicarbonate of soda. When treatments are not sufficient to counter indications, convert to lime mud.
Gypsum/Anhydrite	Increase in calcium Slight increase in rheological and filtration properties	Properly maintained BOREMAX fluid will tolerate up to 600 ppm calcium from anhydrite with minimal effect on rheology or API filtration. Reduce Ca^{++} to 120-180 ppm with soda ash. Replace co-precipitated fluid loss polymers with new product.
Hydrogen Sulfide (H_2S)	Increase in rheological and filtration properties Decrease in pH Presence of hydrogen sulfide as indicated by sulfide indicator test and the Garrett Gas Train	Treat with 2-5 lb/bbl SOURSCAV. If severe, convert to lime-based mud or displace with invert emulsion system.
Low Gravity Solids	Increased: Rheological and filtration properties MBT Filter-cake thickness LGS content	Dilute system with fresh water while maintaining concentration of system's component products. Monitor CLAY GRABBER additions and increase as needed to facilitate solids control equipment. Adjust solids control equipment to improve efficiency. Adjust and maintain dilution rate as required for attainable solids control efficiency.
Salt Formations	Increased: Chlorides concentration Density YP Viscosity Filtrate	Salt stringers: use dilution and PAC-L or FILTER-CHEK Regular to reduce YP and filtrate (after pilot testing for positive results). Massive salt stringers or dome: convert to HYDRO-GUARD system or displace with invert emulsion mud.

Contaminant	Indications	Treatments
	Decrease in alkalinity	
Salt Water Flow	Increased: Pit volume Chloride concentration Filtrate YP Reduced / Changed mud density Decreased alkalinity Well flows with pumps off	Treat YP and filtrate with freshwater dilution and PAC-L or FILTER-CHEK Regular after pilot testing for positive results. Increase density to control the water flow.

1.4. SHALEDRIL Fluids

SHALEDRIL F & B

SHALEDRIL drilling fluid is a cost effective, formation friendly, fresh-water based shale inhibiting fluid. SHALEDRIL is designed to be run with potassium based products while using freshwater for dilution. The system can be used for land rigs that are set up for freshwater dilutions but do not have a tank for whole mud dilutions. As in many of our highly inhibitive WBMs, it is important to remember that the chemistries for shale inhibition should be maintained both from a dilution standpoint and from depletion due to drilling through reactive shale. With SHALEDRIL B, replenishing the polymers that are being lost to reactive shale cuttings will be kept to a minimum if the system is run properly. As this system does not rely on polyacrylamides, the levels of product will be easier to maintain while drilling.

Characteristics to remember include the following:

- The primary inhibiting components are potassium silicate, potassium lignite, potassium Soltex and GEM GP.
- This is a gel free system and should be treated accordingly.
- The fluid can be run at weights up to 16.0 lb/gal.
- The fluid can tolerate contamination up to the tested limits of 10% by volume.
- The fluid can tolerate CO₂ contamination up to 250°F.
- The fluid tolerates cement contamination and can be treated with sodium bicarbonate.
- The system performs optimally when divalent ions are removed, but will perform at an acceptable level under less than optimal pre-treatment conditions.
- BARO-LUBE GOLD SEAL and NXS-LUBE are effective lubricants and can be used up to 3 – 4 % by volume.
- If a softness is observed in the shale and additional potassium ions are needed, K₂CO₃ can be utilized.

SHALEDRIL B is one of the few WBMs that use primarily potassium based products for inhibition. Based on the Slake Durability data generated, this fluid provides high percent recovery values as shown in the table below. The presence of potassium silicate and glycol appears to be very beneficial in the testing.

BARAZAN D Plus Xanthan biopolymer 0.25 – 2.0 ppb

PAC-L Fluid loss control; Polyanionic cellulose 1 -2 ppb

FILTER-CHEK Fluid loss control; Proprietary blend of anionic polymers
1 – 4 ppb

BDF-546 Blend of potassium silicate for shale inhibition 3-7 ppb

Potassium Soltex Potassium laced Soltex 4 – 8 ppb

Potassium Lignite Potassium laced Lignite 4 – 10 ppb

GEM GP Blend of polyglycols to give reactive shale inhibition
and improved wall cake integrity 2 – 10 ppb

Proper maintenance of the above products will reduce the surface area for a given solids content, allowing any amount of dilution to be more effective. Keep in mind that this system will tolerate some fine solids but we should try to maintain low concentrations of sub-micron particles.

Field Guidelines

In the SHALEDRIIL system, inhibition is the key to the system's performance. When inhibition of the system is diminished the rheological profile increases, which gives the system a loss of efficiency.

To run the SHALEDRIIL system effectively, all components must be maintained at the proper concentrations. The mud program has the recommended concentration for each product. Using pre-mixed mud for dilution works well when pit space and rig operations will allow. However, when adding water to the system all products should be maintained in the correct amounts to maintain inhibition of the system. It is imperative to know exactly how much water is being added to the system to maintain correct product concentration.

The MBT should be maintained as low as possible. A rapid jump in the MBT shows that the system's inhibition has been lost. A gradual increase in the MBT shows that not enough inhibition is being utilized to maintain the system at the present drill rate and solids control efficiency. For a gradual increase in the MBT, adjust your chemical concentrations according to your titrations (potassium and silicate), solids control efficiency, cuttings integrity and overall treatment schedule. To combat a rapid increase in the MBT, ***increase the dilution immediately*** and adjust your chemical concentrations upwards. The higher the MBT is allowed to climb, the harder it is to arrest the climb.

Following these guidelines will ensure that the SHALEDRIIL system maintains its inhibition and will provide a good, sound wellbore for our customers. This system will experience elevated rheology due to depleting the chemicals by water additions and not replacing chemicals whose concentrations were reduced through dilution.

Considerations

- Give careful consideration to screen size before drilling out cement.
- Drill cement and discard severely contaminated mud/ cement contaminated water.
- Treat the system with sodium bicarbonate or citric acid before drilling cement to reduce the soluble calcium level to less than 600 mg/l.
- Do not get behind on dilution. Record the dilution rates carefully so that subsequent wells in the area can learn from your experience.
- Maintain MBT, LGS, PV and YP in the recommended ranges for the mud weight in use. Refer to mud program for specific or reduced property ranges.
- FILTER-CHEK and PAC for filtration control.
- Viscosity, YP, gels, and Tau 0 should be maintained with BARAZAN D Plus
- If a deflocculant is needed, THERMA-THIN is very effective.
- BARACOR 95 can be used in place of lime to treat out CO₂ gas contamination. This product is also a free-radical scavenger and can be used to increase the system's thermal stability.
- For particle plugging purposes, this fluid has several products which can exploit the advantages of particle size and competently address any plugging issue.

Solids Control Equipment

- Start with 210 mesh screens on the first interval using SHALEDRIIL. At least two high-speed linear motion shakers should be used. This is the first defense against solids build up so check screens often for blinding, damage/holes and change as soon as the need arises.
- Run the desander and desilter continuously with densities between 10.0 and 10.5 lb/gal. The desander and desilter may be run intermittently for full circulations to help remove low gravity solids when densities are between 10.5 and 11.0 lb/gal. Above 11.0 lb/gal run only as needed.
- Mud cleaners should be used only if their screens are substantially finer than the 210-215 mesh being used at the shakers. Mud cleaners that can process fluid over 325 mesh screens should be used with the underflow

directed into a holding tank to be centrifuged. The forces at work in a small diameter hydro cyclone can break down drilled solids and create ultra-fines, so use these pieces of equipment sparingly and only if required.

- If only one centrifuge is available it should be used to discard solids in mud weights less than 11.0 ppg. The suction line for the centrifuge should be placed into a holding tank for mud cleaner screen underflow, or in a pit receiving processed mud from the desander and desilter.
- Above 11.0 ppg the centrifuge should be used as needed.

Bit Recommendations

Discuss drill bits with the bit vendor and the Operator's representative. The following PDC specifications are recommended when using SHALEDRL:

- Steel-body PDC Bit
- 5- blade bit
- ½ inch cutters
- Vortex nozzles
- Minimum flow rate of 50 gpm per inch of bit diameter

SHALEDRL F & B Formulations

- Density range 16.5 ppg (1.98 sg)
- Temperature tolerance 320°F (160°C)

Table 10 Basic SHALEDRL F&B Formulation

Additive	Description	Function	Typical Concentration, ppb
BARAZAN D PLUS	Xanthan biopolymer	Viscosifier	0.25-2.0
PAC-L	Polyanionic cellulose	Filtration control	1.0-2.0
FILTER-CHEK	Proprietary blend of anionic polymers	Filtration control	1.0-4.0
BDF-546	Blend of potassium silicate	Shale Stabilizer	3.0-7.0
CARBONOX	Powdered Leonardite	Filtration control	4.0-6.0
GEM GP	Blend of polyglycols	Shale Stabilizer	2.0-10.0
KOH	Potassium hydroxide	Alkalinity	0-0.5

Basic Maintenance for the SHALEDRL F & B System

Inhibition is the key to the system's performance. When capacity for inhibition is diminished, the rheological properties increase, causing a loss of efficiency.

Maintaining proper concentrations of the additives shown above reduces the surface area for a given solids content, allowing dilution to be more effective. This system will tolerate some fine solids but concentrations of sub-micron particles should be minimized.

Fluid Property / Operation	Recommended Treatment / Concentration
Dilution	Use pre-mixed mud for dilution when pit space and rig configuration permit. Monitor the dilution rate to maintain the desired properties. It is difficult to "catch up" if the dilution rate is too low. When adding water to the system all products should be maintained in the correct amounts to maintain inhibition of the system. Water additions must be measured accurately to ensure

Fluid Property / Operation	Recommended Treatment / Concentration
	the system is not undertreated.
Rheology	Use BARAZAN-D Plus to maintain viscosity, YP, gel strengths and Tau ϕ at the specified values. THERMA-THIN is an effective deflocculant.
MBT	The MBT should be maintained as low as possible. Rapid increase. A rapid increase in MBT value shows that inhibition has been lost. Increase dilution immediately and adjust chemical concentrations. The higher the MBT is allowed to climb, the harder it is to arrest the climb. Gradual increase. A gradual increase in MBT value shows that inadequate inhibition is present to maintain the system at the current ROP and solids control efficiency. Adjust chemical concentrations according to the potassium and silicate titrations, solids control efficiency, cuttings integrity and overall treatment schedule.
CO₂ Contamination	BARACOR 95 can be used in place of lime to treat out CO ₂ gas contamination.
Drilling Cement	Optimize shaker screen size before drilling cement. Treat with sodium bicarbonate or citric acid before drilling cement to reduce soluble calcium levels to < 600 mg/l. Discard severely contaminated mud / water before drilling ahead.
Filtration Control	Add FILTER-CHEK and PAC as needed.
Thermal Stability	BARACOR 95 is a free-radical scavenger and can be used to increase the system's thermal stability.

SHALEDRIIL H Fluid

SHALEDRIIL H is an environmentally friendly, fresh-water based fluid designed to combat the extreme temperatures of the Haynesville shale. It exhibits effective performance even in geothermal environments. The system can be used for land rigs that are set up for freshwater dilutions but do not have a tank for whole mud dilutions.

As with many high temperature fluids, it is important to remember that chemical depletion and sufficient water additions should be monitored and addressed. The need to replenish products that are being depleted can be minimized if the system is run properly.

- Density range 18.0+ ppg (2.15 sg)
- Temperature tolerance 400°F+ (204°C)

SHALE-DRIL H Formulation

Table 11 Basic SHALEDRIIL H Formulation

Additive	Description	Function	Typical Concentration, ppb
AQUAGEL GOLD SEAL	Premium Wyoming Bentonite	Viscosifier	8.0-10.0
THERMA-THIN	High Temp Polymeric Deflocculant	Deflocculant	2.0-4.0
AQUATONE-S	Wetting Surfactant	Wetting Agent	1.0-3.0
BDF-501	Modified Lignite	Filtration Control	4.0-6.0
BARO-TROL	Modified Shale Stabilizer	Shale Stabilizer	4.0-6.0
BDF-506	HPHT Fluid Loss	Filtration control	1.0-4.0
BARABUF	Alkalinity Agent	pH Control	1.0-3.0
Caustic Soda	Sodium hydroxide	Alkalinity	0.25-0.5

Basic Maintenance for the SHALEDRIIL H System

Inhibition is the key to the system's performance. When capacity for inhibition is diminished, the rheological properties increase, causing a loss of efficiency.

Maintaining proper concentrations of the additives shown above reduces the surface area for a given solids content, allowing dilution to be more effective. This system will tolerate some fine solids but concentrations of sub-micron particles should be minimized.

Maintain MBT, HPHT, LGS, PV and YP in the recommended ranges for the mud weight in use.

Fluid Property / Operation	Recommended Treatment / Concentration
Dilution	Use pre-mixed mud for dilution when pit space and rig configuration permit. Monitor the dilution rate to maintain the desired properties. It is difficult to "catch up" if the dilution rate is too low. When adding water to the system all products should be maintained in the correct amounts to maintain inhibition of the system. Water additions must be measured accurately to ensure the system is not undertreated. Temperature stability can be negatively impacted by failure to maintain proper product concentrations.
Rheology	Use BARAZAN-D Plus to maintain viscosity, YP, gel strengths and Tau ϕ at the specified

Fluid Property / Operation	Recommended Treatment / Concentration
	values. THERMA-THIN is an effective deflocculant.
Filtration Control	Add BDF-506 as needed.
CO₂ Contamination	BARABUF is used in place of lime to treat out CO ₂ gas contamination.
Drilling Cement	Optimize shaker screen size before drilling cement. Treat with sodium bicarbonate or citric acid before drilling cement to reduce soluble calcium levels to < 600 mg/l. Discard severely contaminated mud / water before drilling ahead.
Thermal Stability	BARABUF is a free-radical scavenger and can be used to increase the system's thermal stability.
Bridging / Plugging	Specific formulations of particle plugging material should be designed based on expected pore throat sizes and/or if in a tectonically stressed area the estimated fracture widths. STEELSEAL 50, BARACARB 25 and/or BARACARB 50 are thermally stable.

2. Conventional Water-Based Fluids

2.1. PAC / DEXTRID

PAC/DEXTRID systems are one of the most common non-inhibited, non-dispersed fluid types run. They provide wellbore stability and cuttings integrity through their chemistry of encapsulating polymers. As with any low solids system, good solids control is essential to maintain their low solids performance. Adequate shakers and minimized screen sizes will reduce the requirement for dilution, reducing cost and improving fluid performance.

Formulation

The following table provides guidelines for formulating PAC/DEXTRID systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Treat out calcium in make-up water with soda ash before adding AQUAGEL.

Table 12 Basic PAC/DEXTRID Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
AQUAGEL	Viscosifier in initial formulation	5-8 (14-23)
DEXTRID	Filtration control agent	4-6 (12-17)
PAC	Filtration control agent	1.5-4.0 (4-12)
Caustic soda/ Caustic potash	Alkalinity source	0.5-1.0 (1.4-3)
BAROID	Weighting agent	As needed
*BARAZAN PLUS	Viscosifier	0.25-1.0 (0.7-3)
*Soda ash	Make-up water hardness reducer	As needed
*KCl/NaCl	Reactive shale inhibitor	As needed
*THERMA-THIN	Deflocculant	As needed
*BARASCAV	Oxygen scavenger	As needed
*Lime	CO ₂ scavenger	As needed

Maintenance

Maintain the MBT at less than 20 lb/bbl (57 kg/m³) equivalent bentonite content.

2.2. CARBONOX / QUIK-THIN

Most freshwater systems can be converted to a CARBONOX/ QUIK-THIN system. This is not an appropriate system for saltwater, its performance in that base fluid will be poor.

Formulation

The following table provides guidelines for formulating CARBONOX/QUIK-THIN systems. Products are listed in order of addition. Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 13 Basic CARBONOX/QUIK-THIN Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
AQUAGEL	Viscosifier / Filtration control agent	10-35 (29-100)
QUIK-THIN	Thinner/ Filtration control agent up to 350 F (177 C)	4-12 (11-34)
Caustic soda	Alkalinity source	2-6 (6-17)
CARBONOX	Thinner/Filtration control agent	6-20 (17-57)
BAROID	Weighting agent	As needed
*BARAZAN PLUS BARAZAN D PLUS	Viscosifier up to 275 F (135 C)	0.25-1.5 (0.7-4)
*FILTER-CHEK	Filtration control agent	0.5-2.0 (1.4-6)
*Lime	Alkalinity source	0.25-1.0 (0.7-3)
*PAC-R	Filtration control agent	0.25-1.5 (0.7-4)
*PAC-L	Filtration control agent	0.25-1.5 (0.7-4)
*DEXTRID	Filtration control agent	4-6 (11-17)
*BARODENSE	Weighting agent	As needed
*POLYAC PLUS	Filtration control agent up to 400 F (204 C)	1-6 (3-17)

Pre-hydrate AQUAGEL and AQUAGEL GOLD SEAL in fresh water before using in brackish water.

Treat out any calcium/magnesium.

Maintenance

Increase the pH of makeup water to between 9.0 and 10.5 to precipitate the magnesium.

Add soda ash to treat out the calcium.

Add bentonite.

Add QUIK-THIN.

Add filtration control additives and supplemental viscosifiers.

Add caustic soda to maintain a pH of 9-12.0.

2.3. Gyp / QUIK-THIN

Formulation

The following table provides guidelines for formulating Gyp/QUIK-THIN systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 14 Basic Gyp/QUIK-THIN Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
AQUAGEL	Viscosifier/ Filter cake	10-20 (30-57)
QUIK-THIN	Deflocculant/ Fluid loss control agent	4-12 (11-34)
CARBONOX	Fluid loss control agent	4-20 (11-57)
Caustic soda	Alkalinity source	0.25-3.0 (0.7-9)
Gypsum	Calcium source	4-10 (11-29)
PAC	Fluid loss control agent	0.1-2.0 (0.3-6)
FILTER-CHEK	Fluid loss control agent	2-8 (6-23)
*BARO-TROL	Fluid loss control agent	4-8 (11-23)
BAROID	Weighting agent	As needed

Pre-hydrate AQUAGEL in fresh water.

Ensure that QUIK-THIN is added before caustic soda to prevent the flocculation of bentonite.

Breakover

To convert an existing system to a Gyp/QUIK-THIN system, follow these steps.

1. Dilute the mud to reduce the bentonite equivalent (MBT) to less than 15 lb/bbl (42.75 kg/m³).
2. Add QUIK-THIN.
3. Add caustic soda to adjust the pH to 9.5-10.0.
4. Add gypsum.
Severe flocculation may occur when gypsum is added.
5. Add PAC.
6. Add barite to increase weight as necessary.

Maintenance

Fluid Property / Operation	Recommended Treatment / Concentration
pH	Maintain the pH between 9.0 and 10.5
Calcium Level	Maintain calcium levels between 800 and 1400 mg/L. Calcium levels in excess of 1600 mg/L adversely affect rheology and HTHP fluid loss.
Excess Gypsum Level	Maintain excess gypsum levels at 2-6 lb/bbl (6-17 kg/m³). Excess gypsum, lb/bbl = $0.48 [V_m - (V_f F_w)]$ Excess gypsum, kg/m³ = $1.37 [V - (V_f F_w)]$ An approximation of excess gypsum can be obtained by: Excess gypsum, lb/bbl = $(V_m - V_f) / 2$ Excess gypsum, kg/m³ = $(V_m - V_f) 1.5$ Where: V_f is the versenate endpoint of the filtrate V_m is the versenate endpoint of the mud F_w is the water fraction

2.4. EZ-MUD

The base fluid can be freshwater, seawater, or brine. Add salt (as required) to increase salinity.

Formulation

The following table provides guidelines for formulating EZ-MUD systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 15 Basic EZ-MUD Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
Caustic soda/ Caustic potash	Alkalinity source (pH 9-10)	0.1-1.5 (0.3-4)
Soda ash	Calcium remover	As needed
AQUAGEL	Viscosifier Suspension agent	5-17.5 (14-50.0)
EZ-MUD EZ-MUD DP (Lower concentration than EZ-MUD liquid)	Shale stabilizer	0.5-3 (1.4-9) 0.2-1 (0.6-3)
FILTER-CHEK	Fluid loss control agent	0.2-3.5 (0.6-10.0)
PAC	Fluid loss control agent	0.2-3.5 (0.6-10.0)
BAROID	Weighting agent	As needed
BARAZAN PLUS	Viscosifier	0.1-1.0 (0.3-3)
*DEXTRID	Fluid loss control agent	As needed
*BARO-TROL	Fluid loss control agent	As needed
*ALDACIDE G	Biocide	As needed
*THERMA-THIN	Deflocculant	0.2-3.0 (0.6-9)

Maintenance

Treat out hardness with soda ash and caustic soda being careful not to increase the pH above 10. Pre-hydrate AQUAGEL and AQUAGEL GOLD SEAL before using.

Add EZ-MUD slowly through the hopper. A special shearing device may be helpful.

A viscosity hump will occur when the EZ-MUD is added. With shear, the viscosity should decrease as the system becomes deflocculated.

To obtain the same polymer concentration, use 1/3 as much EZ-MUD DP as liquid EZ-MUD.

NOTE: When building a low solids EZ-MUD system from scratch, the PHPA polymer will have a high affinity for exposed metal. Until enough drill solids have built up, screen blinding may occur. Very coarse screens may have to be run for the first few circulations until enough solids are built up to satisfy the PHPS demand for the metal.

2.5. ENVIRO-THIN

ENVIRO-THIN systems use an environmentally friendly iron lignosulfonate thinner to control rheological and filtration properties in fluid systems with a high solids content.

Formulation

The following table provides guidelines for formulating Low-pH ENVIRO-THIN systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 16 Basic Low-pH ENVIRO-THIN Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
AQUAGEL	Viscosifier	15-25 (43-71)
ENVIRO-THIN	Deflocculant	2-8 (6-23)
Caustic soda	Alkalinity source	As needed
CARBONOX	Fluid loss control agent	2-10 (6-29)
PAC	Fluid loss control agent	0.5-2.0 (1.4-6.0)
BAROID	Weighting agent	As needed
*EZ-MUD	Shale stabilizer	0.25-0.5 (0.7-1.4)
*BARAZAN PLUS	Viscosifier	0.5 (1.4)
*BARO-TROL	Fluid loss control agent	2-6 (6-17)
*BARASCAV	Oxygen scavenger	0.1-0.2 (0.3-0.6)
*Bicarbonate of soda	Hardness control agent	As needed
*THERMA-THIN	Deflocculant	0.5-1.0 (1.4-3)

Maintenance

Maintain the pH at 9.0 and above.

Pre-hydrate all AQUAGEL additions in fresh water.

Pre-hydrate CARBONOX and BARO-TROL additions in caustic water that has a pH of 10 or above.

Maintain total hardness 200 mg/L as calcium.

Use soda ash to treat out calcium to a level at or below 200 mg/l, except for cement contamination when bicarbonate of soda should be used.

Breakover

Most low-solids, non-dispersed (LSND) systems with a low-to-moderate pH range can be converted to an EZ-MUD system.

The conversion procedure is shown below.

1. Check pH, hardness, MBT volume, and low-gravity solids content and adjust the mud, if necessary.
The higher the solids and MBT levels, the longer and more severe the breakover hump will be.
2. Add the recommended concentration of EZ-MUD.
Extreme flocculation of the mud may occur, resulting in water separation. DO NOT add deflocculants at this time. The condition will subside after the EZ-MUD has been sheared.
3. Add FILTER-CHEK, PAC-R, or PAC-L, as required, for filtration control.
The system may become thin after adding a filtration control agent.
4. Add barite to increase mud weight, as required.

Maintenance

Fluid Property / Operation	Recommended Treatment / Concentration
EZ-MUD Concentration	Maintain approximately 0.5 lb/bbl (1.5 kg/m ³) of excess EZ-MUD in the filtrate as determined using the PHPA test.
pH	pH should not exceed 10. Maintain pH 7.5-9.5. Pre-solubilize all caustic materials and add them slowly to the active system. This will prevent the system from getting pH hot spots. Use citric acid treatments to lower the pH, when necessary. Other weak acids can be used to lower pH elevated by cement contamination.
Total Hardness	Maintain a total hardness of less than 200 mg/L for maximum EZ-MUD stability.
Drilling Cement	Use citric acid treatments to lower the pH, when necessary. Other weak acids can be used to lower pH elevated by cement contamination. If ammonia odor is detected while drilling cement, assume EZ-MUD content is zero.

2.6. Saturated Salt

Formulation

The following table provides guidelines for formulating saturated salt systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 17 Basic Saturated Salt Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
ZEOGEL	Viscosifier/Suspension agent	10-20 (29-58)
FILTER-CHEK	Filtration control agent	4-8 (12-23)
Salt (sodium chloride)	Chloride source	As needed
BAROID	Weighting agent	As needed
*AQUAGEL	Viscosifier	As needed
*DEXTRID	Filtration control agent	4-6 (12-17)
*PAC	Filtration control agent	0.25-0.5 (0.7-1.5)
*BARAZAN PLUS	Viscosifier	0.25-2.0 (0.7-6)
*ALDACIDE G	Biocide	As needed

Breakover

If the MBT is greater than 10 lb/bbl (29 kg/m³) equivalent bentonite, dump the system and rebuild. If the MBT is less than 10 lb/bbl (29 kg/m³), add salt, FILTER-CHEK, and ZEOGEL.

2.7. CARBONOX / AKTAFLO-S

Formulation

The following table provides guidelines for formulating CARBONOX/AKTAFLO-S systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 18 Basic CARBONOX/AKTAFLO-S Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
AQUAGEL	Viscosifier / Filtration control agent	8-20 (23-57)
CARBONOX	Thinner / Filtration control agent	10-30 (29-86)
*QUIK-THIN	Thinner up to 350 F (177 C)	2-6 (6-17)
Caustic soda	Alkalinity source	2-6 (6-17)
AKTAFLO-S	Surfactant	4-8 (11-23)
BAROID	Weighting agent	As needed
BARO-TROL	Filtration control agent	4-8 (11-23)
*PAC-L *PAC-R	Filtration control agent up to 300 F (149 C)	0.25-1.5 (0.7-4)
*Lime	Alkalinity source	0.25-1.0 (0.7-3)
*BARODENSE	Weighting agent	As needed

Maintenance

Maintain 1 lb/bbl AKTAFLO-S for every 4 lb/bbl (11.4 kg/m³) bentonite equivalent.

Maintain the pH at 9.5 to 10.5 with caustic soda.

2.8. THERMA-DRIL

THERMA-DRIL system was developed to meet the need for usage of a water-based mud system for high temperature wells. It has been found that the products developed for THERMA-DRIL can also be used to assist other systems to achieve thermal stability above 400 degrees F (200 degrees C).

System Capabilities

- High temperature stability
- Contaminant tolerance
- Can formulate temperature-stable non-dispersed polymer mud system
- Can be used in wide variety of systems for good shale stability
- Minimum dispersion of cuttings and clays
- Flexibility of use

Composition

The THERMA-DRIL system primarily consists of the following products.

1. THERMA-THIN
2. THERMA-CHEK
3. THERMA-VIS
4. AQUAGEL
5. BARANEX/BARABLOK
6. BARACOR 95 carbon dioxide scavenger
7. BARASCAV oxygen scavenger
8. Caustic soda or potassium hydroxide
9. Barite or other weighting agents

THERMA-THIN is a low molecular weight modified polyacrylate deflocculant used to reduce rheological properties of the system. THERMA-THIN is a very efficient deflocculant and is used in small concentrations. THERMA-THIN is most effective at pH values between 8.5 and 11.0. Unlike lignosulfonates which may break down above 320 degrees F, THERMA-THIN is stable to above 400 degrees F.

THERMA-CHEK is an acrylamide copolymer used to control API and HPHT fluid loss. THERMA-CHEK is not affected by salinity or moderate levels of calcium. At higher THERMA-CHEK concentrations, some increase in viscosity will result.

THERMA-VIS is a synthetic inorganic polymer used for viscosity and barite suspension in both fresh and salt water low density mud systems. It can be used as a partial or complete substitute for bentonite or sepiolite clays. THERMA-VIS will not cause high temperature gelation when left static for long periods at high bottomhole temperatures.

Caustic soda and/or potassium hydroxide are alkaline agents used to control the pH of the system. Either is used to maintain the system pH between 8.5 and 11.0.

AQUAGEL bentonite can be added to provide enhanced filtration control by supplying the basis of a filter cake. If seawater is used as the base fluid, then bentonite will have to be prehydrated in freshwater before adding to the system. Add THERMA-THIN to the bentonite mixture before adding it to the seawater.

BARACOR 95 is a highly active inhibitor which combats the effects of carbonate/carbon dioxide (CO₂) and hydrogen sulphide (H₂S) contamination. It has a high thermal stability and provides a pH buffer effect. BARACOR 95 has a pH around 12.

Although the primary components of a THERMA-DRIL system are all stable to above 400 degrees F, for lower temperature wells it may be more cost effective to extend the usage of relatively low temperature polymers such as BARAZAN D+ or PAC L. By adding BARASCAV oxygen scavenger, the temperature limit of these products can be raised by up to 325 degrees F.

Formulation

The following table provides guidelines for formulating THERMA-DRIL systems.

Products are listed in order of addition.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 19 Basic THERMA-DRIL Formulation

Additive	Function	Typical concentrations, lb/bbl
Soda ash	Hardness remover	0-0.2
AQUAGEL	Viscosifier	5-8
THERMA-THIN	Deflocculant	3-5
Caustic soda	Alkalinity source	pH = 8.5-11.0
THERMA-CHEK	Fluid loss control agent	2-6
BAROID	Weighting agent	As needed
BARASCAV D	Oxygen scavenger	0.25-1
THERMA-VIS	Viscosifier	0-3
BARACOR 95	CO ₂ scavenger	0-0.8

A THERMA-DRIL system can be formulated with freshwater, seawater, or various salt types. The product concentrations will depend upon:

- Mud density
- Bottomhole temperature
- Base fluid type
- Required mud properties

To formulate THERMA-DRIL in fresh water; add 5 to 8 lb/bbl AQUAGEL or AQUAGEL GOLD SEAL bentonite, depending on the anticipated mud weight. Add 2 to 6 lb/bbl THERMA-CHEK for filtration control, depending on mud weight and bottom hole temperature. Add caustic soda to adjust the pH to 8.5 – 11.0. Add barite to the required density with small additions of THERMA-THIN if necessary for rheology control. Formulations for higher temperature should include increased concentrations of THERMA-THIN and THERMA-CHEK with lower concentrations of bentonite. High temperatures cause an increased yield of bentonite and this can result in excessive rheology once the fluid has seen the high bottomhole temperature.

When seawater is used as the original makeup water, different formulation techniques are required. First pretreat the seawater to pH 9.5 -10.0 with either caustic soda or potassium hydroxide (0.75 to 1 lb/bbl) to remove the magnesium ions. Prehydrate 25 to 35 lb/bbl AQUAGEL in fresh water and add 1 to 2 lb/bbl of THERMA-THIN to prevent the bentonite from flocculating when it is added to the seawater. Then blend the premixed bentonite with the treated seawater to give a bentonite concentration of 6 – 12 lb/bbl. After drilling has begun, seawater additions should be held to a minimum and will probably not be required if all seawater is pretreated. [Note that THERMA-VIS is not compatible with any salt, it is for fresh water-based systems only.]

The pH of a THERMA-DRIL system is usually controlled in the range of 8.5 to 11.0. Add BARASCAV oxygen scavenger from a chemical barrel directly to the pump suction. THERMA-THIN is added to the active system only if the viscosity increase is due to clay solids build up when adding barite. If high viscosity is caused by polymers, THERMA-THIN will probably have little effect as it works by deflocculating clay solids to lower the overall viscosity. Add 0.5 to 1.0 lb/bbl BARACOR 95 carbon dioxide scavenger if required.

KCl and NaCl salt brines can be used to formulate a THERMA-DRIL system. A new 10% KCl mud containing 1.5 lb/bbl soda ash, 14 lb/bbl AQUAGEL, 27 lb/bbl KCl, 0.5 lb/bbl caustic soda and 0.5 lb/bbl PAC-R filtration control polymer was converted to a 400 degree F THERMA-DRIL system by the addition of 4 lb/bbl THERMA-THIN and 8 lb/bbl THERMA-CHEK.

If formulating a freshwater THERMA-DRIL system, the bentonite concentration and THERMA-THIN concentration can be reduced and at least partially replaced by 1 to 3 lb/bbl THERMA-VIS. However, some prehydrated high quality bentonite is usually necessary for HPHT filtration control. THERMA-VIS will not contribute to a filter cake or HPHT control.

BARANEX or BARABLOK 400 can be used as the primary filtration control additive up to 380 degree F. Above this temperature, THERMA-CHEK should be used exclusively.

THERMAFLOW-500 (0.5-2 lb/bbl) can be added for gel strength control in addition to or instead of THERMA-THIN.

Maintenance

1. Add 6-12 lb/bbl bentonite only in prehydrated form. Use THERMA-THIN to protect the bentonite from flocculation.
2. Add THERMA-THIN to control viscosity increase resulting from clay solids, or when adding barite to increase the density, or when treating for contamination.
3. Avoid adding any lignosulfonates to the system. Unless stabilized with chrome, lignosulfonates are not stable above 320 degrees F. The presence of lignosulfonate or residual lignosulfonate can impede the performance of THERMA-THIN.
4. Prehydrate caustic soda additions and add directly to the pits to maintain the pH at 8.5 – 11.0. Do not overtrear.
5. If pit arrangement allows, avoid making any additions to the suction pit since all mud should be treated before the suction pit.
6. Use all available solids control equipment to maintain the lowest solids content possible and reduce the amount of dilution.
7. Control the API and HPHT filtrate with BARANEX, DURENEX PLUS or THERMA-CHEK. Some viscosity increase will be seen, but it does not require or respond to additions of THERMA-THIN. Additions of THERMA-CHEK will vary from 2 - 6 lb/bbl in fresh water muds to 6 – 8 lb/bbl in highly saline muds.
8. Do not treat for high funnel viscosity. Rely only on the YP, tau₀ and gel strengths for any mud treatments. Maintain tau₀ at ½ to ¾ the hole size while drilling. Keep the PV as low as possible but

- be aware that THERMA-CHEK additions will cause the funnel viscosity to increase. Treat for high PV caused by solids and not from polymer additions.
9. Carbonate contamination should be monitored and treated with BARACOR 95 and/or a calculated amount of lime, gypsum or calcium chloride. Over treatment with soda ash can generate a carbonate problem. Use the Mf/Pf ratio as a guide to the presence of CO₂/carbonate contamination. A Mf/Pf ratio of 3:1 or higher usually indicates contamination. The presence of CO₂ should be confirmed with either a Garrett Gas Train or P1/P2 test kit.
 10. Pilot testing a sample of mud with lime to measure change in rheology can be a simple and effective indicator of carbonate contamination. At least 0.5 lb/bbl should be added and the mixing time must be at least 40 minutes. Shorter mixing times can give erroneous results.
 11. Pretreat the THERMA-DRIL system with 0.5 – 1.0 lb/bbl THERMA-THIN if cement is drilled. Less mud contamination due to cement will be realized and less contaminated mud will have to be dumped.
 12. If the THERMA-DRIL system is to be weighted, pretreat the system with THERMATHIN prior to or during weighting up. More solids tolerance will be exhibited without appreciable viscosity increases. Typically, 0.5 – 1.0 lb/bbl THERMA-THIN will be sufficient.
 13. In high weight systems, solids crowding and the presence of minimal free water will cause high rheology at high bottomhole temperatures.
 14. Some dilution will always be needed in high temperature muds to replace water lost to dehydration and evaporation.
 15. HPHT filtration tests should be undertaken at the anticipated or actual bottomhole temperatures.
 16. It is essential to monitor the condition of the mud whenever bottoms-up is circulated. A complete mud check should be made to ensure the mud is remaining stable under static conditions at actual high bottomhole temperatures.
 17. Inspect rig to eliminate all mechanical sources of oxygen. Run hoppers only when necessary. Treat with small additions of BARASCAV oxygen scavenger whenever the system is circulating.
 18. Optimum pH is 8.5 – 11.0.
 19. Maintain total hardness between 200 – 400 mg/L. Less than 200 mg/L invites a carbonate problem. More than 800 mg/L reduces efficiency of the THERMA-CHEK, especially at high pH. The pH is critical in the presence of calcium.
 20. Avoid an MBT greater than 12-15 lb/bbl equivalent. Run MBT daily in order to determine exactly how much colloidal clay is in the system.
 21. THERMA-CHEK additions should be added at slow steady rates to ensure equal distribution throughout the system. Localized high viscosity due to over-treatments will take a long time to spread out. High shear mixers are helpful. THERMA-CHEK is best added in premix form, preferably without bentonite.
 22. THERMA-CHEK is most efficient when the mud temperature is above 250 degrees F.
 23. At very high mud weights, the YP should be kept low, e.g. in the 4 – 7 lb/100ft² range. The PV should be kept as low as possible with good solids control equipment and treatment with THERMA-THIN to prevent excessive ECD, especially after trips, which may result in induced lost circulation. Gels of 3/5 – 6/10 are desirable at high to very high mud weights.
 24. High temperature gellation of static mud is typically caused by bentonite dehydration and flocculation, resulting in excessive gel strengths and ultimate collapse of the gel structure. This allows rapid settling out of the barite. This may sometimes be seen as high mud weight in the bottoms-up mud sample after trips. Optimum gel strengths should be relatively low and flat but need to prevent barite settling.
 25. THERMA-VIS inorganic viscosifier is stable to above 600 degrees F. It needs to be prehydrated and sheared for as long as possible for optimum yield. THERMA-VIS mixed in fresh water produces a solids free solution. It will not function in water with more than 5000 mg/l chlorides.

Troubleshooting Guide

Problems	Indications	Solutions
Poor hole cleaning	Low viscosity	Additions of prehydrated bentonite, THERMA-VIS or THERMA-CHEK in low weight muds. Use THERMA-CHEK in high weight muds.
Poor suspension	Low gel strengths	Small additions of prehydrated bentonite or THERMA-VIS.
Unstable borehole	Sloughing shale Torque and drag	Increase THERMA-CHEK to lower fluid loss. Add 2-4 lb/bbl Soltex. Use bentonite sweeps. Increase mud weight.
High Viscosity High solids	Increasing MBT High gel strengths	Run solids control equipment. Dilute.
Drilling cement	High pH High viscosity High calcium	Pretreat system with 0.5 – 1.0 lb/bbl THERMA-THIN. Add THERMA-THIN. Dump contaminated mud. Add bicarb to lower calcium.
High viscosity on trips	Mud running off shakers Cuttings Dispersion	If possible, dump badly dehydrated mud. Pretreat with THERMA-THIN (0.5 – 1.0 lb/bbl). Check for carbonate contamination. Dilution.
High drag	High fluid loss	Add THERMA-CHEK. Add BARASCAV to remove oxygen.

2.9. BARASILC

This system can be formulated in either fresh water or monovalent brines. The use of Potassium Chloride brine in a minimum 6% concentration is strongly recommended to maximize the inhibitive properties of BARASILC.

Formulation

The following table provides guidelines for formulating BARASILC systems. Products are listed in order of addition. Products not listed on the formulation table should not be added to the BARASILC system without prior technical approval.

Contingency products are denoted by an asterisk (*); they can be used with the primary products to obtain properties needed for specific situations.

Table 20 Basic BARASILC Formulation

Additive	Function	Typical concentrations, lb/bbl (kg/m ³)
Soda ash	Calcium remover	As needed
Caustic soda/ Caustic potash	Alkalinity source	As needed
BARASIL-S	Formation stabilizer	40-80 (114-228)
DEXTRID	Fluid loss control agent	2-8 (6-23)
PAC	Fluid loss control agent	0.5-4 (1.4-11)
FILTER-CHEK	Fluid loss control agent	2-8 (6-23)
BARAZAN PLUS	Viscosifier	0.2-2.5 (0.6-7)
BAROID	Weighting agent	As needed
*BARACOR 95	CO ₂ scavenger/buffer	0.5-4 (1.4-11)
*BARA-DEFOAM HP	Defoamer	As needed
*BARASCAV D	Oxygen scavenger	0.2-1 (0.6-3)

Maintenance

Treat out hardness in base fluid with soda ash before addition of polymers or BARASIL-S.

Base fluid pH should be between 9.5 and 10.5.

Once BARASILC has been added, maintain pH between 11.5 and 12.5 or silicate will deplete.

Ensure that all lines and tanks are clean and free of divalent cation brines or mud before mixing brines.

Shear polymers thoroughly to obtain optimum yield.

Note: BARASIL-S is an alkali solution which can cause burns to the skin and eyes. Wear appropriate protective gear and avoid breathing mists of the solution when working with BARASIL-S. The active mud should be handled as any high- pH water-based mud system.

Fluid Property / Operation	Recommended Treatment / Concentration
Silicate Level	Maintain SiO ₂ concentration at 40,000 to 50,000 mg/L. Silicate depletion rates can be high. Cement, gypsum, anhydrite, lime, formation surfaces, acid gases, and formation water (containing divalent cations) can severely deplete silicate levels.
pH	Normal operating pH range for BARASILC systems is between 11.5 and 12.5. If pH falls below 11.5, the silicate concentration may be severely depleted. Add BARASIL-S to restore silicate content and pH to appropriate levels.
Dilution	Thin with whole mud dilution or with base fluid.
HPHT Filtration	Nitrogen gas should be used for running HPHT filtration tests. CO ₂ gas will cause silicate depletion and give a waxy filtrate.
Fatty Acid / Acidic Additives	Lubricants or other products containing fatty acid derivatives should not be added to the BARASILC system. Severe foaming can result. Addition of acidic chemicals should be avoided. Acids will cause silicate depletion and mud gelation.

Note that the BARASIL-S is unstable below about 45°F. If the drums reach temperatures below 45°F the silicate will separate from the liquor and become unusable. If this occurs it cannot be re-mixed back into solution and

must be disposed of properly. It is vitally important to maintain the drums above 45°F (7°C). There are currently no liquid lubricants which perform well in BARASILC. The use of solid products like STICK-LESS can provide good lubricant support in moderately deviated holes.

Invert Emulsion Fluids

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1. High-Performance Invert Emulsion Fluids (Oil- and Synthetic-Based Systems)

Overview

For many decades, oil and gas producers have relied on invert oil- and synthetic-based drilling fluid systems as key contributors to trouble-free drilling of high-quality wellbores. These wellbores are in a wide range of subsurface geological formations of interest.

The ability of invert emulsion fluids (IEF) to perform reliably when drilling both porous sandstone and limestones and impermeable shales and clays makes them good fluids for many applications, including complex mineralogical and depositional settings and challenging temperature and pressure regimes.

High-performance invert emulsion fluids (IEF) are free of organophilic clay organophilic lignite and other so called black powder products. The continuous or external phase is a non-aqueous fluid (NAF) such as ester, isomerized olefin (IO), linear alpha olefin (LAO), paraffin, mineral oil, diesel or any combination of these.

The properties of high performance inverts are influenced by the following:

- Oil/Water ratio
- Product concentrations
- Solids content
- Downhole temperature and pressure

High-performance clay-free IEF technology provides significant benefits over conventional IEF systems. Baroid's clay-free IEF design and performance remain unduplicated in the industry.

Traditional organophilic clay, used for controlling rheological properties, and organophilic lignite, used for filtration control, have been replaced with innovative products resulting from advances in fatty acid, tall oil and co-polymer chemistry.

This change resulted in major improvements in performance and has been adopted in a wide range of base fluids. The benefits of high-performance IEF technology are shown below:

- Fragile gel strengths and generally lower ECDs that significantly reduce downhole mud losses and mitigate barite sag
- Minimization of pressure spikes on pipe connections, also reducing down-hole mud losses.
- Superior return permeability with low fluid invasion
- Stable mud properties over a wide temperature range (flat rheology)
- Cold temperature (deepwater riser) rheology can be modified independently of downhole rheology
- Increased tolerance to solids and water contamination
- Fewer product and inventory requirements than a conventional IEF, so that deck space availability and logistical considerations are improved
- Faster treatment response times eliminate need for multiple circulations to increase the viscosity and help prevent over treatment
- Helps decrease overall well construction costs

High-Performance IEF System Classifications

Table 1 High-performance IEF Systems

System	Application & Base Oil
ACCOLADE	Baroid's original ester / isomerized olefin blend provides the highest level of environmental performance along with superior rheological properties. ACCOLADE can be used up to 325°F (163°C).
ENCORE	Developed utilizing the ACCOLADE fluid technology products in 100% isomerized olefin (IO) to meet Gulf of Mexico (GOM) environmental standards. Can be used in most environments, including HPHT capabilities 500°F+ (260°C+).
INNOVERT	Developed utilizing the ACCOLADE fluid technology products in paraffin or mineral oils. Can be used in most environments, including HPHT capabilities of 475°F+ (246°C+).
INTEGRADE	Developed utilizing the ACCOLADE fluid technology products in diesel-based system. Can be used in most environments, including HPHT capabilities of 475°F+ (246°C+).

Products

Baroid's high-performance IEF technology utilizes a unique and proprietary chemical package. To maintain superior performance and quality control no unapproved substitutions are to be made and only the additives below should be used.

No third-party product should be added without adequate laboratory testing, as it may adversely affect environmental compliance and the overall performance of the fluid.

No organophilic clay should be used in these systems either in the lab or in the field. Laboratory testing indicates that a high organophilic clay concentration adversely affects the unique gel structure of the fluid.

Thinners should be added with caution. Overtreatment can negatively impact the gel structure of these fluids. Pilot test prior to use.

Table 2 High Performance IEF: Primary Additives

Additive	Function	SG	PPG
ADAPTA	425°F (218°C) filtration agent	1.03	8.60
BAROID	Weighting agent	4.20	35.05
Calcium Chloride	Salinity source	2.00	16.69
LE SUPERMUL	Primary emulsifier for ACCOLAD and ENCORE systems.	0.91	7.62
RHEMOD L	Viscosifier / Suspension agent	0.96	8.01

Lime should be added only as required for specific products (see Hydrolysis section).

Table 3 Approved Secondary Additives

Additive	Function	SG	PPG
ATC	Thinner	1.03	8.60

OMC 42	Thinner	0.92	7.68
OMC 3	Thinner		
OMC 2	Thinner	0.89	7.43
COLDTROL	Cold temperature thinner	0.95	7.93
EZ MUL (All)	Primary emulsifier for INNOVERT system		
FORTI-MUL	Primary Emulsifier for INTEGRADE system		
PERFORMUL	Emulsifier / Oil Wetting Agent		
FACTANT	Surfactant / secondary emulsifier	0.96	8.01
Lime	Alkalinity for non-ester based fluids	2.20	18.36
DRILTREAT	Oil wetting agent for heavy fluids	1.00	8.35
DEEP-TREAT	Oil Wetting Agent		
TEMPERUS	Temporary viscosifier for all oils	0.99	8.26
VIS-PLUS	Temporary viscosifier for non-ester fluids	0.85	7.09
TAU-MOD	Rheology Modifier		
BDF-489	Rheology Modifier		
BDF-566	Rheology Modifier		
BDF-568	Rheology Modifier		
BDF-570	Rheology Modifier		
LIQUITONE	250°F (121°C) filtration agent / viscosifier	0.98	8.18
ADAPTA 450	450°F (232°C) filtration agent	1.03	8.60
BDF-513	425°F (218°C) filtration agent for low solvency oils		
BDF-454	550°F (288°C) filtration agent		
BAROLIFT	Hole sweeping agent	0.90	7.51
SWEEPWATE	Sweep weighting agent	4.00	33.38
BARACARB	LCM / simulated drill solids	2.70	22.53
BAROFIBRE O	LCM / oil wet fibrous material		
STEELSEAL	Loss circulation material	1.70	14.19
HYDRO-PLUG	Loss circulation material (Severe)	2.00	16.69

Excess lime calculations should not be considered in ester-containing fluids.

1.1. ACCOLADE System

Formulation

The ACCOLADE system uses a blend of isomerized olefin (IO) and a proprietary biodegradable ester. This system offers the best environmental performance.

The ACCOLADE system is stable up to 325°F (163°C).

Excess lime calculations should not be considered in ester-containing fluids. Lime should be added only as required for specific products (see Hydrolysis section).

Table 4 Typical ACCOLADE Formulations

Products	9.0-12.0 ppg (1.08 -1.44sg)	12.0-14.0 ppg (1.44-1.68sg)	14.0-16.0 ppg (1.68-1.92sg)	16.0 + ppg (1.92sg)
Oil/Water ratio	65/35 to 70/30	70/30 to 75/25	70/30 to 80/20	80/20 to 90/10
CaCl ₂ , ppm	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000
ACCOLADE BASE	As needed	As needed	As needed	As needed
LE SUPERMUL, lb (kg/m ³)	6-10 (17-29)	8-12 (20-34)	10-14 (23-40)	12-14 (26-46)
FACTANT, lb (kg/m ³)*	As needed	As needed	As needed	0-5 (0-14)
DRILTREAT, lb (kg/m ³)			As needed	0-4 (0-11)
Lime, lb (kg/m ³)	*	*	*	*
ADAPTA, lb (kg/ m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.5-4 (1.4-11)	1-4 (3-11)
RHEMOD L, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.25-2 (0.7-6)	0.25-2 (0.7-6)
BAROID, lb (kg/m ³)	As needed	As needed	As needed	As needed
BARACARB 5, lb (kg/m ³)	10-25 (29-71)	5-25 (14-71)	As needed	As needed
TEMPERUS (kg/m ³)	As needed	As needed	As needed	As needed
VIS-PLUS*	As needed	As needed	As needed	As needed
OMC 42, lb (kg/m ³)			0 - 0.5 (0-1.4)	0 - 0.5 (0-1.4)

*At the mixing plant, lime should only be added with no more than 1.0 ppb (3.0 kg/m³) total product concentration unless FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used.

If FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used, lime is to be added in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT or VIS-PLUS additives.

Maintenance

Monitor closely to maintain the minimum concentrations of emulsifier and other products. If the minimum product concentrations are not maintained, properties and performance can be affected dramatically.

No third-party products should be added without approval. These products may adversely affect environmental compliance and the overall performance of the fluid.

Most product additions do not require multiple circulations to take effect and many products react immediately. All product additions should be pilot tested before treating the system.

Fluid Property	Recommended Treatment / Concentration
Building New Mud	When building an ACCOLADE system from scratch, BARACARB bridging agent is a vital addition to the system. The addition of BARACARB bridging agent creates the necessary surface area for the viscosifier interaction. If building from a base stock of seasoned fluid (generally 5%-20%), BARACARB bridging agent additions can usually be eliminated. Maintain an accurate product concentration including fluid shipped to the rig.
Oil / Water Ratio	Compared to conventional IEFs, the non-aqueous to water (OWR) ratio of ACCOLADE fluid is generally run 5% to 10% lower to achieve its unique properties.
Water Phase Salinity	Do not saturate the water phase with CaCl ₂ . Emulsion instability and water wetting of solids may occur.
Rheology	FANN 75 tests reflecting actual well temperatures and pressures should be performed during critical operations or when significant changes have occurred in the fluid. Measure 10 second, 10 minute and 30 minute gel strengths. The gel strengths should be high but not excessively progressive between the 10 minute and 30 minute gel strengths.
Emulsion Stability	Use emulsifiers to increase the stability of the emulsion of ACCOLADE fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the electrical stability measures <220 volts or when water is present in the HPHT filtrate. Refer to the table below for emulsifying products.
CO₂ / H₂S	ACCOLADE fluids are run with zero to very small concentrations of excess lime. Treatment of acid gas influx requires a different approach from other invert systems. CO₂ : No treatment is necessary. Lab testing confirms no detrimental effect on the ACCOLADE system. H₂S : Treatment with SOURSCAV hydrogen sulfide scavenger is recommended. NO-SULF hydrogen sulfide scavenger is recommended in ACCOLADE fluid only during zero discharge operations.
HPHT Filtration	Maintain an all-oil HPHT filtrate.
Electrical Stability	Maintain electrical stability above 220 volts.
Commercial Clay / Lignite	Do not add organophilic clay or organophilic lignite to the system.
Weighting Up	Do not add weighting agents when adding water. Add LE SUPERMUL emulsifier or small amounts of DRILTREAT oil wetting agent slowly as weighting agents are added to help oil-wet the additional solids.
Solids Control	Optimize solids control equipment to maintain 2 - 5% low gravity solids in the fluid. Inadequate rheology may occur if low- gravity solids drop below 2%.
Excess Lime	Do not maintain the system based on excess lime calculations. Lime is added to the system <u>only</u> when FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used. VIS-PLUS should be avoided if possible.

ACCOLADE Emulsifiers

Table 5 ACCOLADE Emulsifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
LE SUPERMUL	Primary emulsifier in ACCOLADE. Reduces HPHT filtrate and can impact rheological properties.	Polyaminated fatty acid	0.913	1-3.5 (3-10)
FACTANT*	Secondary concentrated emulsifier for ACCOLADE usually added when temperatures are >275°F (135°C). Improves rheological properties and reduces HPHT filtrate.	Concentrated tall oil derivative	0.96	0-4 (0-11)
DRILTREAT	Improves oil wetting of solids; helps reduces the viscosity when large quantities of solids (barite or other) have been incorporated.	Lecithin liquid dispersion	1	0.25-4 (.7-11)

* If FACTANT additive is used, lime is to be added in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT filtration control agent/emulsifier.

Excess lime calculations should not be considered in ester-containing fluids. Lime should be added only as required for specific products.

ACCOLADE Viscosifiers / Suspension Agents

Use RHEMOD L, LE SUPERMUL, or TAU MOD additives to increase ACCOLADE rheological properties.

TEMPERUS additive can be used for temporary viscosity both at the plant and at the rigsite when bottomhole temperatures do not exceed 150°F (66°C).

VIS-PLUS suspension agent should not be used in ACCOLADE fluid at the rigsite.

Table 6 ACCOLADE Viscosifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
RHEMOD L	Develops viscosity and suspension properties	Modified fatty acid	0.96	0.5-2 (1.4-6)
TEMPERUS	Temporary viscosifier for <150°F (66°C)	Modified fatty acid	0.99	0.1-1.5 (.29-4.3)
VIS-PLUS*	Temporary viscosifier for <175°F (79°C)	Carboxylic acid	0.85	0.1-3 (0.29-9)
TAU MOD	TAU MOD extends RHEMOD L suspension performance in low solids systems	Amorphous / Fibrous material	2.6	05-5.0 (1.4-21.7)
BARACARB	When solids concentrations are low, BARACARB can be added to provide the necessary surface area for the viscosifier interaction	Sized calcium carbonate	2.65	5-25 (14-71)
LIQUITONE	Improves filtration properties and provides gains in rheological properties when bottomhole temperatures are <250°F	Copolymer aqueous	0.98	0.5-3 (1.4-9)

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
	(121°C)	dispersion		

* VIS-PLUS suspension agent should only be used at the mixing plant and not at the rigsite for temporary viscosity. If VIS-PLUS additive is used, lime is to be added in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of VIS- PLUS suspension agent.

ACCOLADE Thinners

To thin the ACCOLADE system, add ACCOLADE base fluid to the mud or treat the mud with approved thinners. Thinners can be used alone or in combination to control viscosity. In most cases blends have proven to be more effective than the individual products.

Example: A blend of 1.5 ppb (4 kg/m³) COLDTROL thinner and 0.25 ppb (0.7 kg/m³) OMC 42 oil mud conditioner can be as effective as 3.0 ppb (9 kg/m³) ATC thinner at lowering both the 40°F (4°C) and 120°F (49°C) rheological properties and is generally more effective than 6.0 ppb (17 kg/m³) COLDTROL thinner at lowering the 40°F (4°C) rheological properties.

Table 7 ACCOLADE Thinners

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ACCOLADE Base	Reduces viscosity and solids content	Ester and isomerized olefin blend	0.82	3-10%
OMC 42	Reduces rheological properties	Polymide surfactant	0.92	0.5-2 (1.4-6)
ATC	Effective at lowering both the 40°F (4°C) properties and the 120°F (49°C) properties	Modified fatty acid ester	1.03	0.25-2 (0.7-6) Do not exceed 2.0 ppb total concentration
COLDTROL	Effective in lowering 40°F (4°C) rheological properties without significantly lowering properties at 120°F (49°C)	Alcohol derivative	0.95	0.1-.5 (.29-1.4) Do not exceed 2.0 ppb total concentration
OMC 2	Reduces rheological properties. Will not lower the 40°F (4°C) properties without additions of COLDTROL	Oligomeric fatty acid	0.89	0.1-1 (.29-3)

ACCOLADE Filtration Control Agents

Table 8 ACCOLADE Filtration Control

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ADAPTA	425°F (218°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)
LIQUITONE	250°F (121°C) Filtration control agent	Copolymer aqueous dispersion	0.98	0-2 (0-6)
FACTANT*	Reduces HPHT filtrate	Concentrated tall oil derivative	0.96	0-4 (0-11)

* If FACTANT additive is used, lime is to be added in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT filtration control agent.

Excess lime calculations should not be considered in ester-containing fluids. Lime should be added only as required for specific products.

Hydrolysis Prevention

Hydrolysis is only a concern when ester is used in the base fluid. Mineral oils, diesel, paraffin's, LAOs and IOs are not susceptible to hydrolysis.

Baroid's ACCOLADE fluid utilizes a blend of IO and the most environmentally responsible ester currently available to the drilling fluids industry. Baroid's ester component is one of the many reasons ACCOLADE fluid has the largest margin of compliance in regards to GOM sediment toxicity testing. The tradeoff to achieve this level of biodegradability is coping with the potential for ester hydrolysis. Baroid has engineered ACCOLADE fluid to negate problematic ester hydrolysis.

Hydrolysis only occurs under certain conditions. Ester hydrolysis is a function of time, temperature and hydroxyl ion. When temperatures approaching 300°F+ (149°C+) are sustained and sufficient amounts of excess hydroxyl ion are present, ester hydrolysis can occur. The source of the hydroxyl ion is primarily from calcium hydroxide (lime) and not calcium silicate (cement). ACCOLADE fluid should not be run by excess lime calculations. Any lime additions should be made with caution. Only 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent should be added.

Hydrolysis breaks down the ester component into carboxylic acid and alcohol. The alcohol is 2-ethyl 1-hexanol and has been used under other conditions as a water-based mud defoamer. If ester hydrolysis were to occur, the alcohol produced is considered stable, and the accompanying smell would require adequate ventilation to evacuate the alcohol. The parent carboxylic acid left in the drilling fluid after hydrolysis can interact with the free calcium to create a soap. This soap inherently increases the viscosity of the drilling fluid. In the event of hydrolysis, environmental testing should be performed to help ensure compliance.

The larger portion of ACCOLADE system base fluid, IO base oil, is not subject to hydrolysis regardless of the conditions. This measure helps ensure there will never be a complete breakdown of the system. Baroid does not recommend ACCOLADE fluid for HPHT applications. The high-performance all-IO ENCORE system or an alternative base fluid is recommended for HPHT applications.

1.2. ENCORE System

Formulations

The ENCORE system consists primarily of isomerized olefin (IO) base oil. This system is designed to have acceptable environmental compliance for the Gulf of Mexico. This system can be used in most drilling environments and is stable to 500°F+ (260°C+).

Table 9 ENCORE Formulations

Products	9.0-12.0 ppg (1.08 -1.44sg)	12.0-14.0 ppg (1.44-1.68sg)	14.0-16.0 ppg (1.68-1.92sg)	16.0 + ppg (1.92sg)
Oil/Water ratio	65/35 to 70/30	70/30 to 75/25	70/30 to 80/20	80/20 to 90/10
CaCl ₂ , ppm	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000
ENCORE BASE	As needed	As needed	As needed	As needed
LE SUPERMUL, lb (kg/m ³)	6-10 (17-29)	8-12 (20-34)	10-14 (23-40)	12-14 (26-46)
FACTANT, lb (kg/m ³) *	As needed	As needed	As needed	0-5 (0-14)
DRILTREAT, lb (kg/m ³)			As needed	0-4 (0-11)
Lime, lb (kg/m ³)	*	*	*	*
ADAPTA, lb (kg/ m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	1-4 (3-11)
RHEMOD L, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.25-2 (0.7-6)	0.25-2 (0.7-6)
BAROID, lb (kg/m ³)	As needed	As needed	As needed	As needed
BARACARB 5, lb (kg/m ³)	10-25 (29-71)	5-25 (14-71)	As needed	As needed
TEMPERUS (kg/m ³)	As needed	As needed	As needed	As needed
VIS-PLUS	As needed	As needed	As needed	As needed
OMC 42, lb (kg/m ³)			0 - 0.5 (0-1.4)	0 - 0.5 (0-1.4)
TAO MOD	0.5-4 (1.4 – 7.6)	0.5-4 (1.4 – 7.6)	0.5-4 (1.4 – 7.6)	0.5-4 (1.4 – 7.6)

* FACTANT and VIS-PLUS additives have lime requirements. When FACTANT or VIS-PLUS additives are used, add lime in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT or VIS-PLUS additives.

Maintenance

Fluid Property	Recommended Treatment / Concentration
Building New Mud	When building an ENCORE system from scratch, BARACARB bridging agent is a vital addition to the system. The addition of BARACARB bridging agent creates the necessary surface area for the viscosifier interaction. If building from a base stock of seasoned fluid (generally 5%-20%), BARACARB bridging agent additions can usually be eliminated. Maintain an accurate product concentration including fluid shipped to the rig.
Oil / Water Ratio	Compared to conventional IEFs, the non-aqueous to water (OWR) ratio of ENCORE fluid is generally run 5% to 10% lower to achieve its unique properties.
Water Phase Salinity	Do not saturate the water phase with CaCl_2 . Emulsion instability and water wetting of solids may occur.
Rheology	FANN 75 tests reflecting actual well temperatures and pressures should be performed during critical operations or when significant changes have occurred in the fluid. Measure 10 second, 10 minute and 30 minute gel strengths. The gel strengths should be high but not excessively progressive between the 10 minute and 30 minute gel strengths.
Emulsion Stability	Use emulsifiers to increase the stability of the emulsion of ENCORE fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the electrical stability measures <220 volts or when water is present in the HPHT filtrate. Refer to the table below for emulsifying products.
CO₂ / H₂S	ENCORE fluids are run with zero to very small concentrations of excess lime. Treatment of acid gas influx requires a different approach from other invert systems. CO₂: No treatment is necessary. Lab testing confirms no detrimental effect on the ENCORE system. H₂S: Treatment with SOURSCAV hydrogen sulfide scavenger is recommended. NO-SULF hydrogen sulfide scavenger is recommended in ENCORE fluid only during zero discharge operations.
HPHT Filtration	Maintain an all-oil HPHT filtrate.
Electrical Stability	Maintain electrical stability above 220 volts.
Commercial Clay / Lignite	Do not add organophilic clay or organophilic lignite to the system.
Weighting Up	Do not add weighting agents when adding water. Add LE SUPERMUL emulsifier or small amounts of DRILTREAT oil wetting agent slowly as weighting agents are added to help oil-wet the additional solids.
Solids Control	Optimize solids control equipment to maintain 2 - 5% low gravity solids in the fluid. Inadequate rheology may occur if low- gravity solids drop below 2%.
Excess Lime	Do not maintain the system based on excess lime calculations. Lime is added to the system <u>only</u> when FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used.

ENCORE Emulsifiers

Use emulsifiers to increase the stability of the emulsion of ENCORE fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the product concentration in WELLSITE indicates a low concentration, when the electrical stability measures <220 volts or when water is present in the HPHT filtrate.

Table 10 ENCORE Emulsifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
LE SUPERMUL	Primary emulsifier in ENCORE. Reduces HPHT filtrate and can impact rheological properties.	Polyaminated fatty acid	0.913	1-3.5 (3-10)
FACTANT*	Secondary concentrated emulsifier for ENCORE usually added when temperatures are >275°F (135°C). Improves rheological properties and reduces HPHT filtrate.	Concentrated tall oil derivative	0.96	0-4 (0-11)
DRILTREAT	Improves oil wetting of solids; helps reduce the viscosity when large quantities of solids (barite or other) have been incorporated.	Lecithin liquid dispersion	1	0.25-4 (.7-11)

* FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

ENCORE Viscosifier / Suspension Agents

Use RHEMOD L or LE SUPERMUL additives to increase the rheological properties. TEMPERUS additive can be used for temporary viscosity both at the plant and at the rigsite when bottomhole temperatures do not exceed 150°F (66°C). VIS-PLUS additive can be used at the plant for shipping viscosity but should not be used in at the rigsite.

Table 11 ENCORE Viscosifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
RHEMOD L	Develops viscosity and suspension properties	Modified fatty acid	0.96	0.5-2 (1.4-6)
TEMPERUS	Temporary viscosifier for <150°F (66°C)	Modified fatty acid	0.99	0.1-1.5 (0.29-4)
VIS-PLUS*	Temporary viscosifier for <175°F (79°C)	Carboxylic acid	0.85	0.1-3 (0.29-9)
TAU MOD	TAU MOD helps extend RHEMOD L suspension performance in low solids systems	Amorphous / Fibrous material	2.6	05-5.0 (1.4-21.7)
BDF-489	Rheology Modifier up to 300°F (79°C) Most effective with 2-3 lb/bbl RHEMOD	Fatty Acid Ester	N/D	1-12 (2.9-34.2)
BDF-566	Rheology Modifier			
BDF-568	Rheology Modifier			
BDF-570	Rheology Modifier	Amine	0.9	1-6 (2.9-17.1)
BARACARB 5	When solids concentrations are	Sized calcium	2.65	5-25 (14-71)

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
	low, BARACARB 5 can be added to provide the necessary surface area for the viscosifier interaction	carbonate		
LIQUITONE	Improves filtration properties and provides some gains in rheological properties when bottom-hole temperatures are <250°F (121°C)	Copolymer aqueous dispersion	0.98	0.5-3 (1.4-9)

* VIS-PLUS lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of VIS-PLUS.

ENCORE Thinners

To thin the ENCORE system, add ENCORE base fluid to the mud or treat the mud with approved thinners. Thinners can be used alone or in combination to control viscosity. In most cases blends have proven to be more effective than the individual products.

EXAMPLE: A blend of 1.5 ppb (4.0 kg/m³) COLDTROL thinner and 0.25 ppb (0.7 kg/m³) OMC 42 oil mud conditioner can be as effective as 3.0 lb/bbl (9.0 kg/m³) ATC thinner at lowering both the 40°F (4°C) and 120°F (49°C) rheological properties and is generally more effective than 6.0 lb/bbl (17.0 kg/m³) COLDTROL thinner at lowering the 40°F (4°C) rheologies.

Table 12 ENCORE Thinners

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ENCORE Base	Reduces viscosity and solids content.	Isomerized olefin	0.82	3-10%
OMC 42	Reduces rheological properties.	Polymide surfactant	0.92	0.5-2 (1.4-6)
ATC	Effective at lowering both the 40°F (4°C) properties and the 120°F (49°C) properties.	Modified fatty acid ester	1.03	0.25-2 (0.7-6) Do not exceed 2.0 ppb total concentration
COLDTROL	Effective in lowering 40°F (4°C) rheological properties without significantly lowering properties at 120°F (49°C).	Alcohol derivative	0.95	0.1-.5 (0.29-1.4) Do not exceed 2.0 ppb total concentration
OMC 2	Reduces rheological properties. Will not lower the 40°F (4°C) properties without additions of COLDTROL.	Oligomeric fatty acid	0.89	0.1-1 (0.29-3)

ENCORE Filtration Control Agents

Table 13 ENCORE Filtration Control

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ADAPTA	425°F (218°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ADAPTA 450	450°F (232°C) Filtration control agent	Copolymer		
LIQUITONE	250°F (121°C) Filtration control agent	Copolymer aqueous dispersion	0.98	0-2 (0-6)
BDF-454	HPHT filtration control up to 500°F (260°C) HPHT filtration control up to 550°F (288°C)	Polymer	1.03	4-7 (11.4-20) 7-10 (20-28.5)
FACTANT*	Reduces HPHT filtrate	Concentrated tall oil derivative	0.96	0-4 (0-11)

*FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

1.3. INNOVERT System

Kinematic Viscosity

While Baroid's high-performance invert technology itself can be adapted to almost any non-aqueous base fluid, some considerations should be made when dealing with very thin kinematic viscosity fluids. Additional viscosifying agents, water phase or solids are needed to achieve similar properties of base fluids with higher kinematic viscosities (i.e., ACCOLADE system). This is of particular interest in low weight systems (<9.0 ppg / <1.08 sg) that have minimal solids.

Formulations

The INNOVERT system is formulated with mineral oils or paraffins to provide the clay-free benefits to a larger market. These systems can be used in most drilling environments and are stable to 475°F+ (246°C+).

Table 14 INNOVERT Formulations

Products	9.0-12.0 ppg (1.08 -1.44sg)	12.0-14.0 ppg (1.44-1.68sg)	14.0-16.0 ppg (1.68-1.92sg)	16.0 + ppg (1.92sg)
Oil/Water ratio	60/40 to 70/30	65/35 to 75/25	65/35 to 80/20	75/25 to 90/10
CaCl ₂ , ppm	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000
Base Fluid (see kinematic viscosity section)	As needed	As needed	As needed	As needed
EZ MUL NT or EZ MUL NS, lb (kg/m ³)	8-10 (17-29)	8-12 (20-34)	10-14 (23-40)	12-14 (26-46)
FACTANT, lb (kg/m ³) *			As needed	0-5 (0-14)
DRILTREAT, lb (kg/m ³)			As needed	0-4 (0-11)
Lime, lb (kg/m ³)*	2-4 (6-11)	2-4 (6-11)	2-4 (6-11)	2-4 (6-11)
ADAPTA, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	1-4 (3-11)
RHEMOD L, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.25-2 (0.7-6)	0.25-2 (0.7-6)
TAU MOD, lb (kg/ m ³)	As needed	As needed	As needed	As needed
BAROID, lb (kg/m ³)	As needed	As needed	As needed	As needed
BARACARB 5, lb (kg/m ³)	10-25 (29-71)	5-25 (14-71)	As needed	As needed
VIS-PLUS*	As needed	As needed	As needed	As needed
TEMPERUS (kg/m ³)	As needed	As needed	As needed	As needed
OMC 42, lb (kg/m ³)			0 - 0.5 (0-1.4)	0 - 0.5 (0-1.4)

* FACTANT and VIS-PLUS additives have lime requirements. When FACTANT or VIS-PLUS are used, add lime in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT or VIS-PLUS.

Maintenance

Fluid Property	Recommended Treatment / Concentration
Building New Mud	When building an INNOVERT system from scratch, BARACARB bridging agent is a vital addition to the system. The addition of BARACARB bridging agent creates the necessary surface area for the viscosifier interaction. If building from a base stock of seasoned fluid (generally 5%-20%), BARACARB bridging agent additions can usually be eliminated. Maintain an accurate product concentration including fluid shipped to the rig.
Oil / Water Ratio	Compared to conventional IEFs, the non-aqueous to water (OWR) ratio of INNOVERT fluid is generally run 5% to 10% lower to achieve its unique properties.
Water Phase Salinity	Do not saturate the water phase with CaCl ₂ . Emulsion instability and water wetting of solids may occur.
Rheology	FANN 75 tests reflecting actual well temperatures and pressures should be performed during critical operations or when significant changes have occurred in the fluid. Measure 10 second, 10 minute and 30 minute gel strengths. The gel strengths should be high but not excessively progressive between the 10 minute and 30 minute gel strengths.
Emulsion Stability	Use emulsifiers to increase the stability of the emulsion of INNOVERT fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the electrical stability measures <220 volts or when water is present in the HPHT filtrate. Refer to the table below for emulsifying products.
CO₂ / H₂S	INNOVERT fluids are run with zero to very small concentrations of excess lime. Treatment of acid gas influx requires a different approach from other invert systems. CO₂ : No treatment is necessary. Lab testing confirms no detrimental effect on the INNOVERT system. H₂S : Treatment with SOURSCAV hydrogen sulfide scavenger is recommended. NO-SULF hydrogen sulfide scavenger is recommended in INNOVERT fluid only during zero discharge operations.
HPHT Filtration	Maintain an all-oil HPHT filtrate.
Electrical Stability	Maintain electrical stability above 220 volts.
Commercial Clay / Lignite	Do not add organophilic clay or organophilic lignite to the system.
Weighting Up	Do not add weighting agents when adding water. Add EZ-MUL NT or EZ MUL NS emulsifier or small amounts of DRILTREAT oil wetting agent slowly as weighting agents are added to help oil-wet the additional solids.
Solids Control	Optimize solids control equipment to maintain 2 - 5% low gravity solids in the fluid. Inadequate rheology may occur if low- gravity solids drop below 2%.
Excess Lime	Do not maintain the system based on excess lime calculations. Lime is added to the system <u>only</u> when FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used.

INNOVERT Emulsifiers

Use emulsifiers to increase the stability of the emulsion of INNOVERT fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the product concentration in WELLSITE indicates a low concentration, when the electrical stability measures <220 volts or when water is present in the HPHT filtrate.

Table 15 INNOVERT Emulsifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
LE SUPERMUL	Primary emulsifier in high-performance inverts. Reduces HPHT filtrate and can impact rheological properties.	Polyaminated fatty acid	0.913	1-3.5 (3-10)
EZ MULNT	Primary emulsifier in high-performance inverts. Reduces HPHT filtrate and can impact rheological properties.	Polyaminated fatty acid	0.958	1-3.5 (3-10)
EZ MUL NS	Primary emulsifier in high-performance inverts. Reduces HPHT filtrate and can impact rheological properties. Best for North Sea applications	Polyaminated fatty acid	0.958	1-3.5 (3-10)
FACTANT*	Secondary concentrated emulsifier usually added when temperatures are >275°F (135°C). Improves rheological properties and reduces HPHT filtrate.	Concentrated tall oil derivative	0.96	0-4 (0-11)
DRILTREAT	Improves oil wetting of solids; helps reduces the viscosity when large quantities of solids (barite or other) have been incorporated.	Lecithin liquid dispersion	1	0.25-4 (.7-11)

* FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

INNOVERT Viscosifiers / Suspension Agents

Use RHEMOD L EZ MUL NT, EZ MUL NS or TAU MOD additives to increase the rheological properties. TEMPERUS additive can be used for temporary viscosity both at the plant and at the rigsite when bottomhole temperatures do not exceed 150°F (66°F). VIS-PLUS additive can be used at the plant for shipping viscosity but should not be used in at the rigsite.

Table 16 INNOVERT Viscosifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
RHEMOD L	Develops viscosity and suspension properties	Modified fatty acid	0.96	0.5-2 (1.4-6)
TEMPERUS	Temporary viscosifier for <150°F (66°C)	Modified fatty acid	0.99	0.1-1.5 (0.29-4)
VIS-PLUS*	Temporary viscosifier for <175°F (79°C)	Carboxylic acid	0.85	0.1-3(0.29-9)
TAU MOD	TAU MOD helps extend RHEMOD L suspension performance in low solids systems	Amorphous / Fibrous material	2.6	05-5.0 (1.4-21.7)

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
BDF-489	Rheology Modifier up to 300°F (79°C) Most effective with 2-3 lb/bbl RHEMOD	Fatty Acid Ester	N/D	1-12.0 (2.9-34.2)
BDF-566	Rheology Modifier	Polymer	0.97-1.0	0.5-3 (1.4-9)
BDF-568	Rheology Modifier	Modified fatty acid	0.9-1.0	0.5-2 (1.4-6)
BDF-570	Rheology Modifier	Amine	0.9	0.5-6
BARACARB 5	When solids concentrations are low, BARACARB can be added to provide the necessary surface area for the viscosifier interaction	Sized calcium carbonate	2.65	5-25 (14-71)
LIQUITONE	Improves filtration properties and provides some gains in rheological properties when bottom-hole temperatures are <250°F (121°C)	Copolymer aqueous dispersion	0.98	0.5-3 (1.4-9)

* VIS-PLUS lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of VIS-PLUS.

INNOVERT Thinners

To thin the INNOVERT system, add base fluid to the mud or treat the mud with approved thinners. Thinners can be used alone or in combination to control viscosity. In most cases blends have proven to be more effective than the individual products.

EXAMPLE: A blend of 1.5 ppb (4.0 kg/m³) COLDTROL additive and 0.25 ppb (0.7 kg/m³) OMC 42 additive can be as effective as 3.0 ppb (9.0 kg/m³) ATC additive at lowering both the 40°F (4°C) and 120°F (49°C) rheological properties and is generally more effective than 6.0 ppb (17.0 kg/m³) COLDTROL at lowering the 40°F (4°C) rheologies.

Table 17 INNOVERT Thinners

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
Base Fluid	Reduces viscosity and solids content.	Varies	0.82	3-10%
OMC 42	Reduces rheological properties.	Polymide surfactant	0.92	0.5-2 (1.4-6)
ATC	Effective at lowering both the 40°F (4°C) properties and the 120°F (49°C) properties.	Modified fatty acid ester	1.03	0.25-2 (0.7-6) Do not exceed 2ppb total concentration
COLDTROL	Effective in lowering 40°F (4°C) rheological properties without significantly lowering properties at 120°F (49°C).	Alcohol derivative	0.95	0.1-5 (0.29-1.4) Do not exceed 2ppb total concentration
OMC 3	Reduces rheological properties. Will not lower the 40°F (4°C)	Fatty acid derivative	0.9	0.1-5 (0.29-1.43)

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
	properties without additions of COLDTROL.			
OMC 2	Reduces rheological properties. Will not lower the 40°F (4°C) properties without additions of COLDTROL.	Oligomeric fatty acid	0.89	0.1-1 (0.29-3)

INNOVERT Filtration Control Agents

For filtration control, add ADAPTA, BDF-513, LIQUITONE, or FACTANT products.

Table 18 INNOVERT Filtration Control

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ADAPTA	425°F (218°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)
LIQUITONE	250°F (121°C) Filtration control agent	Copolymer aqueous dispersion	0.98	0-2 (0-6)
ADAPTA 450	450°F (232°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)
BDF-513	HPHT filtration control up to 350°F (177°C) HPHT filtration control up to 425°F (218°C)	Copolymer	1.03	1-4 (2.9-11.4) 4-6 (11.4-17)
BDF-454	HPHT filtration control up to 500°F (260°C) HPHT filtration control up to 550°F (288°C)	Polymer	1.03	4-7 (11.4-20) 7-10 (20-28.5)
FACTANT*	Reduces HPHT filtrate	Concentrated tall oil derivative	0.96	0-4 (0-11)

* FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

1.4. INTEGRADE System

Formulations

The INTEGRADE system is formulated with diesel base oil to provide the ACCOLADE system technology benefits to a larger market. These systems can be used in most drilling environments and are stable to 475°F+ (246°C+).

Table 19 INTEGRADE Formulations

Products	9.0-12.0 ppg (1.08 -1.44sg)	12.0-14.0 ppg (1.44-1.68sg)	14.0-16.0 ppg (1.68-1.92sg)	16.0 + ppg (1.92sg)
Oil/Water ratio	60/40 to 70/30	65/35 to 75/25	65/35 to 80/20	75/25 to 90/10
CaCl ₂ , ppm	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000	200,000 to 275,000
Diesel	As needed	As needed	As needed	As needed
FORTI MUL, lb (kg/m ³)	8-10 (17-29)	8-12 (20-34)	10-14 (23-40)	12-14 (26-46)
FACTANT, lb (kg/m ³) *	As needed	As needed	As needed	0-5 (0-14)
DRILTREAT, lb (kg/m ³)			As needed	0-4 (0-11)
Lime, lb (kg/m ³)*	2-4 (6-11)	2-4 (6-11)	2-4 (6-11)	2-4 (6-11)
ADAPTA, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	1-4 (3-11)
RHEMOD L, lb (kg/m ³)	0.5-3 (1.4-9)	0.5-3 (1.4-9)	0.25-2 (0.7-6)	0.25-2 (0.7-6)
BAROID, lb (kg/m ³)	As needed	As needed	As needed	As needed
BARACARB 5, lb (kg/m ³)	10-25 (29-71)	5-25 (14-71)	As needed	As needed
VIS-PLUS*	As needed	As needed	As needed	As needed
TEMPERUS (kg/m ³)	As needed	As needed	As needed	As needed
OMC 42, lb (kg/m ³)			0 - 0.5 (0-1.4)	0 - 0.5 (0-1.4)

* FACTANT and VIS-PLUS additives have lime requirements. When FACTANT or VIS-PLUS are used, add lime in a ratio of 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT or VIS-PLUS.

Maintenance

Fluid Property	Recommended Treatment / Concentration
Building New Mud	When building an INTEGRADE system from scratch, BARACARB bridging agent is a vital addition to the system. The addition of BARACARB bridging agent creates the necessary surface area for the viscosifier interaction. If building from a base stock of seasoned fluid (generally 5%-20%), BARACARB bridging agent additions can usually be eliminated. Maintain an accurate product concentration including fluid shipped to the rig.
Oil / Water Ratio	Compared to conventional IEFs, the non-aqueous to water (OWR) ratio of INTEGRADE fluid is generally run 5% to 15% lower to achieve its unique properties.
Water Phase Salinity	Do not saturate the water phase with CaCl_2 . Emulsion instability and water wetting of solids may occur.
Rheology	FANN 75 tests reflecting actual well temperatures and pressures should be performed during critical operations or when significant changes have occurred in the fluid. Measure 10 second, 10 minute and 30 minute gel strengths. The gel strengths should be high but not excessively progressive between the 10 minute and 30 minute gel strengths.
Emulsion Stability	Use emulsifiers to increase the stability of the emulsion of INTEGRADE fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the electrical stability measures <220 volts or when water is present in the HPHT filtrate. Refer to the table below for emulsifying products.
CO_2 / H_2S	INTEGRADE fluids are run with zero to very small concentrations of excess lime. Treatment of acid gas influx requires a different approach from other invert systems. CO_2: No treatment is necessary. Lab testing confirms no detrimental effect on the INTEGRADE system. H_2S: Treatment with SOURSCAV hydrogen sulfide scavenger is recommended. NO-SULF hydrogen sulfide scavenger is recommended in INTEGRADE fluid only during zero discharge operations.
HPHT Filtration	Maintain an all-oil HPHT filtrate.
Electrical Stability	Maintain electrical stability above 220 volts.
Commercial Clay / Lignite	Do not add organophilic clay or organophilic lignite to the system.
Weighting Up	Do not add weighting agents when adding water. Add FORTI MUL emulsifier or small amounts of DRILTREAT oil wetting agent slowly as weighting agents are added to help oil-wet the additional solids.
Solids Control	Optimize solids control equipment to maintain 2 - 5% low gravity solids in the fluid. Inadequate rheology may occur if low- gravity solids drop below 2%.
Excess Lime	Do not maintain the system based on excess lime calculations. Lime is added to the system <u>only</u> when FACTANT filtration control agent/emulsifier or VIS-PLUS suspension agent is used.

INTEGRADE Emulsifiers

Use emulsifiers to increase the stability of the emulsion of INTEGRADE fluid, reduce the HPHT filtrate and reduce the water-wetting tendency of the insoluble solids. Emulsifiers should be added when the product concentration in WELLSITE indicates a low concentration, when the electrical stability measures <220 volts or when water is present in the HPHT filtrate.

Table 20 INTEGRADE Emulsifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
FORTI MUL	Primary emulsifier in high-performance inverts. Reduces HPHT filtrate and can impact rheological properties.	Polyaminated fatty acid	0.913	1-3.5 (3-10)
FACTANT*	Secondary concentrated emulsifier usually added when temperatures are >275°F (135°C). Improves rheological properties and reduces HPHT filtrate.	Concentrated tall oil derivative	0.96	0-4 (0-11)
DRILTREAT	Improves oil wetting of solids; helps reduces the viscosity when large quantities of solids (barite or other) have been incorporated.	Lecithin liquid dispersion	1	0.25-4 (.7-11)

* FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

INTEGRADE Viscosifiers / Suspension Agents

Use RHEMOD L or FORTI MUL additives to increase the rheological properties. TEMPERUS additive can be used for temporary viscosity both at the plant and at the rigsite when bottomhole temperatures do not exceed 150°F (66°F). VIS-PLUS additive can be used at the plant for shipping viscosity but should not be used in at the rigsite.

Table 21 INTEGRADE Viscosifiers

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
RHEMOD L	Develops viscosity and suspension properties	Modified fatty acid	0.96	0.5-2 (1.4-6)
TEMPERUS	Temporary viscosifier for <150°F (66°C)	Modified fatty acid	0.99	0.1-1.5 (0.29-4)
VIS-PLUS*	Temporary viscosifier for <175°F (79°C)	Carboxylic acid	0.85	0.1-3 (0.29-9)
TAU MOD	TAU MOD extends RHEMOD L suspension performance in low solids systems	Amorphous / Fibrous material	2.6	0.5-5.0 (1.4-21.7)
BDF-489	Rheology modifier up to 300°F (79°C) Most effective with 2-3 lb/bbl RHEMOD	Fatty Acid Ester	N/D	1-12.0 (2.9-34.2)
BDF-570	Rheology Modifier	Amine	0.9	0.5-6

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
BARACARB 5	When solids concentrations are low, BARACARB can be added to provide the necessary surface area for the viscosifier interaction	Sized calcium carbonate	2.65	5-25 (14-71)
LIQUITONE	Improves filtration properties and provides some gains in rheological properties when bottom-hole temperatures are <250°F (121°C)	Copolymer aqueous dispersion	0.98	0.5-3 (1.4-9)

* VIS-PLUS lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of VIS-PLUS.

INTEGRADE Thinners

To thin the INTEGRADE system, add base fluid to the mud or treat the mud with approved thinners.

Table 22 INTEGRADE Thinners

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
Base Fluid	Reduces viscosity and solids content.	Varies	0.82	3-10%
OMC 42	Reduces rheological properties.	Polymide surfactant	0.92	0.5-2 (1.4-6)
ATC	Effective at lowering both the 40°F (4°C) properties and the 120°F (49°C) properties.	Modified fatty acid ester	1.03	0.25-2 (0.7-6) Do not exceed 2ppb total concentration
OMC 2	Reduces rheological properties. Will not lower the 40°F (4°C) properties without additions of COLDTROL.	Oligomeric fatty acid	0.89	0.1-1 (0.29-3)

INTEGRADE Filtration Control Agents

Table 23 INTEGRADE Filtration Control

Product	Application	Description	Specific gravity	Treatment, ppb (kg/m ³)
ADAPTA	425°F (218°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)
ADAPTA 450	450°F (232°C) Filtration control agent	Copolymer	1.03	0-2 (0-6)
LIQUITONE	250°F (121°C) Filtration control agent	Copolymer aqueous dispersion	0.98	0-2 (0-6)
BDF-454	HPHT filtration control up to 500°F (260°C) HPHT filtration control up to 550°F (288°C)	Polymer	1.03	4-7 (11.4-20) 7-10 (20-28.5)
FACTANT*	Reduces HPHT filtrate	Concentrated tall oil derivative	0.96	0-4 (0-11)

* FACTANT lime requirement is 0.5 ppb (1.0 kg/m³) of lime to 1.0 ppb (3.0 kg/m³) of FACTANT.

2. High-Performance Invert Emulsion Packer Fluids

A packer fluid is a mud that provides adequate suspension properties and long-term protection from corrosion. Any of the high performance IEFs can be used as a packer fluid.

Viscosify the IEF with RHEMOD L and LE SUPERMUL products to packer fluid specifications before setting. Typical additive treatments are 0.5-1.0 ppb of RHEMOD L and 1.0-2.0 ppb of LE SUPERMUL.

Table 24 Packer Fluid Recommended Properties

Properties	Density, ppg (sg)			
	12.0 (1.44)	14.0 (1.68)	16.0 (1.92)	18.0 (2.16)
Yield point, lb/100 ft ²	30-70	40-70	40-80	50-90
10-second gel, lb/100 ft ²	20-50	20-50	25-60	25-60
10-minute gel, lb/100 ft ²	30-60	35-60	35-60	40-70

Properties		Static Aged, 250F, 24hr	Static Aged, 200F, 2 mos
Hot Rolled at 250F, 16 hr			
PV, cP	21	18	65
Yield point, lb/100 ft ²	11	10	75
LSR	3	4	26
10-second gel, lb/100 ft ²	5	5	37
10-minute gel, lb/100 ft ²	8	10	57
Sag Factor		0.50	0.507

*If remediation is required, the fluid can be washed through and circulated out of the well.

The following additions can be made to existing fluids. The solution remains pumpable until placed in the annulus.

Table 25 Packer Fluid Product Concentrations

Products	Additions	
	ppb	kg/m ³
RHEMOD L	1-2	2.85-5.7
BDF-566	3	8.56
BDF-570	3	8.56
TAU MOD	5	14.27
BARACARB 5	50	142.65

3. XP-07 Synthetic-Based Fluids

XP-07 systems are synthetic based systems in which the continuous or external phase is a pure normal- alkane mixture.

XP-07 fluids can be formulated for stability at temperatures in excess of 450°F (232°C).

The properties of XP-07 systems are influenced by:

- Alkane (synthetic)/water ratio
- Gellant and emulsifier concentrations
- Solids content
- Downhole temperature and pressure

XP-07 System Classifications

Table 26 XP-07 System Classifications

System	Application
XP-07	For deepwater, extended-reach, high-angle, and HPHT drilling where environmental regulations require synthetic based systems.
XP-07 100	For non-damaging coring and drilling where environmental regulations require synthetic based systems.

Formulation

XP-07 systems use emulsifiers, gellants, and fluid loss agents at concentrations based on formation, well geometry, and bottom hole temperature criteria.

Table 27 Basic XP-07 Formulations

Additive	Function	Typical concentrations, lb/bbl(kg/m3) to 325 F (163 C)
XP-07	Continuous phase	As needed
EZ MUL NT	Emulsifier	6-20 (17-57.0)
INVERMUL NT	Emulsifier	1-4 (2.9-11.6)
Lime	Alkalinity source	2-10 (6-29)
DURATONE E	Fluid-loss control agent	2-20 (6-57.0)
Water	Discontinuous phase	As needed
GELTONE II/V	Viscosifier	2-10 (6-29)
SUSPENTONE	Suspension agent	0.5-4 (1.4-11.6)
BAROID, BARODENSE, or BARACARB	Weighting agent	As needed
CaCl ₂	Salinity source	As needed

XP-07 100

XP-07 100 all n-alkane systems are used when maintaining the native state of the geologic formation is a primary concern and environmental regulations require the use of a synthetic based system. These systems are not normally used where water contamination is a known problem.

Table 28 XP-07 100 Product Concentrations

Additive	Function	Typical concentrations, lb/bbl (kg/m ³) to 325 F (163 C)
XP-07 base	Continuous phase	As needed
EZ-CORE	Passive emulsifier	2 (6)
* EZ MUL NT	Emulsifier	0-4 (0-11)
BARABLOK or DURATONE E When using DURATONE HT for filtration control, BARACTIVE must be used as an activator.	Filtration control agent	5-15 (14-43)
GELTONE II/V	Viscosifier	8-16 (23-46)
BARACTIVE	Polar additive	4-12 (11-34)
BAROID, BARODENSE, or BARACARB	Weighting agent	As needed
Lime	Alkalinity source	0-10 (0-29)

* EZ MUL NT may be added when a large amount of water contamination occurs.

For high-temperature applications (350-425°F) see BAROID 100 HT in this section. **XP-**

07 System Maintenance

Fluid Property	Recommended Treatment / Concentration
Building New Mud	Allow several hours at maximum shear when mixing new mud volume.
Oil / Water Ratio	Maintain synthetic/water ratio within the programmed range (see table below).
Water Phase Salinity	Do not saturate the water phase with CaCl ₂ ; emulsion instability and water-wetting of solids can occur.
Rheology	Use OMC 42 when a thinner is required.
HPHT Filtration	Maintain an all-alkane HPHT filtrate.
Electrical Stability	Maintain electrical stability above 400 volts.
Lost Circulation Material	Do not use cellulosic LCM.
Weighting Up	Slowly add EZ MUL NT slowly when weighting agents are added to help oil-wet the additional solids. Do not add weighting agents when adding water.
Solids Control	Use solids control equipment to prevent build-up of low-gravity solids.
Excess Lime	Maintain excess lime at 1 to 3 lb/bbl (3 to 9 kg/m ³).
Products to Avoid	Do not add any materials that contain petroleum hydrocarbons.

Table 29 Recommended Synthetic/Water Ratio for XP-07

Mud Density, lb/gal (sg)	Recommended Synthetic/Water Ratio
9-11 (1.08-1.32)	60/40 - 70/30
11-13 (1.32-1.56)	70/30 - 80/20
13-15 (1.56-1.80)	80/20
15-16 (1.80-1.92)	80/20 - 85/15
16-17 (1.92-2.04)	85/15 - 90/10
17-18 (2.04-2.16)	90/10 - 95/5

XP-07 Emulsifiers

Use emulsifiers to increase the stability of the XP-07 system emulsion, promote alkane wetting of solids, and prevent water-in-filtrate.

Table 30 XP-07 Emulsifiers

Product	Application	Description	Treatment, lb/bbl (kg/m ³)
EZ-CORE	Acts as a passive emulsifier in the XP-07 100 systems	Refined tall oil fatty acid	2-4 (6-11)
EZ MUL NT	Acts as a primary emulsifier and alkane-wetting agent	Polyamide	2-20 (6-57)
EZ MUL NS	Acts as a primary emulsifier and alkane-wetting agent	Polyamide in XP-07 base solvent	2-20 (6-57)
INVERMUL NT	Acts as primary or secondary emulsifier	Modified fatty acid	2-20 (6-57)
DRILTREAT	Alkane-wetting agent	Lecithin dispersion	.5-2 (1.4-6)
PERFOR MUL	Acts as a primary emulsifier and alkane-wetting agent	High performance emulsifier	2-20 (6-57)

XP-07 Viscosifiers / Suspension Agents

Use GELTONE II or V to impart rheological properties to the XP-07 system.

Use SUSPENTONE to minimize barite sag at elevated temperatures.

Use RM-63 to enhance low shear-rate viscosities of the XP-07 systems.

Table 31 XP-07 Viscosifiers

Product	Application	Description	Treatment, lb/bbl (kg/m ³)
GELTONE II/V	Develops viscosity and suspension properties	Organophilic clay	2-16 (6-46)

Product	Application	Description	Treatment, lb/bbl (kg/m ³)
SUSPENTONE	Provides suspension and minimize sag with minimal viscosity build-up	Organophilic clay	1-5 (3-14)
RM-63	Enhances low-shear rheological properties	Polymeric fatty acid	0.5-2 (1.4-6)

XP-07 Thinners

To thin XP-07 systems, add XP-07 base fluid to the system or treat with a polycarboxylic acid or oligomeric fatty acid derivatives.

Table 32 XP-07 Thinners

Product	Application	Description	Treatment, lb/bbl (kg/m ³)
OMC 2	Extreme viscosity reducer	Oligomeric fatty acid	0.2-1 (0.6-3)
OMC 3	Moderate viscosity reducer	Fatty acid	0.5-4 (1.4-11)
OMC 42	Moderate viscosity reducer	Polymer imide surfactant	0.5-4 (1.4-11)

XP-07 Filtration Control Agents

To provide HPHT filtration control in XP-07 systems, add organophilic lignite or various asphaltic materials.

Table 33 XP-07 Filtration Control

Product	Application	Description	Treatment, lb/bbl (kg/m ³)
DURATONE E	HPHT filtration control in XP-07 systems for temperatures to 450 F (232 C)	Organophilic lignite	2-20 (6-57)
AK-70	Controls fluid loss at temperatures up to 275 F (135 C)	Blend of air-blown asphalt and clay with anti-caking agent	1-25 (3-71)
BARABLOK	Controls fluid loss at temperatures up to 385 F (196 C)	Powdered hydrocarbon resin (asphaltite)	1-15 (3-43)
ADAPTA	Controls fluid loss at temperatures up to 425 F (218 C)	Copolymer	0.5-4 (1.4-11)
BDF-513	Controls fluid loss at temperatures up to 425 F (218 C) in low solvency base oils	Copolymer	0.5-4 (1.4-11)

Logging

XP-07 systems do not conduct electric current; therefore, logging tools that require electric conductance to measure resistivity will not function properly.

Table 34 Logging Guidelines for XP-07

Objective	Tool	Notes
Depth control correlation and lithology	Induction/gamma ray log Formation density log Sonic log Neutron log Dipmeter	Use the gamma ray log to determine sand and shale sequences. Use the other logs for identifying complex lithology.
Percent shale in shaley sands	Gamma ray log	The gamma ray log method replaces the sand/shale index found in fresh waters from the SP log.
Net sand (sand count)	Formation density log Gamma ray log	Use the formation density log and/or the caliper log to determine sand count when the sand and shale densities differ.
Detect hydrocarbon-bearing formations	Induction/gamma ray log Sonic log Neutron log	High resistivity values indicate hydrocarbon pore saturation. Use a formation density log in conjunction with neutron and sonic logs to identify hydrocarbons.
Interpretation Water saturation Porosity Permeability Structural formation Productivity	Induction, sonic, density, and neutron logs Formation density, sonic, and neutron logs Sidewall cores Continuous dipmeter Formation tester	Use Archie's equation to compute water saturation.

4. Conventional Oil-Based Muds

Overview

Oil-based muds are muds in which the continuous, or external, phase is oil, such as diesel, paraffin or mineral oil. The properties of oil-based muds are influenced by the following:

- Oil/water ratio
- Emulsifier type and concentration
- Solids content
- Downhole temperature and pressure

Conventional IEF Classifications

Conventional IEFs are classified in four categories. Table 1-53 outlines the primary uses of these different systems.

Table 35 Conventional IEF Classifications

System	Application	Systems - Diesel	Systems – Mineral Oil
Tight-emulsion	For general use as well as high-temperature areas up to 500°F (260°C)	INVERMUL	ENVIROMUL
Relaxed-filtrate (RF)	To provide increased drilling rates	INVERMUL RF	ENVIROMUL RF
All-oil	For use as non-damaging coring and drilling fluid For use as a high-temperature invert emulsion system	BAROID 100	ENVIROMUL 100
High water	To minimize oil retention on cuttings; used primarily in offshore areas that are environmentally sensitive	INVERMUL 50/50	ENVIROMUL 50/50

Either diesel oil or mineral oil is used as the base fluid for conventional oil-based mud systems.

Maintenance

Fluid Property	Recommended Treatment / Concentration
Water Phase Salinity	Do not saturate the water phase with CaCl ₂ ; emulsion instability and water-wetting of solids can occur.
HPHT Filtration	Maintain an all-oil HPHT filtrate.
Electrical Stability	Maintain electrical stability above 400 volts.
Weighting Up	Do not add weighting agents when adding water.
Solids Control	Use solids control equipment to prevent build-up of low-gravity solids.
Lime	Maintain excess lime at 1.5 to 3.0 lb/bbl (4.0 to 9.0 kg/m ³). Add a minimum of 0.5 lb (0.5 kg) of lime for each 1 lb (1 kg) of INVERMUL or INVERMUL NT.

Logging

Conventional oil-based systems do not conduct electric current; therefore, logging tools that require electric conductance to measure resistivity will not function properly.

Table 36 Logging Guidelines for Conventional OBM's

Objective	Tool	Notes
Depth control correlation and lithology	Induction/gamma ray log Formation density log Sonic log Neutron log Dipmeter	Use the gamma ray log to determine sand and shale sequences. Use the other logs for identifying complex lithology.
Percent shale in shaley sands	Gamma ray log	The gamma ray log method replaces the sand/shale index found in fresh waters from the SP log.
Net sand (sand count)	Formation density log Gamma ray log	Use the formation density log and/or the caliper log to determine sand count when the sand and shale densities differ.
Detect hydrocarbon-bearing formations	Induction/gamma ray log Sonic log Neutron log	High resistivity values indicate hydrocarbon pore saturation. Use a formation density log in conjunction with neutron and sonic logs to identify hydrocarbons.
Interpretation Water saturation Porosity Permeability Structural formation Productivity	Induction, sonic, density, and neutron logs Formation density, sonic, and neutron logs Sidewall cores Continuous dipmeter Formation tester	Use Archie's equation to compute water saturation.

4.1. Tight-Emulsion Systems

INVERMUL and ENVIROMUL tight-emulsion systems provide high-temperature stability and tolerance to contaminants. These systems use high concentrations of emulsifiers and fluid-loss agents for maximum emulsion stability and minimal filtrate loss. The volume of the HPHT filtrate is usually less than 15 mL and should be all oil.

Formulations

Table 37 Tight Emulsion Formulations

Additive	Function	Concentrations, lb/bbl (kg/m ³)	
		To 300°F (149°C)	To 400°F (205°C)
Oil	Continuous phase	As needed	As needed
INVERMUL or INVERMUL NT	Primary emulsifier	6-8 (17-23)	8-16 (23-46)
Lime	Alkalinity source	3-4 (9-11)	4-8 (11-23)
DURATONE HT or DURATONE E	Fluid loss control agent	6-8 (17-23)	8-20 (23-57)
Water	Discontinuous phase	As needed	As needed
GELTONE II/V	Viscosifier	0.5-3 (1.4-9)	2-8 (6-23)
EZ MUL or EZ MUL NT	Secondary emulsifier	1-2 (3-6)	2-8 (6-23)
BAROID BARODENSE or BARACARB	Weighting agent	As needed	As needed
CaCl ₂	Salinity source	As needed	As needed

4.2. Relaxed-Filtrate (RF) Systems

INVERMUL RF (diesel base oil) and ENVIROMUL RF (mineral oil) relaxed-filtrate systems have no or very low concentrations of INVERMUL emulsifier and DURATONE filtration control agent. The increased filtrate in these systems promotes faster drilling rates than are possible with tight-emulsion systems. The volume of the HPHT fluid loss is 15 to 20 cm³ with optimized spurt loss. These systems are stable at temperatures up to 325°F (163°C).

Formulations

Table 38 Relaxed Filtrate Formulations

Additive	Function	Concentrations, lb/bbl (kg/m ³) to 300°F (149°C)
Oil	Continuous phase	As needed
EZ MUL or EZ MUL NT	Emulsifier	2-4 (6-11)
Lime	Alkalinity source	2-6 (6-17)
DURATONE HT or DURATONE E	Filtration control agent	0-3 (0-9)
Water	Discontinuous phase	As needed
GELTONE II/V	Viscosifier	2-8 (6-23)
INVERMUL or INVERMUL NT	Emulsifier	0-2 (0-6)
BAROID BARODENSE or BARACARB	Weighting agent	As needed
CaCl ₂	Salinity source	As needed

4.3. All-Oil Drilling / Coring BAROID 100

BAROID 100, an all-oil system (diesel or mineral oil), is used when maintaining the native state of the geologic formation is a primary concern. This system is not used where water contamination is a known problem.

Formulations

Table 39 All-Oil Formulations

Additive	Function	Concentrations, lb/bbl (kg/m ³) to 350°F (177°C)
Oil	Continuous phase	As needed
Lime	Alkalinity source	1-3 (3-9)
EZ- CORE	Passive emulsifier	2.0 (6)
*EZ MUL *EZ MUL NT	Emulsifier	2-4 (6-11)
BARABLOK or BARABLOK 400 or DURATONE HT When using DURATONE HT for filtration control, BARACTIVE must be used as an activator.	Filtration control agent	5-15 (14-43)
AK-70	Filtration control agent	15-25 (43-71)
GELTONE II/V	Viscosifier	6-14 (17-40)
BARACTIVE	Polar additive	2-6 (6-17)
BAROID BARODENSE or BARACARB	Weighting agent	As needed

*EZ MUL, EZ MUL NT may be added when a large amount of water contamination occurs.

4.4. All-Oil BAROID 100 HT

BAROID 100 HT, an all-oil system, is used when circulating and bottomhole temperatures are anticipated in the 350 to 425°F (177 - 218°C) range. The base oil can be diesel, mineral oil or XP-07.

BAROID 100 HT tolerates water contamination at high temperatures with minimal effect on properties.

BAROID 100 HT utilizes both a primary and secondary emulsifier which gives the system greater tolerance to water contamination and the capacity to achieve high mud weights.

Table 40 All-Oil High Temperature Formulations

Additive	Function	Concentrations, lb/bbl (kg/m ³) to 425°F (218°C)
Oil	Continuous phase	As needed
Lime	Alkalinity source	6-10 (17-28)
THERMO MUL	Emulsifier	6-10 (17-28)
THERMO PLUS	Passive emulsifier	2-5 (6-14)
BARABLOK 400 or DURATONE HT or XP-10 When using DURATONE HT for filtration control, BARACTIVE must be used.	Filtration control agent	5-15 (14-43)
GELTONE V	Viscosifier	6-14 (17-40)
BARACTIVE	Polar additive	2-6 (6-17)
BAROID or BARODENSE	Weighting agent	As needed
RM-63	Viscosifier	1-3 (3-9)

4.5. High-Water Systems

INVERMUL 50/50 and ENVIROMUL 50/50 high-water systems were developed for use in areas where discharges of oil are restricted, such as in the North Sea. These systems, which have a 50/50 oil-to-water ratio (diesel or mineral oil), can reduce the oil left on cuttings by as much as 45 percent. High-water systems are not recommended at temperatures greater than 250°F (121°C).

Table 41 High Water Formulations

Additive	Function	Concentrations lb/bbl (kg/m ³) to 250°F (121°C)
Oil	Continuous phase	As needed
INVERMUL or INVERMUL NT	Primary emulsifier	1-2 (3-6)
DURATONE HT or DURATONE E	Filtration control agent	4-8 (11-23)
Lime	Alkalinity source	2-6 (6-17)
Water	Discontinuous phase	As needed
GELTONE II/V	Viscosifier	1-2 (3-6)
EZ MUL or EZ MUL NT	Secondary emulsifier	4-8 (11-23)
BAROID BARODENSE or BARACARB	Weighting agent	As needed
CaCl ₂	Salinity source	As needed

4.6. Special Applications

Because invert emulsion systems are non-corrosive, they are useful for a variety of field applications, including:

- Packer fluids and casing packs
- Arctic casing packs
- PIPE GUARD gelled-oil systems

Packer Fluids and Casing Packs

A packer fluid can be formulated from an INVERMUL or ENVIROMUL mud to provide long-term protection from corrosion.

Casing packs protect the casing from external corrosion and facilitate casing recovery.

Packer fluids are used inside the casing; casing packs are placed in the annular space between the casing and the hole. Viscosify the invert drilling fluid to packer-fluid specifications before setting.

Table 42 Packer Fluid and Casing Pack Recommendations 100°F (38°C).

Properties	Density, lb/gal (sg)			
	12.0 (1.44)	14.0 (1.68)	16.0 (1.92)	18.0 (2.16)
Plastic viscosity, cP	60-80	60-80	70-90	80-100
Yield point, lb/100 ft ²	50-70	50-70	60-80	70-90
10-second gel, lb/100 ft ²	30-50	30-50	40-60	40-60
10-minute gel, lb/100 ft ²	40-60	40-60	40-60	50-70
Alkalinity, mL N/10 H ₂ SO ₄ /mL of mud	3-6	3-6	3-6	3-6
Electrical stability, volts, minimum	600	800	1,000	1,000
Water content, vol%	25-35	20-30	15-25	10-15

Arctic Casing Packs

Arctic casing packs formulated from invert emulsions retard heat loss and prevent permafrost melting. Arctic casing packs also allow casing to expand and contract with temperature changes. An arctic casing pack may be prepared fresh or from an existing mud.

Table 43 Arctic Casing Pack Formulations

Additives	Density lb/gal (sg)		
	10.0 (1.2)	15.0 (1.8)	20.0 (2.4)
Arctic diesel oil, bbl	0.754	0.601	0.444
EZ MUL EZ MUL NT, lb	12.5	12.5	12.5
Water, bbl	0.042	0.034	0.025
GELTONE II/V lb	50	36	25
NaCl, lb	3.0	1.5	1.5
BAROID, lb	21	393	663

To prepare a fresh arctic casing pack:

1. Prepare a premix at 70°F (21°C) or higher.
2. Add half the required amount of GELTONE II/V.
3. Cool the premix to about 40°F (4.5°C).
4. Add the rest of the GELTONE II/V.
5. Pump the pack into position.

To prepare an arctic casing pack from existing mud:

1. Adjust the water content to about 7 percent by volume and the temperature to about 70°F (21°C).
2. Conduct a pilot test to determine the needed concentration of GELTONE II/V.
3. Cool the mud to about 40°F (4.5°C).
4. Add the required GELTONE II/V.
5. Pump the pack into position.

PIPE GUARD Gelled-Oil Systems

PIPE GUARD is designed to prevent corrosion of pipelines that pass under roadbeds and waterways. This system is available in two densities: 9.1 lb/gal (1.09 sg) for under waterways and 19.0 lb/gal (2.28 sg) for under roadbeds.

Mineral oil may be used in place of diesel oil, however the concentration of GELTONE II/V may need to be increased.

Table 44 PIPE GUARD Formulations

Additives	Density, lb/gal (sg)	
	9.1 (1.09)	19.0 (2.28)
Diesel oil, bbl	0.42	0.26
EZ MUL EZ MUL NT, lb	8	8
Lime, lb	5	5
Water, bbl	0.45	0.29
GELTONE II/V lb	8	8
BARACARB, lb	80	—
BAROID, lb	—	598

PIPE GUARD is usually mixed at the plant, but it can also be mixed onsite. Enough PIPE GUARD should be mixed at one time for a number of crossings. After PIPE GUARD has been loaded onto a tank truck, follow these steps at each crossing:

1. Connect the pump from the tank truck to one of the vents.
2. Connect a hose to the outlet vent on the other side of the crossing and run the hose to a small tank for waste collection.
3. Pump PIPE GUARD slowly and steadily into the conduit until clean PIPE GUARD is observed at the outlet vent.
4. Remove the connections and proceed to the next crossing.

Drill-in Fluids

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1. Drill-in Fluids

1.1. Overview

Baroid's reservoir drilling fluids systems have been used for drilling oil and gas development wells worldwide since the early 1970s. Although all these systems are capable of drilling the reservoir section, selection of the optimal fluid for every application is essential to maximize drilling efficiency and achieve the maximum production potential.

All drill-in fluids are designed to protect the reservoir by preventing deep fluid invasion into the rock while drilling. The bridging material particle size distribution is designed to plate out on the face of the rock and plug the pore spaces near the wellbore without traveling deep into it. When formulated correctly, a thin filter cake composed of the sized BARACARB calcium carbonate forms quickly, limiting the invasion of fluid and drilled solids into the rock.

Drill-in fluid filtrate must be chemically compatible with the reservoir rock and the reservoir rock minerals to avoid scale precipitation. Detailed reservoir characterization and sensitivity studies with reliable laboratory test data are required to optimize the design of drill-in and completion fluids systems. Residual damage caused by these fluids can be evaluated in the laboratory using return permeability measurements under the anticipated downhole temperature and pressures. The filter cake is designed to be easily removed with minimum lift-off pressure. Any remaining cake residue can be removed with customized N-FLOW delayed breakers.

Baroid's DRIL-N Fluid Systems

BARADRIL-N	NaCl, NaBr, KCl, CaCl ₂ brine-based with sized CaCO ₃ bridging particles
BRINEDRIL-N	CaCl ₂ , CaBr ₂ , NaCHOO, KCHOO, CsCHOO brine-based with sized CaCO ₃
DRIL-N STIM	BARADRIL-N or BRINEDRIL-N with DRIL-STIM filtrate additives
SOLUDRIL-N	NaCl saturated brine-based with sized salt
COREDIL-N	All oil / synthetic based fluid with sized CaCO ₃ and/ or sized salt
INNOVERT	Engineered high performance invert emulsion fluid with sized CaCO ₃

Dril-N Fluid Selection

Selection of the most appropriate type of drill-in fluid depends on several factors, including the following:

- Reservoir fluids and rock minerals compatibility
- Environmental acceptability
- Drilling performance and economics
- Strategy for removal of filter cake residue

The potential impact of a drill-in fluid on well productivity can be established in the laboratory by exposing the fluid to a section of actual reservoir rock core under simulated downhole conditions for a specific period of time (4-16 hours).

Local environmental legislation and customer environmental policies may preclude the use of certain fluids and should always be considered during the initial selection process.

Well lithology, well profile, completion method and project economics may also influence fluid selection.

Depending upon the well type and completion method, filter cake may be removed by flowing back the well or by using an acid source (live or pre-cursor) to dissolve the filter cake.

- **Injector:** Flowback is likely to be ineffective and filter cake is removed using an acid or acid pre-cursor

- **Flowback filter cake removal:** The fluid selection and design should minimize “pop off” pressure and avoid the risk of screen blockage.
- **Acidizing:** The fluid selection and design should maximize the percentage of the filter cake that will dissolve in acid.

Bridging Optimization

Bridging optimization requires use of the correct amount of bridging solids and the correct particle size bridging solids. Baroid’s preferred bridging material is BARACARB – a sized marble available in the following grades:

- BARACARB 5
- BARACARB 25
- BARACARB 50
- BARACARB 150
- BARACARB 600

The number following the product name represents the approximate median particle size of the material in microns. Products obtained from different approved BARACARB suppliers may exhibit minor differences in particle size.

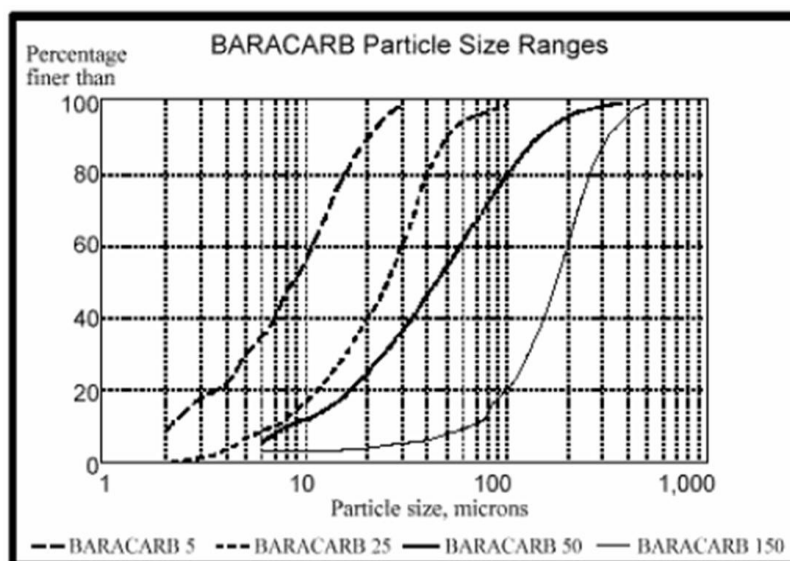


Figure 1 Example BARACARB Particle Size Ranges

Baroid’s WELLSET™ software is used to determine the approximate particle size distribution (PSD) to provide optimum bridging. While there are numerous ways to use the software, good results can be achieved by using 75% to 100% of the D_{50} of the reservoir pore sizes as the D_{50} of the bridging material.

A range of 30 lb/bbl to 40 lb/bbl of BARACARB is recommended to achieve effective bridging. Both the PSD and concentration of BARACARB need to be optimized for each reservoir. Once a bridging plan is made, the fluid must be tested to insure the bridging meets the requirements for the fluid.

Permeability Plugging Apparatus

The permeability plugging test is a fluid loss test that uses porous ceramic discs to more closely simulate the formation than the filter papers used in other fluid loss tests. The suitability of the calculated BARACARB blend

to achieve acceptable bridging should be validated using a permeability plugging test. A variety of pore throat sizes can be simulated using the ceramic discs.

Shale Inhibition in the Reservoir

The effect of a shale inhibitor on reservoir productivity should always be established by testing.

Recommended Shale Inhibitors (after compatibility testing)

- Potassium chloride or other sources of potassium ions
- Polyglycols such as GEM CP, GEM GP, etc.
- Clayseal Plus
- Performatrol

Unacceptable Shale Inhibitors (for most reservoir drilling)

- Sodium silicate
- Potassium silicate
- Polyacrylamides - PAC
- Polyacrylates
- Partially hydrolysed polyacrylamides – PHPA, EZ MUD

Lubricity

For extended reach or highly deviated wells, the addition of a lubricant to WBM or even to OBM drill-in fluids may be required.

Lubricants typically have polar and non-polar groups within their molecular structure. They function as weak surfactants and this can result in the formation of weak, viscous emulsions with crude oil and water (e.g., formation water, WBM filtrate or completion brine). Compatibility tests should be conducted to reduce the risk of emulsion blockage in each scenario.

Many lubricants are ester-based. This class of organic compounds hydrolyzes under acid conditions such as might be experienced during an acid or N-FLOW filter cake breaker job. In many cases, the products of hydrolysis are insoluble fatty acids. Therefore if acid or N-FLOW will be used within the reservoir to remove filter cake then part of the selection process for the lubricant should be to determine the effect of acid exposure.

Laboratory Testing

The first foreign fluid the payzone is exposed to is the drilling fluid used to drill it. This is the first opportunity to damage the reservoir (i.e., change the characteristics of the reservoir rocks and impair the flow of hydrocarbons into the well). The key objective is to minimize changes to permeability: the ability to flow hydrocarbons.

Some of the most common ways of damaging a formation include the following:

- Payzone invasion and plugging by fine particles
- Formation clay swelling
- Commingling of incompatible fluids
- Movement of dislodged formation pore-filling particles
- Changes in reservoir rock wettability
- Formation of emulsions or water blocks

Once one of these damage mechanisms diminishes the permeability of a reservoir, it is seldom possible to restore it to its original condition.

An effective way to preserve the natural reservoir rock permeability is to identify and quantify the complex physical interactions and chemical reactions occurring downhole between the reservoir rock minerals and reservoir fluids, and the drill-in / completion fluids used.

Selecting the most suitable fluid system for drilling into the pay zone or re-completing or working over an existing well **requires a thorough understanding of the reservoir**. All available reservoir characterization tools and methods should be used to identify and quantify the geologic parameters that could influence the producibility of the targeted pay zone.

Special consideration should be given to conducting a reservoir fluid sensitivity study to precisely define attributes of the interval of interest that could be susceptible to formation damage. The fluid sensitivity study should describe the morphological and mineralogical composition of the reservoir rock, based upon data generated by lab testing on core plugs from carefully selected pay zone cores. Natural reservoir fluids should be analyzed to establish their chemical make-up. These data can help determine the reservoir's potential for such problems as particulate dislodging, fines migration, swelling of hydrated clays, or chemical precipitation. The degree of damage that could be caused by anticipated problems can be modeled, as well as the effectiveness of possible solutions for mitigating the risks involved.

Core-plug tests should include thin-section petrography to evaluate pay zone mineralogy, average pore size, pore-throat size, and porosity type and distribution. The porosity and permeability of brine-saturated core plugs should be measured at a specified pressure and temperature.

Table 1 Recommended Analyses / Testing for Reservoir

	Recommended Analyses / Testing
Reservoir Fluid	Water analysis Emulsion tendencies Scaling tendencies
Reservoir Rock	X-Ray Diffraction (XRD) X-Ray Fluorescence (XRF) Define reservoir rock mineralogy and provide semi quantitative analysis of the clay minerals.
	Scanning Electron Microscopy (SEM) Study the reservoir rock morphology and pore size distribution. Define the pore filling material if any.
	Thin section petrography Identify pore filling and grain cementing material
	Capillary flow porometer Measure the average and largest pore throat diameters

Water Analysis

Incompatibility between drill-n fluid filtrate, reservoir fluids and formation water can result in precipitation of salts within the reservoir pore space and subsequent loss of well productivity. Incompatibility between the formation water and the proposed completion brine is also possible.

It is prudent whenever possible to establish the composition of the formation water. Typically this is determined by Inductively Coupled Plasma Emission Spectrometry and/or ion chromatography. Having established the ionic composition and the pH of the formation water and proposed drill-n fluid brine phase, scale prediction programs can be applied to determine the probability of scaling at reservoir temperature.

Crude Oil Analysis

Formation of viscous emulsions within the pore volume (emulsion blockage) can result in reduced well productivity. Emulsions can be formed by mixing aqueous phases such as formation water, completion brine, WBM filtrate with non-aqueous phases such as crude oil and OBM filtrate.

Typically even when mixed under high shear conditions a mixture of alkanes (straight chain hydrocarbons) and water will have a low tendency to form emulsions (i.e., tend to separate into oil and water phases as soon as the shear forces are removed). The presence of certain additional chemical species within any of these phases can promote the formation of emulsions.

Emulsifiers used in OBM can cause formation damage. Therefore emulsifiers should be maintained at the lowest practical concentration needed to maintain a stable fluid while drilling the reservoir section. Other components that promote emulsion formation include corrosion inhibitors and lubricants added to the drill-in fluid or completion brine.

Compatibility Testing

It is good practice to conduct compatibility testing on fluid combination. Typically, this testing will involve mixing the two components together at various ratios (usually 25/75, 50/50 and 75/25). The fluids are mixed using a high shear mixer at the expected reservoir temperature. Once mixed the fluids are placed in an oven or water bath at the reservoir temperature. Separation of the two phases is recorded over a period of several hours. A clean separation of the oil and aqueous phases indicates a low risk of emulsification. Poor separation and a stable blending of the oil and water phases indicate a high risk of emulsification.

1.2. DRIL-N and Drill-In Systems

BARADRIL-N System

BARADRIL-N is a calcium carbonate weighted, clay-free system that is acid-soluble and non-damaging. The system is formulated with freshwater or brine, thermally-stable polymers for suspension and filtration control, and sized calcium carbonate bridging particles. It can also be used for completion and workover operations. Since its introduction in the early 1970s, BARADRIL-N systems have been used to drill different sandstone and carbonate reservoirs in thousands of wells worldwide.

The density range for the system is between 8.6 lb/gal to 14.5 lb/gal.

Advantages of the BARADRIL-N System

- Easy to prepare and maintain in the field
- Provides hole stability and effective fluid loss control while drilling targeted permeable formations
- Can be weighted up quickly with BARACARB for subsurface pressure control
- Acid-soluble and non-damaging to producing formations
- Solids and fluids are prevented from invading the productive zones by selecting the proper
- BARACARB bridging particles size distribution using Baroid's CFG modeling software
- Enables fast penetration rates and provides good lubricating characteristics

BRINEDRIL-N System

BRINEDRIL-N is a high-density, brine-based polymer system designed for drilling, completion, and workover operations. The system uses water-soluble salts such as calcium chloride, calcium bromide, zinc bromide, or sodium, potassium or cesium formate brines and specifically designed polymers for fluid loss control and suspension to achieve a non-damaging, thermally-stable fluid. Carefully sized BARACARB bridging agent can be added to promote a thin, low permeability filter cake for drilling permeable formations.

Densities ranging from 9.5 to 18.0 lb/gal have been used,

Advantages of the BRINEDRIL-N System

- Enables fast penetration rates and ease in discarding drilled solids
- Easy to prepare and maintain the required density with clear brine or dry salts

- Exhibits uniquely high, low shear rate viscosities and shear-thinning capabilities that can ensure stability and effective hole cleaning
- No negative effects on the reservoir rock mineralogy and morphology
- Consistently shows excellent return permeability results
- Provides superior lubricating characteristics

DRIL-N STIM

DRIL-N STIM stimulation while drilling filtrate additive can be added to Baroid's BARADRIL-N and BRINEDRIL-N family of brine based drill-in fluids to enhance reservoir permeability.

DRIL-N STIM is needed because drilling in or completing the reservoir section can cause significant formation damage. This damage may be caused by the fluid introduced to the wellbore and occur through many different mechanisms including emulsion blocks, water blocks, polymer/filtrate/particle invasion, precipitates, and improper wetting of the formation. This damage can be remediated by stimulating the formation through post-drilling treatments that can significantly increase well construction costs.

While a properly engineered reservoir fluid can be designed to lessen formation damage and protect the producibility of the well, filtrate invasion will still occur and can adversely impact producibility by changing the natural permeability of the reservoir.

DRIL-N STIM filtrate additive uses the filtrate to actually help improve the final oil or gas permeability, approaching the natural permeability of the formation. DRIL-N STIM treatments added to the fluid can alter formation wettability and/or eliminate emulsion blocks in oil wells or water blocks in gas wells during the drilling or completion process.

Applying these treatments can greatly reduce formation damage from drilling or completion operations and the invading aqueous fluid can be used as a production enhancement tool to help improve formation producibility. If followed by a treatment with N-FLOW filter cake breaker system it is possible to stimulate the formation in excess of 100%.

COREDRIIL-N SYSTEM

The COREDRIL-N system is an all-oil drill-in and coring fluid system specifically designed for drilling and coring water-sensitive formations.

The water-free COREDRIL-N system helps preserve natural rock wettability characteristics and is the preferred fluid for obtaining native state core samples. Conventional drilling and coring fluids usually contain high concentrations of strong oil-wetting surfactants that could cause dramatic alteration to the core sample. COREDRIL-N prevents this alteration because it contains only a small concentration of latent emulsifier which remains chemically inactive until water is introduced into the system, and then exhibits very weak oil-wetting compared to conventional emulsifiers. The system contains an optimal concentration of sized solids that plug the pores of the reservoir rock but do not penetrate it.

Advantages of the COREDRIL-N System

- Non-damaging to pay zones and ideal for securing native state preserved core samples
- Allows for more reliable core analysis data and more accurate reservoir evaluation
- Can be formulated with effective properties over a wide range of fluid densities and base oils
- Temperature stable and resistant to solids and water contamination
- Bridging particle sizes can be optimized with DFG modeling software to help prevent deep invasion by solids and fluids into the pay zone
- Creates stable, thin, low-permeability filter cake
- Does not contain strong oil-wetting emulsifiers

- Exhibits excellent return permeabilities

SOLUDRIL-N System

The SOLUDRIL-N system is a sized-salt fluid system formulated for drilling, completion or workover operations in horizontal and vertical wellbores.

The SOLUDRIL-N system utilizes a biopolymer and a specially processed, stabilized, non-ionic starch in combination with sized-salt as the bridging particles. Because the bridging material is composed of solid sodium chloride, the base brine used to formulate SOLUDRIL-N must be saturated. If the base brine is not saturated, then the addition of the sized salt bridging materials will only result in the bridging material dissolving in the brine until the brine becomes saturated. Correctly formulated, this produces a non-damaging fluid system that exhibits enhanced rheological properties and excellent filtration control.

Advantages of the SOLUDRIL-N System

- Filter cake is easily removable with freshwater or unsaturated sodium chloride brine
- Excellent return permeability has been demonstrated in laboratory core flood tests compared to conventional water-based systems
- Properly sized salt bridging particles can prevent solids invasion and develop a thin, low permeability filter cake
- Downhole suspension is improved with advanced polymer chemistry that provides low viscosities at high shear
- Fluid is thermally stable up to ~300°F with use of specialty products that allow control of rheological properties at elevated temperatures

INNOVERT and Other High Performance Invert Emulsion Systems

INNOVERT drill-in fluid is a calcium carbonate weighted, clay-free synthetic based system that is acid-soluble and non-damaging.

The systems are formulated with synthetic oil as the external phase and brine as the internal phase. Thermally-stable polymers are used for suspension and filtration control, and sized calcium carbonate is used for bridging. Additional density can be obtained by adding barite. In spite of the addition of barite, results in the lab and in the field show minimal formation damage, due chiefly to the thin, easily removed filter cake.

The density range for the systems is between 7.5 ppg to 18.5 ppg.

Advantages of the INNOVERT Drill-in Systems

- Stable mud properties over a wide temperature and density range (to 450+°F and 18.0+ ppg)
- Fragile gel strengths which can reduce downhole losses to the reservoir
- Unique rheological properties which provide excellent hole cleaning and reduced ECDs
- Increased tolerance to contaminants such as solids and water influxes
- Significantly lower solids content to help increase penetration rates
- No damaging clays or lignitic additives
- Acid-soluble and non-damaging to producing formations
- Fewer products than required for conventional SBMs, improving logistics and rig space usage
- No staging in hole or circulating to stabilize mud weight after long static periods.
- Extremely thin filter cake and low fluid invasion to promote optimal logging conditions, superior return permeability and low lift-off pressures
- Enables fast penetration rates and provides outstanding lubricating characteristics

1.3. Displacement to Drill-In Fluid

Preparation

Prior to bringing the DRIL-N system onboard, all surface pits and lines (including choke manifold, mud manifold etc.) should be cleaned. Everything should be flushed with large volumes of water until no trace of fluid mud or debris remains. To ensure cleanliness, high pressure wash down guns may need to be employed.

It is extremely important that all traces of drilling mud are removed from the casing walls, riser and surface handling system before the DRIL-N fluid is taken on board. Contamination of the DRIL-N with mud solids (such as barite in WBMs) may lead to significant formation damage. For the initial clean-up, the pits and surface lines to be used for mixing the displacement/clean-up pills should be thoroughly cleaned.

Displacement

The drilling mud is often displaced to DRIL-N fluid while drilling the shoe track. Citric acid and sodium bicarbonate should be available to treat any cement contamination while drilling out the shoe. The maximum allowable MBT value is usually 3.0 ppb. Before the MBT reaches 3.0 ppb, active mud must be replaced with clean reserve mud to avoid exceeding the MBT limit.

1.4. Reservoir Drilling and Operational Parameters

Hole Cleaning

The main factors in hole cleaning and cuttings transport are annular velocity and pump rate; the fluid rheology has a relatively minor effect in horizontal sections.

The most effective cleaning is obtained by maintaining a relatively "thin" fluid pumped as fast as possible. An annular velocity of at least 150 fpm is desirable. Working the pipe to agitate the cuttings bed using the tool joint turbulence is helpful when circulating to clean the hole. In addition, pipe rotation at 120 to 160 rpm is recommended while circulating to assist in cleaning the lower side of the hole.

The mud engineer routinely runs DFG hydraulics simulations to assess hole cleaning efficiency. These parameters assist in maintaining a laminar flow profile, good hole cleaning and cuttings support during static conditions.

If hole cleaning becomes inadequate, high density pills should be mixed using large sized BARACARB and pumped while rotating the pipe. Most of the large sized BARACARB will be stripped out at the shakers therefore minimizing effects on the density of the active circulating system. The density of the weighted pill should be decided upon with reference to the fracture gradient. The high density sweep should be modeled in DFG before pumping to ensure that the fracture gradient is not exceeded.

At total depth the following guide should be used to assess the number of circulations required to effectively clean the hole.

Table 2 Circulations Required for Hole Cleaning at TD

Hole Angle (°)	Minimum Wellbore Volume Circulations Required at TD
0 – 30	2
30 – 60	3
60 – 90	4

Correlation of ROP to Cuttings Generated

The trend in the correlation of cuttings generated and seen at surface to ROP can provide another indication of the effectiveness of hole cleaning. The mud engineers should be monitoring cuttings volumes and weight, and the rate

at which cuttings boxes are filled and correlating these with drilling rates. The shaker hands should also be shown what to watch for so that they can provide a speedy warning (e.g., a soft sticky mush means the cuttings are being reground in the well and are not being removed).

Pump Rate

Hole cleaning should be optimized through conventional rheology with pump rates as high as possible within ECD constraints. Bottomhole assemblies must be configured such that pressure limitations allow for these pump rates.

Circulation should be broken periodically when running in the hole to avoid excessive surge pressures on the formation which could destabilize the hole. It is recommended the pumps are started and circulation broken (gently) at intermediate stages to minimize excessive ECD when circulating cold mud.

Losses and Differential Sticking

Increases in mud weight increase the tendency for differential sticking. Bridging agent content should be maintained by hourly additions of the larger sizes of sacked BARACARB through the reservoir when the risk of differential sticking is at its greatest.

DFG hydraulics modeling should be run regularly to simulate static and dynamic overbalance under the respective conditions so that the equivalent mud weight across the reservoir sands is understood at all times. The fast bridging tendency of the mud, due to the presence of correctly sized BARACARB, will ensure a thin filter cake is deposited on newly drilled hole, and differential sticking tendencies are minimized.

Swab and Surge

The DFG hydraulics program can show likely swab and surge pressures when tripping in and out of the hole with the drillstring. The ECD limit for the reservoir sands should not be exceeded while running in the hole, and a swab limit should be set to control tripping speeds. Minimize the risk of inducing hole instability by reducing the equivalent mud weight (i.e., swabbing) by controlling the rate at which the string is pulled out of the hole such that a minimum EMW is maintained throughout the trip. Optimal tripping speeds can vary varied. The table below contains general guidelines for safe tripping speeds.

Table 3 Tripping Speed Guidelines

Depth Range MD BRT (ft)	Maximum Tripping Speed (fpm)	Maximum Minutes per Stand
6500 – surface	180	0.5
8500 – 6500	90	1.0

Solids Control

Good solids control is essential. The shale shakers are the primary means of solids control when running sized bridging material. **Run the shale shakers utilizing the finest mesh screens possible.** This has important implications for the PV of the mud, which if kept as low as possible will help flow rates and ECD values.

Shale shakers have to be attended to at all times, and screens changed as circumstances dictate to keep the optimum screen size on the shakers. Although counterintuitive, if sand blinding occurs going to a finer screen size rather than a coarser size may bring relief.

1.5. Rigsite Quality Control for DRIL-N Fluids

Efforts to optimize the drill-in fluid in the laboratory can be negated by poor quality control in the field. Therefore specific rigsite tests should be performed to ensure that the fluid remains fit for purpose. Changes in the size of BARACARB or BARAPLUG particles during drilling are unavoidable. Without regular testing and maintenance

the fluid will lose its ability to provide suitable bridging and this will result in increased filtrate invasion / potential formation damage.

Testing bridging capability should be conducted at the rigsite on a daily basis (at a minimum). However, if this is not practicable then samples of the fluid should be taken daily and sent to a suitably equipped Halliburton laboratory for testing.

Wellsite Monitoring

Testing should include:

- PSD
- Bridging agent concentration*
- PPA
- MBT
- Filtrate
- Chemistry [pH, K^+ , Ca^{+2} , Mg^{+2} , Fe^{+2} , Cl^- , CO_3^{2-} , HCO_3^- , SO_4^{2-} , etc.]
- Rheology

* BARACARB content can be measured by means of the acid / base titration described in “Completion Fluid Test Methods and Lab Capability”.

Lab Tests for Field Samples

- XRD (total suspended solids)
- Particle size distribution
- Residual damage
- Emulsion and scaling tendencies
- Filtercake clean-up

Permeability Plugging Test

This test confirms that the field mud formulation is still able to achieve the required level of bridging.

The test should be conducted using ceramic discs sized according to the expected pore throat diameters in the reservoir.

The PPT test should be run using the same temperature and overbalance pressure as those downhole. Do not exceed the pressure rating on your PPT cell (psi).

Two filtrate volumes should be recorded and an overall filtrate loss calculated. The two volumes are the spurt loss (i.e., the filtrate volume lost prior to formation of a filter cake) and the total loss (i.e., the total volume of filtrate collected over the 30 minute test period).

Particle Size Analysis

Particle size analysis helps ensure that the active mud system continues to maintain the particle size distribution required to achieve the desired bridging. Typically, particle size analysis is conducted using a laser light scattering instrument.

Particle size analyzers can be used in rigsite mud labs provided that they can be protected from dust and vibration as the instrument is relatively fragile and the replacement cost is significant.

Particle size analysis results are typically expressed as d10, d50 and d90 values. These are the particle sizes below which 10, 50 and 90% of the particles fall. The values are expressed in microns and represent the particle diameter.

1.6. Drill-In Fluid / Reservoir Drilling Questionnaire

Drill-in & Completion Fluids Design Questionnaire

In order to provide custom designed DIF/CF program that will include displacements and filter cake clean up certain questions need to be answered. Please fill out this document as fully as possible; customer input is necessary.

Well Information:

Producer ☐ or Injector ☐ // WBM ☐ or OBM ☐ // Oil producer ☐ or Gas producer ☐ or Water supply ☐

Reservoir rock type:

Carbonate [limestone /dolomite] ☐ Clastic [sandstone] ☐ Other [?] ☐

1. Reservoir permeability _____
2. Reservoir mineralogy _____
3. Shale types present _____

Completion type

Barefoot ☐ Slotted / pre-drilled Liner ☐ Pre-packed screen ☐ Stand alone ☐
Expandable Screen ☐ Open Hole Gravel Pack ☐ Other ☐

Required details (use additional sheets as needed):

☐ Offset well data attach _____	☐ Pore size distribution _____
☐ Hole size _____	☐ Permeability and porosity range _____
☐ Hole angle _____	☐ Fractures encountered / losses _____
☐ Section length _____	☐ Drilling / exposure time _____
☐ Mud density _____	☐ Environmental considerations _____
☐ Static BHT _____	

Lab data (attach additional sheets as needed)

Fluid Interaction Studies

Reservoir water analysis
Filtrate/water compatibility
Filtrate/oil compatibility
Brine/water compatibility
Brine/oil compatibility
Breaker/water compatibility

Available

☐
☐
☐
☐
☐
☐

Need to generate

☐
☐
☐
☐
☐
☐

Shale Studies

XRD
CEC
Dispersion testing
Linear swelling test
SEM analysis
Pore throat distribution
Moisture content
Shale salinity

Available

☐
☐
☐
☐
☐
☐
☐
☐

Need to generate

☐
☐
☐
☐
☐
☐
☐
☐

<u>Other Studies</u>	<u>Available</u>	<u>Need to generate</u>
Return permeability	☐	☐
Cake cleanup optimization	☐	☐
Evaluate cleanup through screens	☐	☐

Lab tests can help to determine the reservoir's potential for such problems as particulate dislodging, fines migration, swelling of hydratable clays, or chemical precipitation. The degree of damage that could be caused by anticipated problems can be modelled, as well as the effectiveness of possible solutions for mitigating the risks involved.

Core-plug tests should include thin-section petrography to evaluate pay zone mineralogy, average pore size, pore-throat size, porosity type, and distribution. The porosity and permeability of brine-saturated core plugs should be measured at a specified pressure and temperature.

Other Relevant Information:

Completion Fluids

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1. Completion Fluids

Completion and workover operations represent the final phase of well construction before production or the maintenance of a well during its productive life.

These operations can be grouped into three categories:

Completion	Referred to as primary or initial completion of the formation in a well bore. Can also be referred to in sidetrack operations when completing the same or different zone aside from the original well bore path.
Re-Completion	Completing a new (usually higher zone) in the original well bore path (Can also refer to sidetrack operations).
Workover	Performing repairs or restoration to the existing producing zone.

The terms re-completion and workover are often used interchangeably.

Information on completion and workover fluids is also found in the following chapters:

- Corrosion
- Displacement
- Lost circulation

Completion and workover fluids provide hydrostatic containment for wells with otherwise direct communication between the reservoir and the atmosphere. With an “open” reservoir they provide a pressure barrier for well control, allowing various well work operations to be safely performed (e.g., running completion tubing, pulling packers and tubing, etc.).

Completion fluids are chosen for the density they provide as well as compatibility with the reservoir rock and fluids to reduce or eliminate certain types of formation damage.

Temperature has a significant effect on the weight of a column of brine fluid, and density calculations should always account for the effect of temperature.

Selection factors also include the following considerations:

If a well is completed...	Then...
In an <i>underbalanced</i> situation, with an underbalanced packer fluid left in the well...	Casing design and cost are the main factors to consider when selecting a brine density and corresponding brine.
In an <i>overbalanced</i> situation, as a workover operation requiring a kill fluid...	The required density is determined by formation pressure, true vertical depth, and temperature gradient.

The most common base fluids are shown below:

- Water-Based
- Brine-Based
- Oil-Based
- Synthetic-Based

These same base fluids are used to formulate drilling fluids, drill-in fluids and completion fluids.

The two most common completion and workover fluids are clear fluid systems and solids enhanced systems.

1.1. Clear Fluid Systems

Clear fluid systems include the following:

- Seawater
- Formation water
- Brine fluid

A clear-fluid system is the preferred completion or workover system because their lack of solids can reduce formation damage. In addition, clear-fluid systems make excellent packer fluids due to the lack of solids to settle out over time.

Seawater

Seawater is readily available in coastal areas. Sodium chloride (NaCl) and potassium chloride (KCl) may be added to adjust the fluid density and to inhibit clay swelling. All fluids should be checked for the following:

- Solids contamination
- Possible introduction of bacteria
- Dissolved mineral or solids which may precipitate as insoluble

Clear-fluid systems are solutions of salts that are classified into two major groups: monovalent and divalent brines.

- Monovalent solutions contain sodium and potassium
- Divalent solutions contain calcium and/or zinc.

Clear-fluid System Selection

To determine whether a fluid will perform effectively in the planned completion or workover operation, consider the following factors:

- Density
- Crystallization point
- Pressure crystallization temperature
- Hydrate inhibition
- Brine / Formation water compatibility
- Corrosion

Density

Clear brines are used in both underbalanced and overbalanced conditions. Frequently, a well is completed in an overbalanced situation and the heavy brine is replaced with a lighter packer fluid.

Table 1 Typical Brine Density Ranges

Monovalent Brines	Density (ppg)
Sodium Bromide	10.0 to 12.7
Sodium Chloride	8.3 to 10.0
Potassium Chloride	8.3 to 9.7
Divalent Brines	Density (ppg)
Zinc Bromide	15.2 to 19.2
Calcium Bromide	11.7 to 15.1
Calcium Chloride	8.3 to 11.6
Formate Brines	Density (ppg)

Cesium Formate	13.0 to 19.5
Potassium Formate	11.1 to 13.1
Sodium Formate	8.3 to 11.0

Crystallization Point

Brines can crystalize into a solid at temperatures above or below the freezing point of fresh water. A brine's crystallization point is the temperature at which salt crystals will begin to fall out of solution given sufficient time and proper nucleating conditions.

Nucleation is the process by which insoluble matter provides a physical platform upon which crystals can form.

The precipitation of insoluble salts can cause a number of problems. For example, when the dissolved salt in the fluid crystallizes and settles in a tank, the fluid density usually drops. Crystallization in brines can also cause lines to plug and pumps to seize.

The following precautions should be taken to ensure that crystallization does not occur in a brine:

- Determine the required crystallization point of the fluid
- Check the actual crystallization point of the fluid
- Adjust the crystallization point of the fluid, as necessary

Determining the Required Crystallization Point

In choosing the lowest-cost formulation for a given density, consider the temperatures at which the brine will be transported, stored, and used. The crystallization point of a fluid should be a minimum of 10 F (6 C) less than the lowest projected temperature of exposure. For deepwater projects, consider the seawater temperature at the ocean floor.

Determining the Actual Crystallization Point

Three temperature values can be used to describe a fluid's crystallization point. These include the:

- First crystal to appear (FCTA)
- Last crystal to dissolve (LCTD)
- **True crystallization temperature (TCT)**

The TCT is the API-prescribed method for describing crystallization point; all temperature values can be determined at the wellsite using the brine-crystallization test kit.

Adjusting the Crystallization Point

Although fluid delivered to a wellsite is formulated to have the correct density and crystallization point for the well and weather conditions, it may be necessary to adjust the crystallization point. This is done by adding dry salts (e.g., CaBr₂ or CaCl₂), stock brines (e.g., 14.2 lb/gal (1.70 sg) CaBr₂ or 19.2 lb/gal (2.30 sg) Ca/ZnBr₂), or water.

Adjusting fluid density using dry salts affects the crystallization point.

For single-salt solutions, the addition of the same type of dry salt lowers the crystallization point of the solution down to the eutectic point, which is the lowest freezing point of a solution obtainable by increasing the concentration of a solute. For example, the addition of dry calcium chloride to water and calcium brines lowers the crystallization point of the brine solution until it reaches a density of 10.8 lb/gal (1.29 sg). Further, the

addition of dry calcium chloride to a 10.8 lb/gal (1.29 sg) brine solution raises the crystallization point, even though the density continues to increase.

For two-salt brines with a crystallization point of 30 °F (-1 °C), the addition of a dry salt in general raises the crystallization point.

The addition of fresh water to a single-salt brine with a density above the eutectic point lowers the density and crystallization point (Figure 2).

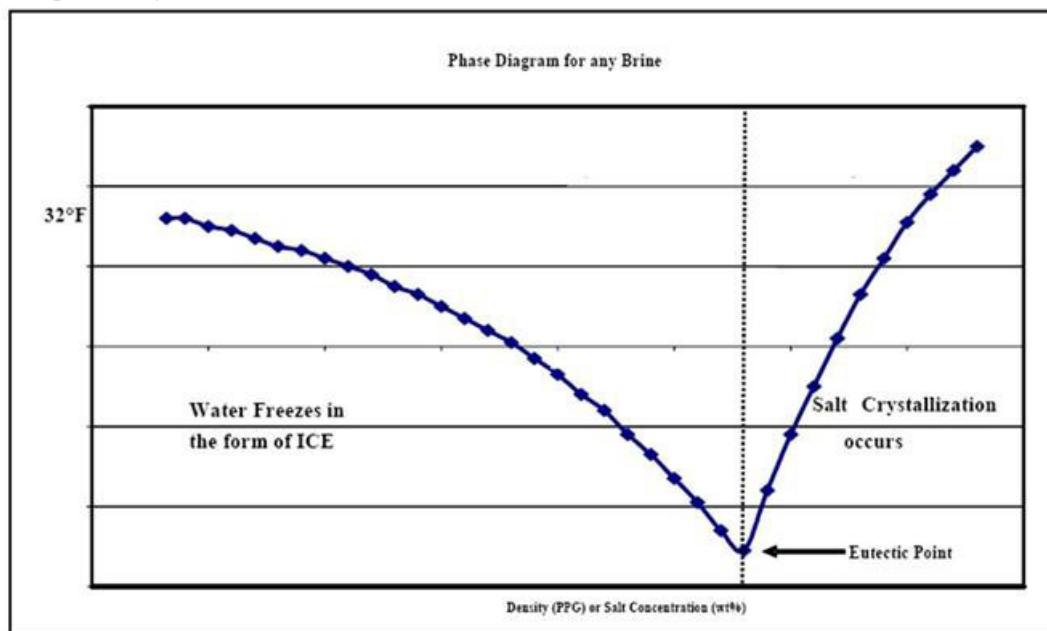


Figure 1 Eutectic point: the lowest freezing point of a solution.

Pressure Crystallization Temperature (PCT)

Applying pressure of 10,000 psi to divalent brines raises their crystallization temperature by as much as 10°F to 20°F. Monovalent brines have less pressure dependency – generally only 1°F to 5°F increases in PCT up to 10,000 psia. Recent tests of certain monovalent blends have shown an interesting decrease in PCT with increasing pressure.

In deeper water, crystallization is most likely to occur at the sea floor – typically the coldest point in the well – in the mud-line wellhead, subsea tree, BOP stack, and choke and kill lines. The choke and kill lines are most vulnerable because they reach seafloor temperature quickly – perhaps within 30 minutes of cessation of circulation. Modeling can be used to predict temperature response.

The accepted practice to avoid crystallization is to choose a salt blend with a PCT that is 10°F below the lowest anticipated temperature at the highest anticipated pressure, which often comes during BOP testing. For a well in 5,000 feet of water, with 11.0 ppg fluid and a BOP test pressure of 7,500 psig (measured at surface), the pressure at the BOP stack is 10,400 psia (7,500 psig + 15 psia atmospheric + 2,860 psi hydrostatic). With a sea floor temperature of 38 °F, the 14-degree TCT brine would be the proper choice to have a 28-degree PCT at the BOP test pressure. The PCT of the selected brine should be confirmed in the lab.

Baroid uses a laboratory Variable Pressure Crystalometer (VPC) to accurately determine the crystallization point under pressure. VPC is capable of measuring pressures from 20,000 to 25,000 psi and -5 °F to + 120 °F and

utilizes proprietary measurement software. The test cell and computer software operate together to produce various pressure/temperature ramps. Multiple runs are made to verify data points. Software algorithms detect super-cooling and run new cycles to avoid incorrect data.

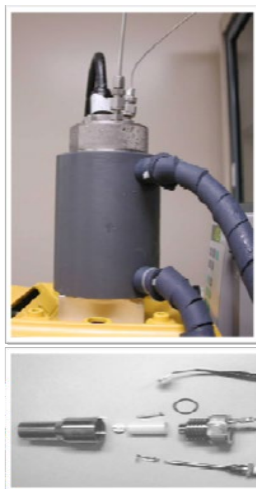


Figure 2 Variable Pressure Crystalometer (VPC)

Changes in Density and Salt Composition

Increasing or maintaining density by adding dry salt or by adding volumes of saturated ‘spike’ brine can change the proportion of salts in a multi-salt blend, which can alter the PCT and hydrate inhibition.

The common practice of using spike fluid to slug the workstring before tripping to prevent U-tubing on trips should be done with caution. Adding water to reduce density will cause the hydrate equilibrium curve to shift, possibly increasing the risk of forming hydrates. Adding lighter salt brine or alternatively, adding drill water along with a hydrate inhibitor, might be a safer option. Density control contingencies should be worked out in advance.

Brine / Formation Water Compatibility

To select the correct brine type, consider the potential interactions of the completion or workover fluid with formation solids, water, and gases. The most common incompatibility problems include:

- Scale production from the reaction of a divalent brine with dissolved carbon dioxide
- Precipitation of sodium chloride from the formation water when it is exposed to certain brines
- Precipitation of iron compounds in the formation resulting from interaction with soluble iron in the completion fluid
- Reaction of formation clays with the clear brine

The following laboratory tests can be used to evaluate the compatibility of a clear fluid with a formation:

- Return permeability
- Formation water analysis
- Formation mineralogy
- Brine/water compatibility

Hydrate Inhibition

Gas hydrates are ice-like solids which form from gas and water under conditions of low temperature and high pressure. Methane, ethane, propane, butane, carbon dioxide and hydrogen sulfide are among the many gasses that can form hydrates.

Gas hydrate formation depends on pressure, temperature and gas/fluid composition. When combined with an unexpected shut-in, failure to prevent hydrate formation can risk plugging of the well and subsea facilities. Plug remediation can be very expensive, especially in the flowlines where vertical intervention (typically washing with coiled tubing) is not possible.

In deepwater operations, the combination of cold sea-floor temperature and substantial imposed pressures results in conditions conducive for hydrate formation. Hydrates can form at the lowest temperature/highest pressure locations in the well. Gas hydrates are often associated with well control and can lead to the following problems:

- Plugging of choke and kill lines, BOP and riser
- Interference with well control operations
- Release of large quantities of gas near the surface as the hydrates melt / decompose
- Plugging of well test equipment during well testing

The fluid density helps determine which brines can be used for hydrate suppression. The candidate brines must be screened for compatibility with the formation water and their PCT values. Completion fluids utilize thermodynamic inhibitors such as salts and alcohols to tie up free water to prevent hydrate formation with the following considerations:

- Hydrate Suppression Design Requirements
- Brine Choices
- Hydrate Modeling*

*Baroid uses the hydrate prediction software PVTsim from CalSep engineering.

Corrosion

The corrosivity of a completion or workover fluid depends on its type.

Monovalent fluids generally show low corrosivity, even at temperatures exceeding 400°F (204°C).

The corrosivity of divalent fluids depends on the density and chemical composition of the fluid. Laboratory data show that, for divalent fluids not treated with corrosion inhibitors, the addition of calcium chloride gives a slower rate of corrosion compared to that of zinc bromide which gives a faster rate of corrosion.

1.2. Conventional Brine Formulations / Solution Requirements

The recommended solution requirements for the following brine formulations are shown in the tables below:

- Sodium chloride
- Potassium chloride
- Potassium bromide
- Potassium nitrate
- Potassium acetate
- Calcium chloride
- Sodium bromide
- Sodium bromide/sodium chloride
- Calcium bromide
- Calcium bromide/calcium chloride
- Zinc Bromide Summer Blend - Mixed From 14.2 ppg CaBr₂, 19.2 ppg ZnBr₂, and dry CaCl₂
- Zinc Bromide Winter Blend - Mixed From 14.2 ppg CaBr₂, 19.2 ppg ZnBr₂, and dry CaCl₂

The formulations are based on LCTD values, not TCT values.

Sodium Chloride

Dry sodium chloride or sodium chloride brine can be used to produce the required crystallization point (CP).

Table 2 Sodium chloride solution requirements to make 1 bbl (42 gal)

Using sacked NaCl (100%)		Brine density at 70°F (21°C), lb/gal	Specific gravity, SG	CP (LCTD) °F (°C)	Using 10.0 lb/gal NaCl brine	
Fresh water, bbl	100% NaCl, lb				Water, bbl	10 lb/gal NaCl, bbl
0.998	4	8.4	1.01	31 (-0.6)	0.96	0.04
0.993	9	8.5	1.02	29 (-1.7)	0.90	0.10
0.986	16	8.6	1.03	27 (-2.8)	0.84	0.16
0.981	22	8.7	1.04	26 (-3.3)	0.78	0.22
0.976	28	8.8	1.05	24 (-4.4)	0.72	0.28
0.969	35	8.9	1.07	22 (-5.6)	0.66	0.34
0.962	41	9.0	1.08	19 (-7.2)	0.60	0.40
0.955	47	9.1	1.09	17 (-8.3)	0.54	0.46
0.948	54	9.2	1.10	14 (-10.0)	0.48	0.52
0.940	61	9.3	1.11	11 (-11.7)	0.42	0.58
0.933	68	9.4	1.13	9 (-12.8)	0.36	0.64
0.926	74	9.5	1.14	6 (-14.4)	0.30	0.70
0.919	81	9.6	1.15	3 (-16.1)	0.24	0.76
0.910	88	9.7	1.16	-1 (-18.3)	0.18	0.82
0.902	95	9.8	1.17	-5 (-20.5)	0.12	0.88
0.895	102	9.9	1.19	5 (-15.0)	0.06	0.94
0.888	109	10.0	1.20	25 (-3.9)	---	1.00

Potassium Chloride

Dry potassium chloride can be added to produce the required crystallization point (CP).

Table 3 Potassium chloride solution requirements to make 1 bbl (42 gal)

Using sacked KCl (100%)		Brine density at 70°F (21°C), lb/gal	Specific gravity, sg	CP °F (°C)	Potassium, ppm	Chloride, ppm	%by weight KCl
Fresh water, bbl	100% KCl, lb						
0.995	4.0	8.4	1.01	31 (-0.6)	005946	005392	1.1
0.986	11.6	8.5	1.02	29 (-1.7)	017041	015452	3.2
0.976	18.9	8.6	1.03	28 (-2.2)	027441	024882	5.2
0.969	26.1	8.7	1.04	26 (-3.3)	037460	033969	7.1
0.960	33.4	8.8	1.05	25 (-3.9)	047392	042976	9.1
0.950	40.7	8.9	1.07	23 (-5.0)	057102	051780	10.9
0.943	47.9	9.0	1.08	22 (-5.6)	066456	060263	12.7
0.933	55.2	9.1	1.09	20 (-6.7)	075743	068684	14.4
0.924	62.4	9.2	1.10	18 (-7.8)	084692	076799	16.1
0.917	69.7	9.3	1.11	16 (-8.9)	093582	084861	17.8
0.907	76.9	9.4	1.13	14 (-10.0)	102151	092631	19.5
0.898	84.2	9.5	1.14	18 (-7.8)	110671	100357	21.1
0.890	91.5	9.6	1.15	40 (4.4)	119013	107922	22.7
0.881	98.7	9.7	1.16	60 (15.6)	127054	115214	24.2

Potassium Bromide

Dry potassium bromide can be added to produce the required crystallization point (CP).

Table 4 Potassium bromide solution requirements to make 1 bbl (42 gal)

Using sacked KCl (100%)		Brine density at 70°F (21°C), lb/gal	Specific gravity, sg	TCT °F	% by weight KCl
Fresh water, bbl	100% KBr, lb				
0.996	4.6	8.4	1.008	31	1.3%
0.994	9.5	8.5	1.020	30	2.7%
0.989	15.4	8.6	1.032	30	4.2%
0.981	22.3	8.7	1.044	29	6.1%
0.978	27.7	8.8	1.056	28	7.5%
0.970	34.6	8.9	1.068	27	9.3%
0.965	40.6	9.0	1.080	26	10.8%
0.960	46.6	9.1	1.092	25	12.2%
0.954	52.7	9.2	1.104	24	13.6%
0.950	58.6	9.3	1.116	23	15.0%
0.944	64.8	9.4	1.128	22	16.4%
0.939	70.8	9.5	1.140	21	17.7%
0.933	76.8	9.6	1.152	20	19.1%
0.928	82.9	9.7	1.164	19	20.4%
0.923	88.9	9.8	1.176	17	21.6%
0.918	94.8	9.9	1.188	16	22.8%
0.913	100.8	10.0	1.200	15	24.0%
0.907	106.9	10.1	1.212	14	25.2%
0.902	113.1	10.2	1.224	12	26.4%
0.897	119.0	10.3	1.236	11	27.5%
0.891	125.4	10.4	1.248	10	28.7%
0.885	131.4	10.5	1.261	8	29.8%
0.880	137.6	10.6	1.273	11	30.9%
0.873	144.3	10.7	1.285	18	32.1%
0.867	150.6	10.8	1.297	23	33.2%
0.860	157.0	10.9	1.309	32	34.3%
0.855	163.1	11.0	1.321	36	35.3%
0.848	169.7	11.1	1.333	42	36.4%
0.842	175.9	11.2	1.345	49	37.4%
0.837	182.0	11.3	1.357	55	38.3%
0.831	188.4	11.4	1.369	68	39.3%
0.824	194.9	11.5	1.381	75	40.4%

Calcium Chloride

Dry calcium chloride or calcium chloride brine can be used to produce the required crystallization point (CP).

Table 5 Calcium chloride solution requirements to make 1 bbl (42 gal)

Using sacked CaCl ₂ (94-97%)		Brine density at 70°F (21°C), lb/gal	Specific gravity, sg	CP (LCTD) °F (°C)	Using 11.6 lb/gal CaCl ₂ brine (38%)	
Fresh water, bbl	CaCl ₂ , lb				Fresh water, bbl	11.6 lb/gal CaCl ₂ , bbl
0.998	3.8	8.4	1.01	31 (-0.6)	0.979	0.021
0.997	8.2	8.5	1.02	30 (-1.1)	0.948	0.052
0.994	13.4	8.6	1.03	29 (-1.7)	0.917	0.083
0.991	18.7	8.7	1.04	27 (-2.8)	0.887	0.113
0.987	24.2	8.8	1.05	25 (-3.9)	0.856	0.144
0.984	29.4	8.9	1.07	24 (-4.4)	0.826	0.174
0.980	35.1	9.0	1.08	22 (-5.6)	0.795	0.205
0.977	40.5	9.1	1.09	20 (-6.7)	0.765	0.235
0.972	46.2	9.2	1.10	18 (-7.8)	0.734	0.266
0.968	52.0	9.3	1.11	15 (-9.4)	0.703	0.297
0.963	57.8	9.4	1.13	13 (-10.6)	0.673	0.327
0.959	63.4	9.5	1.14	10 (-12.2)	0.642	0.358
0.954	69.3	9.6	1.15	7 (-13.9)	0.612	0.388
0.949	75.4	9.7	1.16	4 (-15.6)	0.581	0.419
0.944	81.5	9.8	1.17	0 (-17.8)	0.550	0.450
0.939	87.4	9.9	1.19	4 (-20.0)	0.520	0.480
0.934	93.2	10.0	1.20	9 (-22.8)	0.489	0.511
0.929	99.3	10.1	1.21	13 (-25.0)	0.459	0.541
0.923	105.4	10.2	1.22	18 (-27.8)	0.428	0.572
0.918	111.3	10.3	1.23	23 (-30.6)	0.398	0.602
0.912	117.6	10.4	1.25	29 (-33.9)	0.367	0.633
0.908	123.5	10.5	1.26	36 (-37.8)	0.336	0.640
0.902	129.8	10.6	1.27	43 (-41.7)	0.306	0.694
0.895	136.3	10.7	1.28	51 (-46.1)	0.275	0.725
0.891	142.0	10.8	1.29	57 (-49.4)	0.245	0.755
0.885	148.3	10.9	1.31	35 (-37.2)	0.214	0.786
0.878	155.0	11.0	1.32	19 (-28.3)	0.183	0.817
0.872	161.3	11.1	1.33	6 (-21.1)	0.153	0.847
0.866	167.6	11.2	1.34	7 (-13.9)	0.122	0.878
0.859	174.1	11.3	1.35	19 (-7.8)	0.092	0.908
0.853	180.4	11.4	1.37	27 (-2.8)	0.061	0.939
0.846	186.9	11.5	1.38	36 (2.2)	0.031	0.969
0.840	193.2	11.6	1.39	44 (6.7)	---	1.000

Sodium Bromide

Dry sodium bromide can be used to produce the required crystallization point (CP).

Table 6 Sodium bromide solution requirements to make 1 bbl (42 gal)

Using sacked NaBr (95%)		Brine density at 70 F (21 C), lb/gal	Specific gravity, sg	CP (LCTD) F (C)
Fresh water, bbl	95% NaBr, lb			
0.999	2.1	8.4	1.01	31 (-0.6)
0.996	7.6	8.5	1.02	30 (-1.1)
0.992	13.7	8.6	1.03	29 (-1.7)
0.989	19.2	8.7	1.04	29 (-1.7)
0.984	25.0	8.8	1.05	28 (-2.2)
0.979	31.0	8.9	1.07	26 (-3.3)
0.975	36.7	9.0	1.08	25 (-3.9)
0.970	42.6	9.1	1.09	24 (-4.4)
0.966	48.3	9.2	1.10	23 (-5.0)
0.961	54.2	9.3	1.11	22 (-5.6)
0.956	60.2	9.4	1.13	21 (-6.1)
0.950	66.4	9.5	1.14	20 (-6.7)
0.946	72.0	9.6	1.15	19 (-7.2)
0.941	77.9	9.7	1.16	18 (-7.8)
0.937	83.6	9.8	1.17	16 (-8.9)
0.933	89.2	9.9	1.19	15 (-9.4)
0.927	95.4	10.0	1.20	14 (-10.0)
0.923	101.1	10.1	1.21	12 (-11.1)
0.918	107.1	10.2	1.22	11 (-11.7)
0.914	112.6	10.3	1.23	10 (-12.2)
0.910	118.2	10.4	1.25	8 (-13.3)
0.905	124.1	10.5	1.26	6 (-14.4)
0.900	130.2	10.6	1.27	5 (-15.0)
0.895	136.0	10.7	1.28	4 (-15.6)
0.891	141.7	10.8	1.29	2 (-16.7)
0.886	147.6	10.9	1.31	0 (-17.8)
0.882	153.3	11.0	1.32	2 (-18.8)
0.877	159.2	11.1	1.33	3 (-19.4)
0.872	165.1	11.2	1.34	5 (-20.6)
0.867	171.1	11.3	1.35	7 (-21.7)
0.862	177.0	11.4	1.37	9 (-22.8)
0.857	183.0	11.5	1.38	11 (-23.9)
0.853	188.6	11.6	1.39	14 (-25.6)
0.847	194.8	11.7	1.40	16 (-26.7)
0.844	200.2	11.8	1.41	19 (-28.3)
0.839	206.2	11.9	1.43	10 (-23.3)
0.834	212.0	12.0	1.44	6 (-14.4)
0.831	217.3	12.1	1.45	14 (-10.0)
0.825	223.6	12.2	1.46	27 (-2.8)
0.823	228.5	12.3	1.47	34 (1.1)
0.816	235.1	12.4	1.49	43 (6.1)
0.812	240.7	12.5	1.50	50 (10.0)
0.807	246.7	12.6	1.51	57 (13.9)
0.804	252.0	12.7	1.52	63 (17.2)

Sodium Bromide / Sodium Chloride

Solutions of sodium chloride or sodium bromide can be used to produce the required crystallization point. To achieve the highest crystallization points, use dry sodium bromide.

Table 7 Sodium bromide/sodium chloride solution requirements to make 1 bbl (42 gal)

Using 10.0 lb/gal NaCl, 12.3 lb/gal NaBr, and sacked (95%) NaBr				Brine density at 70°F (21°C), lb/gal	Specific gravity, SG	CP (LCTD) 1°F (°C)
Fresh water, bbl	10 lb/gal NaCl, bbl	12.3 lb/gal NaBr, bbl	95% NaBr, lb			
0.982	---	0.018	---	8.4	1.01	31 (-0.6)
0.957	---	0.043	---	8.5	1.02	30 (-1.1)
0.932	---	0.068	---	8.6	1.03	29 (-1.7)
0.907	---	0.093	---	8.7	1.04	29 (-1.7)
0.882	---	0.118	---	8.8	1.05	28 (-2.2)
0.856	---	0.144	---	8.9	1.07	26 (-3.3)
0.831	---	0.169	---	9.0	1.08	25 (-3.9)
0.806	---	0.194	---	9.1	1.09	24 (-4.4)
0.781	---	0.219	---	9.2	1.10	23 (-5.0)
0.756	---	0.244	---	9.3	1.11	22 (-5.6)
0.730	---	0.270	---	9.4	1.13	21 (-6.1)
0.705	---	0.295	---	9.5	1.14	20 (-6.7)
0.680	---	0.320	---	9.6	1.15	19 (-7.2)
0.655	---	0.345	---	9.7	1.16	18 (-7.8)
0.630	---	0.370	---	9.8	1.17	16 (-8.9)
0.605	---	0.395	---	9.9	1.19	15 (-9.4)
0.579	---	0.421	---	10.0	1.20	14 (-10.0)
---	0.957	0.043	---	10.1	1.21	25 (-3.9)
---	0.913	0.087	---	10.2	1.22	26 (-3.3)
---	0.870	0.130	---	10.3	1.23	26 (-3.3)
---	0.826	0.174	---	10.4	1.25	27 (-2.8)
---	0.782	0.218	---	10.5	1.26	27 (-2.8)
---	0.739	0.261	---	10.6	1.27	27 (-2.8)
---	0.696	0.304	---	10.7	1.28	28 (-2.2)
---	0.652	0.348	---	10.8	1.29	28 (-2.2)
---	0.609	0.391	---	10.9	1.31	29 (-1.7)
---	0.565	0.435	---	11.0	1.32	29 (-1.7)
---	0.522	0.478	---	11.1	1.33	29 (-1.7)
---	0.478	0.522	---	11.2	1.34	30 (-1.1)
---	0.435	0.565	---	11.3	1.35	30 (-1.1)
---	0.391	0.609	---	11.4	1.37	31 (-0.6)
---	0.348	0.652	---	11.5	1.38	31 (-0.6)
---	0.304	0.696	---	11.6	1.39	31 (-0.6)
---	0.261	0.739	---	11.7	1.40	32 (0.0)
---	0.217	0.783	---	11.8	1.41	32 (0.0)
---	0.174	0.826	---	11.9	1.43	32 (0.0)
---	0.130	0.870	---	12.0	1.44	33 (0.6)
---	0.087	0.913	---	12.1	1.45	33 (0.6)
---	0.043	0.957	---	12.2	1.46	33 (0.6)
---	---	1.000	---	12.3	1.47	34 (1.1)
---	---	0.996	6.6	12.4	1.49	43 (6.1)
---	---	0.993	12.2	12.5	1.50	50 (10.0)
---	---	0.989	18.2	12.6	1.51	57 (13.9)
---	---	0.986	23.5	12.7	1.52	63

Calcium Bromide

Dry calcium bromide can be used to produce the required crystallization point (CP).

Table 8 Calcium bromide solution requirements to make 1 bbl (42 gal)

Using sacked CaBr2 (95%)		Brine density at 70°F (21°C), lb/gal	Specific gravity, sg	CP (LCTD) °F (°C)
Fresh water, bbl	95% CaBr2, lb			
0.822	197	11.7	1.40	19 (-28.3)
0.817	203	11.8	1.41	23 (-30.6)
0.811	210	11.9	1.43	25 (-31.7)
0.806	216	12.0	1.44	28 (-33.3)
0.801	222	12.1	1.45	30 (-34.4)
0.795	228	12.2	1.46	34 (-36.7)
0.790	233	12.3	1.47	36 (-37.8)
0.784	240	12.4	1.49	40 (-40.0)
0.778	247	12.5	1.50	44 (-42.2)
0.773	252	12.6	1.51	47 (-43.9)
0.767	259	12.7	1.52	52 (-46.7)
0.762	265	12.8	1.53	55 (-48.3)
0.756	272	12.9	1.55	61 (-51.7)
0.750	277	13.0	1.56	63 (-52.8)
0.746	282	13.1	1.57	66 (-54.4)
0.739	290	13.2	1.58	71 (-57.2)
0.732	298	13.3	1.59	76 (-60.0)
0.728	302	13.4	1.61	79 (-61.7)
0.723	308	13.5	1.62	81 (-62.8)
0.717	315	13.6	1.63	81 (-62.8)
0.711	322	13.7	1.64	81 (-62.8)
0.704	328	13.8	1.65	81 (-62.8)
0.699	334	13.9	1.67	80 (-62.2)
0.692	342	14.0	1.68	50 (-45.5)
0.687	348	14.1	1.69	40 (-40.0)
0.681	354	14.2	1.70	5 (-15.0)
0.676	360	14.3	1.71	10 (-12.2)
0.669	368	14.4	1.73	23 (-5.0)
0.662	376	14.5	1.74	35 (1.7)
0.655	383	14.6	1.75	37 (2.8)
0.651	388	14.7	1.76	44 (6.7)
0.645	394	14.8	1.77	51 (10.6)
0.640	400	14.9	1.79	56 (13.3)
0.637	405	15.0	1.80	60 (15.6)
0.632	410	15.1	1.81	65 (18.3)
0.626	415	15.2	1.82	70 (21.1)
0.621	421	15.3	1.83	76 (24.4)
0.616	427	15.4	1.85	79 (26.1)
0.611	433	15.5	1.86	81 (27.2)

Calcium Bromide / Calcium Chloride

Solutions of calcium chloride brine, dry calcium chloride, and calcium bromide can be used to produce the required crystallization point (CP).

Table 9 Calcium bromide/calcium chloride solution requirements to make 1 bbl (42 gal)

Using 11.6 lb/gal CaCl ₂ , 14.2 lb/gal CaBr ₂ , and sacked CaCl ₂ (94-97%)			Brine density at 70°F (21°C), lb/gal	Specific gravity, sg	CP (LCTD) °F (°C)
11.6 lb/gal CaCl ₂ , bbl	14.2 lb/gal CaBr ₂ , bbl	Sacked CaCl ₂ (94-97%), lb			
0.9714	0.0254	2.86	11.7	1.40	45 (7.2)
0.9429	0.0507	6.06	11.8	1.41	51 (10.6)
0.9143	0.0768	9.09	11.9	1.43	52 (11.1)
0.8857	0.1016	12.13	12.0	1.44	54 (12.2)
0.8572	0.1269	15.15	12.1	1.45	55 (12.8)
0.8286	0.1524	18.18	12.2	1.46	55 (12.8)
0.8000	0.1778	21.22	12.3	1.47	56 (13.3)
0.7715	0.2032	24.24	12.4	1.49	56 (13.3)
0.7429	0.2286	27.28	12.5	1.50	57 (13.9)
0.7143	0.2540	30.31	12.6	1.51	57 (13.9)
0.6847	0.2794	33.34	12.7	1.52	58 (14.4)
0.6472	0.3048	36.37	12.8	1.53	58 (14.4)
0.6286	0.3302	39.41	12.9	1.55	59 (15.0)
0.6000	0.3556	42.44	13.0	1.56	59 (15.0)
0.5714	0.3810	45.47	13.1	1.57	60 (15.6)
0.5429	0.4064	48.49	13.2	1.58	60 (15.6)
0.5143	0.4318	51.53	13.3	1.59	60 (15.6)
0.4857	0.4572	54.56	13.4	1.61	61 (16.1)
0.4572	0.4826	57.59	13.5	1.62	61 (16.1)
0.4286	0.5080	60.62	13.6	1.63	62 (16.7)
0.4000	0.5334	63.66	13.7	1.64	62 (16.7)
0.3714	0.5589	66.69	13.8	1.65	63 (17.2)
0.3429	0.5842	69.72	13.9	1.67	63 (17.2)
0.3143	0.6069	72.75	14.0	1.68	64 (17.8)
0.2857	0.6351	75.78	14.1	1.69	64 (17.8)
0.2572	0.6604	78.81	14.2	1.70	64 (17.8)
0.2286	0.6858	81.84	14.3	1.71	65 (18.3)
0.2000	0.7113	84.88	14.4	1.73	65 (18.3)
0.1715	0.7366	87.90	14.5	1.74	65 (18.3)
0.1429	0.7620	90.94	14.6	1.75	66 (18.9)
0.1143	0.7875	93.97	14.7	1.76	66 (18.9)
0.0858	0.8128	96.99	14.8	1.77	67 (19.4)
0.0572	0.8382	100.03	14.9	1.79	67 (19.4)
0.0286	0.8637	103.06	15.0	1.80	67 (19.4)
0.0000	0.8891	106.10	15.1	1.81	68 (20.0)

Zinc Bromide Summer Blend (calcium bromide/calcium chloride solution requirements)

Solutions of zinc bromide, calcium bromide brine, and dry calcium chloride, can be used to produce the required crystallization point (CP).

Density	Specific Gravity	14.2 ppg CaBr ₂ , bbl	19.2 ppg ZnBr ₂ , bbl	94 - 97% CaCl ₂ , lb	TCT
15.2	1.825	0.867	0.024	103.4	61
15.3	1.837	0.845	0.049	100.8	59
15.4	1.849	0.824	0.073	98.2	59
15.5	1.861	0.802	0.098	95.7	59
15.6	1.873	0.780	0.122	93.1	58
15.7	1.885	0.759	0.146	90.5	57
15.8	1.897	0.737	0.171	87.9	55
15.9	1.909	0.715	0.195	85.3	54
16.0	1.921	0.694	0.220	82.7	53
16.1	1.933	0.672	0.244	80.1	52
16.2	1.945	0.650	0.268	77.6	50
16.3	1.957	0.629	0.293	75.0	50
16.4	1.969	0.607	0.317	72.4	49
16.5	1.981	0.585	0.341	69.8	47
16.6	1.993	0.564	0.366	67.2	46
16.7	2.005	0.542	0.390	64.6	43
16.8	2.017	0.520	0.415	62.0	40
16.9	2.029	0.499	0.439	59.5	36
17.0	2.041	0.477	0.463	56.9	32
17.1	2.053	0.455	0.488	54.3	28
17.2	2.065	0.434	0.512	51.7	31
17.3	2.077	0.412	0.537	49.1	35
17.4	2.089	0.390	0.561	46.5	37
17.5	2.101	0.368	0.585	44.0	41
17.6	2.113	0.347	0.610	41.4	45
17.7	2.125	0.325	0.634	38.8	44
17.8	2.137	0.303	0.659	36.2	44
17.9	2.149	0.282	0.683	33.6	43
18.0	2.161	0.260	0.707	31.0	43
18.1	2.173	0.238	0.732	28.4	42
18.2	2.185	0.217	0.756	25.9	41
18.3	2.197	0.195	0.780	23.3	37
18.4	2.209	0.173	0.805	20.7	35
18.5	2.221	0.152	0.829	18.1	32
18.6	2.233	0.130	0.854	15.5	28
18.7	2.245	0.108	0.878	12.9	25
18.8	2.257	0.087	0.902	10.3	23
18.9	2.269	0.065	0.927	7.8	18
19.0	2.281	0.043	0.951	5.2	18
19.1	2.293	0.022	0.976	2.6	17
19.2	2.305	-	1.000	-	16

Zinc Bromide Winter Blend (calcium bromide/calcium chloride solution requirements)

Solutions of zinc bromide, and calcium bromide brine, can be used to produce the required crystallization point (CP).

Density	Specific Gravity	14.2 ppg CaBr ₂ , bbl	19.2 ppg ZnBr ₂ , bbl	TCT
15.0	1.801	0.840	0.160	-22
15.1	1.813	0.820	0.180	-25
15.2	1.825	0.800	0.200	-27
15.3	1.837	0.780	0.220	-29
15.4	1.849	0.760	0.240	-32
15.5	1.861	0.740	0.260	-34
15.6	1.873	0.720	0.280	-35
15.7	1.885	0.700	0.300	-38
15.8	1.897	0.680	0.320	-40
15.9	1.909	0.660	0.340	-37
16.0	1.921	0.640	0.360	-33
16.1	1.933	0.620	0.380	-30
16.2	1.945	0.600	0.400	-26
16.3	1.957	0.580	0.420	-23
16.4	1.969	0.560	0.440	-20
16.5	1.981	0.540	0.460	-16
16.6	1.993	0.520	0.480	-11
16.7	2.005	0.500	0.500	-8
16.8	2.017	0.480	0.520	-6
16.9	2.029	0.460	0.540	-4
17.0	2.041	0.440	0.560	-4
17.1	2.053	0.420	0.580	-2
17.2	2.065	0.400	0.600	0
17.3	2.077	0.380	0.620	2
17.4	2.089	0.360	0.640	4
17.5	2.101	0.340	0.660	5
17.6	2.113	0.320	0.680	5
17.7	2.125	0.300	0.700	6
17.8	2.137	0.280	0.720	7
17.9	2.149	0.260	0.740	7
18.0	2.161	0.240	0.760	9
18.1	2.173	0.220	0.780	10
18.2	2.185	0.200	0.800	11
18.3	2.197	0.180	0.820	13
18.4	2.209	0.160	0.840	15
18.5	2.221	0.140	0.860	17
18.6	2.233	0.120	0.880	19
18.7	2.245	0.100	0.900	21
18.8	2.257	0.080	0.920	23
18.9	2.269	0.060	0.940	20
19.0	2.281	0.040	0.960	21
19.1	2.293	0.020	0.980	18
19.2	2.305	-	1.000	16

1.3. Solids Enhanced Water-Based Systems

A solids-enhanced fluid is often recommended for completion or workover operations when the use of a clear brine would result in the loss of large fluid volumes to the formation.

BARACARB sized calcium carbonate is used to bridge off the reservoir rock to reduce or eliminate fluid losses. BARACARB is acid-soluble so it can be completely removed from the well by applying acid treatments. Unlike limestone, BARACARB is ground marble, much harder than limestone. Marble retains its size range much better than limestone and is very much preferred over “calcium carbonate” for bridging material.

BARACARB comes in various grind sizes, such as 5, 25, 50 and 150 microns. When formulating a fluid for lost circulation in payzones, the average median particle size of the added solids should be at least one-third the diameter of the pore throat.

Table 10 Sized Calcium Carbonate System Formulations

Additive	Function	Concentration, lb/bbl (kg/m ³)
Brine (monovalent)	Density	As needed
N-VIS	Suspension	0.5-1 (1.4-3)
N-DRIL-HT	Filtration	4-6 (11-17)
BDF445	Filtration	1 (3)
Caustic potash	pH	0.05 (0.15)
BARACARB	Plugging	Minimum of 30 (86)

Drilling fluids can be used in completions and workovers when wellbore clean-out and / or wellbore extensions / side tracks are necessary.

On wells where sand control is not an issue and no specialized zone work is needed, WBM can be used through the completion process and is sometimes specially treated and used as a packer fluid.

Several types of water-based systems can be used:

- Lignosulfonate
- High and low lime
- PHPA
- MMH
- Silicate
- Foam/Aerated
- Cationic Polymer

1.4. Contaminants

Contaminants that can affect completion and workover fluids include:

- Iron
- Solids
- Hardness
- Oil, distillate, grease, and pipe dope
- Polymers
- Surfactants

Contaminant	Discussion
Iron	Iron can be a contaminant in either soluble or insoluble form. Soluble iron is a product of corrosion and is common in zinc fluids. When exposed to fluids present in the reservoir, soluble iron can form a precipitate which can cause formation damage. No brine should be delivered to location with an iron content greater than 75 ppm. Consider displacing a brine when its iron content reaches 625 ppm. At the brine plant, iron should be removed from a fluid by adding hydrogen peroxide to the fluid, flocculating the fluid, then filtering the fluid. On location, treating a fluid for iron is very difficult and is usually successful only in low-density brines such as KCl, NaCl, or CaCl₂. The treatment consists of increasing pH with caustic or lime and removing the precipitated iron by filtering the brine.
Solids	Total solids can be measured at the wellsite using a turbidity meter. Solids that are not added to the system to enhance the performance of a brine are considered contaminants. Contaminants include formation clays, precipitates, polymer residues, etc. These contaminants can be filtered at the wellsite using diatomaceous earth, a plate and frame press, and two-micron absolute cartridges A clear completion fluid should not be sent to the wellsite with an NTU (Nephelometric Turbidity Unit) greater than 40 or a suspended-solids concentration greater than 50 ppm.
Hardness	When a monovalent brine has been selected to minimize calcium and magnesium scale formation, the total hardness content should not exceed 100 mg/L. Brines contaminated in the plant should be treated with soda ash and/or BARASCAV and filtered. To settle the precipitate prior to filtration, a flocculant such as BARAFLOC™ MD may be needed.
Oil Distillate, Grease and Pipe Dope	Produced oils and other hydrocarbons affect brine density and can also blind-off filtration units. Hydrocarbons will form a separate layer above heavy brine and should be removed from the completion fluid. BARASORB™ filter aid/oil absorbent is a completion brine filtration additive that when added can significantly reduce the oil and grease content of completion fluids. Environmental regulations may limit the amount of oil and grease content on fluid discharges in certain regions. Use of BARASORB helps reduce the oil and grease content and thereby reduce environmental risk and disposal costs.
Polymers	Brines contaminated with polymers usually cannot be filtered without chemical and/or special mechanical treatment at the plant site where oxidizers are used to oxidize polymers and permit filtration. This process must be carried out in a controlled access environment due to the possibility that chlorine and bromine gas may be generated. At the wellsite, polymer pills used in displacement should be caught and isolated from the active brine system.

Surfactants	Because surfactants have a high probability of causing formation damage in the reservoir, compatibility tests and formation damage tests should be run with any surfactant package required for completion. This includes soaps, thinners and lubricants.
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Handling Fluids

A completion or workover fluid should be protected from contamination while being prepared, transported, and used at the rig; any contamination can have costly results. All pits, trucks, boats and piping must be inspected for contaminants prior to transferring clean brine.

Some brines are quite corrosive to the skin and eyes. All rig personnel who might come in contact with these fluids must be trained in both handling fluid and personal safety. Proper PPE should be made available and worn.

Transporting Fluids

Major losses of volume often occur because some rig pits and boat tanks do not allow for the transfer of all fluid. When this is the case, consider renting a small, portable pump or modifying the rig pits.

To help maintain the quality of brines during transport:

1. Ensure the boat or truck is clean and dry before loading the brine.
2. Tie the fluid-transfer hose securely and continually monitor the hose for leaks or breaks.
3. Ensure all brine is transferred to the boat or truck, including the brine in trip tanks, sand traps, cement-unit tanks, filter-unit tanks, slugging pits, etc.
4. Strap the boat or truck tanks and check the density of the brine being shipped to help explain any losses in density and/or gains or losses in volume once the material is received.
5. Ensure all hatches and valves on the boat or truck are securely closed before leaving the rig.
6. Instruct the person in charge of transport not to transfer any fluid on board during transport.

Rig Preparation and Housekeeping

Clean and clear completion fluids do not contain solids that might plug a productive formation. Pits and lines must also be clean of solids. A pin hole plugged with mud solids can become unplugged, resulting in the loss of expensive fluid. Immediately investigate any unexplained loss of volume.

Ensuring a successful completion or workover operation requires following certain precautions to help prevent fluid loss due to contamination and equipment leaks.

Before Receiving Fluid	1. Cover all open pits to be used in handling the completion fluid. A solid, raised cover with sufficient overhang is preferable to tarps to keep rain water out of brine. 2. Wash and dry all pits or tanks to be used in handling the fluid. 3. Flush all lines and pumps with sea water or fresh water. 4. Clean and dry the mud-return ditch, shale shaker, possum belly, and sand trap beneath the shale shaker. 5. Seal return-ditch gates, shale-shaker gates, and dump valves by caulking with silicon compound
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	or some other compatible material. 6. Disconnect or plug all water and diesel lines leading to pits. 7. Tie down the fluid-delivery hose to prevent accidents or loss of expensive fluid. 8. Conduct a meeting to establish the methods for emergency communication with boat or truck personnel to allow for rapid shutdown of fluid transfer should problems develop.
While Receiving Fluid	1. Monitor the delivery hose for breaks or leaks. 2. Monitor pits and dump valves for leaks. 3. Maintain communications with the boat or truck for estimated volumes pumped. 4. Allow plenty of time to shut down delivery as soon as pits are full.
After Receiving Fluid	1. Mark the fluid level in pits and monitor for losses. 2. Inspect pits and dump valves for leaks. 3. Use completion fluid to flush sea or fresh water from all lines, pumps, solids-control equipment, and degassers.
During completion / workover operations	1. Monitor fluid level in pits and dump valves for losses. 2. Monitor pits for accidental water additions. 3. Restrict the use of pipe dope to a light coating on pin ends only.

1.5. Filtration

Brine Cleanliness

When performing a completion fluid displacement there is normally a specified level of cleanliness (solids content) required. Suspended solids in a brine can cause severe formation damage and can ruin the well just drilled. Solids in brines are even more damaging than solids in drill-in fluids because they can travel deep into the reservoir to plug off pore throats. Solids in mud will form a filter cake at the face of the wellbore and limit the depth of penetration of damaging solids. It is important to understand the cleanliness requirement to design the displacement appropriately and to measure the performance.

A filtration unit (a pod unit with internal filter elements or a large D.E. press) is used to clean the brine to the required level of cleanliness. This equipment is provided either by Baroid or by a third party. The brine is circulated through the filtration unit prior to being exposed to the formation, and often while the brine is being circulated down hole.

Fluid cleanliness measurements are taken:

- In the suction pit (before pumping downhole)
- Well returns (after pumping downhole)
- Filtration unit in and out

These measurement points can be related to make field decisions, but also serve to provide an estimate of displacement performance. The plan for how, when and where cleanliness measurements are to be taken and reported should be detailed in the displacement program.

Cleanliness Measurements

Nephelometric Turbidity Unit (NTU)	Measured with a turbidity meter, NTU are a measurement of light scattered in a fluid. While not a direct measurement of solids content, because NTU can be used as a relative indicator of cleanliness and clarity for trending purposes. Typical specifications are <20 NTU.
% Solids	Typically measured with a hand crank centrifuge. Samples are taken into the samples tubes, inserted into the centrifuge, and spun. The tubes are taken out and solids content recorded. While this is a direct measurement of solids, accuracy is limited to about 0.05% solids, which is equivalent to 500 ppm. This is acceptable for some operations while other clients have lower specifications.
Total Suspended Solids (TSS)	TSS requires a scale too sensitive for most rig operations. There are field methods available by using pre-weighed filters. A volume of fluid is forced through the filter, the paper collected, and sent in to a lab for measurement. This is not suitable for making real time decisions but can provide data after the operation.
Laser Particle Size Analyzer	Instruments have been used to characterize solids in a completion fluid. While they can provide particle size distribution data, they will not quantify the amount of solids present.

1.6. Personal and Environmental Safety

Safety is important when workers handle completion or workover fluids. To ensure a successful operation, observe the following basic recommendations:

Prior to receiving the fluid, conduct a job-specific safety meeting with all personnel including those (such as production personnel) not directly involved in the completion/workover operation. At this safety meeting, review the safety video tape available through Baroid Fluid Services.

Install eye-wash and shower stations in all areas where contact with the fluid is a possibility.

At a minimum, eye-wash stations should be installed in the following areas:

- Rig floor (two or more locations)
- Mud pit (as needed for easy access)
- Mixing-hopper area
- High-pressure pump service skid unit
- Production deck (under fluid-handling areas)

Provide appropriate eye-protection devices to all personnel working near fluid-handling areas and require the use of eye-protection devices.

Provide slicker suits, rubber gloves, rubber boots, and barrier cream to all personnel who will be working in fluid-handling areas or who might come in contact with the fluid.

If brine comes into contact with eyes or skin, or if ingestion or inhalation is suspected, take the following first-aid measures:

- **Eyes.** Flush eyes promptly with plenty of water for fifteen minutes. Get medical attention.
- **Skin.** Flush skin with plenty of water for fifteen minutes. If necessary, wash skin with soap.
- **Ingestion.** Consult the material safety data sheet for response information and get medical attention.

Environmental regulations vary, and it is important to acquire the specific guidelines for the area where the brine will be used. It is mandatory that compliance with the regulations be carried out.

Displacement

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1. Displacement for Completion

Wellbore displacement prior to completion operations is the process of removing one fluid, usually a drilling fluid, and replacing it with another, usually a clean completion fluid. The process is performed using a series of pills and spacers which provide both chemical and physical cleaning actions. The pills and spacers provide separation, remove drilling fluid, create water-wet surfaces and transport residual solids out of the well. Other displacements, such as replacing one mud with another or displacing seawater with brine, can be achieved using an appropriate high viscosity spacer to maintain separation.

The objectives of drilling fluid displacement include:

- Maximizing the recovery of drilling fluid
- Establishing a clean, non-damaging environment for completion
- Water-wetting all well surfaces
- Minimizing the use of rig time
- Minimizing the production of fluid waste

1.1. Information Gathering

As there are several options for displacement techniques, it is important to obtain accurate information about the components and properties of a well. This information is used to design an effective displacement operation.

Initial data for a design should include the information shown below.

Fluid Data	Drilling fluid Completion fluid Cleaning pills Density Rheology
Well Data	Well construction Riser, casing and string dimensions Cleanup tools Completion design Directional survey Thermal gradient
Installation Data	Pump capabilities Fluid storage facilities BOP configuration Standpipe pressures Equipment limitations
Lab Data	Suitable well cleaner Optimum concentration Fluids compatibility
Hydraulics Simulation	Pump rates Annular velocities Hydraulic horsepower requirements Bypass tool operation

1.2. Displacement Train

The displacement train consists of a series of pills and spacers. The volumes, composition and rheology of the pills should be customized for the well construction, completion type and active fluids.

Water-Based Mud

The train for the displacement of WBM typically consists of the following pills:

- Thinner (optional)
- Push pill
- Wash pill
- Sweep (optional)
- Flocculant (optional)

Oil- / Synthetic-Based Mud

The train for the displacement of OBM typically consists of the following pills:

- Base fluid
- Push pill
- Wash pill
- Sweep (optional)
- Flocculant (optional)

Table 1 Displacement Train Pills

Thinning Pill	A pill containing an appropriate thinner can be used for the displacement of dispersed clay WBM. This spacer is not required for the displacement of polymer WBM.
Base Fluid Pill	A base fluid pill is only required for OBM and SBM displacement. The pill provides separation between the mud and the aqueous cleaning pills. It also reduces the density and the yield point of the mud which assists the displacement.
Push Pill	A weighted, viscosified pill is used to perform a thorough displacement of mud.
Wash Pill	The wash pill performs the main cleaning action in the displacement train. The pill is a solution of well cleaner in an appropriate fluid and the chemical action should be selected based on the fluid to be displaced
Sweep Pill	A sweep pill transports and removes residual solids from the well and separates cleaning chemicals from the completion fluid. It can be used if low annular velocities are likely to compromise solids transport
Flocculant Pill	A flocculant can be added to the completion fluid in order to flocculate clays and fines. This takes these particles out of suspension and makes them easier to remove with filtration.

1.3. Displacement Techniques

A range of operations and techniques may be employed to conduct a displacement. These may be selected and combined in order to provide a customized program.

Table 2 Displacement Techniques

Direct and Indirect Displacement	In a direct displacement the well is displaced to the final completion fluid. This is the most efficient method of displacement with respect to pumping time, waste production and the use of pills and spacers. In an indirect displacement the well is displaced to seawater or fresh water as an intermediate fluid. The intermediate fluid is then displaced with the final completion fluid. This method requires more pumping time, produces more waste and may reduce the hydrostatic pressure. In a variation of this method a volume of intermediate fluid less than the well circulating volume is used to displace the drilling fluid followed by the final completion fluid.
Forward and Reverse Circulation	Most displacement operations use conventional circulation. Some circumstances favor the use of reverse circulation. These include pump rate restrictions, displacement with low density fluid and multi-circulation operations. However, reverse circulation increases the risk of plugging pipes and tools, restricts pipe movement and applies more pressure on the annulus.
Single- and Two-Stage Displacement	In a single-stage displacement the train is pumped down the tubing, through the annulus and returns are taken via the riser. The circulating system is displaced and cleaned in one pumping operation. Single-stage displacements are used in land and shallow water operations and can be used in deep water operations. In a two-stage displacement the well and riser sections are separated by closing the BOP and they are displaced in two separate operations. Two-stage displacements are commonly used in deep water operations.
One- and Two-Part Wash Systems	In a one-part wash system cleaning is performed with one wash pill. One-part wash systems can be used with or without a bypass valve when sufficient annular velocity can be maintained throughout the displacement. If the annular velocity is likely to be compromised during the displacement, the wash pill can be applied in two portions in combination with the use of a bypass valve.
Open Hole Displacement	Open hole sections may be displaced to transition pill or clean fluid. The formation should not be exposed to high concentrations of cleaning chemicals as these are likely to damage the filter cake. Sufficient hydrostatic pressure must be maintained during open hole displacement operations. Open hole and cased sections are usually displaced in separate operations.

1.4. Good Practices

A successful operation depends on employing the best practices during the execution. These are some examples of tasks and practices which should be observed at an installation.

Table 3 Best Practices for Displacement

Fluid Handling	A series of preparations are required for the receipt of clean fluid. In addition, a number of precautions must be exercised in order to maintain the condition of the fluid. Clean fluids must be protected from contamination with debris, mud and water.
Pit Plan	A detailed pit management plan must be prepared for all fluid movements. This must ensure that there is sufficient pit volume available to handle clean fluid, displaced fluid, cleaning pills and waste.
Surface Cleaning	Surface cleaning may represent a substantial part of the preparations. Additional cleaning is required after the displacement. Special attention is required in order to ensure that the BOP, manifold and associated service lines are clean. Cleaning should be carried out systematically and recorded in a checklist.
Field Calculations	Changes to depths, casings and drill pipe may have a significant effect on the well volume and spacer requirements. For these reasons, circulating volumes should be accurately determined and accounted for in the spacer calculations. Spacer calculations should be repeated and verified at the installation.
Mud Conditioning	Mud mobility is essential for a successful displacement. Circulation must be established at the cleanup depth and maintained in order to ensure homogenous density and condition. In some cases, it may be necessary to circulate at least one bottoms up in order to distribute solids. In other cases, it may be necessary to wash through or even drill out settled solids. The opportunity should be taken to minimize the weight and YP of the mud with the addition of base fluid.
Pump Rate Management	Pump rates should be carefully managed during displacement and kept as close to the prepared schedule as possible. This makes the best use of optimized contact times, annular velocities and differential pressures. Pump rates should be increased gradually and steadily. Pump stoppages and abrupt changes in pump rate can promote the production of large interfaces and undesirable buoyancy effects and should be avoided.
Pipe Movement	Pipe rotation and reciprocation serve to remove accumulated solids, break up mud channels and reduce pipe eccentricity. Pipe movement should be managed throughout a displacement operation. However, some completions may restrict pipe movement, activation of a bypass valve may prevent pipe reciprocation and the use of reverse circulation can prevent all pipe movement
Sampling and Testing	The verification of wellbore cleanliness involves a series of samples, observations and tests. No individual observation provides sufficient evidence of cleanliness on its own. However, taken together the records and data provide sufficient proof that the casing is clean. Sampling and testing must be conducted in accordance with the program requirements. Results should be reported promptly in order to prevent delays and NPT.
Contingencies	If there is evidence of an incomplete displacement or residual contamination, additional operations may be required. These may include scraping and brushing, fluid circulation or an additional displacement. Tools, fluids and chemicals must be available to conduct any contingency operations.

Cementing

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1. Cementing Overview

This chapter provides a brief look at cementing types, products, and how Baroid drilling fluids engineers can work in conjunction with Halliburton Cementing engineers to achieve optimum cementing success. Without a good cement job, not only is drilling complicated but the final successful production of the well can be negatively impacted.

The main cementing materials used in oilfield applications are:

- Portland cement, API Classes A, C, H, and G
- Blast furnace slag (BFS)
- Pozzolans (fly ash), ASTM Types C and F

Portland cement is the name used for all cementitious material composed largely of calcium, silica, and aluminum oxides. Blast furnace slag (BFS) is a by-product obtained in the manufacture of pig-iron in a blast furnace. Pozzolans are silica or silica/alumina materials that react with calcium hydroxide (lime) and water to form a stable cement. Pozzolans can be natural or synthetic.

Cementing materials are used in drilling operations to:

- Isolate zones
- Support casing in the borehole
- Protect the casing from collapse, corrosion, and drilling shock
- Plug non-producing wells for abandonment
- Plug a portion of a well for sidetracking

1.1. Additives

Cement slurries prepared with cementing materials are treated with various additives to modify set time, rheological and filtration properties, and density. These additives are classed as follows:

Table 1 Cement Additives

Product Type / Function	Description
Accelerators	Accelerators shorten the slurry set time and allow the slurry to develop necessary compressive strength in a practical time frame.
Retarders	Retarders delay the slurry set time. This delay allows the cement to be placed before hardening occurs. These additives counter the effects of increased temperature on a cement slurry.
Fluid loss control	Excessive losses of water to the formation can prevent cement from hardening correctly. Fluid-loss control additives are used to reduce excessive losses of water to the formation. In addition, these additives: • Increase viscosity • Retard the set time • Control free water in the slurry
Extenders	Extenders lighten the density of the slurry for cementing across weak formations. A lighter slurry lowers the hydrostatic pressure and helps prevent formation damage.
Free water control	Free-water control additives tie up water in lightweight or extended slurries. If this water were not controlled, the slurry properties would change as water was absorbed into the surrounding formations. This absorption affects slurry flow and placement.

Product Type / Function	Description
Weighting materials	Weighting materials can be used to increase the density of the cement or slag and help control formation pressures.
Slag activators	Blast-furnace slag (BFS) is a latent hydraulic cement material that does not readily react with water. Because of this, the hydration process for BFS is initiated by either chemical activators or elevated temperatures. Chemical activators are used as needed in different ratios and concentrations, depending on the expected temperatures to be encountered.
Dispersants	Dispersants reduce slurry viscosity, which is very important for placement and cohesion. Proper dispersion of a slurry results in: • Enhanced early compressive strength • Improved fluid-loss control • Improved free-water control
Strength retrogression preventers	Cement and BFS slurries that remain at temperatures above 200 F (94 C) exhibit a reduction of compressive strength over time. This phenomenon, called strength retrogression, can be minimized or prevented by adding another source of silica, such as silica flour or silica sand, to the slurry. Silica flour requires more mixing water than silica sand to achieve the same viscosity.

1.2. Slurry Design and Applications

Slurries, whether cement or BFS, must be tailored to each different aspect of the drilling operation. Some of the different classifications of slurries include:

Lead Slurry

A lead slurry is designed to cover a large portion of the annulus, either open hole or inside casing. These slurries are lightweight, extended slurries that do not contribute greatly to the hydrostatic head of the cement column.

Tail Slurry

A tail slurry is designed to provide most of the support for the casing or liner being cemented. This slurry is placed over the zone of interest to isolate the zone from contamination. The zone of interest can be a producing formation, a water zone, or some other zone that needs to be closed off. Ideal tail slurry characteristics include:

- High density
- Ability to develop high compressive strength
- Good set time control
- No free water
- Fluid-loss control additives may be required for a tail slurry.

Squeeze Slurry

Squeeze slurries are designed for remedial, or secondary, cementing. These slurries must have good set- time control, good fluid-loss control, and especially good compressive strength development.

Cement Plug

Plugs should be designed to meet the requirements of the specific application, whether kick-off plug, lost circulation plug, plug and abandon, etc. Ideally, plugs should have the following characteristics:

- High compressive strength development to seal the plug zone
- Short set time

1.3. Spacers

The three main functions of spacers are as follows:

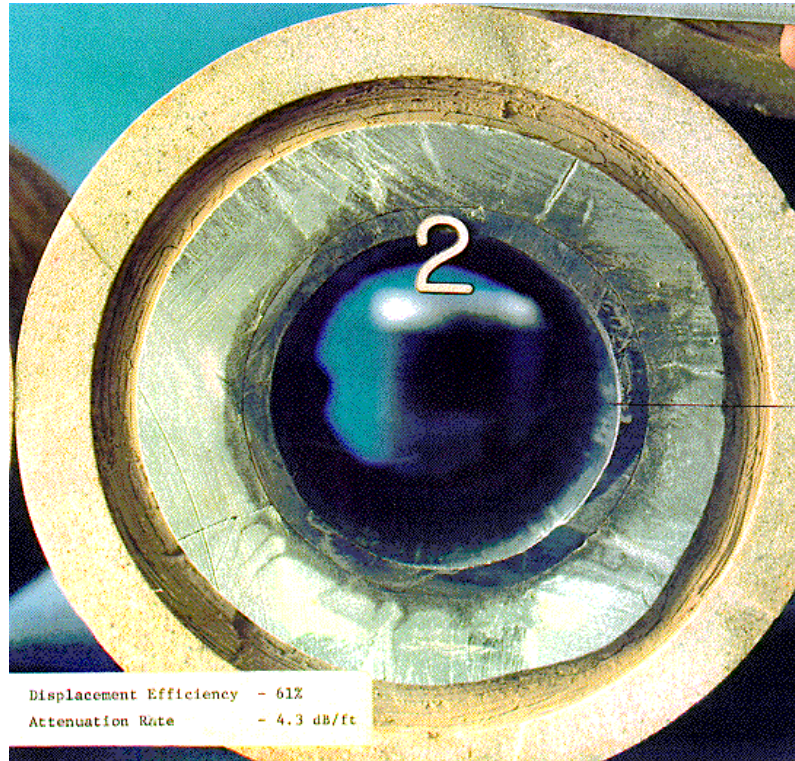
- Serve as a barrier between the drilling fluid and the cement slurry, thus eliminating contamination between the two
- Clean the casing and the formation of drilling fluid that could prevent good adhesion
- Act as a wetting agent that wets the casing and the formations

For a spacer to be effective, it must fall within certain guidelines for density and compatibility. The spacer must be more dense than the mud, but not as dense as the cement slurry. The margin should be 1 to 1.5 lb/gal each way. This range allows the spacer to separate the two fluids (the slurry and the mud) and prevent them from contaminating each other.

The spacer needs to be rheologically compatible with both the mud and the cement. The ideal viscosity of the spacer should fall between the viscosity of the mud and the cement.

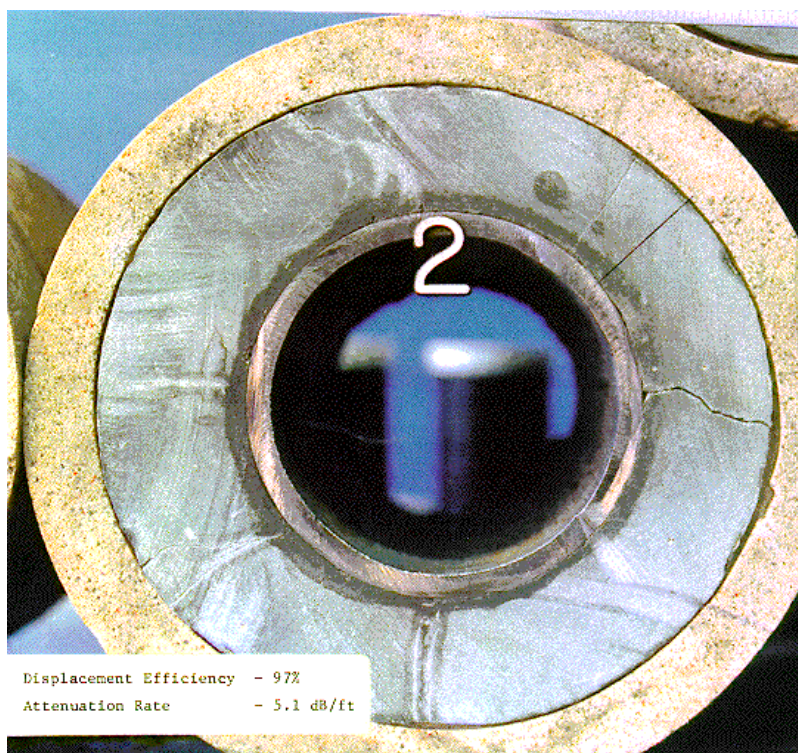
1.4. Successful Displacements

A solid cement sheath between the casing and the wellbore rock is necessary for good zonal isolation and well integrity. Removal of the mud from the casing and the wellbore is essential.



Poor cement bond

The casing string is on the inside, then the cement sheath, then the mud and wall cake between the cement and the wellbore rock. Failure to remove the old mud and wall cake can result in a failed cement job requiring expensive remedial squeezing.



Good cement bond between casing and wellbore rock

A good cement bond between the casing and the wellbore rock will have virtually no mud or wall cake remains.

Halliburton designs the mud-to-cement displacement to remove all the mud and obtain a good bond between the casing and the wellbore rock. Important steps to a good displacement include:

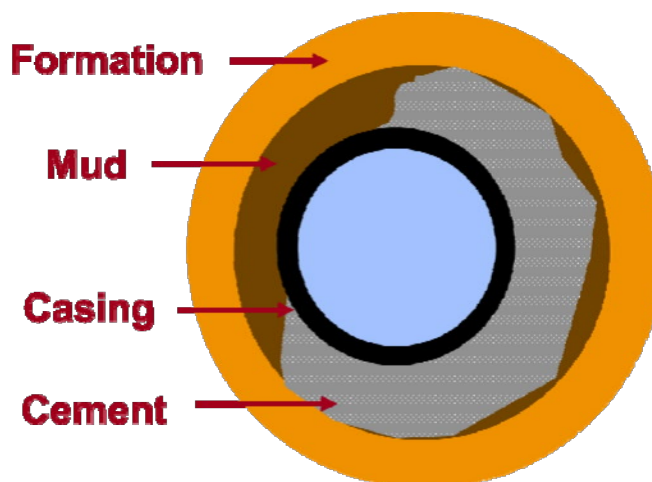
- Mud Conditioning – reduced viscosity and gel strengths, lower filtrate loss
- Pipe Movement – essential for good displacements in high angle holes
- Centralization – mechanical positioning of the casing near center of wellbore
- Fluid Velocity – necessary for any good displacement
- Spacers – separate mud and cement to prevent flocculation, remove wall cake

Mud Conditioning

Numerous studies dating back as far as 1928 show that the foremost factor affecting a cement displacement is the condition of the drilling fluids prior to cementing.

Poor mud conditioning is a serious problem which can prevent a successful mud displacement and defeat the purpose of cementing, which is to surround the casing with a continuous sheath of cement and bond it to surrounding formations. Yet few mud engineers or cementing engineers communicate about this vital factor.

As mud sits in the hole during the time it takes to run casing, the static mud will build gel strength and become difficult to displace. This mud needs to be re-mixed with the rest of the mud in the hole, i.e., conditioned, so that a uniform viscosity profile is achieved in the casing-hole annulus. If the mud is not conditioned the cement will take the flow path of least resistance and leave uncemented, mud-filled channels in the annulus.



A poor displacement can result in mud left behind the casing and poor adhesion of the cement to the casing and the wellbore rock.

The optimum mud properties prior to cementing to help insure the best displacement are as follows:

- Yield point of 10 or less
- Plastic viscosity of 20 or less
- Fluid loss of 15 or less
- Low, non-progressive gel strength profile; a ratio of 10 second to 10 minute gel strengths near 2:3 rather than a 2:10 ratio is required.

Some of these properties, for example the low yield point, are exactly opposite to what is needed when drilling the well.

When it is not possible to achieve all of the above prescribed physical properties of the mud system, the mud engineer should focus on reducing the yield point, plastic viscosity, fluid loss and gels strengths etc. as much as possible.

Pipe Movement

Pipe movement is very important in achieving good displacement efficiency. It aids in mud displacement by breaking up gelled pockets of mud and cuttings that may accumulate in the well. Pipe movement can also help to offset the negative effects of poorly centralized pipe. Mechanical scratchers attached to the casing can further enhance the beneficial effects of pipe movement.

Movement can be either through rotation or reciprocation and may occur before and/or during cementing. Halliburton research indicates that the effects of rotation and reciprocation seem to be equivalent.

Centralization

Centralization is the mechanical positioning of the casing near center of the wellbore. Various mechanical devices are attached to the casing at intervals to provide the necessary stand-off to allow the cement to completely surround the casing.



Centralizers installed on casing prior to running into the well.

Fluid Velocity

Annular fluid velocity directly impacts the efficiency of any displacement, and turbulent flow is often preferred.

While the higher the annular velocity the better for hole cleaning and removal of old mud and wall cake, there are constraints imposed by ECD and wellbore erosion issues. Reductions in the drilling fluid's viscosity allow higher flow rates without exceeding the fracture gradient of the wellbore. Drilling mud with excessive viscosity requires lower pump rates, lower annular velocities and thus reduced displacement efficiency and the risk of a poor cement job.

Spacers

The choice of suitable spacers and flushes requires the same careful consideration that goes into every other aspect of a cementing program. Some important factors to consider are:

Compatibility of the Mud and Cement

Intermixing of the whole drilling fluid with the cement slurry may cause interfacial incompatibility and result in high fluid viscosities. The high viscosity of the cement/mud mixture can cause high pump pressures making it more difficult to place the cement slurry. Extreme cases may lead to job termination or the "breakdown" of a weak formation with the subsequent loss of cement slurry to the formation.

Also, during displacement interfacial intermixing of the drilling fluid and cement slurry may have an adverse effect on the performance of the cement slurry. Contamination of the cement slurry by drilling fluid can adversely affect critical slurry properties like thickening time and compressive strength development as well as fluid loss control. The effect of contamination will depend on the concentration and type of contaminant.

For this reason, use of a spacer or pre-flush designed for compatibility with drilling fluids and cement slurries is necessary to eliminate potential interfacial incompatibility and contamination problems.

General Considerations

Hydrostatic Pressure Requirements of the Well

This will determine fluid density requirements of not only the drilling fluid and cement slurry but also the spacer or pre-flush.

Wetting Characteristics

Oil-based drilling fluids are often preferred over many water based muds since oil-based muds are more compatible with water sensitive formations and improve the drilling rates. While these oil-based drilling fluids help protect these formations, they are incompatible with cement slurries and coat formation and pipe surfaces with an oil film that may not be receptive to bonding with cement.

A spacer and pre-flush formulated with added surfactants can solve this problem by enabling these fluids to leave water wet surfaces upon which an effective cement bond can form.

Mud Displacement

A minimum of 800 to 1000 ft. of annular length or 10 minutes of contact time of the spacer fluid is recommended to help insure optimum mud displacement. For fluid separation, a minimum of 450 ft. of annular length is recommended.

Rate for Turbulence

A high displacement rate is best for turbulent flow. Solids suspension stability is desirable for dynamic and static conditions.

Halliburton has a full range of spacers designed to meet every drilling mud need. These include the traditional turbulent flow spacers, such as Dual Spacer and Dual Spacer E+. Laminar flow spacers like Spacer 500E+ and the new FDP-C543-96 Spacer with adjustable rheological properties can also be used. An oil-based spacer, SAM-4, is also available.

Lost Circulation

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1. Lost Circulation

1.1. Overview

Fluid Selection

Drilling fluids with low non-progressive gels help lower the risk of lost circulation. The ACCOLADE and ENCORE synthetic-based systems and HYDRO-GUARD or BOREMAX water-based systems are examples of fluids formulated with low colloidal content that exhibit desirable gel characteristics.

Baroid offers other systems with similar performance characteristics. Selection depends on conditions such as temperature, shale reactivity, environmental concerns, and solids control efficiencies.

Lost Circulation Indicators

Lost circulation is defined as complete or partial loss of whole mud to the formation that typically occurs when hydrostatic pressure in the annulus exceeds the fracture gradient of the exposed formation or natural fractures are encountered.

When lost circulation occurs, less fluid returns to surface than is pumped downhole. In the event of total loss of circulation, no fluid returns to the surface even though pumping continues. Lost circulation can be detected by monitoring return flow and pit levels with sensors and pit volume indicators. Most sensors are equipped with an alarm set point to alert crews to losses and gains in flow and pit volume.

Surface Losses

Prior to assuming that mud loss to the formation has taken place, all surface equipment should be examined for leaks or breaks (i.e., mud pits, solids control equipment, mud mixing system, riser slip joints, and/or incorrectly lined up pumps or circulating lines). Losses may also occur during a fluid transfer.

Risks and Hazards

Depending on the severity of the rate of mud loss, drilling operations may be significantly impaired. Losses can significantly increase the overall well cost, both in time and in drilling fluid requirements.

If the annulus does not remain full when pumping ceases, the hydrostatic pressure decreases until the differential pressure between the mud column and the loss zone is zero. This may cause formation fluids from other zones, previously controlled by the hydrostatic pressure of the mud column, to flow into the wellbore, resulting in a kick, blowout, or underground blowout (Figure 1).

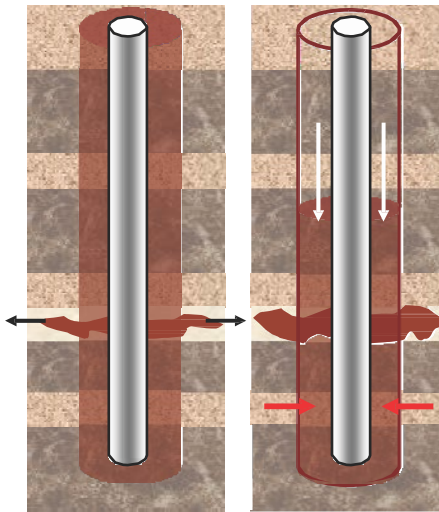


Figure 1 Lost Circulation / Kick Scenario

Loss of hydrostatic pressure may also cause previously stable formations to collapse into the wellbore.

Loss of circulation may lead to differential sticking of the drillstring (Figure 2).

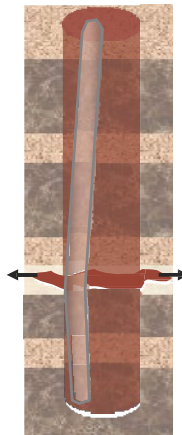


Figure 2 Differential Sticking At or Near Loss Zone

Causes of Lost Circulation

Loss of circulation occurs when the hydrostatic pressure exceeds the fracture gradient (FG) of an intact formation and/or the pore pressure of a formation with open fractures. The most common causes of excessive hydrostatic pressure are as follows:

- Excessive overbalanced mud weight
- Cuttings loading in the annulus due to poor hole cleaning
- Elevated viscosity and rheological properties
- Restricted annular space
- Excessive surge pressure while running the drillstring or casing in the hole

- Combination of the above factors

To help ensure the most appropriate lost circulation treatment(s) are applied in each case, the mud engineer should evaluate not only the characteristics of the loss zone, but all the parameters that may be affecting hydrostatic pressures in the wellbore.

Economic Impact

The economic impact of lost circulation is significant. When unacceptable losses are encountered, normal drilling operations may be delayed indefinitely while attempts are made to regain full returns. Under certain conditions, the operator may decide to “drill blind” (i.e., without returns) in an effort to allow cuttings to seal off the loss zone. In a well with exposed gas- or water-bearing formations, this practice may induce a kick or blowout if the hydrostatic pressure becomes less than the formation pressure.

Lost circulation is a major contributor to non-productive time (NPT) and flat time.

Once well construction begins, a primary goal is the reduction of NPT (i.e., intervals where drilling ceases due to hole problems). Likewise, flat time related to formation evaluation (logging) and setting casing should be minimized by ensuring that hole conditions are at their best for the particular operation.

The cost of a lost circulation incident includes the value of the lost mud, the rig time required to address the problem, the materials added to the mud system to reduce or stop the loss rate, and under very severe circumstances, the abandonment or sidetracking of the well.

Offset well data may indicate where losses may be expected and under what conditions.

Formation Types Associated with Lost Circulation

The following formation types are most commonly associated with lost circulation events:

Table 1 Formation Types Associated with Lost Circulation

Formation Type	Characteristics	Loss Severity
Sandstone	Permeable	Seepage to partial
Sandstone Unconsolidated sand Sub-salt rubble	Highly permeable and/or fractured	Partial to complete
Limestone reef Dolomite bed Chalk	Vugular and/or cavernous	Partial to complete
Shale	Impermeable	Partial to complete

Classification of Losses

The correct treatment of lost circulation depends on the rate of mud loss and the type of loss zone encountered. Historically we have classified losses based on percentage of fluid pumped. The actual values varied between operators and service companies, but examples include the following:

- Seepage losses <10%
- Partial losses 10-50%

- Severe losses 50-100%
- Total losses 100% / no returns

These percentage values provide little guidance in selecting a treatment.

As an example, if the circulation rate is 840 gpm (20 bpm) and the loss is 30%, then the loss rate is 6 bpm.

If the circulation rate is 1260 gpm (30 bpm) and the loss is 30%, the loss rate is 9 bpm – or 50% more.

Whether this is classified as seepage, partial, etc. is of no consequence to the operator. The goal is to reduce the economic impact of losses, and in this case, three more barrels per minute costs 50% more per minute. Consequently, losses are classified based on rate rather than percentage.

Table 2 Example Loss Rates

Seepage Losses	Partial Losses	Severe Losses	Total Losses
<10 bph	10 – 50 bph	50 – 200 bph	>200 bph / No returns
Porous and permeable sands, gravels, shell beds	Small open fractures	Large sections of unconsolidated sands or fractures	Cavernous / large fractures

In addition, the rate of loss in a producing zone is of greater concern than the same loss in a non-productive zone because formation damage can reduce overall productivity and recovery.

1.2. Treatment Options

The main two methods for dealing with lost circulation scenarios are prevention (pre-treatment) and correction (remediation). It is important to have a LCM application matrix prepared for a well prior to drilling so that all personnel are aware and trained on the use of the selected materials, and that these materials are either on location or readily available.

Pretreatment

Key best practices for preventing lost circulation include the following:

- Pre-treat with selected LCM before drilling high risk lost circulation zones, such as depleted sands.
- Add subsequent LCM treatments as sweeps, rather than adding LCM into the active drilling fluid system via the suction pit.
- Base the amount of LCM added on material (ie, normalized by using the specific gravity of the components) volume rather than weight.
- Keep remediation materials on site for immediate application if needed.

Products like STEELSEAL resilient graphitic carbon material, and BARACARB sized calcium carbonate have proven effective when carried as a pre-treatment in the drilling fluid. These products are also generally the primary constituents of corrective lost circulation treatments. BAROFIBRE O is also demonstrating efficient lost circulation mitigation and may be added at a rate of 20% or less of the total LCM volume.

As a rule of thumb, 5.0 to 10.0 ppb STEELSEAL lost circulation material plus 10.0 to 15.0 ppb BARACARB bridging agent are used to pre-treat the active system. A total weight of 15.0 to 25.0 ppb is desirable.

As drilling progresses, additional materials are needed to maintain pre-treatment levels. The amount of LCM lost over the shaker screens depends on the particle size distribution of the LCM, the screen sizes used, and the flow

rate. Wellbore breathing and loss of circulation may be observed in pre-treated systems. The decision whether to use more of the same LCM, go to a different combination of materials, or to change to chemical lost circulation treatments generally depends on the severity of the losses and the potential risk to wellbore stability.

Pre-Mixing vs. On-the-Fly Mixing

Pre-mixing LCM materials before use rather than mixing on the fly helps ensure that the proper amount of materials are added and that the desired particle size distribution can be maintained.

In some cases it may be possible to mix an LCM concentrate that can be diluted with the active mud on location to the desired level. Using a one-sack product that has been engineered for a specific application is another option.

Sweeps

Higher concentrations of materials can aid in fracture tip screenout and help prevent further fracture propagation. This can be achieved by adding LCM in sweeps rather than total system treatments. With sweeps, the wellbore sees a higher concentration of particulate materials in general, and the larger particles in particular. Preventive sweeps should contain a nominal 50.0 ppb of the selected materials.

Treating by Weight or Volume

Conventionally the industry has calculated the amount of LCM to use on a weight basis, i.e., either equal weights of material combinations or a weight ratio based on previous experience.

Treating by material volume rather than weight will help increase the effectiveness of each material added. This is accomplished by using the specific gravity (SG) of the materials to normalize their weights.

Comparing fibers to calcium carbonate is a good example. A nominal SG for many fibers that are used is about 1.1, while calcium carbonate has an SG of 2.7. If equal weights of these materials (1:1 weight ratio) are used, the volume ratio of fibers to calcium carbonate is (2.45):1. Because cellulosic fibers also tend to cause increased viscosity, using a volume calculation brings their use into a more practical range.

Lost Circulation Remediation

Wellbore Breathing / Ballooning

Wellbore breathing, also known as ballooning, is the intermittent loss and recovery of fluid volumes. In this situation, the loss typically occurs while circulating. When the mud is static (pumps off), then all or most of the volume lost re-enters the system.

Wellbore breathing is caused by induced fractures that have not propagated to the far field and can range from an almost complete return of all fluid lost to large losses. Once started the breathing may continue until the interval is cemented behind casing. If not recognized early, continued fracture propagation can increase the severity of the losses and may result in failure to complete the drilling of the well. The time lost waiting for the well to stabilize after each connection can have a major impact on the overall well cost. In areas known for wellbore breathing, controlling the ECD through drilling practices, fluid properties and LCM treatment may prevent the problem.

Annular pressures can continue to open the fractures and increase the severity of the breathing phenomenon if not brought under control. If the fracture gradient is known, DFG modeling and possible real-time PWD can be used to monitor and control the ECD while drilling.

A sufficient flow rate should be maintained in high-angle wells for hole cleaning purpose. Controlling the ROP may be necessary to minimize annular cuttings loading.

Careful drilling practices should be implemented to avoid high surge pressures, including circulating prior to connections, controlling pipe running and pulling speeds, minimizing back reaming on trips, rotating the drill pipe to break gels before starting the pumps, and staging the pump speed on start-up.

STEELSEAL lost circulation material has proven to be one of the most effective products to use for wellbore breathing. In some areas it is the only LCM that has proven effective. STEELSEAL lost circulation material additions can prevent pressure transmission to the fracture tip which could extend the fracture.

A 30-50 ppb STEELSEAL 50 or 100 / BARACARB (50/150) additive blend with the product concentration ratio based upon volume – (1:2 weight ratio), appropriately sized for wellbore coverage – can be circulated across the loss zone. If circulating or spotting STEELSEAL lost circulation material pills alone is not sufficient, then the addition of a background concentration of STEELSEAL 50 or 100 lost circulation material to the active system (minimum 10 ppb is recommended) should be to be considered.

An adequate loading of STEELSEAL or a STEELSEAL / BARACARB lost circulation material blend can produce fracture tip “screen out” the instant the fractures are re-opened as the pumps are brought up to speed.

Seepage

Although seepage losses usually do not impose a significant risk to operations, they should be monitored closely in the event the loss rate increases. If pressure control is critical, safety demands that the losses be cured.

Raising the mud density may cause minor seepage to turn into a more serious loss rate.

General treatment guidelines are shown below:

Surface hole:

- STOP-FRAC D or combinations of BARACARB 25, 50, 150 and BAROFIBRE O

Pretreatment of active system:

- BAROFIBRE O / STEELSEAL / BARACARB combination
- LCM with particle size distribution (PSD) matched to sand being drilled

Water-based muds:

- Increased AQUAGEL viscosifier content (not suitable for DRIL-N fluids)

Oil- and synthetic-based muds:

- AQUAGEL GOLD SEAL viscosifier additions

LCM pills:

- Sweeps pumped frequently while drilling
- Spotted prior to tripping out of hole

Partial Losses

Partial losses are more serious than seepage losses and usually require significant LCM treatments or changes to the current drilling parameters to cure or to reduce the losses.

Often drilling must be slowed or suspended because the drilling fluid cannot properly clean the hole. The cost of the mud and rig time becomes important in deciding the response to partial losses. Logistics and the rig's mud building capabilities may be limited, and it may be necessary to take rig time to cure these losses.

Partial losses may be treated as follows:

STEELSEAL

STEELSEAL additions have been shown to increase fracture initiation pressures.

STEELSEAL lost circulation material can be mixed up to 100 ppb in water-based mud. Best results often obtained by combining STEELSEAL with BARACARB in equal volumes, i.e., 5-bbl STEELSEAL with 5-bbl BARACARB 50.

Combination Pills

- Spot DUO-SQUEEZE H, BDF 551 and/or 562 at 50-80 ppb
- Spot a wide range of particle sizes and a mixture of granular/fiber and flake LCM. Examples are combinations of STEELSEAL, BARACARB, walnut and BAROFIBRE of differing PSD ranges.

HYDRO-PLUG

Fresh-water pill built with approximately 80 ppb, spotted across loss zone and held under gentle squeeze pressure. Supplement with larger BARCARB 1200, STEELSEAL 1000 and/or Walnut M or C as needed.

Can be used in water-, oil- and synthetic-based fluids.

Severe Losses

Severe losses can have a serious impact on drilling operations. Large volumes of expensive mud may be lost in very short periods of time. This can result in a well control situation as the fluid level falls in the annulus and hydrostatic pressure is reduced.

Severe losses can also cause hole stability problems. While experiencing severe losses the hole should be kept full to the equilibrium point with water or base oil. An accurate record of all volumes and pills pumped should be kept so that hydrostatic head can be calculated. The equivalent mud weight and column height when the hole is static after losses can determine the minimum horizontal stress for WellSET modeling.

Severe losses may be treated as follows:

Combination pills

A mixture of coarse materials with a wide size distribution in as high a concentration as the rig equipment will allow to be pumped. Consider a mixture of fiber/flakes/granular material. Use engineered one-sack products individually or in combination.

For non-reservoir use:

- DUO_SQUEEZE H and/SA
- HYDRO-PLUG
- BDF 551 and 562

These can be supplemented with STEELSEAL 1000, BARACARB 1200, Walnut M and C, BAROFIBRE C.

Finally, 0.5 – 1.0 ppb BARO-LIFT may be added if treating open ended or through a treating (e.g., PBL sub).

Chemical sealants are FUSE-IT supplemented with DUO_SQUEEZE H or BDF-562; FlexPlug OBM; DThermaTek RSP (WBM) or ThermaTek LC; shear sensitive cement; gunk or reverse gunk squeeze.

For reservoir use:

Where acid solubility or breakability is required by the operator, use the following:

- EZ-PLUG
- DUO-SQUEEZE R
- The above with N-SQUEEZE/N-Plex

These can be supplemented with BAROCARB 1200 and 600, SEAL. If treating open-ended or through a treating (e.g., PBL sub), add 0.5-1.0 ppb BDF-456 as needed.

Chemical sealants are ThermaTek RSP or LC; shear sensitive cement containing BARACARB.

Complete Losses

Complete lost circulation is indicated by zero returns to surface. The fluid level in the wellbore may drop out of sight.

When a complete loss occurs the annulus should be kept full with monitored volumes of lighter mud and/or water or base oil. The resulting reduction in hydrostatic head should be determined. The density of the active system should be maintained at this calculated equivalent mud weight.

The hole should be monitored very closely for possible well control problems. Some wells are drilled “blind” to the interval TD without no returns to surface at all. This potentially risky operation assumes that all cuttings are transported away from the wellbore through fractures, and that there is no risk of a well control incident.

Total losses may be treated as follows:

A mixture of coarse materials with a wide size distribution in as high a concentration as the rig equipment will allow to be pumped. Consider a mixture of fiber/flakes/granular material. Use engineered one-sack products individually or in combination.

For non-reservoir use:

BDF 551, 562 and HYDROPLUG supplemented with STEELSEAL 1000, BARACARB 1200, Walnut M and C, BAROFIBRE C plus 0.5 – 1.0 ppb BARO-LIFT may be added if treating open ended or through a treating (e.g., PBL sub).

Chemical sealants are FUSE-IT supplemented with DUO_SQUEEZE H or BDF-562; FlexPlug OBM; DThermaTek RSP (WBM) or ThermaTek LC; shear sensitive cement; gunk or reverse gunk squeeze.

For reservoir use:

Where acid solubility or breakability is required by the operator, use the following:

EZ-PLUG, DUO-SQUEEZE R in N-SQUEEZE/N-Plex supplemented with BAROCARB 1200 and BARAFLAKE C, plus 0.5-1.0 ppb BDF-456.

Chemical sealants are ThermaTek RSP or LC; shear sensitive cement containing BARACARB.

For vugular carbonates underbalanced and managed pressure drilling should be considered for next wells.

General Recommendations

Recommendations provided here are general. Actual treatment engineering is based on available information and experience. Treatment variations are also based on whether the losses occur in the producing zone, or in a permeable or impermeable zone.

Table 3 Lost Circulation Treatment Guidelines

Loss Rate Measured at flow rate required to drill ahead.	Producing Formation	Permeable Zone	Impermeable Zone
<10 bph 25-50 ppb treatment	BARACARB 5,25 & 50 BARACARB 25 & 50 + N-SEAL	STEELSEAL 50 or 100 + BARACARB 5, 25 & 50	STEELSEAL 50 or 100 + BARACARB 150 & BAROFIBRE O
10 - 50 bph 40-60 ppb treatment	EZ-PLUG or DUO SQUEEZE RN-SQUEEZE** MAXSEAL** EZ-PLUG / MAX-SEAL	STEELSEAL 400 + BARACARB 150 & 600 + BAROFIBRE OM or DUO-SQUEEZE Hor BDF 551	DUO SQUEEZE H and/or HYDRO-PLUG or BDF-551
>50 - 100 bph 60-80 ppb treatment	DUO SQUEEZE R + N-SQUEEZE or + K-MAX**	DUO SQUEEZE H and/or HYDRO-PLUG* or BDF-562, FlexPlug OBM or FUSE-IT(WBM)	BDF 562 +HYDRO-PLUG or FlexPlug OBM or FUSE-IT (WBM)
>100 - 200 bph 60-80 ppb treatment	Thermatek LC** N-SQUEEZE or K-MAX + DUO-SQUEEZE R plus 1 ppb BDF-456	BDF 562 plus 1 ppb BDF-456 or Thermatek LC FlexPlug OBM or FUSE-IT	BDF 562 or Thermatek LC FlexPlug OBM or FUSE-
>200 bph	ThermTek LC or Low Fluid Loss Acid Soluble Cement	ThermaTek LC or High Fluid Loss Cement	ThermaTek LC or Thixotropic Cement

*HYDRO-PLUG NS for PARCOM regulated countries.

** Check temperature limitations.

LCM Classifications

Types of LCM typically include the following:

- Non-reactive moderate particle size (NRMPSD) material combinations that can be premixed for stand-by service
- Non-reactive large particle size (NRLPSD) material combinations that can supplement the (NRMPSD)
- Reactive Components (RC) used to supplement other combinations
- Reactive swelling material plus large aspect ratio (20-30) fibers (RSMF) to supplement the NRLPSD material combinations. These combinations will generally be applied open ended or through a treating sub such as a PBL sub.
- Chemical sealants that react with the drilling fluid (CSDF).
- Chemical sealants that are stand alone without drilling fluid interaction (CS).

Current Halliburton products that meet these criteria are shown in the following table.

Table 4 LCM Types and Classifications

RMPSD	HYDRO-PLUG	Contains a swelling polymer
NRLPSD	BDF-551	Bimodal PSD without STEELSEAL combinations
NRLPSD	BDF-562	Bimodal Large PSD with STEELSEAL combinations
RC	BDF-tbd	swelling polymer plus large aspect ratio fiber
CSDF	FUSE-IT	swelling polymer in non-aqueous carrier(NAC)
CSDF	FlexPlug OBM	latex base reacts with OBM
CSDF	ThermaTek RSP	ThermaTek materials in NAC.
CS	N-SQUEEZE/N-PLEX	Cross linked polymer
CS	TermaTek	metal oxide/salt produces set acid solid plug
CS	Shear Sensitive Cement	High gel strength thixotropic cement

1.3. Engineered Approach to Lost Circulation

Treating the active system with lost circulation material (LCM) is just one step in the process of reducing or eliminating losses.

Casing Point Selection

Whenever possible, casing should be set in non-porous formations with high fracture gradients. By setting casing as deep as possible, some formations with higher pore pressures may be drilled safely.

A formation of high matrix strength is recognized by one or more of the following:

- Reduction in penetration rates
- Mud logging data
- MWD logging data

Planning

In situations where offset well information indicates a likely encounter with a weak and/or depleted zone, the use of an engineered approach to drilling the zone(s) can help minimize losses, and at times prevent their occurrence completely.

This approach incorporates a number of planning tools:

- Borehole stability analysis
- Equivalent circulating density (ECD) modeling
- Drilling fluid selection
- WellSET™ modeling and lost circulation material (LCM) selection
- Downhole pressure measurement tools
- Connection flow monitoring
- Timing of LCM applications

Borehole stability analysis, hydraulics and WellSET modeling are conducted in advance of the actual drilling operations. The results of these investigations influence drilling fluid selection and help identify the most effective types of LCM for each case. Analysis continues as the well is drilled.

Geomechanical Modeling

The use of geomechanical modeling in well planning can provide the “safe mud weight window” boundaries for ECD. The static mud weights needed to mechanically stabilize the wellbore are influenced by parameters such as in-situ stress, pore pressure gradients, wellbore orientation, and formation material and strength.

Exposure to drilling fluid alters near-wellbore pore pressure, inter-granular stresses and rock strength and can cause progressive wellbore instability. Baroid uses a wellbore stability simulator to evaluate time- dependent mechanical, thermal, and chemical effects.

Hydraulic simulations using Baroid’s proprietary DFG hydraulics modeling software can determine projected ECD levels after the mud weight operating windows are identified in the wellbore stability modeling process.

Baroid Technical Professionals and Senior Service Leaders typically perform DFG hydraulics modeling.

DFG Hydraulics Modeling and ECD

The DFG program accounts for existing fluid properties and drilling parameters such as rate of penetration (ROP), pump rate, pipe rotation speed, wellbore geometries, and hole cleaning efficiency. The user can determine cuttings loading in the wellbore for a given set of conditions and the potential impact on ECD.

Pressure-while-drilling (PWD) values transmitted by the downhole pressure measurement tool help verify the ECD modeling done in the planning stage. During drilling operations, DFG modeling can continue to allow the user to optimize fluid properties and hydraulics. The introduction of the DFG RT (real time) drilling simulator in 2004 provided onshore and wellsite personnel with “ahead of the bit” visualizations related to ECD and hole cleaning efficiency.

Controlling the ECD as fluid properties and wellbore geometries change is a critical factor in preventing lost circulation.

1.4. Wellbore Stress Management

Wellbore Stress Management™ service is Halliburton’s engineered solutions which are designed to improve wellbore strength and help reduce drilling non-productive time due to lost circulation. This fully engineered approach requires both unique planning software and unique materials.

Planning must include means to prevent lost circulation as well as stop losses.

Prevention of Lost Circulation

Conventional loss prevention entails pre-treating the whole system prior to and while drilling permeable formations, or where seepage losses are expected.

Sweeps may also be pumped to prevent fracture propagation or reduce risk of wellbore breathing ballooning.

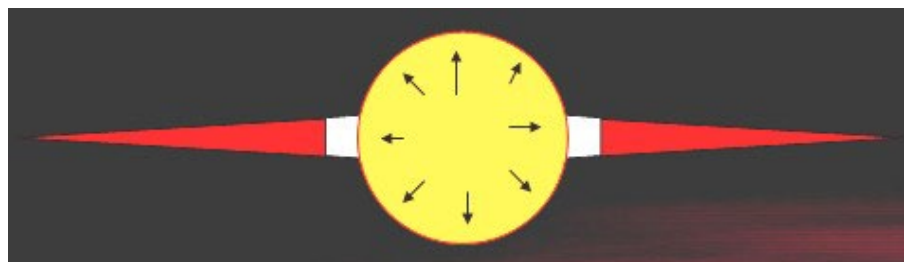
In the last decade, prevention of lost circulation by improving wellbore strength has achieved a successful track record. This is accomplished by designing and applying WellSet treatments that increase the hoop stress around the wellbore.

The goal of all the WellSet treatments is to increase the “hoop stress” (and thus the wellbore pressure containment ability) in the near wellbore region. While drilling, plugging the pores in a permeable sand and plugging microfractures that create wellbore breathing accomplishes this dynamically.

Once an interval has been drilled, a more robust treatment may be applied to more significantly increase the wellbore strength. Though an over simplification, these treatments may be described as placing a designed particle size distribution particulate treating pill across an interval, and then performing an open hole formation integrity test up to the maximum ECD expected while drilling, casing and cementing that interval.

A short fracture (or fractures) is initiated but is plugged immediately by the specially designed particulate treatment (Figure 3) that prevents further pressure and fluid transmission to the fracture tip, while at the same time mechanically propping the fracture to prevent closure. This action increases the hoop stresses around the wellbore, resulting in a strengthened wellbore that can contain a higher fluid pressure (ECD).

Figure 3 Wellbore Strengthening Dynamics



This generally is done by using correctly sized resilient graphitic carbon (e.g., STEELSEAL lost circulation material) and ground marble (e.g., BARACARB 600 bridging agent). Chemical lost circulation treatments that form a deformable, viscous and cohesive material (e.g., FlexPlug sealant) may also have the ability to improve the wellbore pressure containment as long as they can increase compressive stress at the fracture face.

Hydraulics and ECD Modeling

Hydraulic design simulations can be initiated using the DFG hydraulics module to help determine projected ECD levels when the mud weight operating windows have been identified in the wellbore stability modeling process. The principal factors in wellbore hydraulic predictions include:

- Pump rate
- Hole and drill pipe geometry
- Hole cleaning efficiency
- Rate of penetration
- Drill pipe rotation speed

To help obtain ECD predictions within a window of acceptability, operating ranges of each of these major factors should be determined. Hence, the simulation process can be quite lengthy. However, with fine-tuning, the iterative process can produce ECD predictions that can be used with some confidence.

Fracture Modeling

Once the ECDs have been predicted over intervals of interest, another module within DFG can be used to predict a fracture geometry that may be initiated during the well construction process. To do this modeling, the rock elastic properties of Poisson's Ratio (PR) and Young's Modulus (YM) must be known, or at least estimated. Other input parameters for the model are borehole diameter (BD), mud weight (MW), depth, stresses, and a short fracture length.

The fracture width calculated will be dependent on fracture length. Fracture length is possibly determined by fracture toughness based on fracture mechanics theories, as discussed in a previous paper. Rock mechanics theory

Table 5 Wellbore Strengthening Example Data Set

These data are input into the module and a fracture width is calculated (Figure 4).

DFG v4.0 - Fluids Engineering - [025]

All Selected Any Selected Disabled

280 D50 47.28% 40 total LCM, h/ft/s

☒ Use the D value also

1000 0 47.405

2 Search depth

Products | Rules

Let DFG Select

Product Family SS D50

Product	Family	SS	D50
BARACARB 5	BARACARB	2.7	5
BARACARB 10	BARACARB	2.7	50
BARACARB 100	BARACARB	2.7	400
BAROSORB 0F	BAROSORB	1.1	30
BOF 301	BOF	1.75	1133
BOF 302	BOF	1.75	1329
BOF 303	BOF	1.75	564
BOF 306	BOF	1.75	55
BOF 308	BOF	1.75	343
STEELSEAL	STEELSEAL	1.75	75

Particle Volume Fractions | Product Particle Size | Data

12 Hole Diameter, in

6 Crack Length, in

12.5 Mul-height, ft/s

1000 Pts, psi

1000 TPD, h

4000 Horizontal Stress, Sh psi

24 Poissons Ratio

240 Young's Modulus

☐ Step Entry 25 Step 12.5 End Entry, ft/s

Calculate Fracture Select LCM

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BAROID FLUIDS HANDBOOK
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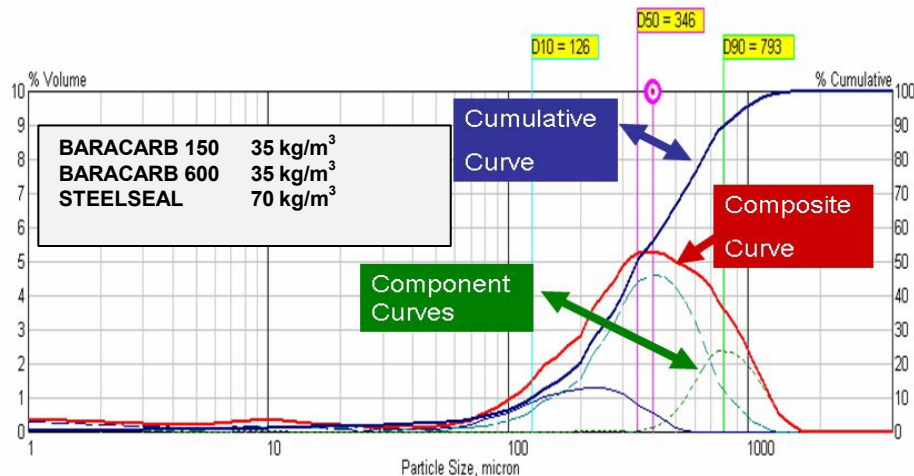
Table 6 Specialty Particulate Materials

Material	D10 microns	D50 microns	D90 microns
BARACARB 1200	300	1200	1489
STEELSEAL 1000	604	1000	1539
BARACARB 600	515	600	1125
STEELSEAL 400	270	400	744
BARACARB 150	70	150	325
BAROFIBRE O	19	90	298
STEELSEAL 100	12	100	182
STEELSEAL 50	12	50	108
BARACARB 50	3	50	125
BARACARB 25	1	25	63
BARACARB 5	1	5	18

An example model solution output is shown in Figure 5.

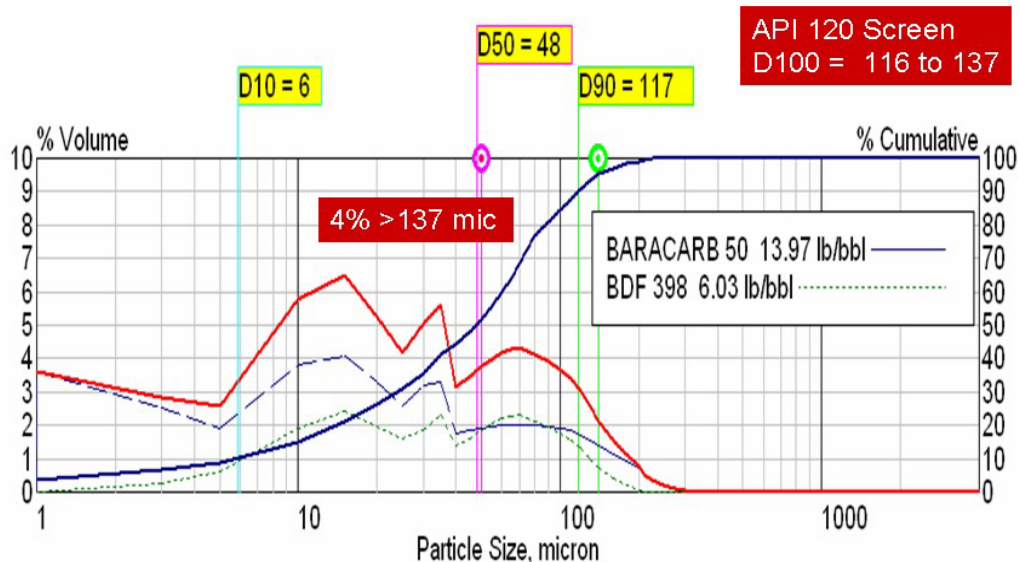
The d10, d50 and d90 of the solution is given, along with a composite curve showing the particle size distribution (PSD) of the mixture of materials as well as the PSD curves for the individual components. In addition, a cumulative curve is shown from which you can determine the volume of materials in the mixture that lies below that micron size by simply placing a cursor at any point along the curve.

Figure 5 Example Material Selection and Particle Size Distribution Solution



A number of engineering scenarios can be evaluated during the planning phase for implementation during the well construction phase. These may be a pretreatment of the entire system (Figure 6) to manage seepage and wellbore breathing issues.

Figure 6 Pretreatment Option for Entire Drilling Fluid System

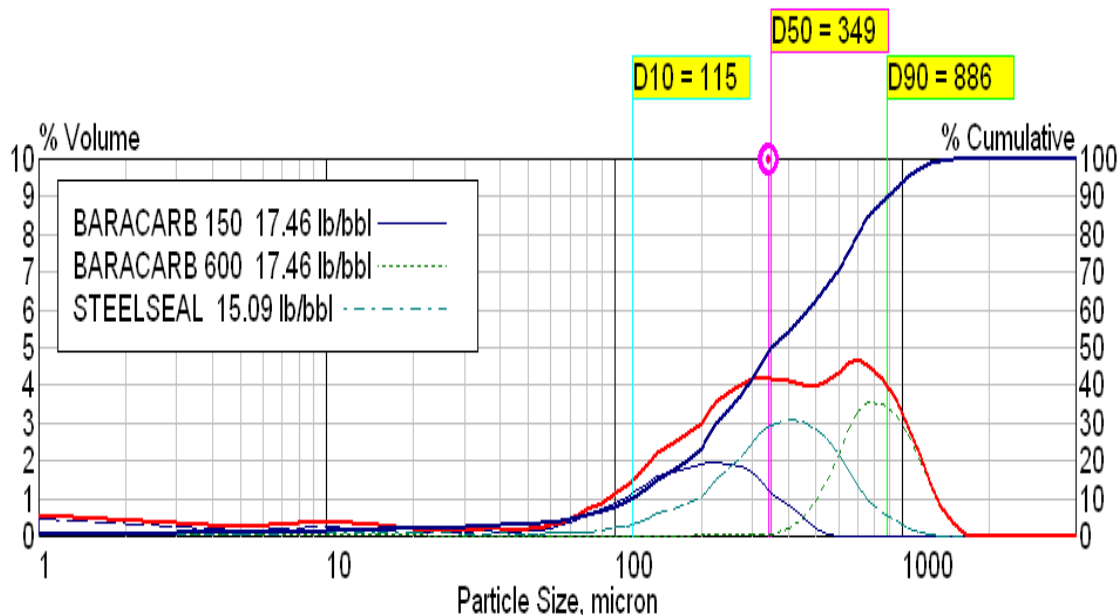


A sweep treatment using larger particles or potential fracture initiation in problem zones (Figure 7).

Figure 7 Sweep Option for Drilling Fluid System

A treating pill can be placed across the problem interval for a borehole stress treatment and/or prior to running casing and cementing (Figure 8).

Figure 8 Open Hole FIT Option– WellSET Treatment



Also shown in these examples is the consideration that is given to what amount of material will be lost from the active system based on solids control screen size.

Rheology Prediction for Invert Emulsion Fluids after the Addition of LCM

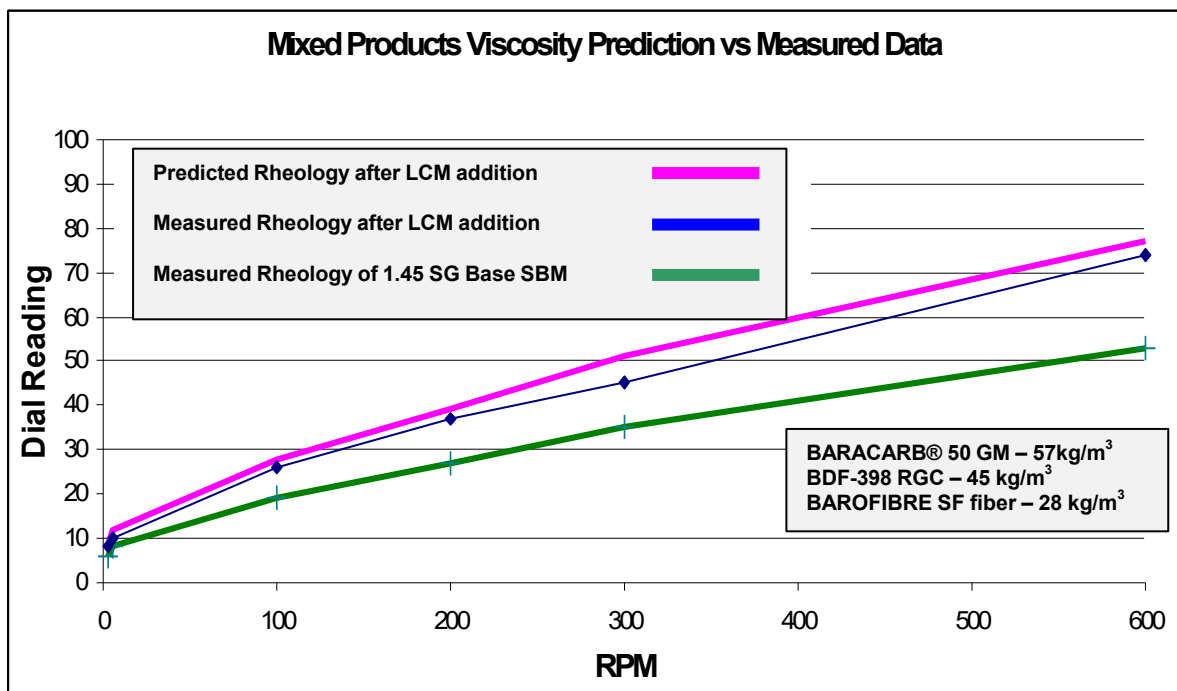
A hydraulically valid model with the resultant viscosity predicting algorithms has been developed for lost circulation material (LCM) addition to invert emulsion drilling fluids (Figure 9).

Figure 9 Rheology Prediction Model Screen Shot

Starting Fluid				Added Products		Predicted Rheology			
17	Mud Weight, lb/gal			Add	Clear	16.935	Mud Weight, lb/gal		
126	600	PV	50			204.7	600	PV	81
76	300	YP	26	*	12 BARACARB	123.4	300	YP	42
58	200			*	22 STEELSEAL	94.2	200		
38	100	n	.794	*	5 BAROFIBRE	61.7	100	n	.794
12	6	k	.502			19.5	6	k	.813
10	3	Tau0	8.952			16.2	3	Tau0	14.545
80	OWR								
4	ASG Solids								
250000	WPS	ppm CaCl2							
.820	Oil SG	ACCOLADE							

Though it does not mimic perfectly the measured performance of all product additions at all concentrations, there is adequate data to support the model. Thus, rheology predictions can be made for LCM additions to invert emulsion drilling fluids with sufficient accuracy that minimize error on ECD predictions (Figure 10).

Figure 10 Effect of LCM Addition on Rheology



The measurement of drilling fluid rheology for fluids that contain LCM is difficult, and sometimes impossible, with a standard bob and sleeve rheometer due to the interference of the particles with the rotation of the sleeve in

the narrow annular gap. The use of a different bob and sleeve with a larger annular gap is likewise problematic since the fundamental assumption of a constant shear rate across the gap is no longer valid. Consequently, the development of a predictive model would not only make the rheology determination easier and more efficient, but it also is likely to be more accurate.

1.5. Treatment Guideline Reference Tables

Less than 10 bph

	Preventive or <10bph losses - Not in reservoir	Pretreatment/Loss
Permeability	Formulation	Total concentration
<500md	STEELSEAL 50 + BARACARB 25	15-25/25-50 ppb
>500md<1000md	STEELSEAL 100 + BARACARB 25	15-25 /25-50 ppb
>1000 md	STEELSEAL 1000 + BARACARB 25 & 50	15-25 /25-50 ppb
Unknown	STEELSEAL 50 + BARACARB 25 & 50	15-25/25-50 ppb
Unknown	STOP-FRAC D	25-50 ppb
Impermeable	STEELSEAL 100	25-50 ppb
	Preventive or <10 bph losses - Reservoir	Preventive/Loss
Permeability	Formulation	Total concentration
<500md	EZ-PLUG	15-20/25-50 ppb
>500md<1000md	EZ-PLUG + BARACARB 5 & 25	20-25 /25-50 ppb
>1000 md	EZ-PLUG + BARACARB 25 & 50 & 150	20-25 /25-50 ppb
Unknown	EZ-PLUG + BARACARB 25 & 50	20-25/25-50 ppb

10-50 bph

Permeable	Corrective treatment – 10-50 bph loss Not in reservoir	Impermeable
Total concentration	Formulation	Total concentration
50 - 60 ppb	DUO-SQUEEZE H	DUO-SQUEEZE H + SA
50 - 60 ppb	BDF-551	50-60 ppb
50 - 60 ppb	BDF-562 bdf-562	50-60 ppb
80 ppb	HYDRO-PLUG	80 ppb
80-120 ppb	HYDRO-PLUG + BDF-551 or 562	80 -120 ppb
Requires cement pumping equipment	FUSE-IT (WBM) or FlexPlug OBM	Requires cement pumping equipment
Permeable	Corrective Treatment – 10-50 bph loss Reservoir	Impermeable
Total concentration	Formulation	Not applicable
50 ppb	EZ-PLUG	
50 ppb	DUO-SQUEEZE R	
50-80 ppb	EZ-PLUG + DUO_SQUEEZE R	
20/4 + DS-R@80	N-SQUEEZE Treatment	
Cementing Equipment required	ThermaTek RSP or LC	Cementing Equipment Required

50-100 bph

Permeable	Corrective treatment –50-100 bph Not in reservoir	Impermeable
Total concentration	Formulation	Total concentration
60 - 80 ppb	BDF-551 or 562 + HYDROPLUG	60-80 ppb
80 - 120 ppb	BDF 551 or 562 + HYDROPLUG + 1.0 ppb BAROLIFT	80-120 ppb
Use cement unit	FUSE-IT with BDF-562 or FlexPlug OBM Supplement with	use cement unit
Requires cement pumping equipment	FlexPlug W or BDF-376 (WBM) – FlexPlug OBM	Requires cement pumping equipment
Permeable	Corrective treatment – 50 - 100 bph Within reservoir	Impermeable
Total concentration	Formulation	Not applicable
80-120 ppb	DUO-SQUEEZE R and EZ-PLUG	

100-200 bph

Permeable	Corrective treatment – 100-200 bph Not in reservoir	Impermeable
Total concentration	Formulation	
80-120 ppb	BDF-551 or 562 + HYDROPLUG + 1.0 ppb BAROLIFT	80-120 ppb
Requires cement pumping equipment	FUSE-IT + BDF 562 or FlexPlug OBM; ThermoTek RSP or LC; Shear sensitive cement	Requires cement pumping equipment
Permeable	Corrective treatment – 100-200 bph Within reservoir	
Total concentration	Formulation	Not applicable
80-120 ppb	DUO-SQUEEZE R and EZ-PLUG + 1.0 [[b BDF-456 in N-SQUEEZE/N-Plex carrier	
Requires cement pumping equipment	ThermoTek LC or ThermoTek RSP	

Greater than 200 bph

Permeable	Corrective treatment – >200 bph or total Not in reservoir	Impermeable
Total concentration	Formulation	Total concentration
120 + ppb	HYDRO-PLUG+ BDF 562 + BAROLIFT	
Requires cement pumping equipment	FUSE-IT or FlexPlug OBM	Requires cement pumping equipment
Requires cement pumping equipment	Shear sensitive Thixotropic Cement	Requires cement pumping equipment
	Corrective treatment – >200 bph or total Within reservoir	
Permeable	Formulation	Not applicable
Requires cement pumping equipment	ThermoTek RSP or LC; Low fluid loss “acid soluble” cement	

Underground Blowout

Underground blowout	Formulation	Total concentration
	FUSE-IT or FlexPlug + Thixotropic cement	Requires cement pumping equipment

Solids Control

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1. Solids Control

Solids control is the process of controlling the buildup of undesirable solids in a mud system. The buildup of solids has undesirable effects on drilling fluid performance and the drilling process. Rheological and filtration properties become difficult to control when the concentration of drilled solids (low-gravity solids) becomes excessive. Penetration rates and bit life decrease and hole problems increase with a high concentration of drilled solids. It is estimated that over 80% of mud treatment costs are directly related to drilled solids 20 μm and smaller.

Solids-control equipment on a drilling operation is intended to be operated like a processing plant. In a perfect world, all drilled solids could be removed from a drilling fluid. In the real world, there are limitations as to the size and volume of the solids that can be efficiently removed.

Under typical drilling conditions, low-gravity solids should be maintained below 6 percent by volume.

Sources and Sizes of Solids

The two primary sources of solids (particles) are commercial solids such as barite, and non-commercial solids such as formation cuttings. Formation cuttings are considered contaminants that degrade the performance of the drilling fluid. Solids that remain in the drilling fluid will be reduced in size until they become difficult to remove from the drilling fluid with normal solids control equipment (SCE).

A large proportion of formation solids can usually be removed by mechanical means at the surface. The smaller the particle, the greater the surface area, the greater the effect on drilling fluid properties and the more difficult they are to remove from the drilling fluid. The particle size of drilled solids incorporated into drilling fluid can range from 0.1 to 250 microns (1 micron equals 1/25,400 of an inch or 1/1,000 of a millimeter).

The table below lists the approximate sizes of contaminating solids.

Table 1 Solids Sizes

Material	Diameter, microns (μm)	API Screen Designation	Diameter, inches
Clay Colloidals Bentonite	1	—	0.00004
	5	—	0.0002
Silt Barite Cement dust	5 - 42	-	0.0002 - 0.0014
Fine sand	43	325	0.0015
	53	270	0.002
	74	200	0.003
API sand	105	140	0.004
	149	100	0.006
Coarse sand	500	35	0.02
	1,000	18	0.04

1.1. Mechanical Solids Removal Equipment

One method of solids control is the use of mechanical solids-removal equipment. Another method, dilution, is discussed later in this chapter. Equipment that removes solids mechanically can be grouped into two major classifications:

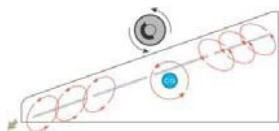
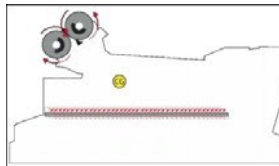
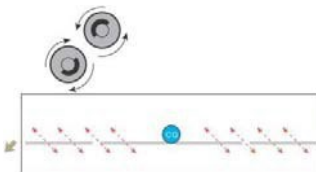
- Screening devices (a form of filtration)
- Centrifugal separation devices

Table 2 Solids Control Equipment and Effective Removal Ranges in Microns

Solids Control Device	Effective Range
Scalping Shaker (coarse screening)	API 18 $\approx$ 1000 microns
Primary Shale Shakers	API 325 $\approx$ 42 microns (in water)
Desanders (10 & 12" hydrocyclones)	$\approx$ 40-45 microns (in water)
Desilters (4" Hydrocyclone)	$\approx$ 20 – 24 microns (in water)
Decanting Centrifuge	$\approx$ 4 microns
Decanting Centrifuge preceded by Chemical Flocculation	$\approx$ Clear water (in Water Based Fluids)

Screening Devices

The most important solids removal device on a drilling rig is the shale shaker. It is the first and best opportunity to remove drilled solids before they become further degraded over time. Shale shakers contain one or more screens, integrated into a vibrating screen frame. Drilling fluid passes through the screen as it is circulated out of the hole. Shale shakers are available in various configurations with one to three decks, varying amounts of G-force at the screen surface and four basic vibratory motions, as shown below.

Circular Motion Uniform round motion for the length of the screen frame	
Elliptical Motion Non-uniform oblong motion for the length of the screen frame.	
Balanced Elliptical Motion Uniform oblong motion for the length of the screen frame.	
Linear Motion Straight line motion (balanced elliptical motion with an aspect ratio of infinity to one)	

Generally speaking, at this time, any shaker that is able to provide 7 G's or more at the screen surface is considered to be a Hi-G shaker.

Hi-G force linear motion is superior in transport of solids and screening with all but very wet, sticky cuttings, in which case balanced elliptical motion would have a slight advantage if Hi-G force Balanced Elliptical Motion Shakers existed.

Screen Effectiveness

Two factors that determine the effectiveness of a screen are mesh size and screen design.

Mesh - Historically, screen mesh meant the number of openings per linear inch as measured from the center of the wire. With the introduction of sandwich screens in the 80's, mesh count became unable to predict how a screen cloth would perform and therefore irrelevant. Variations in how the layers of screen cloth aligned created multiple sizes of screen openings. As a result, the API screen standard was created.



Figure 1 Screen Mesh: variations in hole size due to sandwich screen construction.

API Screens – API rp13c required all screens to be compared against an ASTM test screen and ranked in accordance with how they were able to classify sized Aluminum Oxide grit. This provided a level playing field for all screen manufacturers. It meant for the first time that any screen labeled API 200 for instance, would have a near identical ability to separate solids. The standard also provided a method of testing conductance, the ability of a given screen to allow fluid to pass through it.

Actual separation on the drilling rig will also be influenced by factors such as particle shape, fluid viscosity, g-force, vibratory motion and condition of the screen frame upon which the screens are mounted. The table below lists the range of solids that a given API Screen number is able to remove.

Table 3 API Screens vs Solids Sizes

D ₁₀₀ separation µm	API screen number	D ₁₀₀ separation µm	API screen number
> 3 075,0 to 3 675,0	API 6	> 231,0 to 275,0	API 60
> 2 580,0 to 3 075,0	API 7	> 196,0 to 231,0	API 70
> 2 180,0 to 2 580,0	API 8	> 165,0 to 196,0	API 80
> 1 850,0 to 2 180,0	API 10	> 137,5 to 165,0	API 100
> 1 550,0 to 1 850,0	API 12	> 116,5 to 137,5	API 120
> 1 290,0 to 1 550,0	API 14	> 98,0 to 116,5	API 140
> 1 090,0 to 1 290,0	API 16	> 82,5 to 98,0	API 170
> 925,0 to 1 090,0	API 18	> 69,0 to 82,5	API 200
> 780,0 to 925,0	API 20	> 58,0 to 69,0	API 230
> 655,0 to 780,0	API 25	> 49,0 to 58,0	API 270

D ₁₀₀ separation μm	API screen number	D ₁₀₀ separation μm	API screen number
> 550,0 to 655,0	API 30	> 41,5 to 49,0	API 325
> 462,5 to 550,0	API 35	> 35,0 to 41,5	API 400
> 390,0 to 462,5	API 40	> 28,5 to 35,0	API 450
> 327,5 to 390,0	API 45	> 22,5 to 28,5	API 500
> 275,0 to 327,5	API 50	> 18,5 to 22,5	API 635

Screen Labeling: Every API compliant screen has two tags prominently displayed on the screen and the box in which they came. The minimum information required by API is as per the example below.

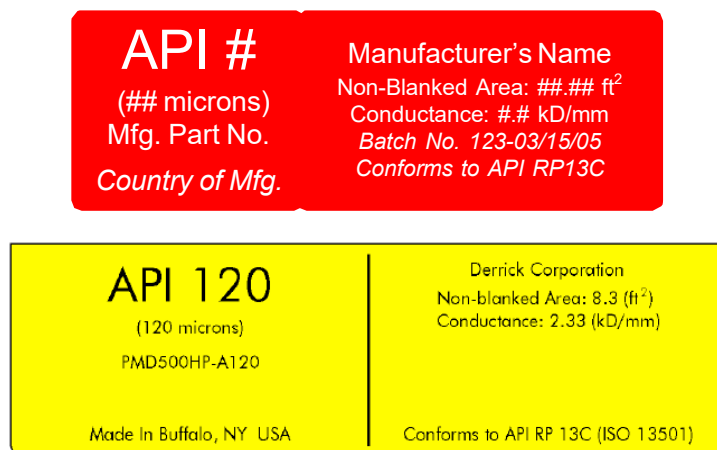


Figure 2 Correct Screen Labeling

Shale Shaker Design

Shale shakers must collect drilling fluid with cuttings from the flow line, and then meet the following design objectives:

- Distribute the flow of drilling fluid evenly across the available shakers
- Distribute the flow of drilling fluid evenly across the screen
- Hold the screens tightly to a screen frame without screen movement or flexing
- Distribute the vibration of the screen frame to the screen surface
- Provide efficient solids transport off the screen
- Collect drilling fluid that has passed through the screen and direct it toward a pit or trough

Screen Construction

Screens are available in two- and three-dimensional designs (i.e., flat and corrugated). Two-

dimensional screens include:

- Panel screens: with two or three layers of screen cloth on a backing screen, bound on each side by a one-piece, double-folded hook strip
- Perforated plate screens: two or three layers of screen cloth bonded to a perforated, metal plate that provides support and is repairable.

- Pre-tensioned screen panels: Stiff screen panel with 2 or more layers of screen cloth bonded to a metal or plastic panel.

Three-dimensional Screens include Pyramid™ and Pyramid Plus™:

Three-dimensional screens use a perforated plate to provide structural strength with a corrugated surface running parallel with the flow of fluid. Three dimensional screens provide more usable (non-blanked) screen area than two-dimensional screens. This can be confirmed by comparing the conductance figures on the API screen labels.

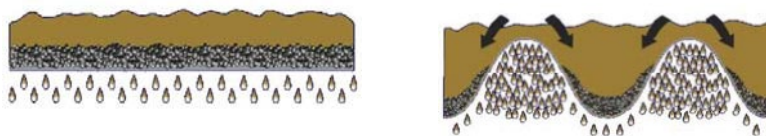
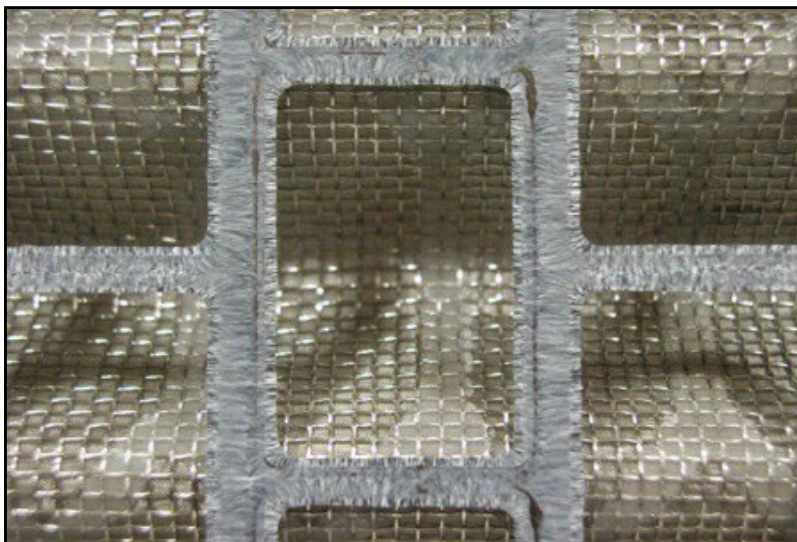


Figure 3 Two-dimensional vs Three-dimensional Screens

Vibration forces solids into the trough on three-dimensional screens allowing improved conductance while flat screens allow a uniform bed of solids which impedes fluid throughput.

High Temperature Screens

Water-based fluids with relatively high glycol concentrations and/or high temperatures can be prone to delamination. Although more pronounced in three-dimensional screens due to their relatively higher mass, delamination also occurs in flat panel screen construction. This screen type is designed for fluid temperatures up to 110C and/or glycol fluid systems.



1.2. Centrifugal Separation Devices

All centrifugal separations are governed by Stokes Law.

Stokes Law - Determination of settling rates (for spherical particles)

$$U_t = (g D_s^2 (s-l)) / 18\mu$$

Where:

U_t = terminal, or settling, velocity

ρ_s = density of the solid

g = acceleration of gravity

ρ_l = density of the liquid

D_s = diameter of the solid

μ = viscosity of the liquid

Settling rates and therefore centrifugal separation efficiencies are dependent upon the variables above. We may be unable to change the density of the solid, the density of the liquid or the diameter of the solids but in many devices, G-forces and the retention time can be adjusted to our benefit.

The two types of centrifugal separation devices most commonly used for removing drilled solids are:

- Hydrocyclones
- Decanting centrifuges

Hydrocyclones

Hydrocyclones, like all centrifugal devices separate solids by their relative mass. These conical solids separation devices in which hydraulic energy is converted to centrifugal force are fed by a centrifugal pump through the feed inlet tangentially into the feed chamber. The centrifugal forces thus developed multiply the settling velocity of the higher mass solids, forcing them toward the wall of the cone. The 'beach' is the area of the cone in which particles come into contact with the side wall of the cone. The lighter particles move inward and upward in a spiraling vortex to the overflow opening at the top. The discharge at the top is the overflow or effluent; the discharge at the bottom is the underflow. The underflow should be in a fine spray with a slight suction at its center. A rope discharge with no air suction is undesirable. Figure 11-4 illustrates the hydrocyclone process. The ability to adjust the vortex and thereby change the pressure equilibrium inside the cone provides a level of adjustment to assist in optimizing the performance of a cone.

The diameter and length (think residence time inside the cone) plus the fluid velocity determines the cut obtained. Lower feed manifold pressures result in coarser separation and reduced capacity. Most hydrocyclones are advertised as being effective with 75 feet of head. (Manifold pressure in psi should be roughly 4 times the mud weight in pounds per gallon).

Hydrocyclone Solids Separation Process

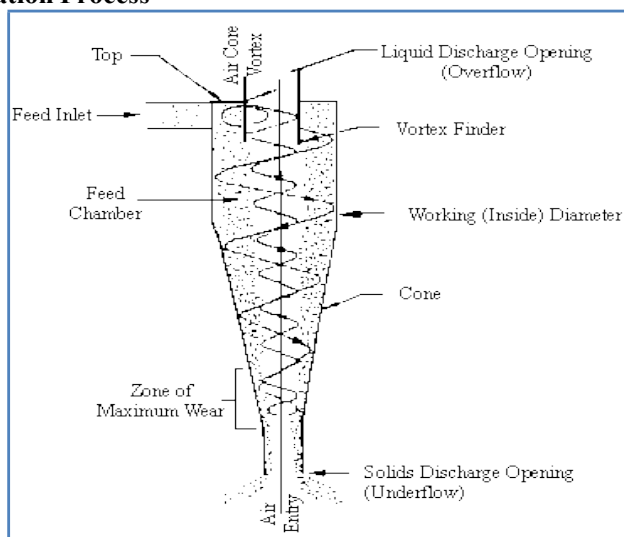


Figure 4 Hydrocyclone solids-removal process

Hydrocyclone separation is tested with solids in water. The actual size of the solids separated will be highly dependent upon the viscosity of the drilling fluid in addition to the velocity of the fluid entering the cone.

Hydrocyclone Plugging

Hydrocyclones are prone to plugging with drilled solids. The smaller the cone, the more prone to plugging they will be. Bypassing the shale shakers for a screen change while drilling will quickly demonstrate the accuracy of this statement. Once large solids are distributed throughout the pit system, unplugging the desilter cones becomes a full time occupation. Overloaded cones will move solids that cannot pass through the apex back into the active fluid system.

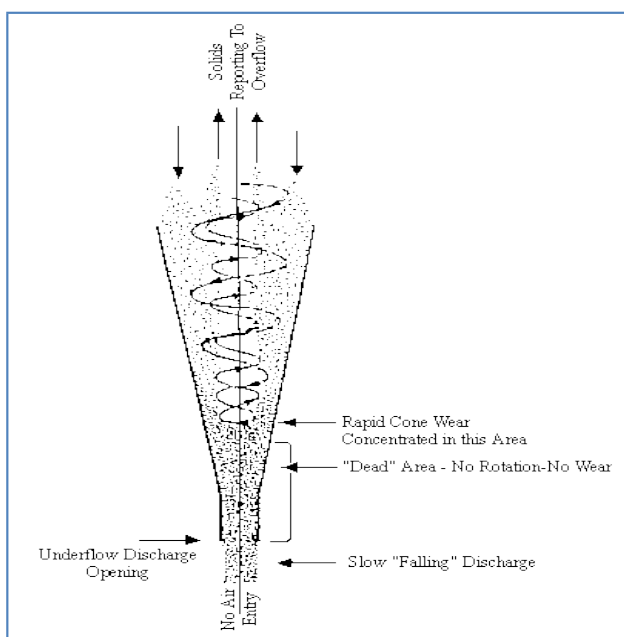


Figure 5 Hydrocyclone in 'Rope' discharge

Hydrocyclones are classified as either desanders or desilters.

Desanders

Desanders consist of a battery of 6, 10 and 12-inch cones. Even though desanders can process large volumes of mud per single cone, the minimum size particles that can be removed are in the range of 40 microns (with 6-inch cones).

Desilters

Desilters consist of a battery of 4-inch or smaller diameter cones. Depending on the size of the cone, a particle size cut between 6 and 40 microns can be obtained. Micro-clones (1" diameter cones) are used to remove colloidal solids from low density fluids. They are known chiefly for their propensity to plug with solids. Current solids control equipment seldom provides hydrocyclones for drilling fluid that are smaller than 4."

Even though hydrocyclones are effective at removing solids from a drilling fluid, their use is not recommended for fluids that contain significant amounts of weighting materials or fluids with expensive fluid phases. When hydrocyclones are used with these fluids, not only will undesirable drilled solids be removed, but also the weight material along with base fluid, which can become cost-prohibitive.

Table 4 Nominal Cone Processing Capacities and Cut Points

Cone Size	Ft. of Head*	D ₅₀	GPM
12"	75	45	500
10"	75	38	500
4"	75	24	50
3"	100	10	35

Mud Cleaners

The mud cleaner is a solids separation device that combines a desilter manifold with a screening device. The combination of anywhere from 8 to 20 desilter cones processing a nominal 50 gpm per cone (for 4" cones) allows the screening of up to 1000 gpm over fine screens with a single device. Modern drilling rigs are usually equipped with shale shakers that can screen as fine as mud cleaners. The mud cleaner is increasingly seen as a back up solids removal device for when the shale shakers are bypassed or when shale shaker screens get holes in them.

When recovering weighting material with a mud cleaner, any fine solids that pass through the screen will be retained in the mud and, over time, will lead to a fine-solids buildup that can only be addressed by a decanting centrifuge.



Figure 6 Mud Cleaner – Courtesy of Derrick Equipment Corporation

Decanting Centrifuges

A decanting centrifuge consists of a conical, horizontal steel bowl that rotates at high speed using a double helix-type conveyor. The conveyor rotates in the same direction as the outer bowl but at a slightly different speed.

These can be either a leading or a following conveyor. An important aspect of centrifuge operation is the dilution of the slurry being fed into the unit. The slurry dilution reduces the feed viscosity and increases separation within the bowl. The higher the viscosity of the base mud, the more dilution is needed (2 to 4 gallons of water per minute is common). The effluent (liquid output from the centrifuge) viscosity should be 35 to 37 seconds per quart for efficient separation. Manufacturers' recommendations concerning mud-feed rates and bowl speeds should be followed closely.

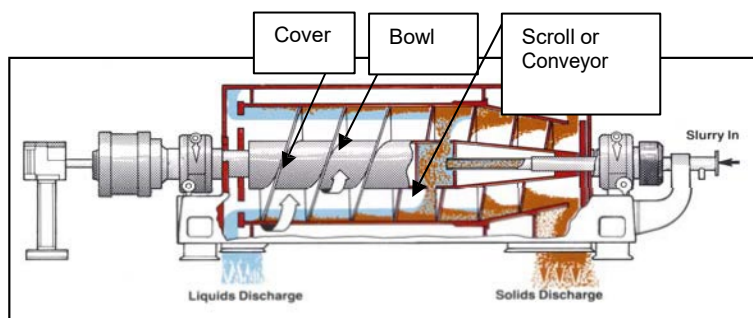


Figure 7 Cross-section of a decanting centrifuge

In this diagram, open arrows indicate the path of liquids; solid arrows indicate the path of solids.

Centrifuges are available with different drive systems and several different bowl dimensions.

- Belt Drive Systems require a pulley change and/or belt adjustment to change bowl speeds with one or more different conveyor (scroll) speeds based on the design of the gearbox on the unit.
- FHD (full hydraulic drive) systems have variable speed bowl and scroll speeds
- VFD (Variable Frequency Drive) systems have variable speed bowl and scroll speeds

The most common diameters for drilling fluids are 14", 16" and 21". Larger diameter bowls provide the same G-Force as smaller diameter bowls, turn at a lower rpm and process fluids at higher rates. They also require more space for installation and power to operate. The most common centrifuge in the industry is a 14" bowl unit, typically set to no more than 3400 rpm with a maximum G-force of 2298 G's. A 21" bowl can reach the same 2298 G's at 2776 rpm and process at 2-3 times the rate.

$$\text{G-Force} = 0.0000142 \times \text{bowl diameter (inches)} \times \text{RPM}^2$$

Ideally centrifuges are fed with a positive displacement pump such as a progressive cavity pump or a lobed pump. These pumps provide a known volume of fluid per rpm, thus the feed rate can be known. Many of the VFD centrifuges also have a VFD for the feed pump allowing changes in feed rate, bowl rpm and scroll speed to be adjusted from a single control panel. Basic belt drive centrifuges typically have no rpm meters and all speed changes require shutting down the unit for speed changes.

Historically, most rig centrifuges are configured set to minimize the operators work load and down time rather than maximize the volume of solids removed. This is due in large part to the amount of time and labor required to 'unplug' a plugged centrifuge as a result of a sudden surge in solids content in the feed.

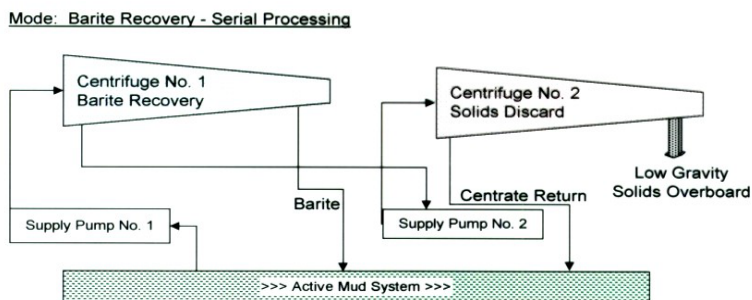
A single centrifuge set for solids discard should be used for low-density (sub 14ppg) systems. The primary function of a centrifuge is not to control total percent solids in a system, but rather to maintain acceptable and desirable flow properties in that system. Two centrifuges operating in series are recommended for these systems:

- Invert emulsion (i.e., synthetic and oil-based systems)
- High-density, water-based systems
- Water-based systems in which base fluid is expensive (i.e., brines)
- Closed loop
- Zero discharge

In Serial Processing, the first centrifuge unit is run at low speeds to separate high mass materials, such as barite and return them to the active mud system. The second centrifuge operating at the highest possible speed processes the liquid overflow from the first unit, discarding all removable solids while returning the liquid 'dilution fluid' (CENTRATE) to the mud system.

Centrifuge solids discharge: The solids discharged from a centrifuge (CAKE) should be relatively creamy and wet. Centrifuges remove fine solids. Fine solids per unit volume have significantly more surface area than larger solids (e.g. shale shaker discharge) and will accordingly, have more liquid. Trying to make the centrifuge discharge as dry as possible is almost always a poor solids control strategy. The cake should not be able 'stand up' on its own as would a pile of cuttings from the shale shaker. Not only does it place a high load on the scroll but it also reduces the rate at which solids are removed.

Centrifuge Theory & Installation Barite Recovery



Separation Theory

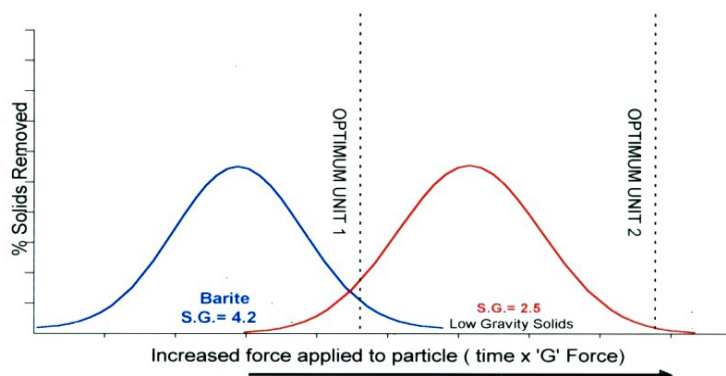


Figure 8 Simple schematic for serial processing (Barite Recovery)

The goal in serial processing is to return a relatively clean dilution fluid to the active system while maintaining the high mass solids – barite and larger diameter drilled solids.

1.3. Pit Systems

The arrangement and installation of equipment, piping and valves on a pit system can have a profound influence on solids control efficiency. This includes agitation, piping, weir settings, pump sizing and installation and sizing of solids control equipment.

Pit systems are designed to:

- Contain enough usable mud to fill the hole at all times
- Satisfy operational requirements (displacements, well control, etc)
- Mix new fluid as required by drilling process (new hole, losses)
- Provide segregated residence time for effective products application
- Perform Required processes (solids control) to maintain mud properties
- Provide sufficient surface area for mud cooling and entrained gas release
- Provide contingency
- Provide flexibility

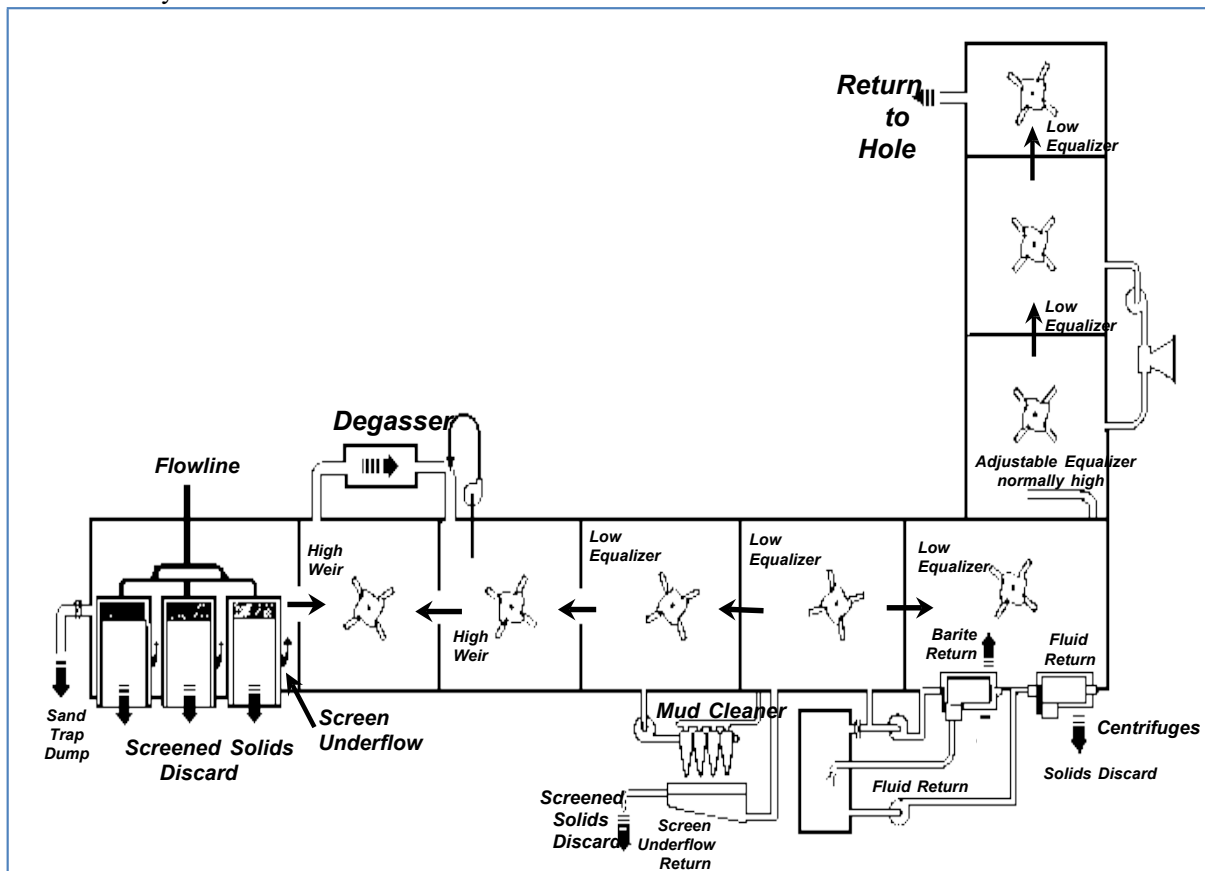


Figure 9 Example Pit System

1.4. Dewatering: Chemically Enhanced Solids Control

Solids control *systems* on drilling rigs are based on removing progressively smaller solids with each piece of equipment in the process.

- Scalping shakers, gumbo chains, etc. (Coarse Screening)
- Shale Shakers (Fine screening - to about 70 μ)
- Sand Trap – the pit system design aids the process
- Desanders (45 μ) & Desilters (24 μ)—(Hydrocyclones)
- Decanting Centrifuges (down to $\approx 10\ \mu$)
- ‘Dewatering’ (10 μ to submicron) i.e. the goal is clear water

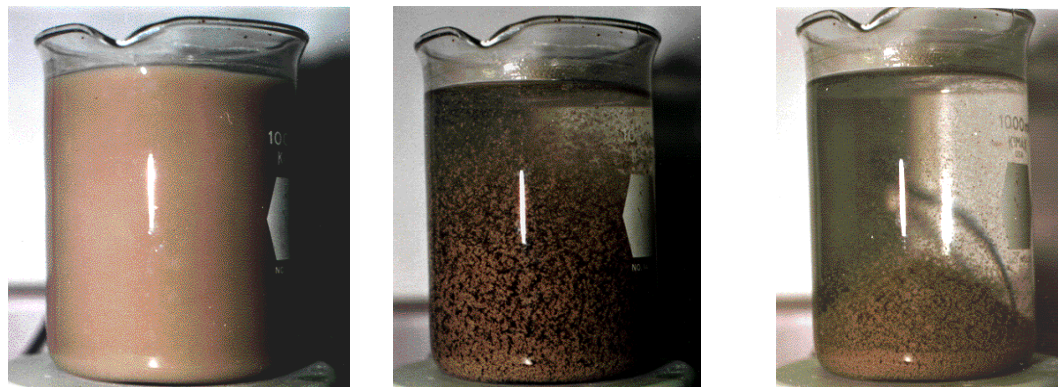
Mechanical solids control processes alone are unable to remove sub-10 μ m solids – often called ‘*ultra-fines*’

Dewatering uses flocculation / coagulation reactions to assist centrifuges in removing 10 micron and smaller solids.

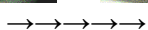
When fine solids join together, they are called ‘flocs’. Because flocs have higher mass than the colloidal solids they are composed of, they are easily removed by a centrifuge, provided the floc is not easily dispersed inside the centrifuge.

Dewatering is normally used for the following:

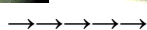
- Solids control processing to remove colloidal - ultra-fines - providing a clean ‘dilution fluid’ back to the active system
- Recovery and recycling of the liquid phase of a suspension
- Zero-Discharge or Pit-less locations with no pit construction allowed
- Clear water drilling (i.e., faster drilling)



FROM THIS



TO THIS



TO THIS

1.5. Dilution

Dilution, the addition of water, base oil, or new drilled solids free drilling fluid to a mud system, serves to:

- Reduce concentration of solids left by mechanical solids-removal equipment
- Replenish liquids lost when using mechanical solids-control equipment

Dilution can generate excessive volumes that have to be dealt with, normally by the ‘dump & dilute process.

Example I (idealized): 80% Solids Removal Efficiency

- Drilling 1000’ of 12¼” hole → 145 bbls drilled solids
- If solids removal efficiency is 80% → 29 bbls of drilled solids remain in the system.
- To maintain 5% drilled solids requires $29/.05 = 580$ bbls dilution.
- Less 145 bbls mud to maintain volume = 435 bbls additional new volume drilling fluid required.

Example II (idealized): 90% Solids Removal Efficiency

- Drilling 1000’ of 12¼” hole → 145 bbls drilled solids
- Assuming 90% removal efficiency; $145 \text{ bbls} \times (1-.90) = 14.5$ bbls solids remaining in system
- $14.5/.05 = 290$ bbls dilution required to maintain 5% drilled solids in active system
- 290 bbls less 145 bbls to maintain volume → 145 bbls additional dilution volume
- $145/435 = 1/3$ the dilution required to maintain 5% drilled solids with 10% improvement in separation efficiency.

In other words: A small improvement in solids removal efficiency has a large impact upon dilution rates and mud costs.

Depending upon the location, environmental laws and the type of drilling fluid, disposal and clean-up costs can be very expensive. It is not uncommon for the waste disposal costs to exceed the cost of drilling fluids for the well.

For calculating the efficiency of solids-control equipment, see API RP13C. Access this through Chapter 2 in this Guide.

1.6. Terms and Definitions

For the purposes of this document, the following terms and definitions apply.

Addition section	Compartment(s) in the surface drilling fluid system, between the removal section and the suction section, which provides (a) well-agitated compartment(s) for the addition of commercial products such as chemicals, necessary solids and liquids
Agitator - mechanical stirrer	Mechanically driven mixer that stirs the drilling fluid, by turning an impeller near the bottom of a mud compartment to blend additives, suspend solids and maintain a uniform consistency of the drilling fluid
Aperture	Screen cloth opening between the wires in a screen cloth Screen surface opening in a screen surface
Apex	Opening at lower end of a hydrocyclone
API sand	Physical description: particles in a drilling fluid that are too large to pass through a 74 μm sieve (API 200 screen)
API screen number	Number in an API system used to designate the D100 separation range of a mesh screen cloth - Both mesh and mesh count are obsolete terms, and have been replaced by the API screen number. - The term "mesh" was formerly used to refer to the number of openings (and fraction thereof) per linear inch in a screen, counted in both directions from the centre of a wire. - The term "mesh count" was formerly used to describe the fineness of a square or rectangular mesh screen cloth, e.g. a mesh count such as 30 $\times$ 30 (or, often, 30 mesh) indicates a square mesh, while a designation such as 70 $\times$ 30 mesh indicates a rectangular mesh.
Backing plate	Support plate attached to the back of screen cloth(s)
Baffle	Plate or obstruction built into a compartment to change the direction of fluid flow
Barite	Natural barium sulfate (baso4) used for increasing the density of drilling fluids The standard international requirement is for a minimum specific gravity of 4.20 or 4.10 for two grades of barite, but there is no specification that the material must be barium sulfate. Commercial API Spec 13A barite can be produced from a single ore or a blend of ores, and can be a straight-mined product or processed by flotation methods.
Blinding	Reduction of open area in a screening surface caused by coating or plugging
Bonding material	Material used to secure screen cloth to a backing plate or support screen
Capture	Mass fraction of incoming suspended solids that are conveyed to the reject stream
Centrifugal pump	Machine for moving fluid by spinning it using a rotating impeller in a casing with a central inlet and a tangential outlet. The path of the fluid is an increasing spiral from the inlet at the centre to the outlet, tangent to the impeller annulus. In the annular space between the impeller vane tips and the casing wall, the fluid velocity is roughly the same as that of the impeller vane tips. Useful work is produced by the pump when some of the spinning fluid flows out of the casing tangential outlet into the pipe system. Power from the motor is used to accelerate the fluid entering the inlet up to the speed of the fluid in the annulus. Some of the motor power is expended as friction of the fluid in the casing and impeller.
Centrifuge	Device, rotated by an external force, for the purpose of separating materials of various masses (depending upon specific gravity and particle sizes) from a slurry to which the rotation is imparted primarily by the rotating containing walls NOTE In a weighted drilling fluid, a centrifuge is usually used to eliminate colloidal solids.

Check section - suction section	Last active section in the surface system which provides a location for rig pump and mud hopper suction, and ideally is large enough to check and adjust drilling fluid properties before the drilling fluid is pumped downhole
Clay mineral	Soft, variously colored earth, commonly hydrous silicate of alumina Clay minerals are essentially insoluble in water, but disperse under hydration, grinding, heating or velocity effects. Particle sizes of clay mineral can vary from sub-micrometre to larger than 100 μm .
Clay particle	Colloidal particles of clay mineral having less than 2 μm equivalent spherical diameter
Coating	Substance: material adhering to a surface to change the properties of the surface. See blinding. Physical process: procedure by which material forms a film that covers the apertures of the screening surface
Colloidal solid	Particle of diameter less than 2 μm Note: this term is commonly used as a synonym for clay particle size.
Conductance	Permeability per unit thickness of a static (not in motion) shale shaker screen Conductance is expressed in units of kilodarcies per millimetre ²
Cuttings	Formation pieces dislodged by the drill bit and brought to the surface in the drilling fluid Field practice is to refer to all solids removed by the shaker screen as "cuttings", although some can be sloughed material.
D100 separation	Particle size, expressed in micrometres, determined by plotting the percentage of aluminum oxide sample separated by the test screen on the plot of cumulative mass fraction (expressed as a percentage) retained versus US sieve opening (expressed in micrometres) for the sieve analysis of the aluminum oxide test sample 100 % of the particles larger than the D100 separation are retained by the test screen.
Decanting centrifuge	Centrifuge that removes solids from a feed slurry by rotating the liquid in cylindrical bowl at high speed and discharges the larger particles as a damp underflow Colloidal solids are discharged with the liquid overflow or light slurry. The decanting centrifuge has an internal auger that moves solids that have settled to the bowl walls out of a pool of liquid and to the underflow.
Density	Mass divided by volume In si units, density is expressed in kilograms per cubic metre; in usc units, it is expressed as pounds per gallon or pounds per cubic foot. Drilling fluid density is commonly referred to as "drilling fluid weight" or "mud weight".
Desander	Hydrocyclone with an inside diameter of at least 152 mm (6 in) that removes a high proportion of the particles with a diameter of at least 74 μm from a drilling fluid
Dilution	Method of decreasing the drilled-solids content of a slurry by addition of (a) material(s) other than drilled solids, usually a clean drilling fluid
Dilution factor	K -ratio of the actual volume of clean drilling fluid required to maintain a targeted drilled-solids concentration to the volume of drilling fluid required to maintain the same drilled-solids fraction over the same specified interval of footage with no drilled-solids removal system
Drilled solids	Formation solids that enter the drilling fluid system, whether produced by the drill bit or from the side of the borehole
Drilled solids fraction	Average volume fraction of drilled solids maintained in the drilling fluid over a specified interval of footage
Drilled-solids removal system	Equipment and processes used while drilling a well that remove the solids generated from the hole and carried by the drilling fluid These processes include settling, screening, desanding, desilting, centrifuging and dumping.

Drilled-solids removal system performance	Measure of the removal of drilled solids by surface solids-control equipment The calculation is based on a comparison of the dilution required to maintain the desired drilled-solids content with that which would have been required if none of the drilled solids were removed.
Drilling fluid	Liquid or slurry pumped down the drill string and up the annulus of a hole during the drilling operation
Eductor	Fluid stream: device using a fluid stream that discharges under high pressure from a jet through an annular space to create a low-pressure region When properly arranged, it can evacuate degassed drilling fluid from a vacuum-type degasser or pull solids through a hopper. Pressure jet: device using a high-velocity jet to create a low-pressure region which draws liquid or dry material to be blended with the drilling fluid The use of a high-velocity jet to create a low-pressure region is known as the Bernoulli principle.
Effluent	Discharge of liquid, generally a stream, after some attempt at separation or purification has been made
Equalizer	Opening for flow between compartments in a surface fluid-holding system, which allows all compartments to maintain the same fluid level
Flow capacity	Rate at which equipment, such as a shaker, can process drilling fluid and solids It is a function of many variables, including shaker configuration, design and motion, drilling fluid rheology, solids loading, and blinding by near-size particles.
Flow line	Piping or trough which directs drilling fluid from the rotary nipple to the surface drilling fluid system
Flow rate	Volume of liquid or slurry that moves through a pipe in one unit of time Flow rate is expressed as cubic metres per minute, gallons per minute, barrels per minute, etc.
Foam	Phase system: two-phase system, similar to an emulsion, in which the dispersed phase is air or gas Floating material: bubbles floating on the surface of the drilling fluid The bubbles are usually air-cut drilling fluid, but can be formation gasses.
Gumbo	Cuttings that agglomerate and form a sticky mass as they are circulated up the wellbore
Head	Height that a fluid column would reach in an open-ended pipe if the pipe were attached to the point of interest The head at the bottom of a 300 m (1 000 ft) well is 300 m (1 000 ft), but the pressure at that point depends upon the density of the drilling fluid in the well.
High specific gravity solids	Solids added to a drilling fluid specifically to increase drilling fluid density Barite (specific gravity = 4.2 or 4.1) and haematite (specific gravity = 5.05) are the most common.
Hook strip	Hooks on the edge of a screen section of a shale shaker which accept the tension member for screen mounting
Hopper - mud hopper	Large, funnel-shaped or coned-shaped device, into which dry components are poured to mix the components uniformly with liquids or slurries that are flowing through the lower part of the cone
Hydrocyclone-cone-cyclone	Liquid-solids separation device using centrifugal force for settling Note: fluid enters tangentially and spins inside the hydrocyclone. The heavier solids settle to the walls of the hydrocyclone and move downward until they are discharged at the hydrocyclone apex. The spinning fluid travels part way down the hydrocyclone and back up to exit out the top of the hydrocyclone through a vortex finder.
Impeller	Spinning disc in a centrifugal pump with protruding vanes, used to accelerate the fluid in

	the pump casing
Manifold	Length of pipe with multiple connections for collecting or distributing drilling fluid
Marsh funnel viscosity - funnel viscosity	Viscosity measured with the instrument used to monitor drilling fluid A marsh funnel is a tapered container with a fixed orifice at the bottom so that, when filled with 1 500 cm ³ of fresh water, 946 cm ³ (one quart) will drain in 26 s. It is used for comparison values only and not to diagnose drilling fluid problems. See API RP 13B-1 or API RP 13B-2.
Mud	Slurry of insoluble and soluble solids in either water or a synthetic or oil continuous-phase fluid See drilling fluid
Mud balance	Beam-type balance used in determining drilling fluid density See API RP 13B-1 or API RP 13B-2
Mud cleaner	Combination of hydrocyclones and screens in series with the underflow of the hydrocyclones The hydrocyclone overflow returns to the drilling fluid, while the underflow of the hydrocyclones is processed through a vibrating screen. The screen is usually of size API 150 or finer. The screen solids discharge is discarded while the liquid and solids passing through the screen are returned to the drilling fluid.
Mud compartment	Subdivision of the removal, addition or check/suction sections of a surface system
Mud gun	Submerged nozzle used to stir drilling fluid with a high-velocity stream
Near-size particle	Particle whose size is close to the size of the openings in the screen through which its passage is under evaluation
Oil-based drilling fluid	Drilling fluid in which the continuous phase is not miscible with water, and water or brine is the dispersed phase Oil-based drilling fluids are usually referred to as non-aqueous drilling fluids, or NAF.
Overflow – centrate	Discharge stream from a centrifugal separation that contains a higher percentage of liquids than the feed does
Particle	Discrete unit of solid material that consists of a single grain, or of any number of grains stuck together
Particle size distribution	Mass or net volume classification of solid particles into each of the various size ranges, as a percentage of the total solids of all sizes in a fluid sample
Plastic viscosity	Measure of the high-shear-rate viscosity, which depends upon the number, shape and size of solids and the viscosity of the liquid phase Plastic viscosity is calculated by subtracting the 300 r/min concentric cylinder viscometer reading from the 600 r/min concentric cylinder viscometer reading. See API RP 13B-1 or API RP 13B-2. In SI units, plastic viscosity is expressed in pascal seconds; in USC units, plastic viscosity is expressed in centipoises.
Plugging	Wedging or jamming of openings in a screening surface by near-size particles, which prevents the passage of undersize particles and leads to the blinding of the screen
Possum belly	Compartment or back tank on a shale shaker, into which the flow line discharges and from which drilling fluid is either fed to the screens or is bypassed, if necessary
Removal section	First section in the surface drilling fluid system, consisting of a series of compartments to remove gas and undesirable solids
Retort	Instrument used to distil oil, water and other volatile material in a drilling fluid The amount of volatile fluid is used to determine oil, water and total solids contents as volume fraction percent, expressed as a percentage.

	See API RP 13B-1 or API RP 13B-2.
Sand trap	First compartment in a surface system, and the only compartment that is unstirred or unagitated, which is intended as a settling compartment
Screen cloth	Type of screening surface woven in square, rectangular or slotted openings
Screening	Mechanical process that results in a division of particles on the basis of size, based on their acceptance or rejection by a screening surface
Shale shaker	Mechanical device that separates cuttings and large solids from a drilling fluid The separation methods can include vibrating screens, rotating cylindrical screens, etc.
Sieve	Laboratory screen with wire-mesh or electronically-punched holes of known dimensions
Sieve analysis	Classification by mass of solid particles passing through or retained on a sequence of screens with decreasing aperture sizes Sieve analysis can be carried out by wet or dry methods
Slug tank	Small compartment, normally adjacent to the suction compartment, used to mix special fluids to pump downhole Slug tanks are most commonly used to prepare a small volume of weighted drilling fluid before a drillstring trip out of the borehole.
Suction compartment	Area of the check/suction section that supplies drilling fluid to the suction of the drilling fluid pumps In general terms, a suction compartment is any compartment from which a pump removes fluid.
Sump	Pan or lower compartment below the lowest shale shaker screen
Tensioning	Stretching of a screening surface of a shale shaker to the proper tension, while positioning it within the vibrating frame
Total dilution	Volume of drilling fluid that would be built to maintain a specified volume fraction of drilled solids over a specified interval of footage, if there were no solids removal system
Total non-blanked area	Net unblocked area that permits the passage of fluid through a screen Total non-blanked area is expressed in square metres (square feet). Some screen designs can eliminate as much as 40 % of the gross screen panel area from fluid flow due to backing-plate and bonding-material blockage.
Trip tank	Gauged and calibrated vessel used to account for fill and displacement volumes as pipe is pulled from and run into the hole Close observation allows early detection of formation fluid entering a wellbore and of drilling fluid loss to a formation.
Underflow	Centrifugal separator: discharge stream from a centrifugal separator that contains a higher percentage of solids than the feed does Screen separator: discharge stream from a screen separator that contains a lower percentage of solids than the feed does
Unoccluded	Unobstructed area of a screen opening
Unweighted drilling fluid	Drilling fluid that does not contain commercial suspended solids added for the purpose of increasing the density of the drilling fluid
Viscosity	Ratio of shear stress to shear rate In SI units, viscosity is expressed in pascal seconds; in USC units, viscosity is expressed in centipoises. If the shear stress is expressed in the centimeter-gram-second (CGS) system of units (dynes per square centimeter) and the shear rate is expressed in reciprocal seconds, the viscosity is expressed in poises (P). $1 \text{ P} = 1 \text{ dyn}\cdot\text{s}/\text{cm}^2 = 1 \text{ g}\cdot\text{cm}^{-1}\cdot\text{s}^{-1} = 10^{-1} \text{ Pa}\cdot\text{s}$; $1 \text{ cP} = 1 \text{ mPa}\cdot\text{s}$.

Volume of solids drilled	Volume of solids drilled over a specified interval
Vortex	Cylindrical or conical shaped core of air or vapour that lies along the central axis of the rotating slurry inside a hydrocyclone
Water-based drilling fluid	Drilling fluid in which water is the suspending medium for solids and is the continuous phase, whether oil is present or not
Weighted drilling fluid	Drilling fluid to which solids have been added in order to increase its density
Weighting material	Solids used to increase the density of drilling fluids This material is commonly barite or hematite; in special applications, it might be limestone.

1.7. Symbols and Abbreviations

A	cross-sectional area, expressed in square centimetres
C	conductance of screen cloth, expressed in kilodarcies per millimetre
h	head, expressed in meters
hT	head for testing, expressed in millimetres (inches)
k	dilution factor
K	constant of proportionality, or permeability, expressed in darcies
L	length of the porous medium, expressed in centimetres
$m1$	mass of empty container, expressed in grams
$m2$	mass of container plus sample, expressed in grams
$m3$	mass of dried/retorted container, expressed in grams
mS	sample mass
p	pressure, expressed in kilopascals
Δp	pressure differential, expressed in atmospheres
q	flow rate through a porous medium, expressed in cubic centimetres per second
Va	volume of total drilling fluid system, expressed in cubic metres (gallons)
Vb	volume of base fluid added to drilling fluid system, expressed in cubic metres (gallons)
Vc	volume of drilling fluid built, expressed in cubic metres (gallons)
Vd	volume of solids drilled, expressed in cubic metres (gallons)
Ve	volume of total dilution, expressed in cubic metres (gallons)
V/t	flow rate (volume per time), expressed in m ³ /h, (gal/min)
w	mass fraction, expressed as a decimal fraction
wa	mass fraction of suspended solids removed, expressed as a percentage

w1	mass fraction of suspended solids in the feed to a piece of separator equipment, expressed as a decimal fraction
w2	mass fraction of suspended solids in the overflow from a piece of separator equipment, expressed as a decimal fraction
w3	mass fraction of suspended solids in the underflow from a piece of separator equipment, expressed as a decimal fraction
w4	mass fraction of weighting material, expressed as a decimal fraction
w5	mass fraction of low-gravity solids, expressed as a percentage
η	efficiency, drilled-solids removal system performance
a	base fluid volume fraction of total drilling fluid system, V_a , determined by retort and salinity measurement, expressed as a percentage
b	drilled-solids volume fraction of total drilling fluid system, V_a , determined by retort, salinity and bentonite measurement, expressed as a percentage
μ	fluid viscosity, expressed in centipoises
ρ	density of oil or drilling fluid, expressed in kg/m ³ (lb/gal, lb/ft ³)
ρ_1	specific gravity of separated solids
ρ_2	specific gravity of weighting material

Stuck Pipe

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1. Stuck Pipe

In drilling operations, the drill pipe is considered stuck when it cannot be raised, lowered, or rotated. Stuck pipe can be caused by several different mechanisms. Typical stuck pipe situations include the following:

- Differential-pressure effects
- Packing off
- Undergauge hole
- Keyseating

1.1. Differential Sticking

Most incidents of stuck pipe are caused by differential-pressure effects. Excessive differential pressures across lower-pressure permeable zones can cause the drillstring to push into the wellbore wall where it becomes stuck. (Figure 1).

Differential sticking may be identified by the following characteristics:

- Pipe sticks after remaining motionless for a period of time
- Pipe cannot be rotated or moved when circulation is maintained

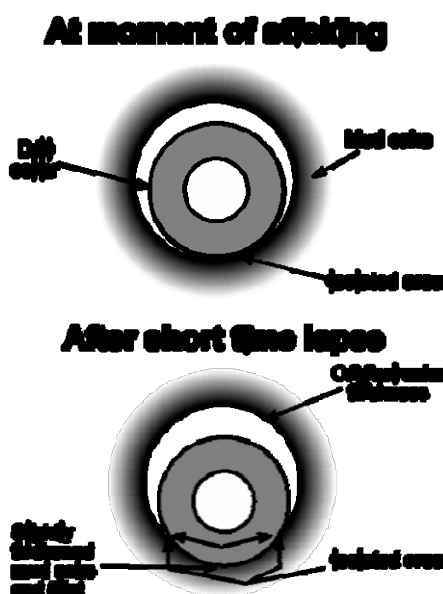


Figure 1 Differential-pressure effect

The difference in pressure between the hydrostatic head pressure and the formation pore pressure forces the drillpipe into the wallcake and sticks the pipe.

To avoid or minimize the risk of differential sticking, follow these guidelines:

- Drill with the lowest practical mud weight.
- Maintain a low filtration rate.
- Keep low-gravity solids to a minimum.
- Never allow the drillpipe to remain motionless for any period of time.

- Ream any undergauge section.
- Add appropriate bridging agents.
- Change to an oil/synthetic-based mud

1.2. Spotting Fluids

When differential sticking occurs, spotting fluid can sometimes free the drillpipe. It is critical to have a spotting fluid readily available and apply it within six hours of the stuck pipe occurrence

Spotting fluids are designed to penetrate and break up the filter cake.

EZ-SPOT Spotting Fluid

EZ-SPOT is a good all-purpose spotting fluid suitable for use in many different drilling regions. To mix the EZ-SPOT spotting fluid, start with the required volume of oil and add EZ-SPOT, water, and barite in that order. Base fluids can be diesel, mineral oil, water, etc.

Table 1 EZ-SPOT Formulation.

EZ-SPOT [^] spotting fluid formulation for 100 bbl						
Weight, lb/gal (sg)	7.3 (0.87)	10.0 (1.20)	12.0 (1.44)	14.0 (1.68)	16.0 (1.92)	18.0 (2.16)
Oil, bbl (m ³)	64 (10.3)	58 (9.2)	54 (8.6)	49 (7.8)	51 (8.1)	44 (7.0)
EZ-SPOT, 55 gal drum	6 (.98)	6 (.98)	6 (.98)	6 (.98)	6 (.98)	6 (.98)
Water, bbl (m ³)	28 (4.5)	26 (4.1)	22 (3.5)	21 (3.3)	11 (1.7)	10 (1.6)
BAROID, lb (kg)	n/a	14,000 (6,350)	25,000 (11,340)	35,000 (15,876)	46,500 (21,092)	57,000 (25,855)

*Not registered for use in the United Kingdom or Norway.

North Sea Spotting Fluids

Due to stringent environmental and discharge regulations across the various North Sea sectors, alternative spotting fluid products and procedures are used.

Typically, North Sea spotting pills are formulated with drill water or calcium chloride brine, containing concentrations of acid (acetic or citric) and mutual solvent (EGMBE).

North Sea stuck pipe pill formulations and procedures are provided locally, in accordance with specific country guidelines. These should be provided by the supervising Technical Professional during the well planning phase.

QUIK-FREE Pipe-Freeing Agent

For unweighted pills, add QUIK-FREE in slugging pit for and mix for 15 minutes.

For weighted pills, add 2 bags of GELTONE® V viscosifier per 26 bbl of QUIK-FREE and then add weighting material to the desired weight.

Displacement

1. Place the QUIK-FREE pill across the stuck zone.
2. Pump QUIK-FREE into the drill string at 1 bbl/hr (0.16 m³/hr).

Soak Time

1. QUIK-FREE should have 4-6 hr soak/exposure time.
2. After pipe is free, incorporate the pill into the system which can improve lubricity.

For North Sea, use QUIK-FREE NS or alternative North Sea spotting fluid formulation and procedures provided by the Technical Professional assigned to the project.

1.3. Determining Depth to Stuck Zone

Measure the drillstring stretch to estimate the depth that pipe is stuck. The following formula locates the depth at which the pipe is stuck. The length of free pipe is based on the drillstring dimensions and the measured amount of stretch.

$$L = \frac{E e W}{40.8 P}$$

Where

L is the length of free pipe (ft)

E is the modulus of elasticity (30×10^6) (psi)

e is the stretch (in)

W is the weight of pipe (per ft)

P is the amount of tension applied (lb/ft)

1.4. Packing Off

Drilling-fluid systems with poor suspension characteristics exhibit strong packing-off tendencies. Also, if cuttings beds are present and junk slot area is not sufficient to allow cuttings to pass through, pack off will occur (Figure 2).

2). Factors that can lead to pack off of the drillstring include the following:

- Formation collapse
- Shale hydration
- Poor hole cleaning
- Poor BHA design (junk slot area)

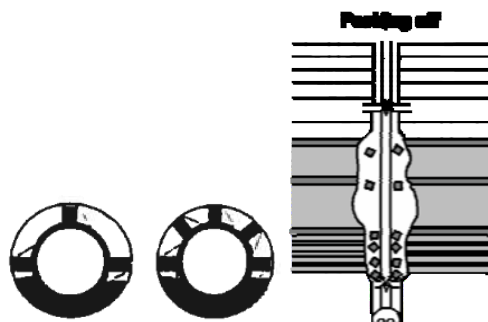


Figure 2 Packing off

1.5. Undergauge Hole

Undergauge hole is a condition where the borehole is smaller than the bit diameter used to drill the section. Undergauge hole can result from any of the following causes:

- Plastic flowing formations
- Wall-cake buildup in a permeable formation
- Swelling shales

Plastic Flowing Formations

A plastic flowing formation is a formation that is plastic (easily deformable when stressed) and can flow into the borehole. When these types of formations are penetrated by the bit, the hole is at gauge. However, when the hydrostatic pressure exerted by the column of drilling fluid is less than the hydrostatic pressure of the formation, underbalance results, the formation flows, and hole diameter decreases.

Undergauge hole is a common problem when drilling a thick salt section with an oil mud. The salt can flow into the borehole and make the section undergauge. When plastic salt formations exist, they are usually below 5,000 feet. Spotting fresh water is the best way to free the pipe from a plastic salt formation.

Wall-Cake Buildup

Wall-cake buildup occurs when the drilling fluid has poor filtration control across a permeable zone. Excessive wall-cake buildup can also be caused by:

- High percentage of low-gravity solids
- Keyseating

Keyseating is a situation frequently encountered in deviated or crooked holes when the drillpipe wears into the wall. The normal drilling rotation of the drillstring cuts into the formation wall in deviated areas where the drillpipe tension creates pressure against the sides of the hole.

Keyseating is diagnosed when the drillpipe can be reciprocated within the range of tool joint distances or until collar reaches the keyseat, while pipe rotation and circulation remain normal. See Figure 13-3 for an example of a keyseat effect in a crooked hole.

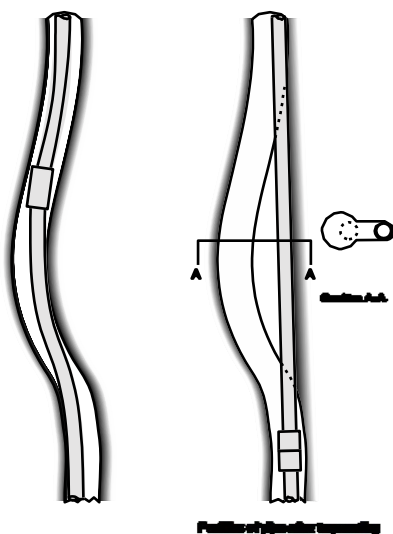


Figure 3 Keyseating

The friction generated by drillpipe rotation against the borewall cuts a narrow channel, or keyseat, into the formation.

A preventive measure is to carefully control upper hole deviation and dogleg severity throughout the well path. This action will eliminate the force that leads to keyseat creation.

Once a keyseat is formed, the best solution is to ream out the small-diameter portions of the hole with reaming tools (Figure 4). This action will solve the immediate stuck-pipe problem, but the keyseat can be formed again unless preventive steps are taken.

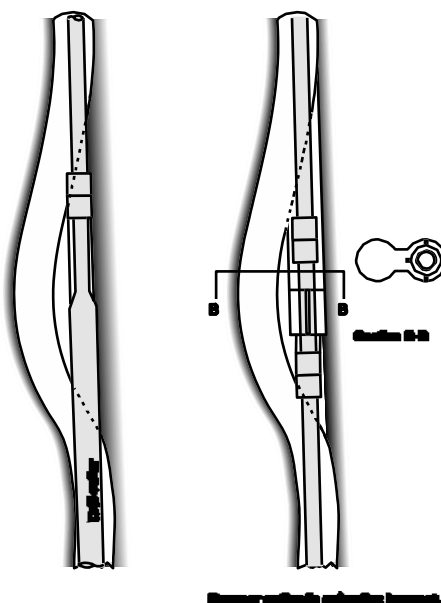


Figure 4 Reaming the keyseat

1.6. Freeing Stuck Pipe

The following guidelines can be used to free stuck pipe:

Table 2 Guidelines for Freeing Stuck Pipe

Cause	Procedure
Differential sticking	Reduce mud weight. Spot lighter fluid in the stuck zone Use spotting fluid.
Packing off	Do not jar up on the string Attempt to go down one stand and try and clean up the hole with rotation and flow. Back off and wash over.
Undergauge hole	Increase mud weight. Underream.
Keyseating	Ream the keyseat.

Well Control

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1. Well Control Basics

1.1. Overview

This section explains kicks, warning signs, and kick control. Shut-in procedures and common kill methods are explained, and the steps to accomplish each one are provided. Common kill problems are identified, and the solutions for these problems are given.

1.2. Kicks

A kick is an influx of formation fluids into the wellbore. Some of the conditions that can induce a kick are:

- Drilling into an abnormally pressured formation
- Failure to keep the hole full during trips
- Insufficient mud weight
- Lost circulation
- Swab/surge pressures
- Warning signs of a kick include:
 - Drilling breaks
 - Increase in pit volume
 - Increase in mud-return flow rate
 - Flow with the mud pumps off
 - Pump-pressure decrease and stroke-rate increase
 - Drilling reversal
 - Hole not taking proper fluid volume during a trip

Controlling a Kick

Follow this procedure to control a kick:

1. Pull off bottom.
2. Shut off pumps.
3. Check for flow.
4. Shut-in the well.
5. Record pressures.
6. Kill the well.
7. Verify that the well is dead.

Note: The best indication that a well has been killed is when the choke is open 100 percent and there is no flow.

1.3. Shut-in Procedures

A shut-in procedure can either be soft or hard. When conducting a soft shut-in the choke is partially to fully open when the annular preventer is closed. When conducting a hard shut-in, the choke is fully closed when the annular preventer is closed.

Kill Methods

Two basic techniques used to kill a well are:

- Wait-and-weight method
- Driller's method

Wait-and-Weight Method

The most widely used kill method is wait-and-weight. In this method, the well is shut-in, and the surface system is weighted-up to the required kill weight. The weighted mud is pumped into the well, and the kick is killed in one complete circulation. This method is also called the engineer's or one-circulation method.

Driller's Method

In the second kill method, the influx is pumped out of the wellbore after recording the shut-in pressures and pit volume increase, but before weighting-up the drilling fluid. Once the influx has been pumped out of the well, the well is shut-in and the surface mud system is weighted up to the required kill weight. This procedure is also called the two-circulation method.

1.4. Example Kill Sheet

The top half of a kill sheet is a work sheet necessary for kill calculations.

PRERECORDED INFORMATION							
Original Mud Weight (OMW), ppg		Surface to Bit	bbl	stk			
True Vertical Depth, ft		Bit to Surface	bbl	stk			
Measured Depth, ft		Totals	bbl	stk			
Pump 1, bbl/stk							
KRS	spm	KRP	psi	KRS	spm	KRP	psi
Pump 2, bbl/stk							
KRS	spm	KRP	psi	KRS	spm	KRP	psi
Shoe Test							
ppg	Depth	ft	MACP	psi	Bit to Shoe		
RECORDED INFORMATION							
SIDPP	psi	SICP	psi	PIT GAIN	bbl		
KILL CALCULATIONS							
Kill Weight Mud (KWM)	$= (SIDPP / 0.052 / TVD) + OMW$						
Initial Circulating Pressure (ICP)	$= SIDPP + KRP$						
Final Circulating Pressure	$= KRP \times KWM / OMW$						

Where:

- OMW* is the original mud weight (ppg)
- TVD* is the true vertical depth (ft)
- MD* is the measured depth (ft)
- stk* is strokes
- spm* is strokes per minute
- KRS* is the kill rate speed (spm)
- KRP* is kill rate pressure (psi)
- MACP* is the maximum allowable casing pressure (psi)
- SIDPP* is the shut in drillpipe pressure (psi)
- SICP* is the shut in casing pressure (psi)
- KWM* is kill weight mud (ppg)

See Example Wait and Weight Kill Sheet at the end of this chapter.

1.5. Kick Control Problems

While controlling a kick, some of the problems that can occur include:

- Lost circulation
- Plugged jets
- Washed-out choke
- Plugged choke
- Washout in the drillstring
- Gas migration
- Off-bottom bit
- Gas hydrate formation

These problems can result from the increased pressure and heavy kill-weight mud. In some cases, more than one problem can occur. The table below can help determine the source of problems during a kill procedure.

Table 1 Kill procedure problem indicators.

	Indication		
Situation	Drillpipe pressure	Casing pressure	Pump rate
Total circulation loss	Major decrease	Major decrease	Increase
Partial circulation loss	Major decrease	Decrease	Increase
Choke plugging	Major increase	Major increase	Decrease
Jet plugging	Major increase	No change	Decrease
Choke washout	Major decrease	Major decrease	Increase
Drillstring washout	Major decrease	No change	Increase

Solutions for dealing with a lost circulation problem are detailed in Chapter 10 - **Lost Circulation**.

[illegible]

SUBSEA STACK CALCULATIONS

CHOKE LINE VOLUME

$$\boxed{} \times \boxed{} = \boxed{}$$

Choke Line Capacity (bbls/ft) Length of Choke Line (ft) Volume in Choke Line (Bbls) [Refer to Line F, Block "E" on Front]

If both choke and kill lines are being used, multiply barrels in choke line by two.

If ID or length of kill or choke line is different, use choke line formula and calculate for kill line.

ANNULAR VOLUME BETWEEN DRILL PIPE AND RISER

$$\begin{array}{|c|} \hline \boxed{} \\ \hline \end{array} - \begin{array}{|c|} \hline \boxed{} \\ \hline \end{array} - \begin{array}{|c|} \hline \boxed{} \\ \hline \end{array} = \begin{array}{|c|} \hline \boxed{} \\ \hline \end{array} \times \begin{array}{|c|} \hline \boxed{} \\ \hline \end{array} = \begin{array}{|c|} \hline \boxed{} \\ \hline \end{array}$$

Riser Capacity (bbls/ft)	Drillpipe Capacity (bbls/ft)	Drillpipe Metal Displacement (bbls/ft)	Capacity Between DP & Riser (bbls/ft)	Length of Riser (RKB to Sea Floor)/MD-ft	Volume Between DP & Riser (bbls)
-----------------------------	---------------------------------	--	--	--	--

STROKES TO DISPLACE RISER

$$\frac{\text{Volume Between DP \& Riser (bbbls)}}{\text{Pump Output bbbls/stk}} = \text{Strokes to Displace Riser}$$

Gas in BOP stack must be swept and mud in riser must be weighted to the same weight as the annulus before the stack is opened to avoid a possible reduction in bottom hole pressure.

CHOKER LINE FRICTION PRESSURE (CLFP)

CLFP = (Circulating Pressure through Choke and / or Kill line) - (Circulating through open BOP) @ _____ PPG

	Kill Rate Speed (spm)	Pump Output (bbls/stk)	Kill Rate (bbls/min)	KRP BOP Open (psi)	KRP thru Choke Line (psi)	KRP thru Choke & Kill Line (psi)	CLFP	
							Choke Line (psi)	Choke & Kill Line (psi)
Pump #1								
Pump #2								

CORRECTED CHOKE LINE FRICTION PRESSURE (IF MUD WEIGHT CHANGES)

$$\frac{\text{CLFP (psi)}}{\text{Different Mud Weight (ppg)}} \div \frac{\text{Mud Weight When C/L Pressure Measured (ppg)}}{\text{Corrected CLFP (psi)}}$$

TOTAL VOLUME IN ACTIVE SYSTEM

$$\boxed{} + \boxed{} + \boxed{} + \boxed{} = \boxed{}$$

Vol. in Active Pits (bbls) Drill String Vol. (bbls) Total Annular Vol. (bbls) Vol. Between DP & Riser (bbls) Subsea only Total Vol. in Active Sys. (bbls)

SACKS PER 100 BARRELS

$$\boxed{35} - \boxed{} = \boxed{}$$

Weight of 1 Gallon of Barite in lbs. Kill Mud Weight (ppg)

$$\boxed{} - \boxed{} \times \boxed{14.7} \div \boxed{} = \boxed{}$$

Kill Mud Weight Present Mud Weight Sxs Per bbl Sxs Per bbl

TOTAL BARITE REQUIRED

$$\boxed{\text{Total Vol. in Active system (bbls)}} \times \boxed{\text{Sacks Per bbl.}} = \boxed{\text{Total Barite Required (sxs)}}$$

VOLUME INCREASE DUE TO BARITE ADDITION

$$\boxed{} \div \boxed{14.7} = \boxed{}$$

Total Barite Sxs Per bbl Volume
Required (sxs) Increase (bbls)

REQUIRED MIXING RATE

$$\boxed{} \times \boxed{} = \boxed{}$$

Sxs Per bbl Circulating Rate Required Mixing Rate
(bbls/min) (sxs/min)

DILUTION OF RESERVE MUD WITH WATER

$$\boxed{} - \boxed{8.33} = \boxed{}$$

Kill Mud Weight (ppg) Weight of 1 Gallon of Fresh Water in lbs.

$$\frac{\text{Mud Weight in Reserve Pit (ppg)} - \text{Kill Mud Weight (ppg)} \times \text{Volume in Reserve Pit (bbls)}}{\text{Volume of Water to Add (bbls)}} = \text{Volume of Water to Add (bbls)}$$

SUBSEA STACK CALCULATIONS

WELL CONDITIONS vs. CLFP

CLFP is always present while circulating (Dynamic Conditions) through the choke line. When bringing the pump on line, if CLFP is not considered, loss of circulation or formation breakdown may occur.

STATIC

SICP (psi)

MAASP (psi)

DYNAMIC

Estimated Csg.
Press. at kill rate
speed.

New MAASP (psi)

PUMP START-UP (CHOKE PRESSURE SCHEDULE)

$$\frac{\text{CLFP}}{4} = \text{Drop Per Step (psi)}$$

		Shut-in	0	stks/min
	psi	SICP		
		.5 X Kill Rate Speed		stks/min
	psi			
		.75 X Kill Rate Speed		stks/min
	psi			
		.875 X Kill Rate Speed		stks/min
	psi			
		Kill Rate Speed		stks/min
	psi			

A 20x20 grid for drawing strokes. The vertical axis is labeled 'PRESSURE' and the horizontal axis is labeled 'STROKES'.

Corrosion

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1. Corrosion

Corrosion is the destruction of metal through electrochemical action between metal and its environment. Corrosion can be costly in terms of damage to pipe and well parts and can even result in the loss of an entire well. About 75 to 85 percent of drill pipe loss can be attributed to corrosion. Other areas affected by corrosion include pump parts, bits, and casing.

Factors Affecting Corrosion

Factors affecting corrosion include:

- **Temperature:** Generally, corrosion rates double with every 55°F (31°C) increase in temperature.
- **Velocity:** The higher the mud velocity, the higher the rate of corrosion due to film erosion (oxide, oil, amine, etc.).
- **Solids:** Abrasive solids remove protective films and cause increased corrosive attack.
- **Metallurgical factors:** Mill scale and heat treatment of pipe can cause localized corrosion.
- **Corrosive agents:** Corrosive agents such as oxygen, carbon dioxide, and hydrogen sulfide can increase corrosion and lead to pipe failure.

The categories of corrosion range from uniform corrosion to mechanical damage.

Table 1 Categories of Corrosion

Category	Explanation
Uniform corrosion	Even corrosion pattern over surfaces
Localized corrosion	Mesa-like corrosion pattern over surfaces
Pitting	Highly localized corrosion that results in the deep penetration of surfaces

1.1. Drilling Fluid Corrosive Agents

Corrosive agents found in drilling fluids include the following:

- Oxygen
- Hydrogen sulfide
- Carbon dioxide
- Bacteria
- Dissolved salts
- Mineral scale

Oxygen

Oxygen causes a major portion of corrosion damage to drilling equipment. Oxygen acts by removing protective films; this action causes accelerated corrosion and increased pitting under deposits.

Primary Sources of Oxygen

- Water additions
- Actions of mixing and solids-control equipment
- Aerated drilling fluids

- The atmosphere

Water Additions

Water added to drilling mud during normal drilling operations can contain dissolved oxygen. Very small concentrations of oxygen (<1 ppm) can cause severe corrosion by setting up differential aeration cells that can show preferential attack with pitting under barriers or deposits. The primary corrosion by-product of low oxygen concentrations is magnetite.

The products recommended for the removal of dissolved oxygen are:

- BARASCAV L
- BARASCAV D

Actions of Mixing and Solids Control Equipment

Mixing and solids-control equipment cause aeration of the drilling fluid during drilling operations.

For example, aeration occurs as mud falls through the shaker screen or when hopper or mud guns are discharged above the surface of the mud in the pits. To reduce the amount of oxygen introduced into drilling fluid by mixing and solids control equipment, follow these guidelines:

- Use a premix tank to mix mud, when possible.
- Operate mud-mixing pumps, especially the hopper, only when mixing mud.
- Keep the packing tight on centrifugal pumps.
- Ensure the mud in the suction pit is deep enough to keep the mud pump from pulling in air.
- Keep discharge below the mud surface when moving mud from the reserve pit.
- Ensure guns discharge below the mud surface; do not allow the mud-stirring device to create a vortex.
- Ensure the degasser and desander discharges are below the mud surface.

The products recommended for treating drilling fluid containing oxygen because of mixing and solids-control equipment are:

- BARASCAV L
- BARASCAV D

Aerated Drilling Fluids

While conventional drilling fluids require the removal of oxygen, aerated drilling (foam and mist drilling) fluids require the use of passivating (oxidizing) inhibitors to combat corrosion due to oxygen.

The product recommended for inhibiting oxygen in aerated drilling fluids is BARACOR 700.

The Atmosphere

The atmosphere is another source of oxygen, and hence corrosion. The main by-product of atmospheric corrosion is iron oxide rust. To prevent atmospheric corrosion, wash the pipe free of all salts and mud products and then spray or dip the pipe in an atmospheric corrosion inhibitor.

The product recommended for inhibiting atmospheric corrosion is BARAFILM.

Hydrogen Sulfide

Hydrogen sulfide can enter the mud system from the following sources:

- Formation fluids containing hydrogen sulfide

- Bacterial action on sulfur-containing compounds in drilling mud
- Thermal degradation of sulfur-containing drilling-fluid additives
- Chemical reactions with tool-joint thread lubricants containing sulfur

Hydrogen sulfide is soluble in water. Dissolved hydrogen sulfide behaves as a weak acid and causes pitting. Hydrogen ions at the cathodic areas may enter the steel instead of evolving from the surface as a gas. This process can result in hydrogen blistering in low-strength steels or hydrogen embrittlement in high-strength steels. Both the hydrogen and sulfide components of hydrogen sulfide can contribute to drillstring failures.

Hydrogen sulfide corrosion is mitigated by increasing the pH to above 9.5 and by using sulfide scavengers and film-forming inhibitors.

The products recommended for combating corrosion due to hydrogen sulfide are:

- BARACOR 700
- NO-SULF (zinc carbonate)

Hydrogen sulfide and carbon dioxide are often encountered in the same geologic formation; therefore, treatments should be designed to deal with both contaminants simultaneously. Ensure that well hydrostatic pressures are sufficient to prevent further influxes of gases.

Carbon Dioxide

Carbon dioxide is found in natural gas in varying quantities. When combined with water, carbon dioxide forms carbonic acid and decreases the water's pH, which increases the water's corrosivity. While carbon dioxide is not as corrosive as oxygen, it can cause pitting.

Maintaining the correct pH is the primary treatment for carbon dioxide contamination. Either lime or caustic soda can be used to maintain pH. The use of BARACOR 95 can also be beneficial for carbon dioxide treatment.

Treatment with caustic soda produces sodium carbonate, which is soluble and can create mud problems. Treatment with lime, on the other hand, produces an insoluble calcium carbonate precipitate and water.

Treatment with BARACOR 95 has few downsides.

Bacteria

Microorganisms can cause fermentation of organic mud additives, changing viscosity and lowering pH. A sour odor and gas are other indicators that bacteria are present. Degradation of mud additives can result in increased maintenance costs.

The by-products of bacteria are carbon dioxide and hydrogen sulfide. The presence of aerobic bacteria is determined by the phenol-red serum test. The presence of anaerobic bacteria is determined by the marine anaerobic serum test. Microbiocides are used to control bacteria in drilling environments. The products recommended for controlling bacteria are:

- ALDACIDE G
- Isothiazolone-based biocide powder

Dissolved Salts

Dissolved salts increase corrosion by decreasing the electrical resistance of drilling fluids and increasing the solubility of corrosion by-products. Some of these by-products can cause a scale or film to form on the surface of the metal. The products recommended for combating the effects of dissolved salts are:

- BARACOR 700

Mineral Scale

Mineral scale deposits set up conditions for local corrosion-cell activity. The continuous addition of a scale inhibitor can control the formation of scale deposits.

1.2. Packer Fluid Treatments

When using a drilling fluid as a packer fluid, the drilling fluid must be conditioned to minimize corrosion under long-term, static conditions. Recommended treatments for various packer fluid systems are shown below.

Table 2 Packer Fluid Treatment Recommendations

Packer-Fluid Systems	Recommended Treatment
Water-based mud	Increase pH to between 9.5-11.5. Add 2 to 4 lb/bbl (6-11 kg/m ³) NO-SULF to control hydrogen sulfide. Add a biocide to control bacteria. Add 0.25 to 1.4 lb/bbl BARACOR 95 to control carbon dioxide. Add 0.1 to 0.3 lb/bbl OXYGON to control dissolved oxygen.
Clear fresh water Clear salt water	Add BARACOR 100 (0.5-1% by volume). Add 0.25 to 1.4 lb/bbl BARACOR 95
Oil-based mud (diesel, mineral)	Add 2 to 10 lb/bbl (6-29 kg/m ³) primary emulsifier and 2 to 10 lb/bbl (6-29 kg/m ³) GELTONE II/V. Add 4 to 6 lb/bbl (11-17 kg/m ³) lime.
Heavy brine (CaCl ₂ , CaBr ₂ , ZnBr ₂ , or blends of the three)	Add BARACOR 100, 0.5-2% by volume or BARACOR 450, 0.2-0.4% by weight.

1.3. Completion/ Workover Fluids

The corrosivity of a given completion or workover fluid depends on its brine type. Brines fall into two categories: monovalent and divalent.

Monovalent Brines

Monovalent brines contain salts that have monovalent cations; these salts include sodium chloride, potassium chloride, potassium bromide, sodium bromide, sodium formate, and potassium formate. Potassium bromide and sodium bromide are especially effective in calcium-sensitive formations and in formations where carbon dioxide gas might react with calcium brines to create a calcium-carbonate precipitate.

Monovalent brines generally show low corrosivity, even at temperatures exceeding 400°F (204°C).

Divalent Brines

Divalent brines contain salts that have divalent cations; these salts include calcium chloride, calcium bromide, and zinc bromide. A divalent brine might consist of a single salt or a blend of salts, depending on the required brine density and crystallization point.

The corrosivity of these brines depends on their density and chemical composition. Laboratory data show that the addition of calcium chloride lowers the rate of corrosion, while the addition of zinc bromide rapidly increases the rate of corrosion.

Corrosive Agents

When working with completion or workover fluids, the two corrosive agents to monitor are oxygen and hydrogen sulfide.

Oxygen

The oxygen content of fluids is difficult to determine, and most engineers in the field do not have access to the proper equipment. Because the dissolved oxygen content varies as conditions change during the day, it is difficult to select a set feed rate of oxygen scavenger to remove a known concentration of oxygen.

Laboratory tests show that the oxygen content of calcium chloride, calcium bromide, and zinc bromide brines is very low. The solubility of gases in a liquid is directly related to the total dissolved-solids concentration of that liquid. The higher the dissolved-solids content, the lower the solubility of gases in the liquid.

In a well at elevated temperatures, the oxygen content should be much lower.

The table below lists oxygen concentrations measured in stock brine at room temperature.

Table 3 Oxygen Concentrations Measured In Stock Brine At Room Temperature

Brine density, lb/gal (sg)	Oxygen concentration, ppm
11.6 (1.39) CaCl ₂	0.1-0.2
14.2 (1.70) CaBr ₂	0.05-0.1
19.2 (2.30) CaBr ₂ / ZnBr ₂	0.4-0.6

Some products used as oxygen scavengers contain sulfites that react with the dissolved oxygen in fluids to form sulfates, eliminating the corrosive effects of the dissolved oxygen. Calcium brines should not be treated with oxygen scavengers containing sulfides because chemicals could precipitate calcium scale and cause problems. In a packer-fluid application where there is a static system with no aeration of the fluid, the dissolved oxygen content is so low that an oxygen scavenger usually is not required.

Hydrogen Sulfide

In solids-enhanced systems, the most often used hydrogen-sulfide scavenger is zinc carbonate. The zinc reacts with the soluble sulfide ions to form zinc sulfide, which is insoluble and precipitates as an unreactive compound. In solids-free systems, soluble zinc bromide salt serves the same function and absorbs the hydrogen sulfide. In operations where hydrogen-sulfide contamination is expected, offset the hydrogen sulfide's acidic nature by maintaining a proper pH in the brine, as outlined the table below.

Table 4 Proper Brine pH

Brine	Recommended pH	Treatment
Non-zinc	7.0	Caustic soda or lime
Calcium	7.0-10.5	Caustic soda or lime
Zinc	3.0-5.0	Lime

Corrosion Inhibitors

A corrosion inhibitor is a chemical product that substantially reduces metallic loss when it is added in small concentrations to a corrosive environment. Chemicals used as corrosion inhibitors include both inorganic and organic compounds.

The products recommended for treating corrosive agents in completion and workover fluids are:

- BARACOR 100
- BARACOR 450

1.4. Corrosion Test

The best and most direct method for testing for the presence of corrosion is the use of a drillstring coupon. A drillstring coupon is a ring made from a section of tubing. The coupon, which has a smooth surface, is placed at a predetermined depth during a round trip. Later, it is removed and inspected. The coupon is weighed both before and after downhole exposure. A high metal loss after exposure indicates corrosion is taking place. The coupon surface is another indicator of corrosion. When there is evidence of pitting on the coupon, pitting is also most likely occurring on the drill pipe.

Coupons in a wide range of drill pipe sizes can be ordered from the FANN Instrument Company.

Table 5 Corrosion Coupons

Drill Pipe Size and Type	Recommended Coupon
2 7/8-in internal flush and 3 1/2-in slim hole	No. 636-18 2 1/2-in OD x 0.250-in wall
3 1/2-in extra hole and 3 1/2-in full hole	No. 636-19 2 3/4-in OD x 0.188-in wall
3 1/2-in internal flush and 3 1/2-in extra hole	No. 636-20 3-in OD x 0.313-in wall
4-in full hole	No. 636-21 3 1/4-in OD x 0.250-in wall
4-in internal flush and 4 1/2-in extra hole	No. 636-23 3 1/4-in OD x 0.3125-in wall
4 1/2-in full hole and 4 1/2-in extra hole and 4-in internal flush	No. 636-24 3 %-in OD x 0.375-in wall
4 1/2-in internal flush and 5-in extra hole	No. 636-25 4 1/2-in OD x 0.1325-in wall
5 9/16-in or 5 1/2-in API regular or full hole and 6 %-in API regular	No. 636-26 4 %-in OD x 0.500-in wall

Drill Pipe Size and Type	Recommended Coupon
4 ½-in extra hole	No. 636-29 3 13/16-in OD x 0.200-in wall
5-in x H tool joint	No. 636-31 4 3/16-in OD x 0.2185-in wall

Plastic coated corrosion coupons are available.

Coupons are weighed to 0.1 milligram and the weight and ring number are permanently recorded at the FANN Instrument Company. For shipment, the rings are placed in a plastic bag containing an inert desiccant, such as silica gel, and are sealed in a sturdy envelope. The coupon's size, number, and weight are recorded on the envelope.

Handling Coupons

Follow these steps when handling coupons in the field:

1. Remove the coupon from its package and place the coupon in the tool joint box.

Handle the coupon carefully to prevent damage to the coupon.

2. Save the envelope and plastic bag for shipping the coupon to the laboratory.
3. Make up the joint.
4. Leave the coupon in the pipe string for the desired number of bit runs (usually 50 hours).

A visual inspection of the coupon, or previously determined corrosion rates, determine the actual length of exposure.

5. Remove the coupon, wipe it dry, and smear it with grease or heavy oil.
6. Pack the coupon in the plastic bag and envelope along with a copy of the mud report.
7. Ensure the following information appears on the envelope: Mud properties, such as salt content, pH value, inhibitor treatments in effect.
8. Ship the coupon to the laboratory by the fastest means possible.

Test Results

At the laboratory, the coupon is cleaned and weighed, and the corrosion rate is determined. Corrosion rates are reported as weight loss in pounds per square foot per year according to the following formula:

$$\text{Weight loss, lb/sqft/yr} = \frac{\text{Weight loss grams} \times \text{ring factor}}{\text{Exposure time, hours}}$$

Uniform corrosion rates below 2.0 lb/sq ft/yr are considered acceptable.

1.5. Troubleshooting

Table 6 Corrosion Troubleshooting Chart

Oxygen from water additions	
Source: Water additions	By-product: Oxides of iron
Indication: Concentration cell pitting under barrier or deposits and pits filled with black magnetic corrosion by-products	Tests: Black to red dust Some by-product insoluble in 15% HCl Some by-product attracted to magnet
Treatment: - Treat with an oxygen scavenger having a range equivalent of 2.5 to 10 lb/hr of sodium sulfite. - Maintain 20 to 300 mg/L sulfite residual.	
Oxygen from mixing and solids control equipment	
Source: Mixing and solids-control equipment	By-product: Oxides of iron
Indication: Concentration cell pitting under barrier or deposits and pits filled with black magnetic corrosion by-products	Tests: Black to red dust Some by-product insoluble in 15% HCl Some by-product attracted to magnet
Treatment: - Coat pipe with film-forming inhibitors to reduce atmospheric attack and cover concentration cell deposits. - Reduce air entrapment in pits. - Defoam drilling fluid.	
Oxygen from aerated drilling fluids	
Source: Aerated drilling fluids	By-product: Oxides of iron
Indication: Severe pitting	Tests: Black to red dust Some by-product insoluble in 15% HCl Some by-product attracted to magnet
Treatment: - Maintain a high pH and keep drillpipe free of mineral scale deposits with scale inhibitor. - Coat pipe with filming inhibitors.	
Oxygen from the atmosphere	
Source: Atmosphere	By-product: Oxides of iron
Indication: Generalized to localized corrosion	Tests: Black to red dust Some by-product insoluble in 15% HCl Some by-product attracted to magnet
Treatment: Wash equipment free of salts and mud products.	

Spray equipment with atmospheric filming inhibitors.	
Hydrogen sulfide	
Source: - Formation - Thermally degraded mud products	By-product: Iron sulfide
Indications: - Localized to sharp pitting - Dark blue-to-black film on equipment - Sulfide stress corrosion cracking (SSCC) failures - Rotten-egg odor	Tests: - Acid arsenic solution produces a bright yellow precipitate, soluble in 15% HCl - Lead acetate test
Treatment: - Maintain high pH with caustic soda. - For 0-100 ppm sulfide, add 3-5 lb/bbl (9-14 kg/m³) iron oxide and/or 0.1-0.5 lb/bbl (0.3-1.4 kg/m³) zinc carbonate/zinc oxide to remove sulfide ions. <i>The combined treatments of iron oxide and zinc compounds should provide lower sulfide ion contamination in most drilling fluids.</i>	
Carbon dioxide	
Source: - Formation - Thermally degraded mud products	By-product: Iron carbonate
Indications: - Localized corrosion to pitting - Dark brown-to-black film	Test: Slow effervescence in 15% HCl
Treatment: Maintain basic pH with caustic soda or lime to neutralize the acid-forming gas.	
Bacteria	
Source: Bacteria	By-product: Carbon dioxide; hydrogen sulfide
Indications: - Fermentation of organic mud additives - Change in viscosity - Lower pH - Sour odor - Gas formation	Tests: - Phenol-red serum test (aerobic bacteria) - Marine anaerobic serum test (anaerobic bacteria)
Treatment: Add biocides.	
Dissolved salts	
Source: Dissolved salts	By-product: Oxides of iron
Indications: - Localized corrosion - Pitting	Test: Black to red rust
Treatment: Add film-forming inhibitors.	
Mineral scale deposits	
Source: Formation and mud materials	By-product: Iron products beneath mineral deposit
Indication: Corrosion cell pits beneath deposit	Test: White mineral scale: calcium, barium and/or magnesium compounds

Treatment:

- Slowly and continuously, add scale inhibitor at 5-15 mg/L.
- Reduce treatments of scale inhibitor when phosphate residual exceeds 15 mg/L.
- Use 1 gal/1,000 bbl (0.25 L/m³) mud/day for maintenance treatment under normal drilling conditions.

1.6. Product Information

Table 7 Product Information

Product	Function	Description	Treatment
ALDACIDE G	Microbiocide	Glutaraldehyde solution	0.2-0.5 lb/bbl (0.6-1.4 kg/m ³)
BARACOR 95	Corrosion inhibitor	Low toxicity amine-based	0.25-1.4 lb/bbl (0.7 to 3.9 kg/m ³)
BARACOR 100	Corrosion inhibitor	Film-forming amine	Clear fresh water or salt water: 21-42 gal/100 bbl fluid. Heavy brine: 0.5-2.0% by volume (5-10 L/m ³).
BARACOR 450	Corrosion inhibitor	Cyanogen-based inorganic compound	0.2-0.4% by weight
BARACOR 700	Corrosion inhibitor	Blend of phosphonates and alkyl phosphates	0.5-1.5 lb/bbl (1.4-4 kg/m ³)
BARASCAV D	Oxygen scavenger Thermal extender for polymers	Powdered sodium sulfite	0.5-1 lb/gal of fresh water (1.4-2.9 kg/m ³)
BARASCAV L	Oxygen scavenger Thermal extender for polymers	Liquid ammonium bisulfite	Initially 0.1-0.5 lb/bbl (0.3-1.4 kg/m ³)
OXYGON	Oxygen scavenger Thermal extender for polymers	Erythrosorbate based	0.1-0.3 lb/bbl (0.3-0.9 kg/m ³)
NO-SULF	Hydrogen sulfide scavenger	Blend of zinc compounds	Pretreat with 0.1-5 lb/bbl (0.3-14 kg/m ³)

Foam and Aerated Fluids

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1. Foam and Aerated Fluids

In situations where normal drilling fluids are not appropriate, air, foam, and aerated muds are effective alternatives. These fluids can be used when drilling the following:

- Extremely porous formations
- Subnormally pressured formations
- Cavernous formations
- Large diameter boreholes

Table 1 Air, Foam, and Aerated Mud Drilling Fluids Applications

Drilling Fluid	Description	Application
Air / Gas	Air / Gas is the continuous phase. Large volumes of air / gas are required.	Extremely low formation pressure Competent formations No water-bearing formation exposed
Foam	A mixture of water and/or bentonite or polymer slurry and foaming agents is added to compressed air.	Larger annular spaces than air drilling Water-bearing formations exposed Stable and unstable formations can be drilled
Aerated mud	Drilling fluid is the continuous phase. Air is added to reduce the hydrostatic pressure.	Weak formations Unstable formations with subnormal pressures (6 to 8 lb/gal EMW) (0.72-0.96 sg)

Air drilling techniques use air velocity and air volume to drill formations that present major problems for drilling fluids.

Foam is a combination of water or polymers / bentonite slurry mixed with a foaming agent; air from a compressor combines with the foaming agent to form bubbles that act as carrying agents for cuttings removal. Modified foam systems, water / polymer / bentonite and foaming agent, increase carrying ability and increase borehole stability.

Aerated mud can be virtually any water-based mud to which air is added. This type of mud has less hydrostatic pressure and less tendency to fracture weak formations. Foam systems and aerated muds are useful in situations where air drilling is not possible and drilling fluids are not efficient.

1.1. Air Drilling

Air drilling uses compressed gas for hole cleaning. Air is the most commonly used gas, but natural gas and other gases can also be used.

Problems that can be encountered with gas drilling include the following:

- Regulation of gas pressure
- Influxes of formation fluids
- Erosion of the wellbore

As the stream of gas and cuttings erodes the wall and widens the annulus, a greater increase in gas volume is required to maintain gas velocity. Sometimes water or mud is misted into the well to inhibit shales and reduce torque and drag.

The most important aspect of gas drilling is maintaining adequate annular velocity.

If the annular velocity falls below the point where it can clean the hole, the cuttings will accumulate and cause stuck pipe. A minimum annular velocity of 3,000 ft/min is normally required for air drilling. Foam drilling will lower the necessary annular velocity to between 200 -400 ft/min for effective hole cleaning.

A useful reference for air and gas drilling is "Volume Requirements for Air and Gas Drilling" by R.R. Angel, Gulf Publishing Company. This small handbook contains charts showing volume requirements for various hole size combinations and penetration rates for both natural gas and air. "Air and Gas Drilling Manual" by William C. Lyons, McGraw Hill Publishing Co. deals in depth with the theory and practice of air drilling methods.

1.2. Foam Drilling

Foam drilling uses foam as the carrying agent for cuttings removal instead of air velocity. Foam drilling requires less air volume than air drilling and relies on bubble strength to remove cuttings, while air and mist drilling depend on extremely high air flow rates.

An indication of effective foam drilling is a continued and regular flow of foam at the blooey line. A pulsating, irregular flow (heading) can indicate problems with the flow columns.

In addition to hole cleaning, the foam deposits a thin filter cake on the walls of the hole to improve borehole stability. To stiffen foam and improve hole cleaning and water tolerance, polymers and/or bentonite are used to mix a slurry for injection.

Determining Air and Fluid Volumes

In foam drilling, the injected air controls the amount of foam. Air volume requirements are calculated using the following formula:

$$\text{Velocity ft/min} = \frac{(183.4)\text{cfm}}{D_h^2 - D_p^2}$$

Where

cfm = cubic feet per minute

D_h = diameter of the wellbore in inches

D_p = diameter of the drillpipe in inches

Controlling the Foam Drilling Fluid

During the drilling operation, changes to foam injection rates are made based on the following indications:

- Changes in the character of the foam at the blooey line
- Changes in torque
- Changes in pressure

Surface Injection Pressure

Foam drilling is most effective when the lowest possible standpipe pressure is maintained. Pressure on the standpipe can range from 80 to 350 psi.

Changes in the standpipe pressure are the best means of detecting problems. As pressure changes are identified, adjust the foam injection rate and the gas volume percentage to deal with the change. Table 8-2 provides corrective adjustments for different types of pressure changes.

Table 2 Surface Injection-Pressure Adjustments

Pressure Change	Probable Cause	Treatment
Quick drop	The gas has broken through the foam mix, preventing the formation of stable foam.	Increase the liquid injection rate and/or decrease the air injection rate.
Slow, gradual increase	There is an increase in the amount of cuttings or formation fluid being lifted to the surface.	Increase the gas/air injection rates slightly.
Rapid increase	The bit is plugged or the formation is packed off around the drill pipe.	Stop drilling and attempt to regain circulation by moving the drill pipe.

Foam Condition at the Blooey Line

Under normal drilling conditions, foam at the blooey line should be similar in appearance and texture to shaving-cream foam. If the foam is not thick or does not hold its shape, adjust the rates of gas and foam- solution injection.

Table 3 Blooey Line Foam Conditions

Foam Condition at Blooey Line	Probable Cause	Treatment
Gas blowing free with fine mist of foam	The gas has broken through the liquid foam mix, preventing the formation of stable foam.	Increase the liquid injection rate and/or decrease the gas injection rate.
Foam thin and watery (salt-cut)	Saltwater from the formation is diluting the foam.	Increase the liquid and gas injection rate. If necessary, increase the percentage of chemical foaming agent.
Foam thin and watery (oil-stained)	Oil from the formation is contaminating the foam.	Increase the liquid and gas injection rates. Use AQF-2 foaming Agent

Heading / Regularity of Foam Return at the Blooey Line

For optimal removal of cuttings, foam returns at the blooey line should be continuous. Heading and unloading can indicate problems with the foam column.

If the hole is...	Then...
Unloading at regular intervals while drilling...	Continue drilling as long as unloading intervals are regular and short.
Heading (irregular intervals)...	Increase the foam concentrate to improve foam quality.

Foam Drilling Formulations and Applications

QUIK-FOAM, Baroid's principal agent for foam-drilling systems, is nontoxic and biodegradable. It should be used at concentrations of 0.25%-2.0% by volume of injection fluid, or approx. 1 to 8 pints per barrel of injection fluid for foam injection.

Stiff Foams

Drilling fluid additives can be added to the foam when specific problems, such as water influx, occur. For severe water influx, the following modified QUIK-FOAM systems can be used:

- Water influx QUIK-FOAM
- AQUAGEL/QUIK-FOAM
- KCl/QUIK-FOAM
- Di-ammonium phosphate (DAP)/QUIK-FOAM
- HEC/QUIK-FOAM

The Marsh funnel viscosity test is the only control test for the foam-injection mixture. A test result of 40 to 50 seconds/qt is standard. Check the funnel viscosity before adding QUIK-FOAM.

Water influx QUIK-FOAM

The following QUIK-FOAM formulation is applicable in cases of severe water influx. For best results, this fluid should have a Marsh funnel viscosity of 40 to 50 sec/qt before QUIK-FOAM is added.

Table 4 Water Influx QUIK-FOAM. Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m ³)
Soda ash	Improves foaming ability and maximizes bentonite yield	1.0 (3)
AQUAGEL	Gives foam stability and is the primary component of the filter cake	12.0 (36)
PAC-R	Polymer additive that adds stiffness and stability to the foam and reduces the permeability of the filter cake	1.0 (3)
QUIK-FOAM	Foaming agent	0.25-2% by volume injection fluid

When formulating a QUIK-FOAM system:

- Add materials in the order listed.
- Add QUIK-FOAM after the initial mixing and stir gently to prevent foam formation before injection.

KCI / QUIK-FOAM

The following QUIK-FOAM formulation is for cases of severe water influx with water-sensitive shales. This mixture is especially effective for controlling water influx with water-sensitive shales exposed.

Table 5 KCI/QUIK-FOAM Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m ³)
AQUAGEL (optional)	Pre-hydrated; functions same as water influx QUIK-FOAM	6.0-8.0 (17-23)
Potassium chloride (KCI)	Helps prevent caving in water-sensitive shales	10.0-25.0 (29-71)
PAC-R	Functions same as water influx QUIK FOAM	0.75-1.5 (2.1-4)
QUIK-FOAM	Foaming agent	0.25-2% by volume injection fluid
BARACOR 700	Corrosion inhibitor	1.0-2.0 (3-6)

DAP/QUIK-FOAM

The following QUIK-FOAM formulation is for cases of severe water influx, corrosion and water-sensitive shale problems in environmentally sensitive areas.

This foam mixture has proven useful in shale formations with severe water influx where sensitive shales are exposed and in environmentally sensitive areas.

Table 6 DAP/QUIK-FOAM Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m ³)
DAP (Diammonium phosphate)	For corrosion only For shale stability	2.0 (6) 6.0 (17)
PAC-R	Stiffness and hole stability	1.5-2.5 (4-7)
EZ-MUD	Additional hole stability or stiffness; can also be substituted for PAC-R	1.0-2.0 (3-6)
QUIK-FOAM	Foaming agent	0.25-2% by volume injection fluid
BARACOR 700	Corrosion inhibitor	1.0-2.0 (3-6) BARACOR 700 may not be needed in this system.

HEC/QUIK-FOAM

The following QUIK-FOAM formulation is used where an acid soluble polymer is needed to avoid formation damage. This foam mixture can be acidized to remove polymer from sensitive formations.

Table 7 HEC/QUIK-FOAM Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m ³)
BARAVIS	Viscosifier	1.5-2.5 (4-7)
Potassium chloride (optional)	Inhibits shale swelling	10.0-25.0 (29-71)
QUIK-FOAM	Foaming agent	0.25-2% by volume injection fluid
BARACOR 700	Corrosion inhibitor	1.0-2.0 (3-6)

1.3. Aerated Mud

Aerated mud systems reduce lost circulation in areas with very low fracture gradients. At the same time, shale hydration and corrosion are minimized. Effective mud weights of 4 to 6 pounds per gallon (0.48-0.72 sg) are possible with an aerated system. These weights substantially reduce differential pressure in the wellbore. Because of the lower pressure, the driller can reach a higher penetration rate than is possible with normal drilling fluids.

Equipment Requirements

The following equipment is needed for an aerated mud system:

- An air compressor capable of 850 SCFM
- A back-up compressor capable of 850 SCFM

When comparing compressor ratings, remember that ratings are made at sea level. Adjust the ratings as necessary to allow for the altitude at the drilling site.

- An air bypass (or other means of limiting the air volume) when the total compressor capacity is not required, as with a surface hole
- A Barton recorder for gauging the actual CFM of air injected
- A rotating head to direct the air and mud flow out of the flowline instead of up through the rotary table or over the drilling nipple into the cellar

The rotating head should be maintained to prevent mud loss at the head. If the drilling crew is not paying close attention, an undetected loss at the head can be mistaken for lost circulation in the hole.

- An air-mud separator (gas buster) at the flowline

The separator is typically a cylindrical tank 3 to 6 feet in diameter and 8 to 10 feet high with baffles to help break the air out of the mud.

- An air vent on the top of the tank aimed toward the reserve pit

This vent also accommodates overflow when the return is hard.

- A mud flow discharge on the bottom of the tank for discharge into the possum belly

Lime / IMPERMEX Fluid Formulation / Applications

A lime / IMPERMEX mud system is used when corrosion and/or reactive formations may be a problem. The following table provides formulations for the lime/IMPERMEX mud system.

Table 8 Lime/IMPERMEX System Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m³)
AQUAGEL	Provides suspension and borehole stability	3.0-5.0 (9-14)
ENVIRO-THIN	Reduces gel strength	As needed
IMPERMEX	Controls the filtration rate	2.0-5.0 (6-14)
Lime	Inhibits corrosion and shale swelling	0.8-1.5 (2.3-4)
X-CIDE 207	Controls bacterial growth	As needed

The lime/IMPERMEX mud will have the following properties:

- Mud weight 8.6-8.8 lb/gal
- Funnel viscosity 28-32 sec/qt
- Plastic viscosity 1-9 cP
- Yield point 0-2 lb/100 ft²
- Gels 0/0 lb/100 ft²
- API filtrate 8-10 mL
- pH 11.5-12.5
- Calcium 240-450 mg/L
- Solids 1-3 % by vol

DAP/PAC Fluid Formulation / Applications

A DAP/PAC mud system can be used for additional inhibition and corrosion protection. The system is run at a low pH and the phosphate ion provides corrosion protection while the ammonium ion provides shale inhibition.

DAP/PAC mud is not recommended for carbon dioxide (CO₂) or hydrogen sulfide (H₂S).

Table 9 DAP/PAC Fluid Formulation

Additive	Function	Typical concentrations lb/bbl (kg/m ³)
AQUAGEL	Provides viscosity and wall cake	8-12 (23-34)
DAP	Provides shale stability and corrosion control	2-6 (6-17)
EZ-MUD	Provides viscosity and shale stability	0.50-1.50 (1.4-4)
PAC-R	Controls fluid loss	0.50-1.50 (1.4-4)

The DAP/PAC mud will have the following properties:

- Mud weight 8.6-8.9 lb/gal
- Funnel viscosity 35-40 sec/qt
- Plastic viscosity 1-12 cP
- Yield point 6-8 lb/100 ft²
- Gels 2-5 lb/100 ft²
- API filtrate 8-10 mL
- pH 7-8
- Solids 1-3 % by vol

Do not add caustic soda or lime because ammonia will be liberated.

Recommended Operating Procedures for Aerated Mud

Plumbing / Piping	Inject air into the standpipe and arrange the piping so air can be bypassed at the floor for making connections, etc. Plumb the piping so mud can be pumped downhole while air is bypassed.
Bit	Run the bit with open water courses (no jets) to prevent excessive air pressure requirements. With the reduced bottomhole pressure, jet impact is not as critical for cleaning the bottom of the hole.
Drill Pipe	Larger drill pipe sizes of 4 ½ or 5 inches are recommended to reduce compressor volume requirements.
Tripping	Filling the hole between trips is not necessary with aerated mud.
Circulation Rate	Circulate the mud system at a constant rate of 6 to 8 bbl per minute and treat it as a normal mud system. Do not vary pump output to maintain constant bottomhole pressure or to control gains and losses; instead, regulate the airflow to correct these problems.
Air Injection Volume	Use the aerated mud chart to determine the amount of air to inject to achieve a specific reduction in bottomhole pressure.
Float Valves	Install float valves in the drillstring approximately every 200 feet (61 meters) to prevent backflow on connections.

Determining Hydrostatic Loss Caused by Gas-Cut Mud

1. To find bottomhole (BHP) loss due to gas-cut mud:
2. Find hydrostatic pressure of uncut mud.
3. Start with hydrostatic pressure at bottom of chart (Figure 1)
4. Proceed up to intersect percent gas in mud.
5. Read on right the BHP loss due to gas content.
6. Subtract loss from original BHP to find new effective head of gas-cut mud.

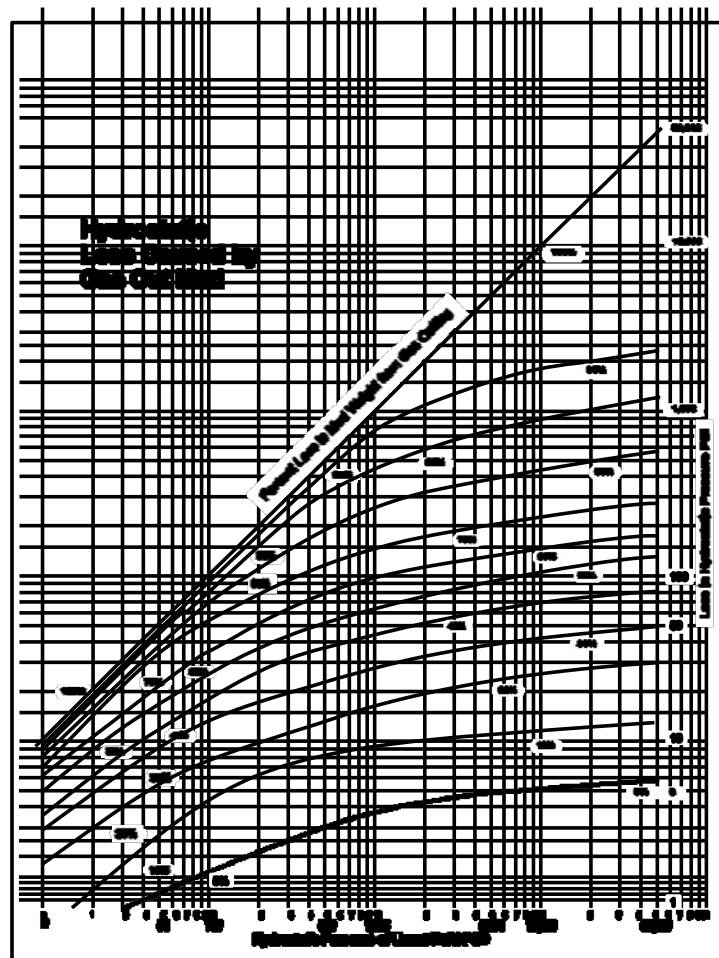


Figure 1 Hydrostatic loss caused by gas-cut mud.

Moderate gas cutting reduces measured mud weights on the surface, but, due to gas behavior under pressure, produces little effect on the effective hydrostatic head at depth.

When minimum overbalances are being used, or gas cutting becomes severe, an accurate method of determining the BHP reduction is needed. This graphical solution disregards the effect of gas density and thus provides a tool

useful for either gas or air. As such, it becomes useful for determining air injection volumes required for a desired reduction of hydrostatic pressure.*

*White, R. J. "Bottom-Hole Pressure Reduction Due to Gas Cut Mud," Journal of Petroleum Technology, July 1957.

1.4. Corrosion

Foam and aerated fluids can be corrosive. The injected air contains carbon dioxide and oxygen that promote corrosion. Inhibitors are needed to counter the effect of these gases. The products in the following table are recommended for corrosion problems.

Table 10 Corrosion Products

Additive	Application	Treatment
BARACOR 700	Inhibits corrosion by treating mud slurry	Treat mud initially at 1,500 ppm, then 0.5-1.5 lb/bbl (1.4-4 kg/m ³).
STABILITE	Inhibits scale	Add to mud at 10-100 ppm, then 1 gal/tour to 1 gal/hr.

Examine corrosion coupons and rings to ensure that enough inhibitors are being used.

For more information on how to treat for carbon dioxide and oxygen contamination, see Corrosion.

Troubleshooting

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1. Troubleshooting

The troubleshooting tables include a list of contaminants or operational problems, as well as indications of and treatments for the contaminants or operational problems.

1.1. Completion / Workover Fluids

Completion/Workover Fluids—Contaminants		
Contaminant	Indications	Treatments
Dilution from water or lower-density brine	Loss in density	Identify source of influx. Add a compatible solid salt to the brine. Blend the brine with a compatible, higher-density (spiking) brine. Note: Blending brine is usually more cost effective than adding salt to brine.
Iron	Change in brine color to chartreuse, green, green/brown, or rust/red Iron content of brine exceeds the operator's specified limit	For monovalent brines, raise pH by adding caustic soda or caustic potash and filter. Displace brine with uncontaminated brine and return it to stock point for chemical treatment and filtration.
Solids	Loss of brine clarity/ increase in turbidity Particles suspend in or settle out of brine	Filter brine using plate and frame unit. As an option, filter brine using 2 angstrom pore-size cartridge unit. Treat brine with VERSAFLOC M341 or VERSAFLOC M441 to facilitate the filtration process.

1.2. Foam / Aerated Drilling Fluids

Foam/Aerated Drilling Fluids—Maintenance / Operational Issues		
Problem	Indications	Treatments
Inadequate hole cleaning	Fill on trips/connections Increase in torque and drag Sporadic returns	Adjust the volume of injected air.
Influx of formation water (air drilling)	Water present at the return (blooey) line	Increase the rate of air injection. Convert to foam or mist drilling.

1.3. Oil-Based Fluids

OBM—Contaminants		
Contaminant	Indications	Treatments
Acid gas	Depletion of alkalinity	Increase mud density if possible. Add lime. Add NO-SULF H ₂ S scavenger.
Salt	Salt crystals on the shaker and in the mud Drop in electrical stability Increase in chloride content in water phase	Add water to dissolve the salt, then add primary/secondary emulsifier, and lime. Add new mud containing no salt.
Solids	Increase in solids (retort analysis) Increase in plastic viscosity Decrease in electrical stability	Verify shaker screen integrity Reduce shaker screen size. Optimize mud cleaner/centrifuge use. Dilute mud with oil and maintain density with weight material. Use optimum solids control.
Water	Change in mud weight Change in oil/water ratio Water in HTHP filtrate Increase in funnel viscosity Decrease in electrical stability Increase in mud volume	Add oil, primary/secondary emulsifier, DRILTREAT, and weight material. Increase fluid density, if possible
Formation hydrocarbons	Decrease in mud weight Increase in oil/water ratio Increase in HTHP filtrate Change in luminescence fingerprint	Add emulsifier. Add water and salt. Add weight material. Increase fluid density, if possible

OBM—Maintenance / Operational Issues		
Issue	Indications	Treatments
Emulsion breaking	Water in HTHP filtrate Low electrical stability Water-wet solids	Add primary/secondary emulsifier, or DRILTREAT if using conventional fluids Add DURATONE HT (Conventional fluids). Add lime.
High yield point and gel strengths	Excess organophilic additives Solids build-up Water-wet solids	Use optimum solids control. Dilute with oil. Add emulsifier. Add PERFORMUL if fluid is solids laden Add OMC.
Hole instability	Cavings Shale slivers on shaker	Adjust water phase salinity. Add DURATONE HT/BARABLOK to reduce filtrate (in Conventional fluids).

OBM—Maintenance / Operational Issues		
Issue	Indications	Treatments
	Excessive torque and drag	Add sized BARACARB and STEELSEAL Add primary/secondary emulsifier to tighten the emulsion. Consider increase in mud density.
Inadequate hole cleaning/suspension	Increase in torque and drag Inadequate gel strengths Residue in cup Few cuttings on shaker Fill on trips/connections	Add GELTONE II/V SUSPENTONE, or RM-63 (Conventional Fluids). Add TAU MOD Add RHEMOD L Increase pipe rotation to stir up cuttings beds
Insoluble salt	Low electrical stability Water in HTHP filtrate	Add water to solubilize salt.
Lost circulation *See also Lost Circulation Section	Whole mud losses Decrease in pit volume Drop in circulating pressures	Add lost-circulation material or set a soft plug. Lower the mud weight and the equivalent circulating density when possible. Set a cement squeeze. Reduce pump speed. Use a GELTONE II/V squeeze or a high- solids squeeze when there is major mud loss. Add MICATEX lost-circulation material, WALL-NUT seepage-loss control, BAROFIBRE seepage-loss control, or calcium carbonate when there is minor mud loss. Note: Do not add cellophane or BARO- SEAL lost-circulation material.
Water wetting	Mud appears dull/grainy Large barite flocs Aggregation of solids Settling in cup Over-saturation with calcium chloride	Add oil. Add secondary emulsifier, DRILTREAT, or primary emulsifier. Dilute mud with fresh mud. Adjust the shaker screen to remove aggregated solids. Add water to solubilize excess salt.
Weight material settling	Weight material settles in the viscometer cup Mud weight varies when circulating after trips	Add GELTONE II/V SUSPENTONE, X-VIS, or RM-63 (conventional fluids) Add TAU MOD and RHEMOD L (Clay-free).

1.4. Synthetic-Based Fluids

Synthetics—Contaminants		
Contaminants	Indications	Treatments
Acid gas	Depletion of alkalinity	Increase mud density if possible. Add lime. Add NO-SULF H ₂ S scavenger.
Formation hydrocarbons	Decrease in mud weight Increase in oil/water ratio Increase in HTHP filtrate Change in luminescence fingerprint	Add emulsifier. Add water and salt. Add weight material. Increase mud density if possible.
Salt	Salt crystals on the shaker and in the mud Drop in electrical stability High chloride content in water phase	Add water to dissolve the salt, then add primary/secondary emulsifier. Add new mud containing no salt.
Solids	Increase in solids (retort analysis) Increase in plastic viscosity Decrease in electrical stability	Verify shaker screen integrity Reduce shaker-screen size. Optimize mud cleaner/centrifuge use. Dilute with base fluids. Add weight material. Use optimum solids control.
Water	Drop in mud weight Change in S/W ratio Water in HTHP filtrate Increase in funnel and plastic viscosity Decrease in electrical stability	Add base fluids, primary/secondary emulsifier, and weight material.

Synthetics—Maintenance / Operational Issues		
Issue	Indications	Treatments
Emulsion breaking	Water in HTHP filtrate Low electrical stability Water-wet solids	Add primary/secondary emulsifier, or DRILTREAT if using conventional fluids. Add DURATONE HT (Conventional fluids). Add lime.
High yield point and gel strengths	Excess organophilic additives Solids build-up Water-wet solids Low S/W ratio for mud weight	Use optimum solids control. Dilute with base fluids. Add PERFORMUL if fluid is solids laden Add emulsifier. Add OMC 42 or OMC 2
Hole instability	Cavings Shale slivers on shaker	Adjust water phase salinity. Add sized BARACARB and STEELSEAL Add DURATONE HT/BARABLOK to

Synthetics—Maintenance / Operational Issues		
Issue	Indications	Treatments
	Excessive torque and drag	reduce filtrate for ENVIROMUL fluids. Add primary/secondary emulsifier to tighten the emulsion. Consider increase in mud density.
Inadequate hole cleaning/suspension	Increased torque and drag Inadequate gel strengths Residue in cup Few cuttings on shaker Fill on trips/connections	Add GELTONE II/V SUSPENTONE, or RM-63 for ENVIROMUL fluid. Add TAU MOD and RHEMOD L for Clay-free fluids. Test yield point and gel strengths at elevated temperature. Consider raising S/W ratio.
Insoluble salt	Low electrical stability Water in HTHP filtrate Increase in funnel viscosity and water-wet solids	Add water to solubilize salt. Add emulsifier
Lost circulation *See also Lost Circulation Section	Whole mud losses Decrease in pit volume Drop in circulating pressures	Add lost-circulation material or set a soft plug. Lower the mud weight and the equivalent circulating density when possible. Set a cement squeeze. Use a GELTONE II/V squeeze or a high-solids squeeze when there is major mud loss. Add MICATEX lost-circulation material or BARACARB when there is minor mud loss. Note: Do not add cellophane or BARO-SEAL lost-circulation material.
Water-wetting	Mud appears dull/grainy Large barite flocs Aggregation of solids Settling in cup Over saturation with calcium chloride	Add base fluids. Add primary/secondary emulsifier or DRILTREAT. Dilute mud with fresh mud. Adjust the shaker screen to remove aggregated solids. Add water to solubilize excess salt.
Weight material settling	Weight material settles in the viscometer cup Mud weight varies when circulating after trips	Add GELTONE II/V SUSPENTONE, or RM-63 for ENVIROMUL fluids. Add TAU MOD and RHEMOD L for Clay-free.

1.5. Water-Based Fluids

This table provides generalized treatments for water-based mud contaminants. For treatments specific to certain water-based muds, see Chapter 4 - **Water-Based Fluids**.

Water-Based Muds—Contaminants		
Contaminants	Indications	Treatments
Carbonates/carbon dioxide (CO ₂)	Presence of bicarbonates and carbonates Increase in rheological and filtration properties Increase in the spread between Pf and Mf High and progressive gel strengths (ash gels)	Treat the mud with lime or gyp. Increase the mud weight if there is an influx of carbon dioxide.
Cement	Increase in rheological and filtration properties Increase in calcium concentration Increase in pH	Add soda ash or sodium bicarbonate. Optimize solids control equipment. Treat with thinners if appropriate. Convert to a system that tolerates high cement levels (e.g., POLYNOX) when treatments are not sufficient to counter indications.
Gypsum/anhydrite	Increase in calcium concentration Increase in rheological and filtration properties Thick/spongy filter cake	Treat with soda ash to maintain acceptable calcium levels. Convert to a system that tolerates high calcium levels when treatments are not sufficient to counter indications.
Hydrogen sulfide (H ₂ S)	Increase in rheological and filtration properties Decrease in pH Presence of hydrogen sulfide, as indicated by the sulfide indicator test and the Garrett Gas Train Rotten-egg odor	Treat the mud with hydrogen sulfide scavengers. Adjust the pH with caustic soda. If H ₂ S is detected proper PPE must be worn.
Low-gravity solids	Increase in rheological and filtration properties Increase in bentonite content as determined by MBT Increase in low-gravity solids content	Optimize solids control equipment. Treat with thinners Dilute with base fluid. If there is not sufficient free base fluid, thinners will not function.
Salt formations	Rapid increase in chloride concentration Increase in mud weight Rapid decrease in alkalinity Increase in filtrate Thicker/spongy filter cake Increase in or inversion of rheological properties	Convert to a saturated saltwater system or displace to an oil-based mud system or synthetic systems when treatments are not sufficient to counter indications.
Saltwater flow	Increase in pit volume	Increase density to control the water flow.

Water-Based Muds—Contaminants		
Contaminants	Indications	Treatments
	Increase in chloride concentration Change in mud density Decrease in alkalinity Decrease in MBC Increase in filtrate Thicker/spongy filter cake Increase in or inversion of rheological properties Well flows with pumps shut off	

Water-Based Muds—Maintenance / Operational Issues		
Issue	Indications	Treatments
Air entrapment	Decrease in mud weight Air bubbles encapsulated in mud Increase in plastic viscosity Hammering of pumps	Thin fluid with chemical treatment or water. Minimize surface air entrapment.
Bacterial degradation	Decreasing hydroxyl alkalinity Increasing carbonate alkalinity Increase in filtration and rheological properties	Add biocide. Add lime. Treat with fluid loss additive if required. Treat with rheological control agents if required.
Bit balling	Reduced drilling progress Balled bit and string Swabbing on trips Packed bits that show little wear	Maintain appropriate viscosity and gel strengths to keep drilling assembly clean. Make CON DET or LIQUI-DRIL additions to coat pipe. Optimize hydraulics. Use inhibitors to prevent hydration of clays.
Corrosion	External and/or internal pitting on drillpipe Drillpipe failure Washouts	Raise pH to between 11 and 11.5 if possible. Lime may be used in some applications. Add a compatible Baroid corrosion inhibitor.
Differential sticking	Partial or full circulation String against porous zone No keyseats High fluid loss in muds with a high solids content Cannot rotate or reciprocate drill pipe	Cover drillstring at the stuck zone with a Baroid spotting fluid, keeping some in the pipe to move at 10-minute intervals. Use stretch equation to help locate stuck region. Reduce mud weight where possible. Treat with sized particulate materials. Lower HTHP filtrate to minimize cake build-up.
Foaming	Decrease in mud weight Foam on surface of mud pits	Add a Baroid defoamer to the surface of the mud.

Water-Based Muds—Maintenance / Operational Issues		
Issue	Indications	Treatments
	Decrease in pump pressure Hammering of pumps	Spray water on the pits to break up the foam. Add AQUAGEL to salt or low-solid muds.
Gas influx	Increase in pit volume Appearance of gas-cut mud Well does not flow after shutting down pump Decrease in mud weight at flow line	Increase mud weight. Operate degasser.
Gas kick	Increase in pit volume Well flows after shutting down pump	Shut-in well. Follow proper kill procedures.
Keyseating	Can rotate but cannot reciprocate drillpipe more than one joint Partial or full returns Well is dog-legged	Backoff and wipe out keyseat.
Lost circulation *See also Lost Circulation Section	Decrease in pit volume Loss of returns Whole mud losses Decrease in circulating pressures	Add lost-circulation material or set a soft plug. Lower the mud weight and the equivalent circulating density when possible. Set a cement squeeze. Reduce pump speed.
Mechanical sticking	Cannot rotate or reciprocate drillpipe Reduced or no circulation Packing off	Backoff and wash over. Improve hole cleaning.
Plastic salt	Salt sections undergauge after trips Tight connections Stuck pipe	Increase mud weight. Spot water pill. Make regular check trips back through salt. Decrease mud salinity. Use water to dissolve salt at stuck point.
Sloughing shale	Excessive shale slivers at shaker Tight connections	Reduce fluid loss. Increase mud weight if possible. Convert mud to an inhibitive fluid. Increase mud viscosity if possible. If drilling through bentonitic shale, increasing the mud viscosity is not necessary. Add BAROTROL or BARABLOK. Reduce pressure surges. Reduce drill pipe whipping.
Thermal instability	Bottoms-up mud has high viscosity and gel strengths Difficulty in breaking circulation Difficulty in running tools to the bottom	Add water and use optimum solids control. Treat mud with thinners, dispersants, or deflocculants. Consider conversion to a THERMA-DRIL

Water-Based Muds—Maintenance / Operational Issues

Issue	Indications	Treatments
	Decrease in alkalinity Increase in fluid loss	system. Add lime if carbonate level is increasing. Add oxygen scavenger Add CO2 scavenger

DFG Hydraulics Modeling Software

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1. DFG Hydraulics Modeling Software

DFG is Baroid's Drilling Fluids Graphic Simulation, a software package that includes calculators for versatile hydraulic scenarios. DFG can be used as a planning tool for Technical / Field Professionals or as the daily solution for Drilling Fluid Engineers.

DFG provides the most accurate hydraulics modeling software available to the industry. This tool can allow the user to simulate proposed drilling conditions, allowing optimization of not only the fluid properties but also the drilling parameters. Accurate modeling allows the user to look ahead of the bit, avoid future problems and reduce NPT.

Output from drilling simulations focuses on annular cuttings concentrations, including average cuttings concentration as well as localized cuttings distribution in the annulus, and equivalent circulating density (ECD). Documented variance between DFG software-predicted and actual ECD, as measured by PWD tools, is consistently less than 1%. The DFG software is the only hydraulics modeling program that accurately accounts for effects of drilling fluid compressibility and thermal expansion on fluid density, including the reduction in temperature while the mud returns to surface through a long deepwater riser that causes an increase in density, viscosity and rheological properties. DFG software modeling is critical to operations where very narrow margins between pore pressure (PP) and fracture gradient (FG) exist.

Example modeling capabilities are shown below.

Hydraulics

The hydraulics design offers straightforward calculations for the wellbore.

DRILLAHEAD

DrillAhead Hydraulics offers simulation and optimization over a range of cutting sizes.

Surge and Swab

The surge and swab design computes the EMW corresponding to a selectable range of pulling/running speeds, in order to maintain swab above a certain density and surge below a certain density.

1.1. General Guidelines on How to Use DFG

Navigation

Navigation is simple and intuitive. There are three major parts to a DFG screen:

1. The Toolbar and Menu.
2. The Navigation Pane, including different toolsets to use for Hydraulics, Surge & Swab, or DrillAhead designs. The Data toolsets are located at the bottom of the Navigation Pane. When selected a toolset from the list, the menu for the tool is displayed above. The Navigator Pane can be hidden, if more room for workspace is needed just by clicking the Auto Hide button (Pushpin).
3. The Working Pane, which displays input and output information for DFG. You can navigate through designs using tabs in order of right-to-left and top-to-bottom.

Data Import

DFG can import existing hydraulics files from earlier DFG versions and edit them.

It is also possible to import hydraulics files from:

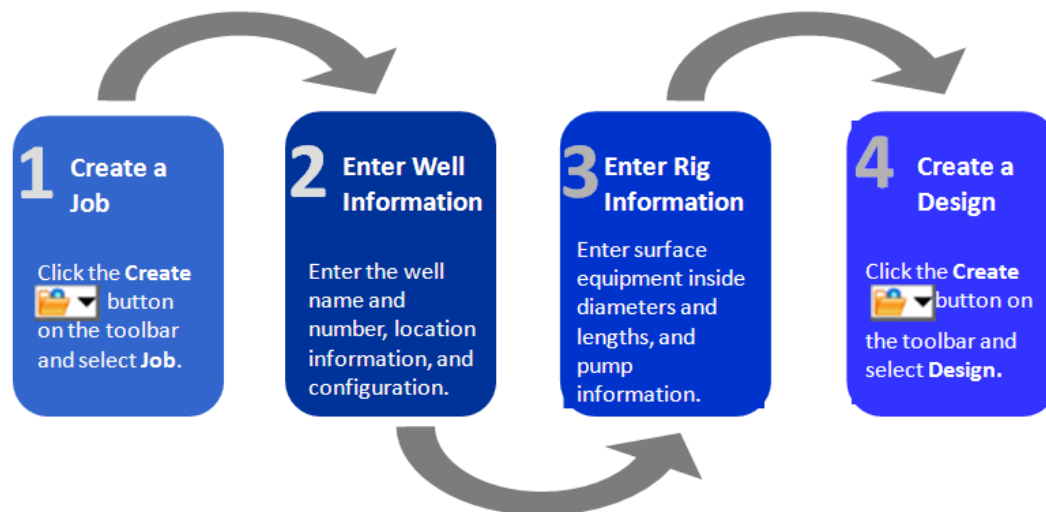
- Wellsight 2000,
- Wellsight Express,

- WS2000 Fluid Check,

To import files, select File > Import from the DFG menu; from the Import menu, select Import .hyd; locate your file and click Open.

Data Input

The scheme below should help to achieve a proper data entry.



1. Create a Job

Click the Create button on the toolbar and select Job.

2. Edit Well Information

Enter the well name and number, location information, and configuration. After you create the well, it appears in the Navigator Pane.

Enter as much information as possible to help you identify and describe the well.

Save your entries.

3. Enter Rig Information

Enter surface equipment inside diameters and lengths, and pump information.

From the Navigator Pane, find your well and expand the folder. Highlight the Rig under your well to display the Rig Equipment page. Enter the rig name, and enter the inside diameter and lengths of the surface equipment, if known. You can also enter the pumps you will be using; either manually or using the DFG database.

Save your entries.

4. Create a Design

Click the Create button on the toolbar and select Design.

You can create as many designs as you wish for this particular well and rig. With the well highlighted in the Navigator Pane, select Create > Design from the menu.

The New Design dialog box appears, so enter a name for your design. Then click the drop-down arrow in the **Design Type** field to choose a Hydraulics, DrillAhead, or Surge & Swab design.

When creating a design, you can either:

- Create one from scratch (i.e., input all new information)
- Create one in reference to an existing design (i.e. copying the data over into the new design)

In case you want to copy information from another well, click the drop-down arrow in the **Job/Well** field to select from a list of wells that exist in the database.

If you want to create your design from scratch, leave this **Job/Well** field blank.

Once you have selected the well you want to use, click the drop-down arrow in the

Design field to choose the design which has the data you want to copy. You can choose the specific areas of data you want to copy over (Fluid Properties, Geometry, Survey etc.). Click the **Create** button.

If you want to create your design from scratch, leave this **Design** field blank.

The new design appears under your well and rig in the Navigator Pane.

Save your entries.

Entering Data

TIP: In certain areas in DFG (e.g., BHA and Pumps), you can select equipment from the DFG database. Doing this allows DFG to automatically populate the properties of that equipment. If you do not want to use equipment from the database, or you want to customize the equipment, you also have the option to enter the information manually.

The following **example** shows how to select a BHA component from the database:

In the Working Pane's first section "Wellbore", choose the vertical tab "Drillstring".

Click in the **Description** field. Select the type of component you want to add.

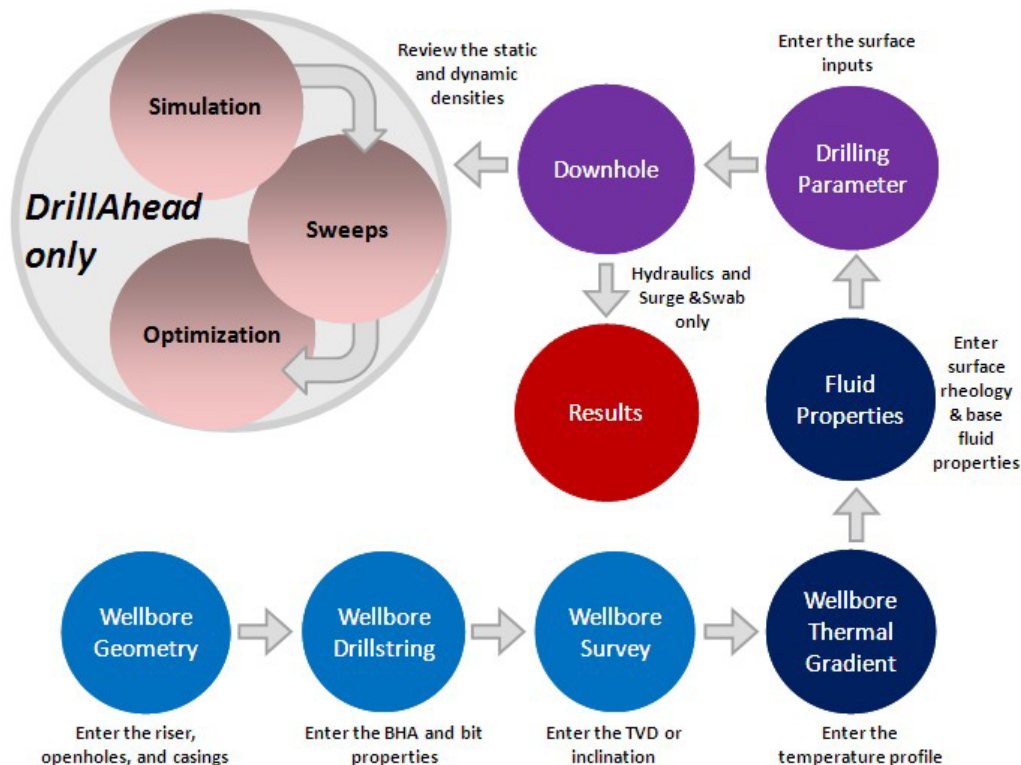
For this example, choose Drill Pipe. Then select the grade/type of the drill pipe you want to use. For this example, choose grade E.

Scroll down until you find the drill pipe you are seeking. You can sort the list in ascending or descending order by OD (Outer Diameter) or Nominal Weight.

Click on the line of the drill pipe you want to use.

OD, ID (Inner Diameter) and Component Types are populated automatically with information from the database.

If you want to delete a line of entry in DFG, click the box to the left of the line to highlight the line. Press the **Delete** button on your keyboard. There is no other shortcut available to delete a line.



The described way above is the logical way to create a design in DFG – follow the recommended order of steps - and save your work after each step!

Entering Wellbore and Geometry

On the **Wellbore Geometry** tab, enter the riser, open holes, and casings. As you enter the riser, open holes, and casing information, DFG draws the well in the Drawing Pane.

Enter the dimensions and depth of the riser, if applicable. For riserless drilling, just skip this point.

Enter measured depths and hole size of the open hole(s).

Enter the measurements and depths for the casing(s), if applicable.

NOTE: For more complete and precise modeling, entry of all open holes (even those that are cased) is encouraged.

The second tab above the Drawing Pane beside the Drawing tab brings you to the Volumes tab which shows the numerical values of:

- Volumes in each section of the annulus
- Total annulus volume
- Volumes in each section of the tool string
- Total tool string volume

Entering Wellbore and Drillstring

On the **Wellbore Drillstring** tab, enter BHA and bit properties. The Bit is entered directly as a tool string component.

Enter the BHA (bottom hole assembly), such as bit, bit sub, mud motor, stabilizers, drill collars, heavy-weight drill pipe, jars and crossovers for various thread forms.

Note that you can either enter the measured depth (MD) of a component, and let DFG compute the length, or you can enter the length of the component, and let DFG compute the MD.

As you enter the bit and BHA, the drill components appear in the Drawing Pane.

Entering Wellbore and Surveys

On the **Wellbore Survey** tab, enter the true vertical depth or inclination.

The Wellbore Survey tab is based on a modified version of directional drilling software.

It is set with constraints to allow angles of 0° to 180° and an azimuth of 360°.

You can enter either the inclination at any given depth and let DFG compute the true vertical depth (TVD), or you can enter the TVD for any given depth, and let DFG compute the inclination. After you enter the survey, TVDs are automatically interpolated.

Your survey inputs are graphed as you enter each line.

Did you save your entries lately?

Entering Wellbore and Thermal Gradient

On the **Wellbore Thermal Gradient** tab, enter the formation temperature profile.

The temperature profile provides the data that DFG uses to calculate compensated density for all fluid types.

Enter the temperatures for each TVD you entered in the survey.

The temperature gradient is graphed as you enter each line.

Entering Fluid Properties

On the **Fluid Properties** tab, just beside the wellbore tab, is the place to enter the drilling fluid properties.

Under the surface rheology section, first enter the rheological model you want to use for calculations and then enter the dial readings. DFG calculates the viscosity parameters automatically.

Select the base fluid type under the Fluid Properties section and enter its properties as exactly as possible.

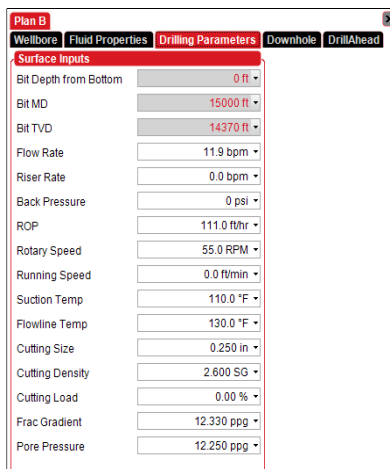
Now enter the gel strengths measured at 10-second, 10-minute, and 30-minute intervals.

The rheology curve is graphed as you enter data.

Entering Drilling Parameter

On the third tab you will find the **Drilling Parameters**.

Enter the surface inputs right there.



Surface Inputs	
Bit Depth from Bottom	0 ft
Bit MD	15000 ft
Bit TVD	14370 ft
Flow Rate	11.9 bpm
Riser Rate	0.0 bpm
Back Pressure	0 psi
ROP	111.0 ft/hr
Rotary Speed	55.0 RPM
Running Speed	0.0 ft/min
Suction Temp	110.0 °F
Flowline Temp	130.0 °F
Cutting Size	0.250 in
Cutting Density	2.600 SG
Cutting Load	0.00 %
Frac Gradient	12.330 ppg
Pore Pressure	12.250 ppg

Entering Downhole Data

On the **Downhole** tab, review the static and dynamic densities based on the actual temperature profile and stored base fluids profile.

The **Static Density** tab displays the measurements of the fluid density, assuming that the fluid is at rest in stable equilibrium.

The **Dynamic Density** tab displays the measurements of the fluid density, taking the hours pumping into consideration. You can increase/decrease the hours in the Hours Pumping field, and the data recalculates accordingly.

NOTE: From DFG 5 on, **Downhole Properties** has its own tab. Here, you can select the density calculations as static or dynamic, and the basis for the prediction (for example generic or Fann data).

Select the **Use Downhole Properties** checkbox if you want to use the downhole properties for the annular and pipe sections. When you select this checkbox, data for the annular and pipe sections appear below.

You can use one of the following rheology predictions:

- None
- Generic
- Fann 75

For example, if you select Fann 75 rheology prediction, you can select a Fann-file to load by clicking the Open File button.

The **Rheology Prediction** calculator opens. Here, you can calculate predicted readings.



Close the window to return to the **Downhole** tab.

Select the Use **Dynamic Thermal Gradient** checkbox if you want to use the dynamic thermal gradient in the calculations for densities.

The graphs on the **Downhole** tab show four different types of information:

- Static temperature gradient

When you have the **Static Density** tab and temperature graph selected simultaneously, the graph shows the static temperature gradient.

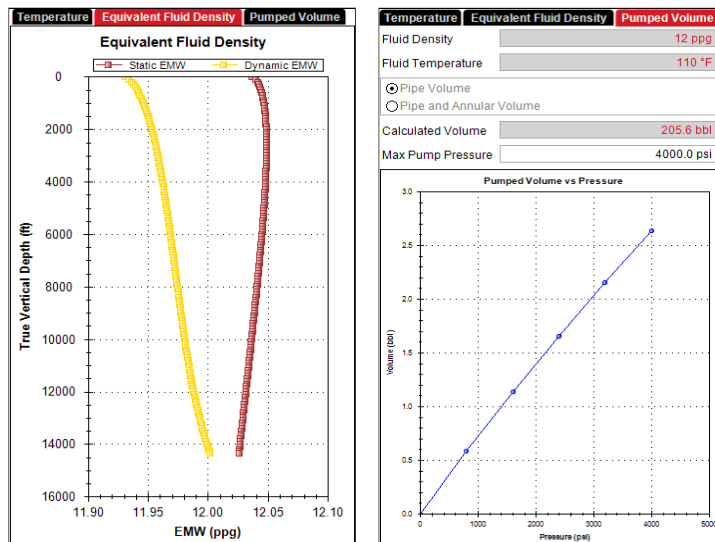
- Dynamic temperature gradient

When you have the **Dynamic Density** tab and Temperature graph selected simultaneously, the graph shows the temperature gradient, pipe temperature and annulus temperature.



- Equivalent fluid density

The Equivalent Fluid Density graph shows the static and dynamic EMW's based on TVD.



- Pumped volume

The Pumped Volume graph shows pumped volume versus pressure.

The Data entry should be completed now, so there are 2 ways to proceed with calculating the results, depending on the well design:

- If you are creating a **DrillAhead** design, go to the DrillAhead tab.

Wellbore Fluid Properties Drilling Parameters Downhole **DrillAhead**

- If you are creating a **Hydraulics design** or **Surge & Swab design**, click the Calculator button on the toolbar to display the Results tab.

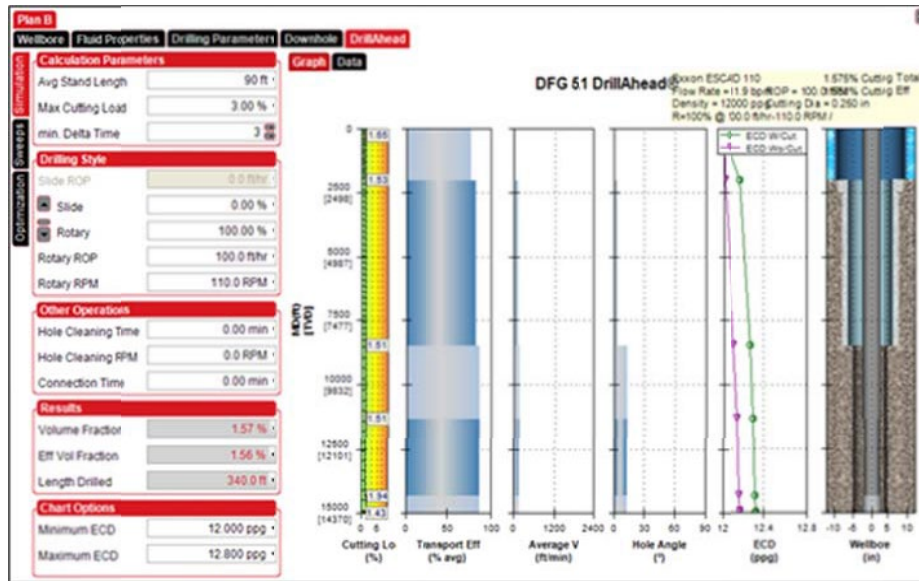
Wellbore Fluid Properties Drilling Parameters Downhole **Results**

1.2. DRILLAHEAD Design

DrillAhead Simulation

On the **DrillAhead Simulation** tab, enter calculations to simulate the impact of cuttings and drilling on ECD.

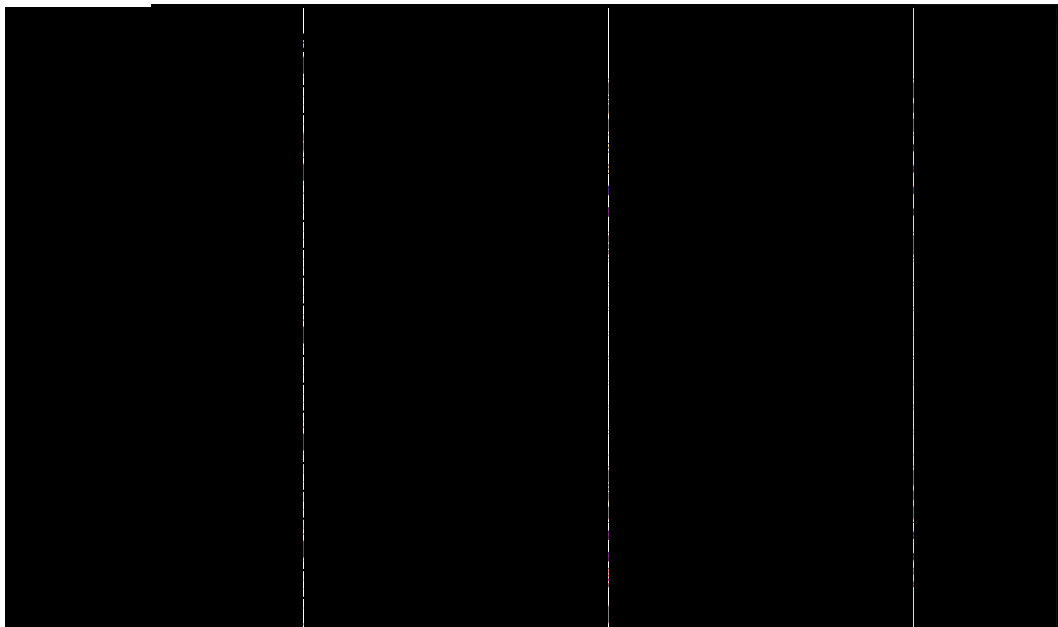
Enter the data you want to use in the simulation and click the Calculate button on the toolbar to run the simulation (left side of screen).



The graphs show the following information, per MD and TVD, based on your simulation data (right side of screen):

- Percentage of cutting load
- Average transport efficiency
- Average velocity
- Angle of the hole
- ECD with and without cuttings

The **Data** tab beside the Graph function, lists the data for each section of the annulus.

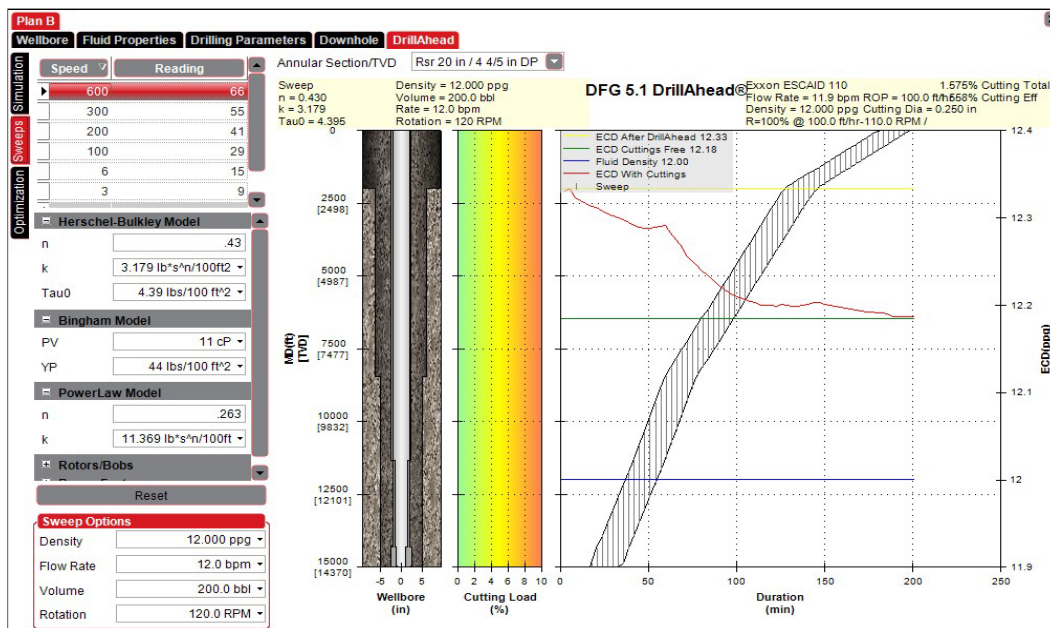


DrillAhead Sweeps

On the **DrillAhead Sweeps** tab, you can design a sweep to clean out cuttings and predict maximum ECD. The purpose of this simulation is to determine the ECD consequences of using a sweep to help clean the hole; it is not to prove that sweeps help to clean the hole.

Enter the dial readings of the Sweep Fluid. The values for the three models are calculated and populated automatically, based on the dial readings.

Enter the sweep options and click the **Calculator** button on the toolbar to run the sweeps simulation.

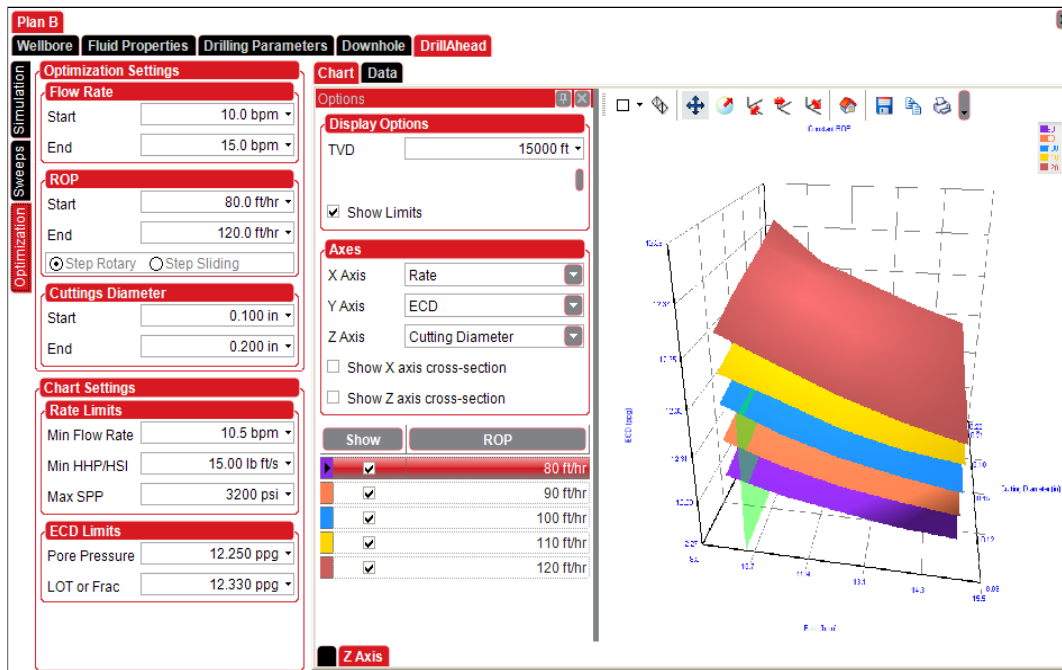


The sweeps simulation shows the sweep fluid moving through the wellbore, as well as the cutting load percentage.

DrillAhead Optimization

On the **DrillAhead Optimization** tab, input the parameters for the DrillAhead optimization.

DFG optimization will do 125 DrillAhead ECD simulations for ranges of flowrate, rotary or sliding ROP, and cuttings diameter.



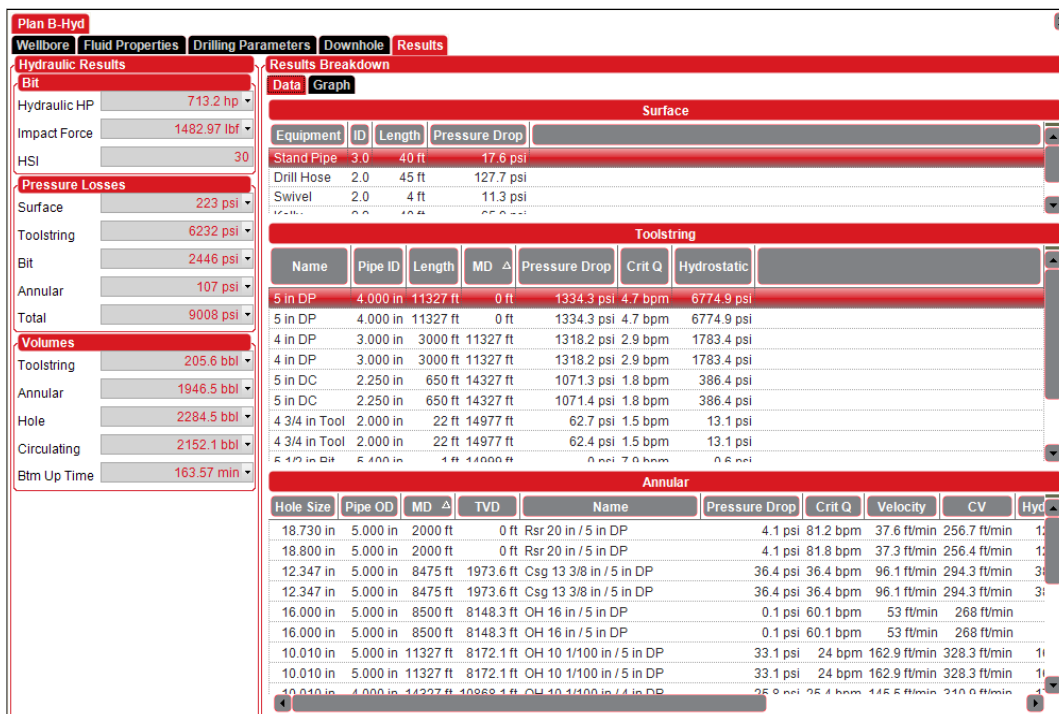
The middle pane Chart Options is the area supposed to customize the chart, regarding display options, area or axis configuration.

Click the **Calculator** button to run the results. The **DrillAhead Progress** box appears, indicating the calculations being made in the background. The **DrillAhead Progress** box closes automatically when the calculations are complete, and the results are graphed in the **Chart** section.

The screenshot shows the 'DrillAhead Progress' dialog box. It contains input fields for 'Rate' (11.00 bpm), 'ROP' (80 ft/hr), 'Cutting Diameter' (.15 in), 'Depth of Cuttings' (ft), and 'Stand Drilled' (3). A 'Cancel' button is located at the bottom. A progress bar at the bottom indicates '78 of 125'.

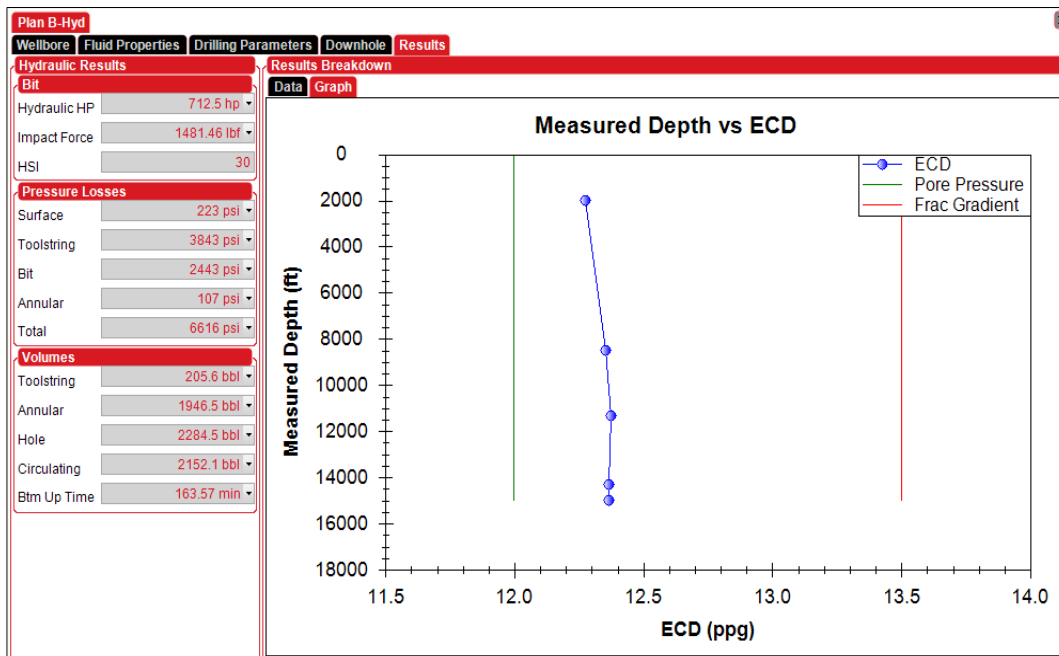
Hydraulics Design

If you are on a Hydraulic Design, you must click the **Calculate** button before you can see any data on the Hydraulics Results tab.



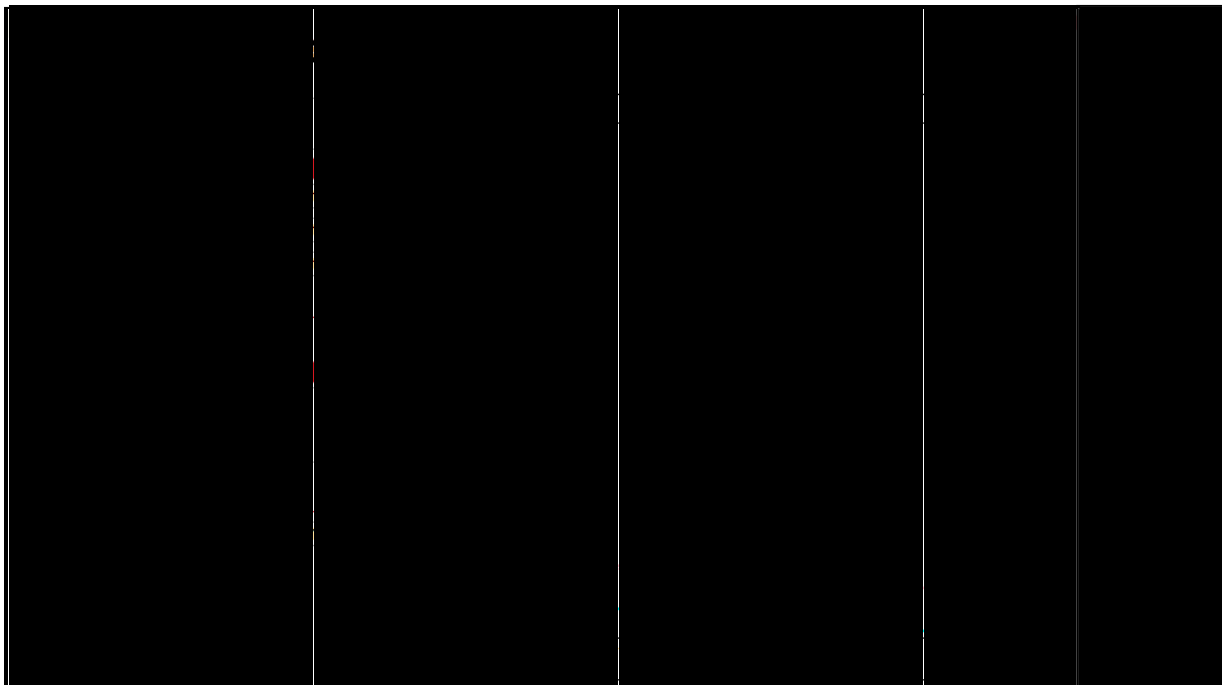
View the hydraulic results for the bit, pressure losses, and volumes. Further results are broken down by surface equipment, toolstring, and annulus.

The graph in the **Results Breakdown** section shows the ECD, pore pressure, and frac gradient by measured depth.



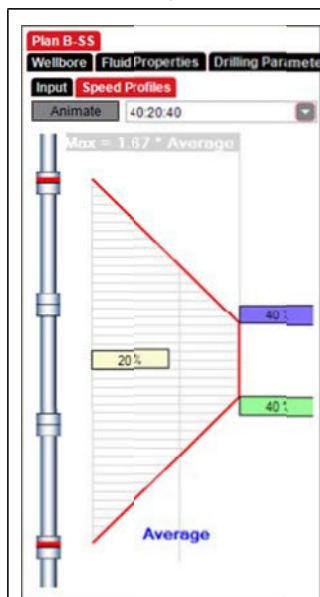
Surge and Swab Design

Choose between **Specify Running Speed** and **Specify ECD and Depth**. The choices affect what is shown on the graph, as well as the input fields.



Enter the running speed or ECD inputs. Click the **Calculator** button to show the graph to show the graph for the running speed and ECD.

On the **Speed Profiles** tab, you can choose one of the pre-selected profiles from the drop-down box, or you can click and drag the numbers to match what you want to show.



After choosing the speed profiles, click the **Animate** button to view the drill pipe speed

The number in the purple box represents the percentage of the stand's trip that is accelerating.

The number in the green box represents the percentage of the stand's trip that is decelerating.

Report Builder

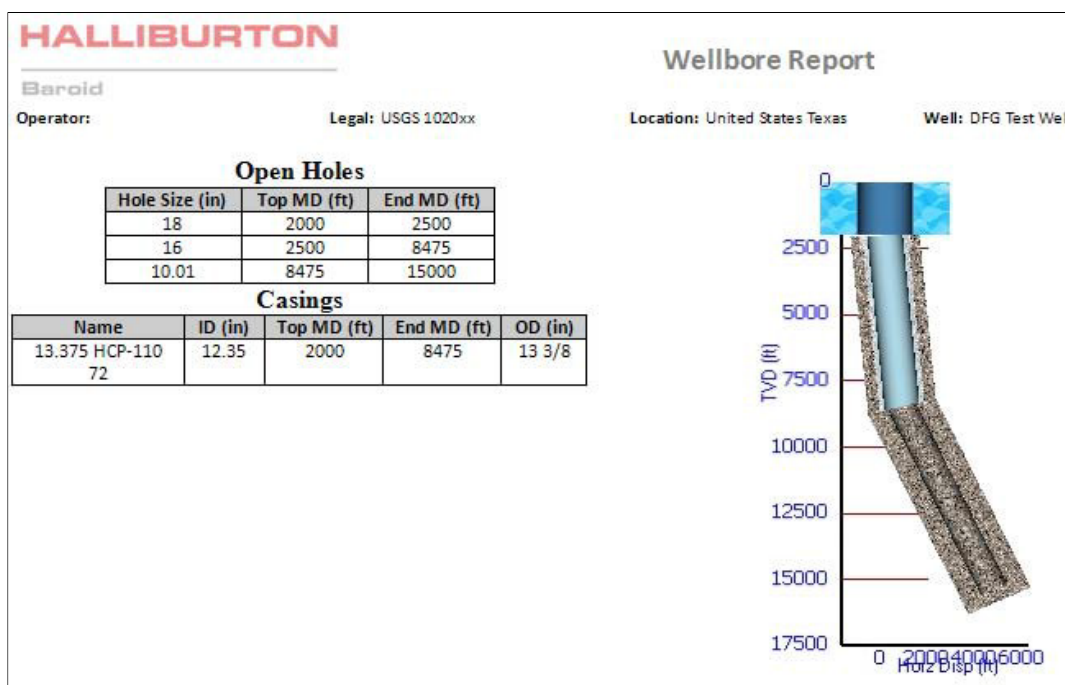
You can export data from a design in DFG to a Microsoft® Word document.

NOTE: You must have Report Builder installed on your computer to perform this function.

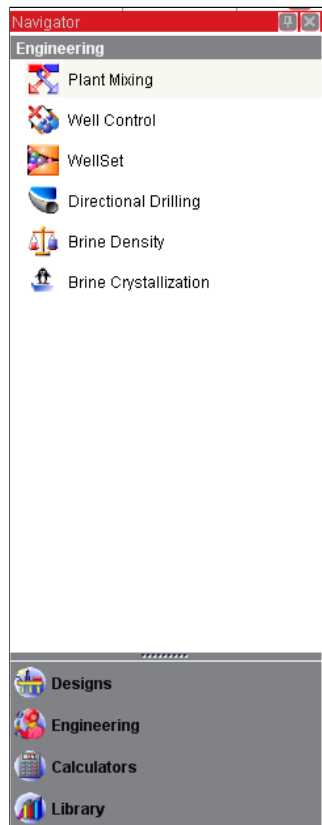
The **Run Report** button appears on the Toolbar when this function is available.



Click the **Run Report** button to export the data that is currently on the screen. Microsoft® Word will automatically open with the report.



1.3. Engineering



The Engineering section offers special solutions from Liquid Mud Plant Operations over WellSet and Wellcontrol to a Horizontal Drilling solution and detailed brine calculations.

Plant Mixing

This tool provides a selection of basic mixing procedures of your particular inventory, including a solution creation and cost breakdown.

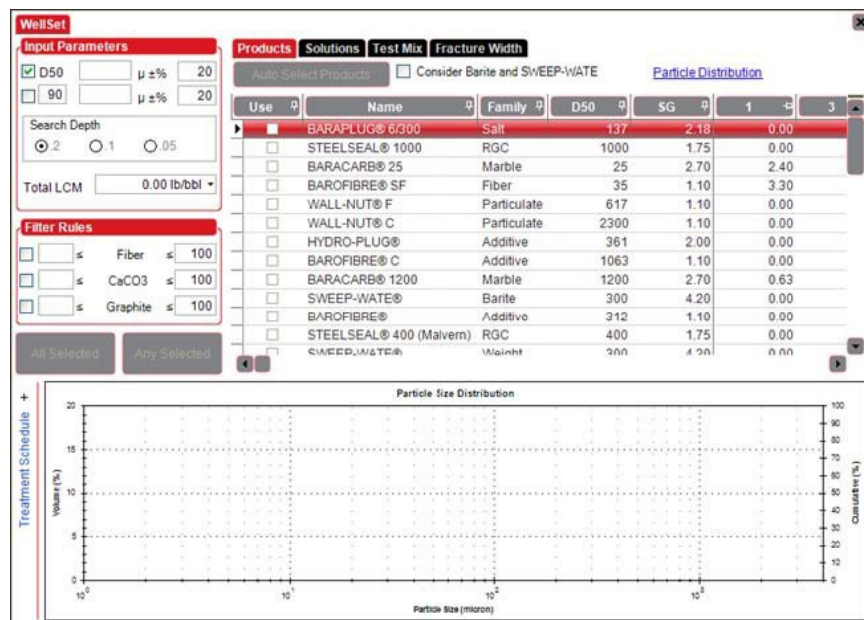
Well Control

The Well Control section provides a quick solution in case of an intervention with graphical output. Please mind the report builder for creating a Baroid Well Control Report.

WellSET

WellSET™ Lost Circulation Treatment Prevention of lost circulation by improving the wellbore strength is accomplished by designing and applying WellSET treatments that increase the hoop stress around the wellbore.

The goal of WellSET treatments is to increase the hoop stress (and thus the wellbore pressure containment ability) in the near wellbore region. This is accomplished by placing a plugging material in an induced fracture that prevents further pressure and fluid transmission to the fracture tip, while at the same time widening and propping the fracture. Chemical lost circulation treatments that form a deformable, viscous and cohesive material (e.g., FlexPlug® sealant) also may have the ability to improve the wellbore pressure containment as long as they can increase compressive stress at the fracture face.



Input Parameters

Complete the following fields:

Field	Description
D50	Enter the desired D50 of the LCM treatment.
D90	Enter the desired D90 of the LCM treatment.
Search Depth	Determines the number of possible product combinations.
Total LCM	Enter the total loss-control material needed.

Filter Rules

You can filter which products are selected by specifying the parameters for fiber, calcium carbonate, and/or graphite. Select the type(s) of filter(s) you want to use and enter the minimum and maximum percentages the product should contain.

Completion Fluids

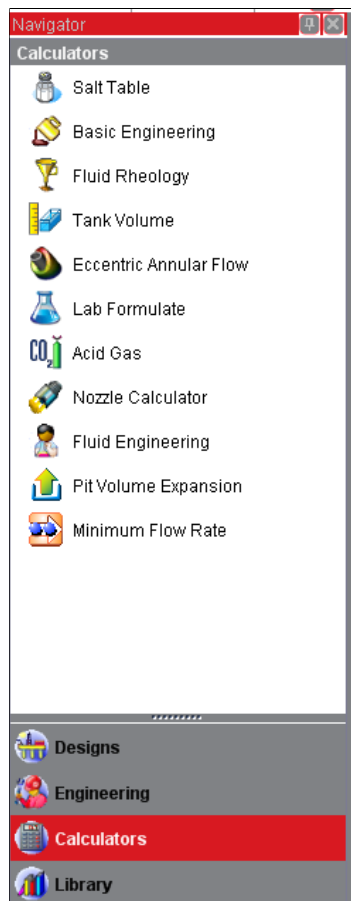
Brine Density

Brine densities of Sea Salt, Sodium Chloride, Sodium Bromide, Potassium Chloride and more can be computed with graphical output, based on certain temperatures, or temperature and pressure profiles.

Brine Crystallization

Calculations with different salts, covering True Crystallization Temperature (TGT) or Last Crystal To Dissolve (LCTD) scenarios. An NH_4Cl – TCT simulation is included also.

1.4. Calculators



This entire section provides selected solutions on basic oil field related calculations for quick daily usage and easy to handle.

Salt Table

The Salt Tables Calculator is a brine calculator. In this form, you can specify various properties of a brine such as weight percent salt, mg/liter salt, specific gravity (SG), etc., and DFG calculates all the properties of that brine. Additionally, DFG calculates the required amount of dry salt and water required to mix the solution. In the case where the salt is not pure, DFG assumes that impurity is water and automatically account for it as such.

Based on the brine you select in the Salt Properties section, DFG populates a graph and table, showing the weight, specific gravity, and salt concentration of the brine.

Basic Engineering

This tool is designed to calculate basic drilling fluid scenarios, consisting of three parts: Basic Fluid Engineering, Hydrostatic Pressure and Pump Calculations.

Fluid Rheology

Mud rheology is measured on a continual basis while drilling and adjusted with additives or dilution to meet the needs of the operation. In water-base fluids, water quality plays an important role in how additives perform. Temperature affects behavior and interactions of the water, clay, polymers and solids in a mud. Downhole pressure must be taken into account in evaluating the rheology of oil based drilling fluids.

Because most drilling fluids exhibit yield stress, the Herschel-Bulkley model describes the rheological behavior of drilling muds more accurately than any other model. Refer to Chapter 10: Rheology and Hydraulics of the Baroid Fluid Services Handbook for more information. Beside Herschel-Bulkley modeling this calculator can be used for calculating rheology models after Bingham or Power-Law.

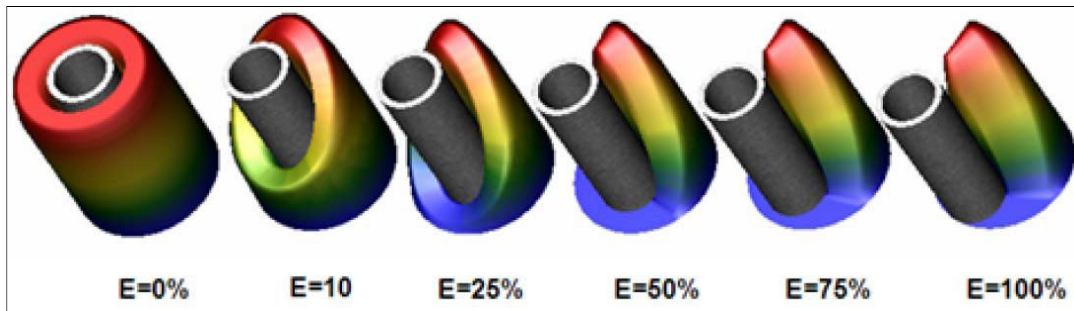
Tank Volume

This is a versatile pit volume calculation for regular and specific shaped pits.

Eccentric Annular Flow

DFG allows you to model eccentric annular flow. In this model, pressure drop is calculated as well as cuttings bed height using either a rolling or bed yield stress model 1. When DFG performs DrillAhead Hydraulics calculations, the rolling model is preferred, but DFG automatically uses the Bed Yield Stress model if it experiences convergence problems with the rolling model.

Hydraulics - As the eccentricity changes from 0 to 100 percent, the fluid will preferentially flow to the wider gap side of the annulus. Eccentricity at 0% means that the pipe is concentric to the annulus. Eccentricity at 100% means that the pipe is touching the wall of the hole. Eccentricity can have a dramatic impact on cuttings bed height as well.



Eccentricity has dramatic effect on the efficiency of sweeps, pills and cementing. In addition to where the fluid flows, it is important to know that the pressure drop through the eccentric annulus is less than the pressure drop through a concentric one. In DFG, for hydraulics purposes, we normally assume the annulus is concentric because this will produce the more conservative ECD results. The exception is when DFG calculates the bed heights for sliding in DrillAhead Hydraulics.

Rolling Bed Height -In the rolling model the cutting is modeled as a single particle. The model particle has a highly angular geometry and is modeled fixed on one end with rotation permitted. The algorithm then calculates the bed height when equilibrium of rolling or not occurs. At this point the equilibrium bed height is determined.

Generally, larger cuttings are more difficult to transport and thus create thicker cuttings beds.

Bed Yield Stress - The bed yield stress method assumes the cutting drilling fluid mixture yield stress. In this case DFG calculates the boundary yield stress and at what bed height the shear stress of the flow equals the yield stress of the cuttings bed.

Lab Formulate

Use this tool to calculate the formulation of water or oil muds. Input the desired properties and components of an oil or water mud. DFG will calculate the amount of oil, brine and weighting material to mix together to get the appropriate properties.

Acid Gas

Use this tool to calculate the acid gas contamination in carbon dioxide. You can calculate the acid gas contamination according to the concentration or the pressure volume. Selecting either method disables the other's field.

Nozzle Calculator

Advanced TFA calculations, based on various nozzle sizes and with the possibility of managing a wellsite related inventory of bit nozzles.

Fluid Engineering

This Calculator covers the basic calculations of drilling fluid engineering with tabs for fluid adjustment, ASG Oil, ASG Water and a Formulate section.

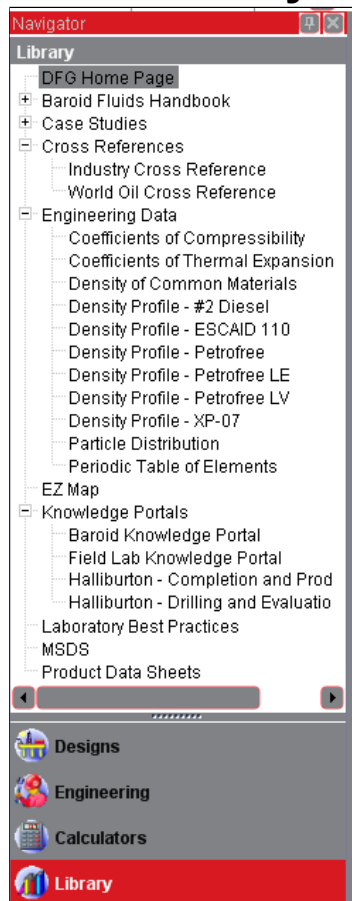
Pit Volume Expansion

Use this tool to calculate the expansion and/or contraction of drilling fluids due to changes in temperature.

Minimum Flow Rate

Use this tool to calculate the minimum flow rate (BP model) recommended for cuttings transport.

1.5. Library

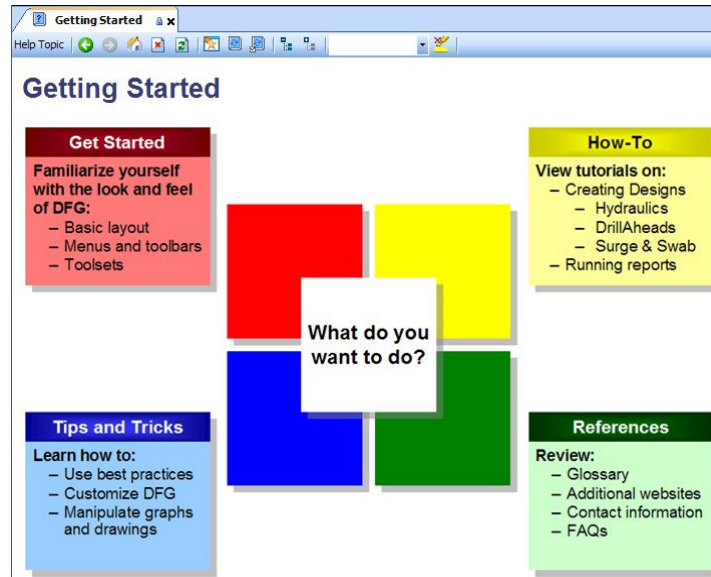


The Library section provides an universal access to most versatile support information on drilling fluid engineering when being online, such as:

- The Baroid Fluid Handbook
- Case Studies
- Cross References
- Engineering Data & Density Profiles of selected base fluids
- Coefficients of Fluids Compressibility and Thermal Expansion,
- Particle Distribution
- Basic Chemistry
- Direct access to Baroid Knowledge Portal
- Easy access to the DFG 5 support share point site
- Laboratory Best Practices
- Material Safety Data Sheets - Search Tool
- Product Data Sheets

1.6. Online Help

A very detailed online help is available for the DFG 5 application. Press the F1 button to launch the help files while DFG is open.



The online help contains detailed, current information on creating designs, using calculators, and running reports.

Digital Solutions

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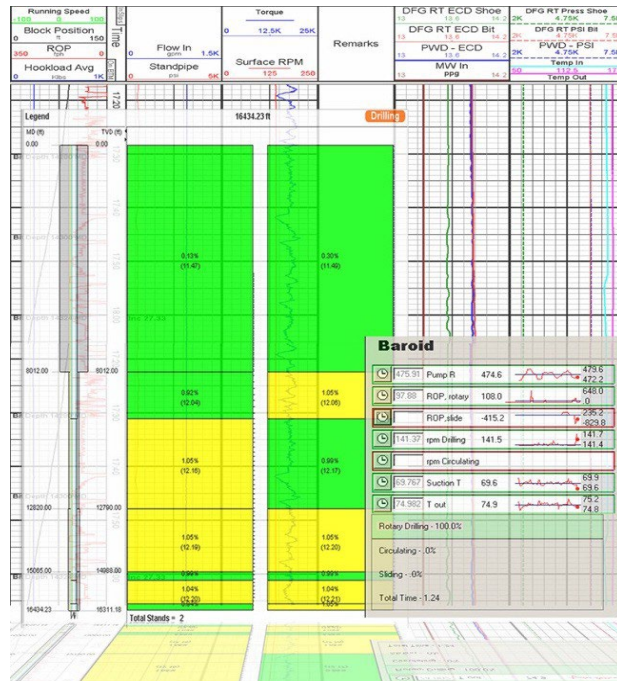
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1. Applied Fluids Optimization (AFO)

Baroid's Applied Fluids Optimization (AFO) optimizes the drilling process utilizing our digital tools and workflows to avoid impending downhole issues. AFO is comprised of two services; AFO Modeling and AFO Monitoring.



AFO Modeling Service

AFO modeling provides fluids expertise and engineering tools in the collaborative well planning process. This service adds fluids knowledge and technology to client drilling teams to improve front end planning and overall well design. Modeling and consulting services use the Baroid planning software such as DFG, WellSet, P Bore 3D.

AFO Monitoring Service

AFO monitoring combines fluid experts, workflows and technology to monitor drilling activity and fluids performance in real time. The service uses DFG RT™ integrated with INSITE and INSITE Anywhere.

AFO utilizes a modeling and a monitoring service in order to accomplish our goal in assisting our customers where is counts, downhole.

With better planning and real time optimization of fluids as part of the overall drilling effort AFO can reduce NPT, lower drilling costs and accelerate time to first production. This is possible through AFO's proactive use of the digital environment:

- **Model:** Better collaborative planning on the front end enables the team to design around problems, reduce risk and increase planning efficiency.

- **Measure:** The combination of real time drilling practice data, predictive models and workflows enable Baroid to identify and validate emerging problems, predict upcoming drilling events and provide intervention recommendations.
- **Optimize:** Collaborative well planning allows better choices to be made before drilling starts. Dynamic monitoring allows (1) decisions to be made early enough to avoid or mitigate emerging problems (2) optimize drilling practices to take advantage of opportunities presented by the actual drilling environment.

AFO Services is able to enhance the client's existing operations by providing recommendation to optimize the drilling process, detecting significant events, and also by providing options to mitigate the events before the turn into problems.

AFO Monitoring Specialists

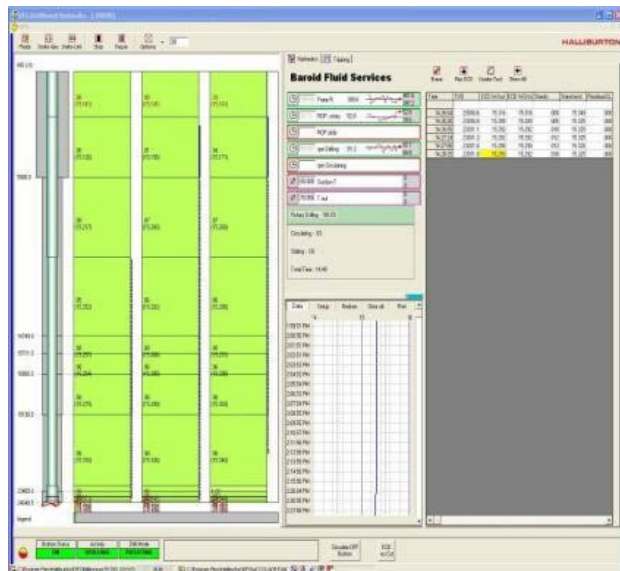
The AFO specialist, typically at a customer's real time center or a Halliburton Real Time Operation Center (ROC), will monitor and analyze DFG RT data for up to three rigs, making fluids optimization recommendations. The specialist serves as an interface between the operations team, fluid engineers at the rig, Baroid's technical professionals and INSITE system support.

An AFO Specialist is expected to be at or near the same competency level of a Technical Professional. Therefore, it is recommended that AFO Specialists be recruited from senior field service representatives (FSR III and higher) or Technical Professionals. The service is to provide drilling practice recommendations from a fluids perspective, so an AFO Specialist must have a very strong knowledge of both drilling fluids and sound drilling practices.

AFO Software

DFG RT

DFG RT is a module in the DFG 4.5.x software application. It is the real time implementation of the DFG DrillAhead Drilling Simulator. The simulator will calculate the downhole pressures due to cuttings loading and frictional pressure losses. It provides more than just a model comparison of ECD to PWD. It provides cuttings transport and hydraulics modeling several stands ahead of the bit by analyzing the current drilling process and simulates drilling ahead of the bit.



DFG RT ECD predictions are typically within 0.02 ppg and 0.1 ppg of PWD measured ECD. Deviations between calculated ECD and PWD ECD outside of this accepted window usually indicate an unanticipated problem downhole that can then be acted upon to prevent any further impacts on the operation.

DFG RT Data Routing Scenarios

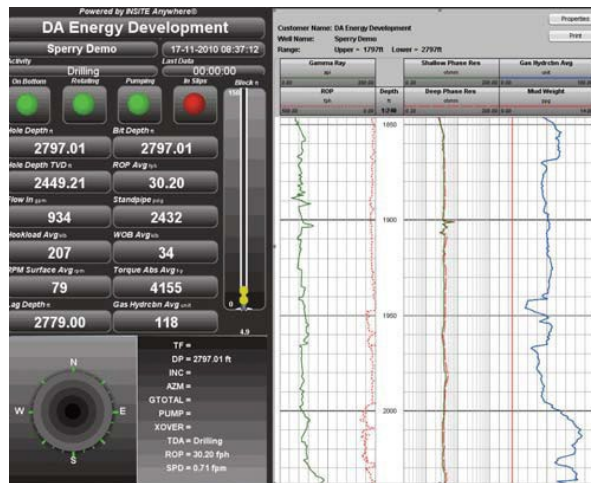
Scenario	Description
Rig=>RTOC=>IA Server & ROC w/ DFG RT VMS	Data Exchange starts at the rig, then goes to EORTOC, with DFG RT run on a VM Server pointed at the receiving EORTOC server, then to an IA server and the ROC.
RIG=>Customer RTOC=>EORTOC=>ROC w/ DFG RT run in Customer RTOC	Data is routed from the rig to the customer's RTOC, then to the EORTOC, then to the Lafayette ROC with DFG RT being run in the customer's RTOC/Office.
Rig=>ROC=>EORTOC=>IA Server	Data is routed from the rig to the ROC, (with DFG RT workstation pointed at the ROC server), to EORTOC, then to Insite Anywhere server. This is a low-cost approach when DFG RT data needs to be displayed in IA/IAD.
Rig=>Customer RTOC=>ROC & ADT Workstation in Customer's Office	Data is routed from the rig, to the customer's RTOC, then from there it is routed to ROC and IA server and a second stream is routed to a workstation in the customer's local office through the customer's network. There was also an Insite workstation installation made on the rigsite supervisor's computer pointed at the ADT workstation back in the office. This allowed the Co. Rep to view DFG RT ECD on the rig. This scenario was used in the DFG RT North Sea trials with ConocoPhillips.
Rig(WITS)=>DFG RT Workstation=>DFG RT IA Server	Rig-in-a-box deployed on rig sends DEX to DFG RT Workstation and DFG RT workstation sends DEX to DFG RT Group-maintained Insite Anywhere server.

INSITE Integrated System for Information Technology and Engineering.

INSITE® data management service can be a key enabler, allowing drilling and other relevant rigsite data to be collected, transmitted, replicated and managed in real time.

The INSITE service is a common platform for replicating wellsite data generated and acquired by Sperry. The wellsite instance of the INSITE service communicates with a remote-site INSITE system server, creating a duplicate database which can be used for remote-site analysis, as well as for providing data to a corresponding INSITE Anywhere® web server.

INSITE Anywhere



INSITE Anywhere® service was the first to provide quick, easy and secure access to real-time well information via a web browser. Today, as part of Halliburton's real-time strategy, we are still leading the way. You get easier access, simpler operation and more capabilities... the capability to share knowledge with every member of your team and to integrate all of Halliburton's solutions throughout the life of your well.

Among the key benefits of our newest version of INSITE Anywhere service are:

- Improved security
- Greater flexibility
- More functionality

Typical INSITE Anywhere displays used for the AFO specialist are shown below.



Figure 1 AFO Drilling Activity Screen



2. Real-Time Density and Viscosity (RTDV)

The Real Time Density and Viscosity (RTDV) instrument is a fully automated unit that measures the density and rheological properties of drilling fluids. The RTDV also has LAN connections and software for obtaining, sorting and viewing data. Automating routine fluid property tests provides access to more precise and dependable drilling fluids properties in real time and data that becomes vital when crucial decisions are required.

Critical decision making occurs daily in the oilfield, and operators depend on accurate drilling fluid data to determine the appropriate course of action for the success of their drilling program.

Theory of Operation

The RTDV instrument is an automated unit that measures the density and rheological properties of a drilling fluid. It is designed to reside at the drilling rig location near the fluid tanks.

The unit has three valves: V1, V2, and V3. These valves control fluid flow in and out of the unit and through tubing that contains the density sensor and the viscometer. Drilling fluid flows from the fluid tanks to the RTDV through a connection for mud supply and flows through the internal stainless steel tubing to the pneumatic 2-way valve, V2. The V2 valve is usually in the open position, allowing the fluid to flow to other tubing sections. Closing the V2 valve is necessary only when multiple RTDV units are used and share a common fluid supply line. Closing the V2 valve stops fluid from entering one RTDV and forces full fluid flow to the next RTDV in line. Closing the V2 valve stops fluid from entering one RTDV and forces full fluid flow to the next RTDV in line.

Density Measurement

Under normal flow conditions, the fluid flows past the V2 valve, and is blocked from the viscosity flow loop at the pneumatic, 2-way valve, V1. The V1 valve position is normally closed. This diverts the fluid into the tubing section which contains an online density sensor as shown below. As the fluid flows through the density sensor, its density and temperature are measured and the data is recorded in the RTDV control computer. The fluid flows past the density sensor and enters the pneumatic, 3-way valve, V3. One port of the V3 valve is closed to fluid flow in the viscosity flow, and the other ports are open to fluid flow for the density flow loop, allowing flow through the valve and out of the RTDV.

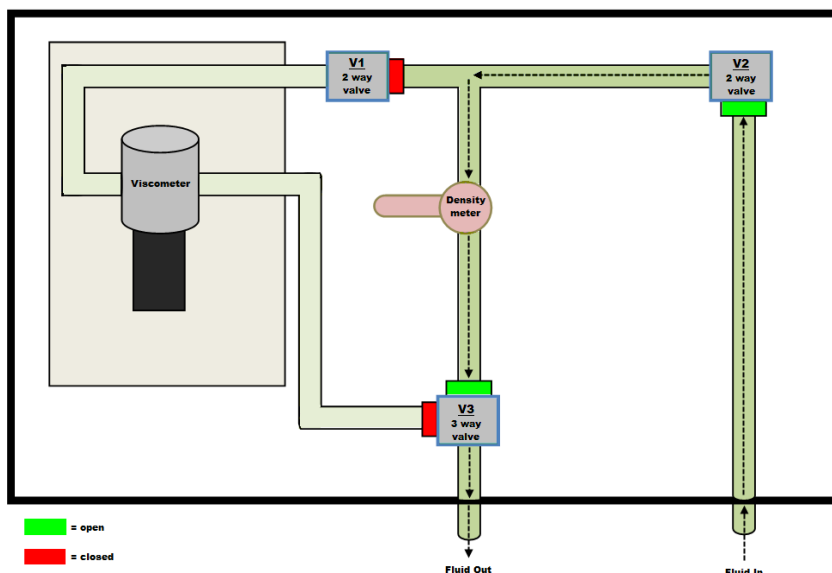


Figure 3 RTDV Density Flow Loop

Viscosity Measurement

To measure viscosity, the fluid flow is diverted from the density flow loop into a secondary tubing loop that can be isolated and pressurized. This secondary loop is the viscosity flow loop and includes an in-line viscometer.

Fluid flows into the viscosity flow loop through the actuating valve, V1. Concurrently, the V3 valve is pneumatically actuated to block flow from the density flow loop and permit fluid flow from the viscosity flow loop through the V3 valve and out of the RTDV as shown below. Fluid flows through the viscosity flow loop, and fluid fills the viscometer measurement chamber. By using the RTDV computer interface, the user can set and adjust the time that the fluid flows through the viscosity flow loop. When the time expires, flow reverts to the density flow loop.

The remaining fluid in the measurement chamber is effectively isolated within the viscosity flow loop. It is pressurized to approximately 80 psi (551 kPa) in order to collapse any entrained air bubbles. Then the fluid is agitated and heated to the user defined temperature, typically 120°F/150°F (49°C/66°C). Heating is accomplished by dual 100 watt cartridge heaters located in an aluminum block. This aluminum block contacts the viscometer's measurement chamber and allows heat transfer from the cartridge heaters to the chamber holding the sample.

When fluid temperature in the viscometer chamber is reached, the viscometer measures the shear stress at six speeds. The American Petroleum Institute (API) recommends speeds at 600, 300, 200, 100, 6 and 3 rpm. After a rheology measurement, the flow is again diverted from the density flow loop into the viscosity flow loop. This new flow of fluid flushes the previous fluid sample from the measurement chamber. Air from the RTDV pneumatic system is injected into the measurement chamber for one to five seconds to also flush out the previous fluid sample. After the viscometer is filled with a new drilling fluid sample, the process is repeated.

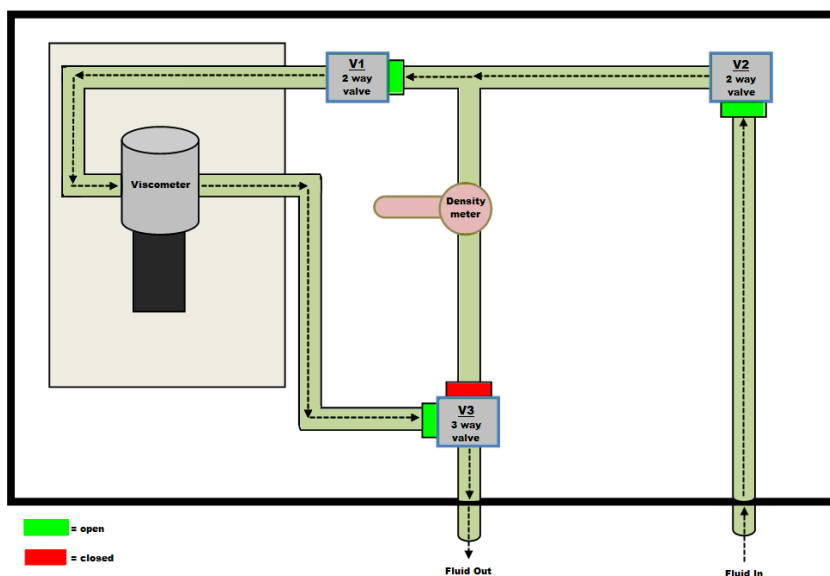


Figure 4 RTDV Viscosity Flow Loop

Safety

Positive Pressurization Purge System

In order to permit RTDV operation in a hazardous location, the RTDV is equipped with a positive pressure purge system. Typically, the RTDV is purged with an inert gas (nitrogen) and the purge control unit continually

monitors the interior of the RTDV enclosure for pressure greater than external, atmospheric pressure. This over-pressurization is generated by continuous flow of inert gas into the RTDV enclosure. In the event of a loss of internal over-pressurization, the purge control unit will turn off electrical power to the RTDV. Over-pressurization must be reestablished in order to return electrical power into the RTDV. In the event of a power shut down, or a loss of purge gas supply, a spring loaded valve will shut off fluid flow into the RTDV. The RTDV purge control unit is equipped with an exhaust port which vents the purge gas from the RTDV enclosure. If the RTDV is installed in a closed environment or a confined space, the exhaust port **MUST** be routed as to direct the exhausted purge gas to vent in an open air environment.

Specifications

Table 1 RTDV Specifications

Category	Specification
Power Requirements	110/220 VAC, 50/50 Hz
Power Consumption, Maximum	6 amps
Air Requirements	80 -180 psi (551-689 kPa)
Nitrogen Requirements	80 -100 psi (551-689 kPa) 19.5 L/min
Purge Time, Minimum	80 minutes
Operating Temperature	-20oC – 50oC oF - oF
RTDV Dimensions (Length x Depth x Height)	41.24 x 13.96 x 43.22 inches x x centimeters
RTDV Weight	326 lb (xx kg)
RTDV Enclosure Material	Type 316L Stainless Steel (enclosure & fasteners) Enclosure and attachments contain < 7.5 wt % magnesium or titanium.
ATEX Rating	Certified to Standard IEC 600079-0:2007, 5th Edition. II 2 G Ex px IIC T4 Gb

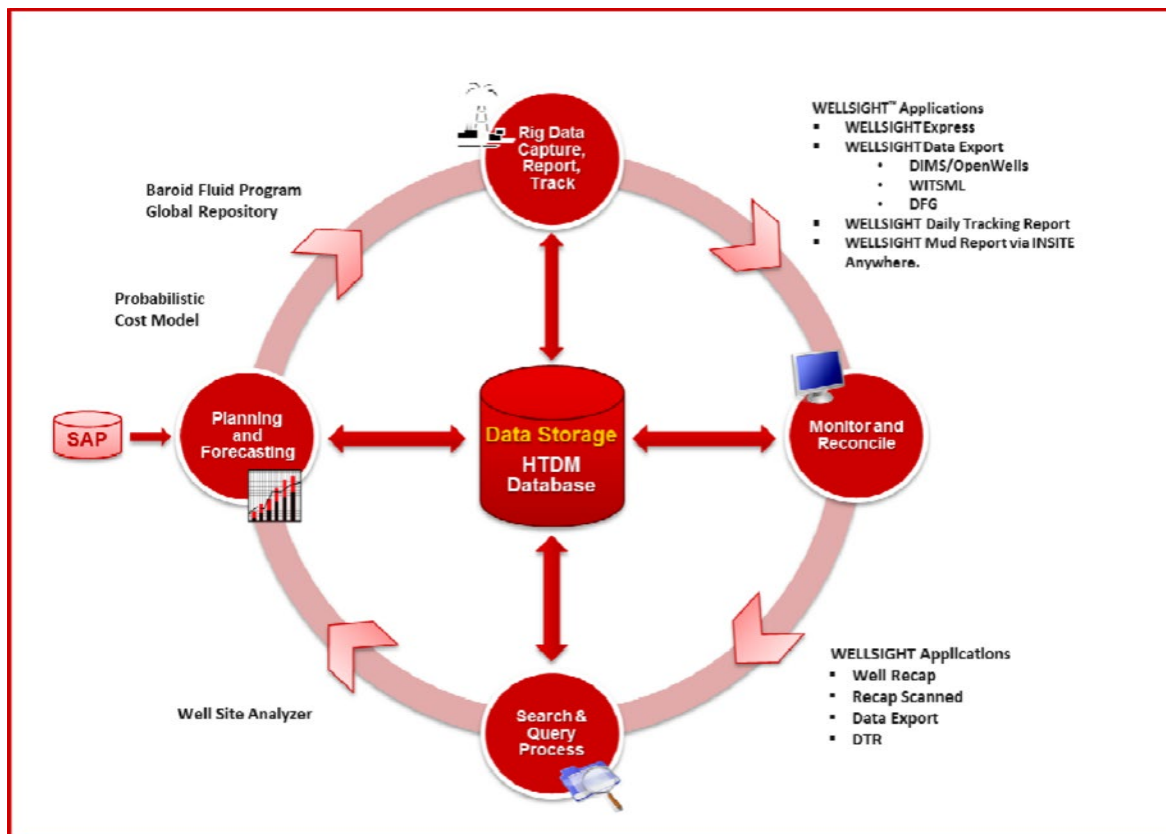
Installation

The RTDV is installed at the rig site, near the mud tanks, and can be skid or wall mounted. The unit is ATEX certified for zone 1 use. Zone 1 installation requires the use of an inert gas for the RTDV positive pressure purging system. A self-generating nitrogen source is recommended for the purge gas supply. The nitrogen generator should be installed adjacent to the RTDV with sufficient space between the units to connect the nitrogen, fluid and compressed air lines

3. Data Management

The mission of the Baroid Data Management system is to provide software, applications and IT tools that facilitate the 'cradle to grave' tracking of all stages of the fluids service related data with planning, capturing, organization, analysis, evaluation, benchmarking and value creation functionality.

The output documentation is to focus on internal and external performance presentation and in clear and concise graphical format, wherever feasible.



Capabilities	Applications:
- Planning and Forecasting - Data Capture, Report, Track, Monitor and Reconcile. - Data Export - Data storage - Search and Query	- WELLSIGHT™ Express - Well Site Analyzer - Baroid Probabilistic Cost Model

3.1. WELLSIGHT™ Express

Data Capture, Reporting, Monitoring, Tracking, Reconciliation and Export

WELLSIGHT is a software application utilized to manage the technical information captured during Baroid field operations.

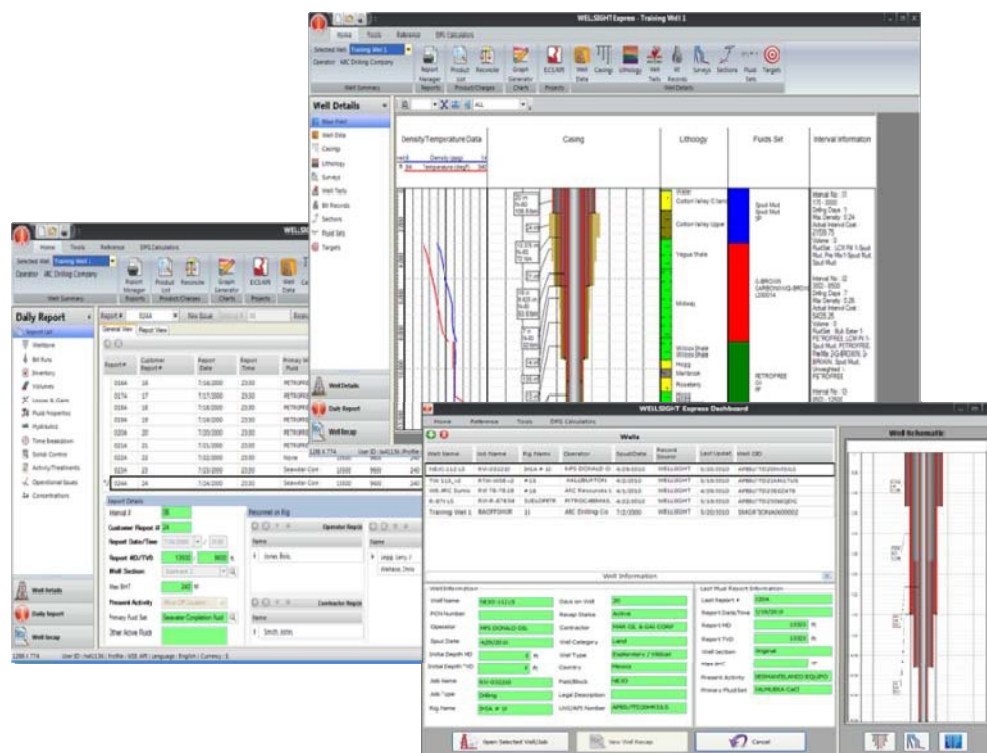
The information is collected and entered by drilling fluids engineers from the different operations in the field and is continuously transferred Halliburton office locations to be monitored, reconciled, and stored on a secure database in Houston, Texas.

WELLSIGHT data entry programs have been around since 1990, and they have gone through many transformations in order to adapt to with the operation of Baroid in the field and the advancement of technology.

WELLSIGHT was introduced under the concepts of Right information, Right Place, Right format, and Right time and we will continue to apply this philosophy throughout the lifecycle of this product.

The last version of this application is **WELLSIGHT Express**.

WELLSIGHT Express allows the user to record and track all relevant drilling fluids information about the well on a daily basis. The system is designed to allow the field engineer full access to relevant past well histories at the rig site and to have data and reports sent via email and also using INSITE Anywhere.



Benefits

- Toggles between multiple input sections easily
- Simplifies printing graphics and saving them to file

- Improves data entry time
- Handles complex field scenarios with ease
- Provides information to aid in decision making
- Presents the ability to globally capture, store, share, and graph Baroid well data
- Standardized high-quality reports and graphics on a global level
- Captures issues and actions data better
- Prepares **WELLSIGHT** for future integration with other Halliburton strategic business applications and decisions (i.e. SAP, INSITE, etc.)
- Responds faster to customers
- Improved employee productivity
- Has fewer technical limitations

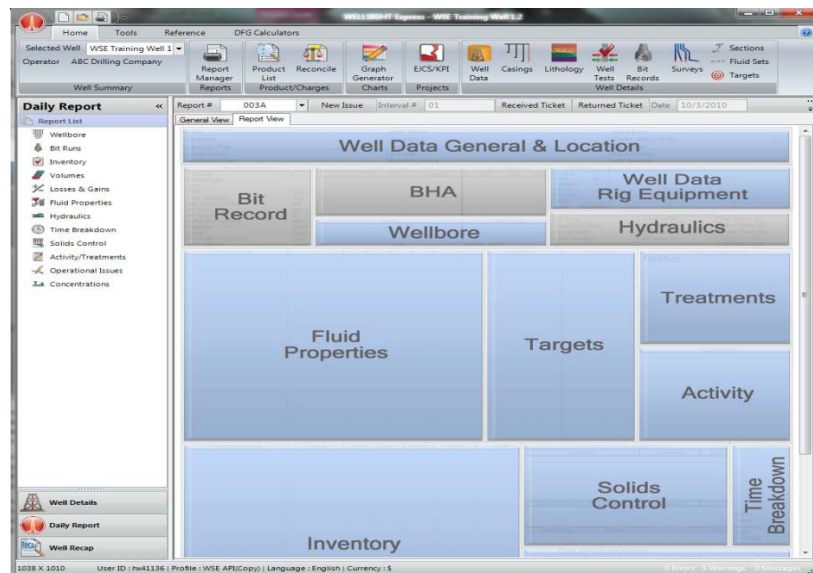
Daily Information and End of Well Recap

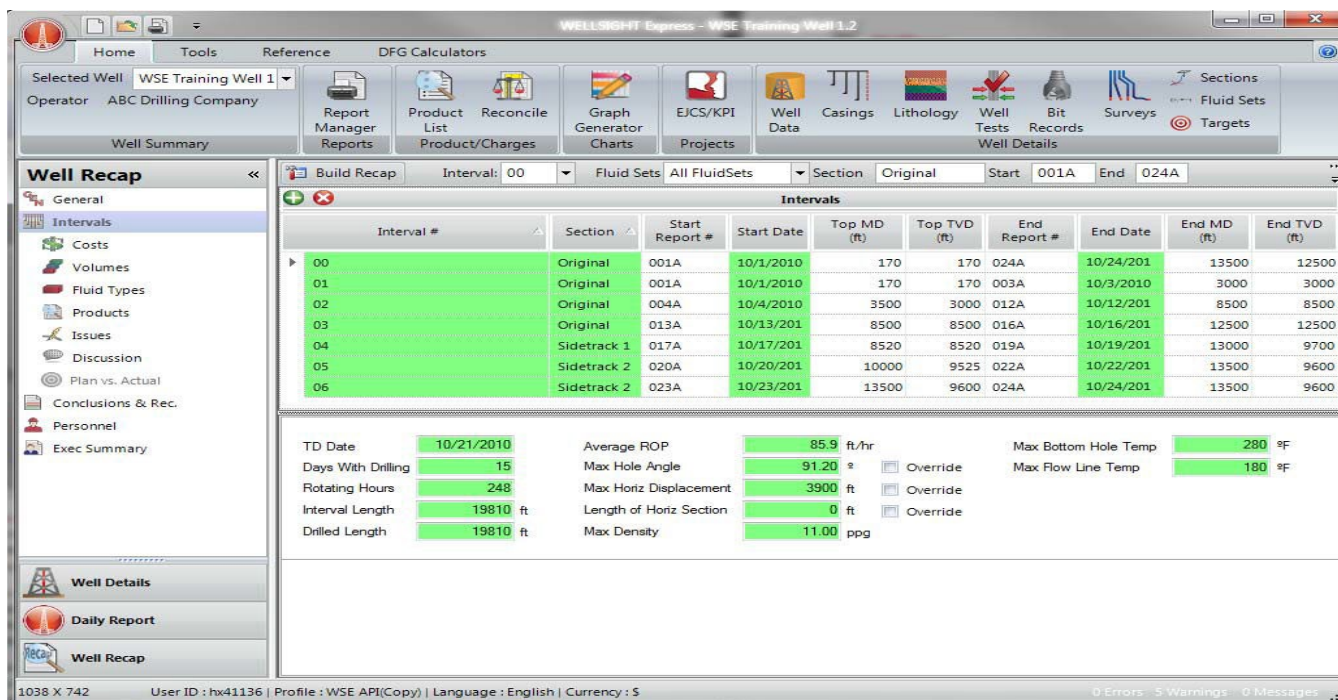
Information is recorded on daily basis, includes Lithologies, Surveys, Wellbore Geometry, Bit and BHA information, Mud Property checks, Volume accounting, hydraulics calculations, Inventory tracking, cost allocation, solids control equipment, chemical concentrations, and treatment discussions.

Daily Reports in any of the above topics are available which could be shared with the parties involved for easy decision making.

Information captured during the entire well is classified identifying well intervals based on the drilling activity in order to produce the end of well recap.

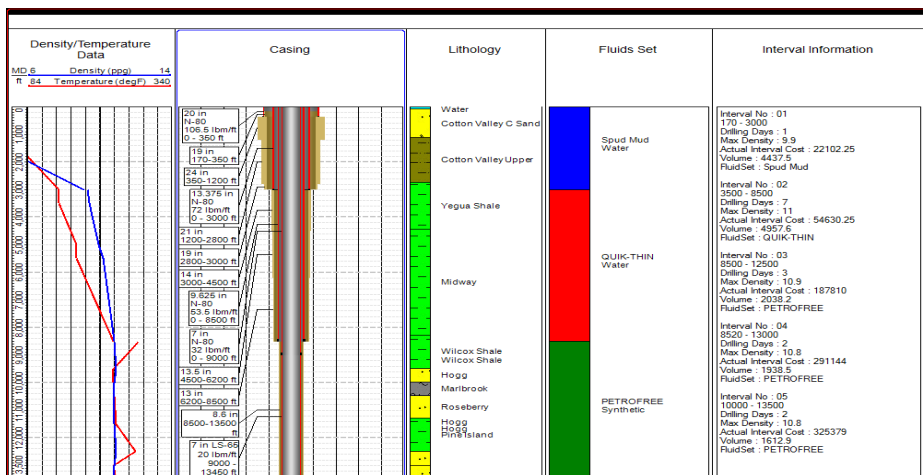
After a final review of the information in seconds, **WELLSIGHT** generates the recap report calculating volumes, costs, product consumption, fluid types used in any of the defined intervals. A comprehensive and accurate set of reports are available to be part of the recap report based on the complexity of the well.





Generator for Trend Analysis

The graph designer allows both standard and customized graphs to be built. Multiple parameters including the lithology column can be displayed together to aid in visual trend analysis.



Report Manager

The system is able to provide both field and office reports. Field engineers can provide both daily and summary reports as well as draft copies of the Post Well Audit (Recap). Specific customer orientated reports can be generated from the database for multi well analysis as required. In conjunction with the customer, Baroid is able to design formal reports of performance measurements that can be printed as required including daily, monthly, quarterly and yearly.

Features of Report Generator

- 5 Languages reports (English, Spanish, Portuguese, German, and Russian)
- Reports can be generated in the following formats: PDF, Word, Excel, RTF
- Globally standardized high quality reports and graphics
- Customer Reports capabilities
- Template managements
- Customize report setting

Tracking, Monitoring and Reconciling

WELLSIGHT was developed to capture and filter the necessary technical data for better decision making. WELLSIGHT provides the right information, in the right place, the right format, and the right time - as WELLSIGHT is used day in and day out. It will grow with well information and the tighter the cycle will become, reducing risks and providing valuable right time solutions.

Other Material Charges	
Description	Cost (\$)
Drilling Cost	0.00
Completion Filterate Cost	0.00
SCE Waste Cost	0.00
Lost Damaged Cost	0.00
Cement Cost	1187.50
Other Cost	0.00
Total Other Material Charges in Fluid...	0.00
Total Other Material Charges Not in...	1187.50
Total	1187.50
Charges	
Drilling Fluids Engineer	13200.00
Engineering Laboratory	1200.00
Second Fluids Engineer	3600.00
Total Charges in Fluid Cost	0.00
Total Charges Not in Fluid Cost	18000.00
Total	18000.00
Fluid Cost per vol of fluid	70.35 \$/bbl
Fluid Cost per drilled ft	44.10 \$/ft
Fluid Cost per vol of hole drilled	288.88 \$/bbl

Fluid Material Charges	
Description	Cost (\$)
Previously Uncharged Carry Over	
Later Interval Chargeback	0.00
Mixed Products	171718.00
Charges to Fluid Set	0.00
Whole Fluids	701905.00
Total Fluid Material Charges in Fluid Cost	873623.00
Total	873623.00
+ Total Fluid Material Charges in Fluid Cost 873623.00 \$ + Total Other Material Charges in Fluid Cost 0.00 \$ + Total Charges in Fluid Cost 0.00 \$	
(=) True interval fluid cost 873623.00 \$	
(=) Actual interval fluid cost 873623.00 \$ + Total Fluid Material Charges Not in Fluid Cost 0.00 \$ + Total Other Material Charges Not in Fluid 1187.50 \$ + Total Charges Not in Fluid 18000.00 \$	
(=) Actual interval cost 892810.50 \$ Planned Costs \$ Planned Fluid Costs \$	

Performance indicators include Cost/bbl, Cost/ft, Cost/bbl hole drilled, bbl/ft, bbl/ bbl hole drilled, bbl lost surface/ft, bbl lost downhole/ft, losses/bbl hole drilled on a defined recap interval for a defined fluid set and for the entire well. Well costs can be broken down and separated to ensure an accurate cost/bbl is obtained excluding charged items not mixed into the fluid set.

The Inventory Reconciliation form is used to update the cost or movements of inventory items for a selected well normally done before completing the well recap. Tracking fluid movements and volumes can become very difficult. WELLSIGHT will track volume additions and losses and categorize per specified destination. Downhole losses or gains are linked back to the lithology application. All fluid movements and net volume changes are calculated and reported in the daily fluids report.

Reconciliation

Chemical Products | Other Consumable | Whole Fluids | Charges

Product: **BAROID - 100 LB BULK** List Price: Discount Price:

☒ Display List Price ☐ Display Discount Price

Apply Prices from Pricelist for Current Product Apply Prices from Pricelist for All Product

Report #	Report Date	Start Qt	Receive Ticket #	Receive Qty	Receive List Price	Receive Discount Price	Used Qty	Return Ticket #	Return Qt	Report Unit	Returned List Price	Returned Discount Price	End Qty	Report Cost
001A	10/1/2010 11:30:30 PM		SO-001	1000	7.00	7.00				EA	0.00	0.00	1000	
002A	10/2/2010 11:30:00 PM	1000			0.00	0.00				EA	0.00	0.00	1000	
003A	10/3/2010 11:30:00 PM	1000			0.00	0.00	60			EA	0.00	0.00	940	420
004A	10/4/2010 11:30:00 PM	940			0.00	0.00	300			EA	0.00	0.00	640	2,100
005A	10/5/2010 11:30:00 PM	640			0.00	0.00	500			EA	0.00	0.00	140	3,500
006A	10/6/2010 11:30:00 PM	140	SO-004	2000	7.00	7.00	600			EA	0.00	0.00	1540	4,200
007A	10/7/2010 11:30:00 PM	1540			0.00	0.00	700			EA	0.00	0.00	840	4,900
008A	10/8/2010 11:30:00 PM	840	SO-007	1000	7.00	7.00	400			EA	0.00	0.00	1440	2,800
009A	10/9/2010 11:30:00 PM	1440			0.00	0.00	325			EA	0.00	0.00	1115	2,275
010A	10/10/2010 11:30:00 P	1115			0.00	0.00	800			EA	0.00	0.00	315	5,600
011A	10/11/2010 11:30:00 P	315			0.00	0.00	250			EA	0.00	0.00	65	1,750
012A	10/12/2010 11:30:00 P	65	SO-010	2000	7.00	7.00				EA	0.00	0.00	2065	
013A	10/13/2010 11:30:00 P	2065			0.00	0.00	675			EA	0.00	0.00	1390	4,725
014A	10/14/2010 11:30:00 P	1390			0.00	0.00	500			EA	0.00	0.00	890	3,500

Apply OK Cancel

Data Export

WELLSIGHT™ Data Export has different options to extract data from WELLSIGHT™ database present on the local machine in the following industry standard formats or customer specific formats:

DIMS/OpenWells Export

Exports the Drilling fluids related data into a specifically formatted file (Landmark's Data Management applications); data is coming from wells present in the WELLSIGHT™ local machine's database. Includes the following data objects:

- Mud Properties
- Volumes
- Inventory

WITSML™ Export

WITSML™ (Wellsite Information Transfer Standard Markup Language) is an oil industry initiative that set a new standard for drilling information transfer. The Halliburton **WITSML™ Export application exports in WITSML™ version 1.3.1**. So you can deliver this information to your customers and co-workers applications. Provide these files to drilling data management applications users like **WellView**, **OpenWells**, or other that are compatible with **WITSML™** version 1.3.1 to import drilling fluids data. **WITSML™ Export** exports the data objects in **WITSML™** format (*.xml).

Includes the following data objects:

- Wells
- Wellbores
- Fluids Reports
- Inventory
- Pit Volumes
- Mud Volumes
- Mud Pumps
- Shaker and shaker screens

- Discharges
- Personnel

Daily Tracking Report

Daily Tracking Report, exports the Drilling fluids related data into a specifically formatted excel file, data is coming from wells present in the WELLSIGHT™ local machine's database.

Provide these excel files to a customer who needs to see this data.

Purpose

Use Daily Tracking Report to quickly generate reports in Microsoft Excel (*.xls) format that can be delivered to customers and coworkers.

Data Exchange

Exchanging data with the **WELLSIGHT Express** is the process by which well files are uploaded to the **WELLSIGHT** server centrally located in Houston, TX. In addition, new reference table data such as product information and office locations is also uploaded. In order to initiate this process the user level registered in the application must be Technical Representative or higher.

Note: To be able a user to send file to the server, needed a Technical Professional level or higher, are granted permission to post to the server. The user's password file needs to be at a Technical Professional or higher prior to proceeding. If this level is not in place the user will not have the "Upload wells to the server" option listed.

Data Storage

A complete database of all of the information from all of the wells on which Baroid provides technical services and products is of benefit because.

- **Technical Resource.** A database of all of the issues encountered on all of the wells we have worked on and how we have overcome them. When proposing a technical solution to a customer it is useful to be able to highlight examples of how similar problems have been overcome previously.
- **Commercial Advantage.** Ability to provide better technical solutions we can sell our services better. Ability to provide the customer with evidence of our successes we can sell our services better – if you could for example, in five minutes, generate a list of all wells globally on which STEELSEAL has been used to cure losses where the differential pressure was greater than 5000 psi the customer is more likely to accept our proposed solution and use purchase

When putting together a well listing for a tender – if you can only find 14 wells on which a Sat Salt mud has been used that is all you can put in the tender response – if we have in actual fact worked on 50 Sat Salt jobs we are selling ourselves way short!

In order for this to work it is critical that everybody involved in the process does their part. Firstly, it is essential that all of the daily information is entered correctly at the rig site – not just what appears on the reports that are being handed in to the operator – all of the information from that well WILL BE USEFUL to someone in Baroid at some time. Secondly, it is essential that the recap is completed and reviewed properly by the field and office personnel. Finally the well needs to have been uploaded to the server in Houston.

A significant advantage of the WELLSIGHT™ database is improved integration between WELLSIGHT™ applications. For example, WELLSIGHT™ data allows for improved plan vs. actual comparisons and complete store of design iterations from prototype to Plan to actual.

Licensing and Installation

Licenses

User can request licenses via TSOders or by calling the Help Desk (IT Services Center)

Machine License

Machine licenses authorize the use of a software package on a specific machine. Users need machine license for each application.

User Licenses

User licenses authorize individuals to access the software that is installed on a machine within their specific security level. Users need license for each application (tied to user's HAL id)

Installation

- Installation is available via SMS (Software Management System)
- ISO image also is available via TSOders. You can request to burn CD/DVD with the installation.

Note Complete information about the installation process and Licenses is available in the **WELLSIGHT Express Installation Notes**. The installation notes are available in the Baroid Data Management SharePoint Site:

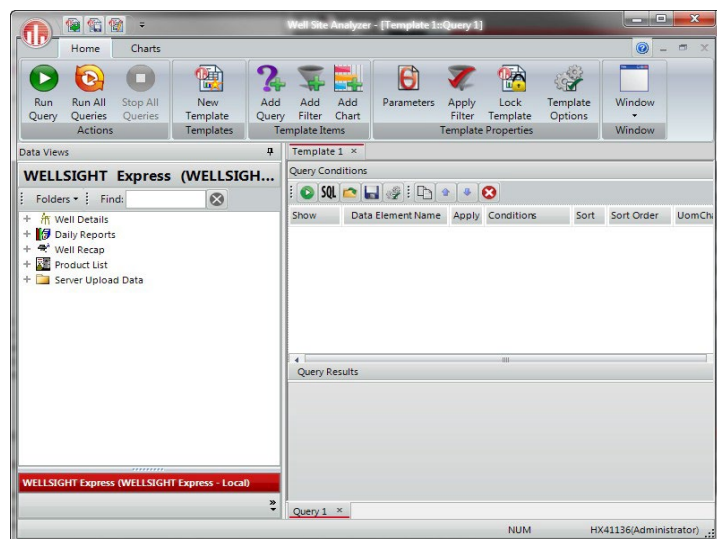
<http://sphou/sites/KMBaroid/DataMgt/default.aspx>

Note: See full instructions Manual in the Baroid Data Management SharePoint site:

<http://sphou/sites/KMBaroid/DataMgt/default.aspx>

3.2. Well Site Analyzer Advanced Search and Query Tool

Well Site Analyzer is an advanced search and query tool that provides efficient access to all of the data that is captured in your database. This application provides users a quick and easy access point to data and provides a method to display that data graphically for the benefit of research and business development.



This second version of **Well Site Analyzer** introduces many new and exciting improvements that are designed to assist you. A few of these improvements included are:

- Employs a more stable development platform that allows for easier expansion and maintenance.
- Uses a familiar, **Microsoft® Office 2007**-like user interface that should be familiar with a broad spectrum of users.
- Improves upon the previous query engine and delivers faster query results.
- Provides information to aid in decision making
- Responds faster to customers
- Improved employee productivity
- Enhanced graphics
- Integration between Well Site Analyzer and WELLSIGHT™ Express
- Integration of WS SQT capabilities

Important – If you would like to perform queries on a local database, you must have the HTDM database installed and configured on your local machine. The HTDM database is used with WELLSIGHT™ Express and HIPS|SUMMIT and is installed and configured when you install either of these products.

This software is available to any employees with Halliburton network access.

You will be required to register for this product.

Licensing and Installation

Licenses

User can request licenses via TSOders or by calling the Help Desk (IT Services Center)

Machine License

Machine licenses authorize the use of a software package on a specific machine. Users need machine license for each application.

User Licenses

User licenses authorize individuals to access the software that is installed on a machine within their specific security level. Users need license for each application (tied to user's HAL id)

Installation

- Installation is available via SMS (Software Management System)
- ISO image also is available via TSOders. You can request to burn CD/DVD with the installation.

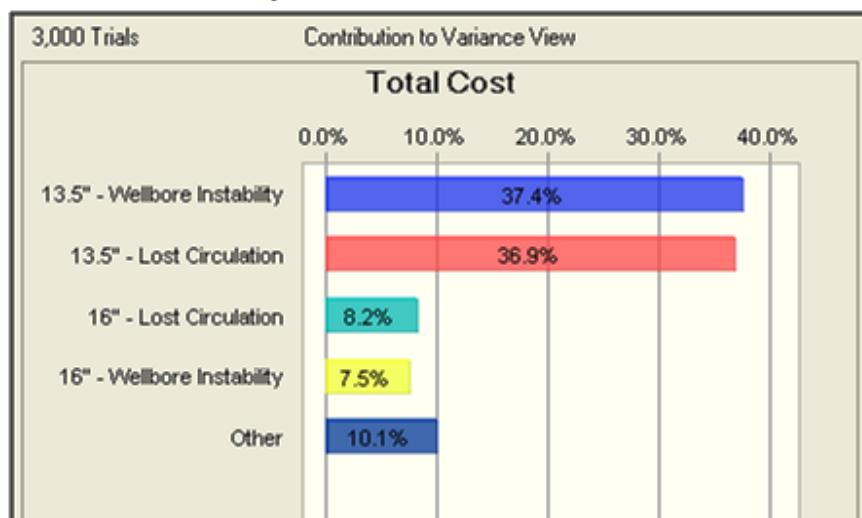
Note Complete information about the installation process and Licenses is available in the **Well Site Analyzer Installation Notes**. The installation notes are available in the Baroid Data Management SharePoint Site: <http://sphou/sites/KMBaroid/DataMgt/default.aspx>

Note: See full instructions Manual in the Baroid Data Management SharePoint site:
<http://sphou/sites/KMBaroid/DataMgt/default.aspx>

3.3. Baroid Probabilistic Cost Model Risk Analysis & Cost Forecasting

Baroid Probabilistic Cost Model is a risk analysis method and forecasting tool used to quantify major risk during the wellbore construction and with this we can provide a more accurate and realistic cost in the planning of the Drilling Fluids Jobs.

Total Cost Sensitivity Chart



Risk analysis is the science of risks and their probability and evaluation. Give Halliburton and the customer a heads up on potential problems likely to be encountered during the drilling process. Having that information means they can approach these areas more cautiously during normal operations.

End result is a better and more accurate operational analysis by interval; more realistic cost estimates and is time saving, easy and fast use. Customers expect a higher level of interaction and proactive participation from us in managing their major investment decisions.

Justifying investment cost using a probabilistic forecasting solution is standard practice for some of Halliburton's biggest customers. Some customers have been quoted as saying that every major investment decision made for wells is run through a forecasting solution to determine the risk involved.

The model utilizes historical data from the WELLSIGHTTM Database to help the user identify potential risk after narrowing down the data. Once all risks have been identify, the model will use Decisioneering's Crystal Ball software which utilizing the Monte Carlo simulation, to generate the cost forecast for the well and by interval.

Note: A part from HIPS, the Crystal software is required to be on the machine before you can run probabilistic cost models for proposals.

The model provides the guidelines shown below.

Cost Percentiles

As desired (i.e. P10, P50, P90) by interval and total well cost. These statistical results of the simulation are uniquely determined and correspond to three parameters, the lowest possible cost, the most likely cost and the highest possible cost so the well can be tracked based on them against the actual numbers. Normally the P50

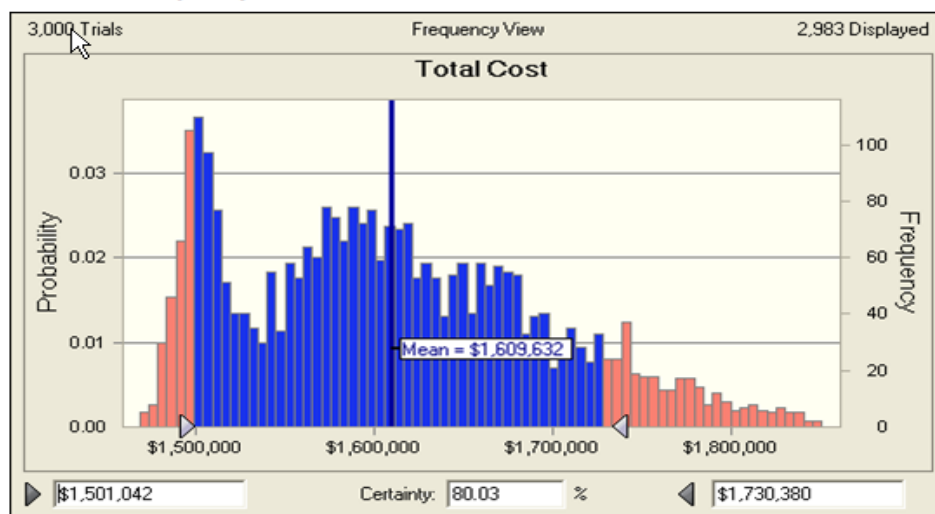
should be the value to take into account as the “drilling fluids program value” or lower once it has been optimized after a number of wells.

Cost Percentiles				
	P Low	P Mid	P High	
	10	50	90	Mean Std Dev
17.5" Total	\$0	\$0	\$0	\$1,483 \$9,642
12.25" Total	\$90,685	\$90,685	\$90,685	\$91,893 \$8,719
8.5" Total	\$54,437	\$54,437	\$54,437	\$57,585 \$36,052
Total	\$145,122	\$145,122	\$154,914	\$150,961 \$38,164

Frequency Chart

Displays the number of values (frequency - right hand side of the chart) that contains a given interval and how possible (probability – left hand side of the chart) is the value to fall into that number. This is available per hole interval and total well.

Total Cost Frequency Chart



Sensitivity Chart

Displays a bar graph of which risks have the most impact (variance) on the model with respect to interval and total well cost.

Identification of the major risks allows for a more pro-active approach to risk management.

Tables, Charts and Calculations

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1. Formulas for Adjusting Drilling Fluid Properties

1.1. Mud Weight

Calculating Material Requirements to Increase Mud Weight

Weight-up calculations (volume increase tolerated)

Use the following formulas to calculate the amount of weight material required to increase the density of a drilling fluid when a volume increase can be tolerated.

$$B = \left[\frac{(350.5)(\rho_{WM})(W_F - W_I)}{(8.3454)(\rho_{WM}) - W_F} \right] \times V_I$$

$$V = \frac{B}{(350.5)(\rho_{WM})}$$

Where

B is the weight material to add, lb

V_I is the starting volume of mud, bbl

ρ_{WM} is the specific gravity of the weight material

W_F is the desired mud weight, lb/gal W_I is the starting mud weight, lb/gal V is the volume increase, bbl

Weight-Up Calculations (Final Volume Specified)

Use the following formulas to calculate a starting volume of mud and amount of weight material required to increase the density of a drilling fluid when the final volume is specified.

$$V_I = \frac{8.3454 \frac{W}{W_I}}{8.3454 \frac{W}{W_I} - W_F} V_D$$

$$B = (V_D - V_I) (W_I) (350.5)$$

Where

V_I is the starting volume of mud, bbl

ρ_{WM} is the specific gravity of the weight material

W_F is the desired mud weight, lb/gal

W_I is the starting mud weight, lb/gal

V_D is the desired final volume, bbl

B is the weight material to add, lb

Calculating Material Requirements to Decrease Mud Weight

Decrease mud weight (volume increase tolerated)

Use the following formula to calculate the volume of dilution fluid required to decrease the density of a drilling fluid when a volume increase can be tolerated.

$$V_{DF} = V_1 \frac{W_1 - W_F}{W_F - 8.3454 \rho_{DF}}$$

Where

V_{DF} is the volume of dilution fluid required, bbl

V_1 is the starting volume of mud, bbl

W_1 is the starting mud weight, lb/gal

W_F is the final mud weight, lb/gal

ρ_{DF} is the specific gravity of the dilution fluid

Decrease Mud Weight (Final Volume Specified)

Use the following formula to calculate the starting volume of mud and a volume of dilution fluid required to decrease the density of a drilling fluid when the final volume is specified.

$$V_1 = \frac{8.3454 \rho_{DF} W_1}{8.3454 \rho_{DF} W_F - W_1} V_D$$

$$V_{DF} = V_D - V_1$$

Where

V_1 is the starting volume of mud, bbl

ρ_{DF} is the specific gravity of the dilution fluid

W_F is the desired mud weight, lb/gal

W_1 is the starting mud weight, lb/gal

V_D is the desired final volume, bbl

V_{DF} is the volume of dilution fluid to add, bbl

1.2. Oil / Water Ratio

Use the following formulas to calculate the volume of oil or water required to change the oil/water ratio of a mud when a volume increase can be tolerated.

Increase Oil/Water Ratio

Increase the oil/water ratio by adding oil using the following formulas.

$$P_w = \frac{R_w}{R_w R_o V_o}$$

Or

$$V = \frac{R_w}{P_w} \cdot R_o$$

$$W_R = \frac{W_i \cdot 8.3454 \cdot V_o}{1 \cdot V_o}$$

Where

V_o is the volume of oil to be added, bbl/bbl mud

R_o is the % oil from retort, decimal equivalent

R_w is the % water from retort, decimal equivalent

P_w is the new % by volume water in the liquid phase, decimal equivalent

W_R is the resulting mud weight, lb/gal

W_i is the starting mud weight, lb/gal

is the specific gravity of the oil

Decrease Oil/Water Ratio

Decrease the oil/water ratio by adding water using the following formulas.

$$P_o = \frac{R_o}{(R_o R_w V_w)}$$

Or

$$V = \frac{R_o}{P_o} \cdot R_w$$

$$W_R = \frac{W_i \cdot 8.3454 \cdot V_w}{1 \cdot V_w}$$

Where

V_w is the volume of water to be added, bbl/bbl mud

R_o is the % oil from retort, decimal equivalent

R_w is the % water from retort, decimal equivalent

P_o is the new % by volume oil in the liquid phase, decimal equivalent

W_R is the resulting mud weight, lb/gal

W_I is the starting mud weight, lb/gal

Calculate the amount of weight material required to increase density back to original density.

2. Formulas for Calculating Area and Volume

All dimensions expressed in feet.

2.1. Calculating Pit and Tank Volume

Rectangular Tank

$$\text{Volume (bbl)} = \frac{\text{length} \times \text{width} \times \text{height}}{5.6146}$$

$$\text{Volume (bbl/ft)} = \frac{\text{length} \times \text{width}}{5.6146}$$

$$\text{Volume (bbl/in)} = \frac{\text{length} \times \text{width}}{67.375}$$

Vertical Cylindrical Tank

$$\text{Volume (bbl)} = \frac{\text{diameter}^2 \times \text{height}}{7.1486}$$

$$\text{Volume (bbl/ft)} = \frac{\text{diameter}^2}{7.1486}$$

$$\text{Volume (bbl/in)} = \frac{\text{diameter}^2}{85.7833}$$

Horizontal Cylindrical Tank (Half Full or Less)

$$\text{Volume (bbl)} = \frac{0.3168d \times 1.403h^2 \times 0.933h^3}{5.6146} \times \text{length}$$

Where

h is the height of the fluid level, ft

d is the diameter of the tank, ft

Horizontal Cylindrical Tank (More Than Half Full)

$$\text{Volume (bbl)} = \frac{\text{diameter}^2 \times \text{length}}{7.1486}$$

$$\frac{0.3168d \times h \times 1.403h^2 \times 0.933h^3}{5.6146} \times \text{length}$$

Where

h is the height of the empty portion of the tank, ft

d is the diameter of the tank, ft

2.2. Calculating Hole Volume

All diameters are expressed in inches; section lengths are expressed in feet.

Hole Volume (No Drillstring in the Hole)

$$\text{Volume (bbl)} = \text{Section length} \times \frac{\text{Hole diameter}^2}{1029.4}$$

$$\text{Volume (bbl/ft)} = \frac{\text{Hole diameter}^2}{1029.4}$$

Annular Volume (Capacity)

$$\text{Volume (bbl)} = \text{Section length} \times \frac{\text{Hole diameter}^2 - \text{Pipe diameter}^2}{1029.4}$$

$$\text{Volume (bbl/ft)} = \frac{\text{Hole diameter}^2 - \text{Pipe diameter}^2}{1029.4}$$

Drill Pipe or Drill Collar Capacity and Displacement

$$\text{Capacity (bbl/ft)} = \frac{\text{Inside diameter}^2}{1029.4}$$

$$\text{Displacement (bbl/ft)} = \frac{\text{Outside diameter}^2 - \text{Inside diameter}^2}{1029.4}$$

Calculations (Metal Only with Couplings)

$$0.002 \times (\text{weight of pipe/ft with couplings}) \times (\text{depth, ft}) = \text{Displacement of pipe, ft}^3$$

$$0.000367 \times (\text{Weight of pipe/ft with couplings}) \times (\text{depth, ft}) = \text{Displacement of pipe, bbl}$$

3. Dimensions

3.1. Casing Dimensions

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
4	3.732	5.56
	3.550	9.26
	3.550	9.50
	3.480	11.0
	3.430	11.60
	3.364	12.60
4 1/2	4.216	6.75
	4.090	9.50
	4.052	10.50
	4.030	10.98
4 1/2	4.026	11.00
	4.000	11.60
	3.990	11.75
	3.958	12.60
	3.960	12.75
	3.920	13.50
	3.826	15.10
	3.826	16.60
	3.640	18.80
	3.500	21.60
	3.380	24.60
	3.240	26.50
4 3/4	4.364	9.50
	4.082	16.00
	4.070	16.50
	4.000	18.00
	3.910	20.00
	3.850	21.00
5	4.696	8.00
	4.560	11.50
	4.500	12.85
	4.494	13.00
	4.450	14.00
	4.408	15.00
	4.276	18.00
	4.184	20.30
	4.154	21.00
	4.044	23.20
	4.000	24.20
5 1/4	4.944	8.50
	4.886	10.00
	4.768	13.00
	4.650	16.00
	5.044	13.00
	5.012	14.00
	4.974	15.00
	4.950	15.50
	4.892	17.00
	4.778	20.00
	4.670	23.00
	4.580	25.00
	4.548	26.00
	4.276	32.30
	4.090	36.40

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
5 3/4	5.290	14.00

	5.190	17.00
	5.090	19.50
	5.090	20.00
	4.990	22.50
	4.990	23.00
	4.890	25.20
6	5.672	10.50
	5.524	15.00
	5.50	16.00
	5.450	17.00
	5.424	18.00
	5.352	20.00
	5.240	23.00
	5.140	26.00
	6.287	12.00
	6.260	13.00
	6.135	17.00
	6.049	20.00
	5.980	22.00
	5.921	24.00
	5.880	25.00
	5.855	26.00
	5.837	26.80
	5.791	28.00
	5.761	29.00
	5.675	31.80
	5.675	32.00
	5.595	34.00

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
7	6.652	13.00
	6.538	17.00
	6.456	20.00
	6.398	22.00
	6.366	23.00
	6.336	24.00
	6.276	26.00
	6.214	28.00
	6.184	29.00
	6.168	29.80
	6.154	30.00
	6.094	32.00
	6.048	33.70
	6.004	35.00
	5.920	38.00
	5.836	40.20
	5.820	41.00
	5.736	43.00
	5.720	44.00
	5.540	49.50
	7.263	14.75
	7.125	20.00
	7.025	24.00
	6.969	26.40
	6.875	29.70
	6.765	33.70
	6.760	34.00
	6.710	35.50
	6.655	38.00
	6.625	39.00
	6.445	45.00
	6.435	45.30
7 ¾	6.560	46.10
8	7.528	20.00
	7.386	26.00
8	7.485	28.00
	7.385	32.00
	7.285	35.50
	7.285	36.00
	7.185	39.50
	7.185	40.00
	7.125	42.00
8	8.191	20.00
	8.097	24.00
	8.017	28.00
	7.921	32.00
	7.825	36.00
	7.775	38.00
	7.725	40.00
	7.651	43.00
	7.625	44.00
	7.537	48.00
	7.511	49.00

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
9	8.290	34.00
	8.196	38.00
	8.150	40.00
	8.150	41.20
	8.032	45.00
	8.032	46.10
	7.910	50.20
	7.810	54.00
	7.812	55.20
9	9.063	29.30
	9.001	32.30
	8.921	36.00
	8.885	38.00
	8.835	40.00
	8.799	42.00
	8.755	43.50
	8.750	44.30
	8.681	47.00
	8.680	47.20
	8.535	53.50
	8.450	57.40
	8.435	58.40
	8.375	61.10
10	9.384	33.00
	9.200	41.50
	9.120	45.50
	9.016	50.50
	8.908	55.50
	8.790	61.20
	8.780	60.00
10 ¾	10.192	32.75
	10.140	35.75
	10.050	40.50
	9.950	45.50
	9.950	46.20
	9.902	48.00
	9.850	49.50
	9.850	51.00
	9.784	54.00
	9.760	55.50

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
10 ¾	9.660	60.70
	9.560	65.70
	9.450	71.10
	9.350	76.00
	9.250	81.00
11	10.552	26.75
11 ¾	11.15	38.00
	11.084	42.00
	11.00	47.00
	10.950	50.00
	10.880	54.00
	10.772	60.00
	10.770	61.00
	10.682	65.00
12	11.514	31.50
	11.384	40.00
12 ¾	12.250	33.38
	12.188	37.42
	12.126	41.45
	12.130	43.00
	12.090	43.77
	12.062	45.58
	12.000	49.56
	11.970	53.00
13	12.438	40.00
	12.360	45.00
	12.282	50.00
	12.200	54.00
	12.715	48.00
	12.615	54.50
	12.515	61.00
	12.415	68.00
	12.347	72.00
	12.275	77.00
	12.175	83.00
	12.175	83.50
	12.159	85.00
	12.031	92.00
	11.937	98.00
14	13.448	42.00
	13.344	50.00

Outside diameter, in	Inside diameter, in	Wt/ft with coupling, lb
15	14.418	47.50
16	15.396	52.50
	15.375	55.00
	15.250	65.00
	15.198	70.00
	15.124	75.00
	15.010	84.00
	14.688	109.00
	14.570	118.00
18	17.180	80.00
18	17.855	78.00
	17.755	87.50
	17.655	96.50
20	19.190	90.00
	19.124	94.00
	19.000	106.50
	18.730	133.00
	18.376	169.00
21 ½	20.710	92.50
	20.610	103.00
	20.510	114.00
24 ½	23.850	88.00
	23.750	100.50
	23.650	113.00
30	29.376	98.93
	29.250	118.65
	29.000	157.53
	28.750	196.08
	28.500	234.29
	28.000	309.72
	27.750	346.93
	27.500	383.81
	27.000	456.57

3.2. Cylinder Capacities

Capacity of a Long Cylinder

$$\text{bbl}/100 \text{ ft} = 0.0972$$

$$D^2 \text{ bbl}/\text{inch} = 0.000081$$

$$D^2 \text{ bbl}/1,000 \text{ ft} = 0.972$$

$$D^2 \text{ ft}/\text{bbl} = 1029 \div D^2$$

Where D is the diameter of the cylinder, in

Inside Diameter of a Steel Cylinder

$$ID = \sqrt{OD^2 - 0.3745W}$$

Where

OD is the outside diameter, in

W is the weight, lb/ft

3.3. Drill Pipe Dimensions

Outside diameter, in	Inside diameter, in	Wt/ft with tool joints, lb
1.9	1.5	3.75
2	2.00	4.80
	1.995	4.85
	1.815	6.65
2	2.469	6.45
	2.441	6.85
	2.323	8.35
	2.151	10.40
3 ½	3.063	8.50
	2.992	9.50
	2.900	11.20
	2.764	13.30
	2.602	15.25
	2.602	15.50
3	3.181	14.50
4	3.500	10.40
	3.476	11.85
	3.382	12.50
	3.340	14.00
	3.244	15.30
	3.240	15.70
4 ½	4.00	12.75
	3.958	13.75
	3.826	16.60
	3.754	18.10
	3.640	20.00
4 ¾	4.00	19.08
5	5.00	14.20
	4.408	15.00
	4.408	16.25
	4.276	18.35
	4.276	18.35
	4.214	20.50
	4.00	25.60
5 ½	4.778	21.90
	4.670	23.25
	4.670	24.70
5 ¾	5.00	23.40
5 9/16	4.975	19.00
	4.859	22.20
	4.733	23.30
	4.733	25.25
6	6.065	22.20
	5.965	23.30
	5.965	25.20
	5.761	31.90

Outside diameter, in	Inside diameter, in	Wt/ft with tool joints, lb
7	6.965	28.75
	6.969	29.25
8	7.825	40.00
	7.625	46.50

3.4. Tubing Dimensions

Outside diameter, in	Inside diameter, in	Weight/ft, lb
0.75	0.636	0.42
1.00	0.866	0.67
1.050	0.824	1.14
	0.824	1.20
	0.742	1.55
1.315	1.125	1.30
	1.097	1.43
	1.065	1.63
	1.049	1.70
	1.049	1.72
	1.049	1.80
	1.049	1.90
	0.957	2.25
	0.957	2.30
1.660	1.410	2.10
	1.380	2.30
	1.380	2.40
	1.278	3.02
	1.264	3.24
	1.264	3.29
1.900	1.650	2.40
	1.610	2.75
	1.610	2.90
	1.500	3.64
	1.462	4.19
2	1.670	3.30
	1.670	3.40
2 ¹ / ₁₆	1.813	2.66
	1.750	3.25
	1.613	4.50
2	2.125	3.10
	2.107	3.32
	2.041	4.00
	1.995	4.60
	1.995	4.70
	1.947	5.00
	1.939	5.30
	1.867	5.80
	1.867	5.95
	1.853	6.20
	1.703	7.70

Outside diameter, in	Inside diameter, in	Weight/ft, lb
2	2.579	4.36
	2.563	4.64
	2.469	5.90
	2.441	6.40
	2.441	6.50
	2.323	7.90
	2.259	8.60
	2.259	8.70
	2.195	9.50
	2.151	10.40
	2.091	10.70
	2.065	11.00
	1.995	11.65
3 ½	3.188	5.63
	3.068	7.70
	3.018	8.50
	3.018	8.90
	2.992	9.20
	2.992	9.30
	2.992	10.20
	2.992	10.30
	2.900	11.20
3 ½	2.750	12.70
	2.764	12.80
	2.750	12.95
	2.764	13.30
	2.602	14.90
	2.602	15.50
	2.548	15.80
	2.480	16.70
	2.440	17.05
4	3.548	9.25
	3.548	9.40
	3.548	9.50
	3.476	10.80
	3.476	10.90
	3.476	11.00
	3.428	11.60
	3.340	13.30
	3.340	13.40
	3.000	19.00
	2.780	22.50

Outside diameter, in	Inside diameter, in	Weight/ft, lb
4 ½	4.026	11.00
	3.990	11.80
	3.958	12.60
	3.958	12.75
	3.920	13.50
	3.826	15.40
	3.826	15.50
	3.754	16.90
	3.640	19.20
	3.500	21.60
	3.380	24.60
	3.240	26.50

4. Formulas for Calculating Pump Output

4.1. Duplex Pump

Pump output

$$\frac{\text{Efficiency}}{100} \times \frac{2 \text{ liner}^2 \times \text{rod diameter}^2 \times \text{stroke}}{6176.4}$$

4.2. Triplex Pump

Pump output (bbl/stroke) = (liner inside diameter)² × 0.000243 × stroke length

5. Pump Capacities

5.1. Duplex Pumps

The capacities of duplex pumps are given in barrels per cycle at different liner bores and strokes. No allowance is made for the volume occupied by the pump rods.

Liner bore, in (mm)	Stroke, in (mm)	Volume, bbl/cycle (L) at 100% pump efficiency
4.00 (102)	10 (254)	0.0518 (8.24)
4.50 (114)	10 (254)	0.0656 (10.4)
5.00 (127)	10 (254)	0.0810 (12.9)
5.25 (133)	10 (254)	0.0893 (14.2)
5.50 (140)	10 (254)	0.098 (15.6)
5.75 (146)	10 (254)	0.107 (17.0)
6.00 (152)	10 (254)	0.117 (18.6)
6.25 (159)	10 (254)	0.127 (20.2)
6.50 (165)	10 (254)	0.137 (21.8)
6.75 (171)	10 (254)	0.148 (23.5)
7.00 (178)	10 (254)	0.159 (25.3)
7.25 (184)	10 (254)	0.170 (27.0)
6.00 (152)	12 (305)	0.140 (22.3)
6.25 (159)	12 (305)	0.152 (24.2)
6.50 (165)	12 (305)	0.161 (25.6)
6.75(171)	12 (305)	0.177 (28.1)
7.00 (178)	12 (305)	0.190 (30.2)
7.25 (184)	12 (305)	0.204 (32.4)
6.00 (152)	14 (356)	0.163 (25.9)
6.25 (159)	14 (356)	0.177 (28.1)
6.50 (165)	14 (356)	0.192 (30.5)
6.75 (171)	14 (356)	0.207 (32.9)
7.00 (178)	14 (356)	0.222 (35.3)
7.25 (184)	14 (356)	0.238 (37.8)
6.25 (159)	16 (406)	0.202 (32.1)
6.50 (165)	16 (406)	0.219 (34.8)
6.75 (171)	16 (406)	0.236 (37.5)
7.00 (178)	16 (406)	0.254 (40.4)
7.25 (184)	16 (406)	0.272 (43.2)
6.00 (156)	18 (451)	0.210 (33.4)
6.25 (159)	18 (451)	0.228 (36.3)
6.50 (165)	18 (451)	0.246 (39.1)
6.75 (171)	18 (451)	0.266 (42.3)
7.00 (178)	18 (451)	0.286 (45.5)
7.25 (184)	18 (451)	0.306 (48.7)
7.50 (191)	18 (451)	0.328 (52.2)
7.75 (197)	18 (451)	0.350 (55.7)
6.00 (156)	20 (508)	0.233 (37.0)
6.25 (159)	20 (508)	0.253 (40.2)
6.50 (165)	20 (508)	0.274 (43.6)

Liner bore, in (mm)	Stroke, in (mm)	Volume, bbl/cycle (L) at
6.75 (171)	20 (508)	0.295 (46.9)
7.00 (178)	20 (508)	0.317 (50.4)
7.25 (184)	20 (508)	0.340 (54.1)
7.50 (191)	20 (508)	0.364 (57.9)
7.75 (197)	20 (508)	0.389 (61.9)
8.00 (203)	20 (508)	0.414 (65.8)
7.00 (178)	22 (559)	0.349 (55.5)
7.25 (184)	22 (559)	0.374 (59.5)
7.50 (191)	22 (559)	0.401 (63.8)
7.75 (197)	22 (559)	0.428 (68.1)
8.00 (203)	22 (559)	0.456 (72.5)
8.25 (210)	22 (559)	0.485 (77.1)
8.50 (216)	22 (559)	0.515 (81.9)
8.75 (222)	22 (559)	0.545 (86.7)
9.00 (229)	22 (559)	0.577 (91.7)
9.25 (235)	22 (559)	0.610 (97.0)
8.00 (203)	24 (610)	0.497 (79.0)
8.25 (210)	24 (610)	0.529 (84.1)
8.50 (216)	24 (610)	0.562 (89.4)
8.75 (222)	24 (610)	0.595 (94.6)
9.00 (229)	24 (610)	0.630 (100.2)
9.25 (235)	24 (610)	0.665 (105.7)
9.75 (248)	24 (610)	0.739 (117.5)
10.00 (254)	24 (610)	0.777 (123.5)

5.2. Triplex Pumps

7-inch stroke triplex pump, bbl/cycle

Diameter, in (mm)	Stroke, in (mm)	Displacement, bbl/cycle (L)
7.00 (178)	7 (178)	0.083 (13.25)
6.50 (165)	7 (178)	0.072 (11.43)
6.00 (152)	7 (178)	0.061 (9.73)
5.00 (140)	7 (178)	0.051 (8.18)
5.00 (127)	7 (178)	0.043 (6.78)
4.50 (11)	7 (178)	0.035 (5.49)

8-inch stroke triplex pump

Diameter, in (mm)	Stroke, in (mm)	Displacement/cycle, bbl (L)
6.25 (159)	8 (203)	0.076 (12.07)
6.00 (152)	8 (203)	0.070 (11.13)
5.50 (140)	8 (203)	0.059 (9.35)
5.00 (127)	8 (203)	0.049 (7.72)
4.50 (114)	8 (203)	0.039 (6.25)
4.00 (102)	8 (203)	0.031 (4.96)

9-inch stroke triplex pump

Diameter, in (mm)	Stroke, in (mm)	Displacement/cycle, bbl (L)
7.00 (178)	9 (229)	0.107 (17.03)
6.50 (165)	9 (229)	0.092 (14.69)
6.25 (159)	9 (229)	0.085 (13.55)
6.00 (152)	9 (229)	0.079 (12.49)
5.50 (140)	9 (229)	0.066 (10.48)
5.00 (127)	9 (229)	0.055 (8.66)
4.50 (114)	9 (229)	0.044 (7.04)

11-inch stroke triplex pump

Diameter, in (mm)	Stroke, in (mm)	Displacement/cycle, bbl (L)
7.00 (178)	11 (279)	0.130 (20.82)
6.50 (165)	11 (279)	0.113 (17.94)
6.00 (152)	11 (279)	0.096 (15.29)
5.50 (140)	11 (279)	0.081 (12.83)

6. Chemical Properties

The following table identifies the chemical properties of some elements used in the oilfield.

Element	Symbol	Atomic weight	Atomic number
Aluminum	Al	26.98	13
Arsenic	As	74.92	33
Barium	Ba	137.36	56
Bromine	Br	79.916	35
Calcium	Ca	40.08	20
Carbon	C	12.011	6
Cesium	Cs	132.91	55
Chlorine	Cl	35.457	17
Chromium	Cr	52.01	24
Copper	Cu	63.54	29
Fluorine	F	19	9
Hydrogen	H	1.008	1
Iodine	I	126.91	53
Iron	Fe	55.85	26
Lead	Pb	207.21	82
Lithium	Li	6.94	3
Magnesium	Mg	24.32	12
Manganese	Mn	54.94	25
Mercury	Hg	200.61	80
Nitrogen	N	14.008	7
Oxygen	O	16	8
Phosphorous	P	30.975	15
Potassium	K	39.1	19
Silicon	Si	28.09	14
Silver	Ag	107.873	47
Sodium	Na	22.991	11
Sulfur	S	32.066	16
Titanium	Ti	47.9	22
Tungsten	W	183.86	74
Zinc	Zn	65.38	30

6.1. Periodic Table of the Elements

Chemical elements are physically related to one another.

The table below shows elements with similar chemical behavior in vertical groups.

PERIODIC TABLE OF THE ELEMENTS

KEY

1 — Atomic Number
 H — Symbol
 Hydrogen
 1.00797 — Atomic Weight (use, if no parenthesis, it is the Mass Number of the most stable isotope)

1	2																	18	19	20																	36	37	38																	54	55	56																	82	83	84																	118	119	120																	154	155	156																	202	203	204																	278	279	280																	352	353	354																	430	431	432																	503	504	505																	578	579	580																	654	655	656																	729	730	731																	810	811	812																	894	895	896																	970	971	972																	1044	1045	1046																	1122	1123	1124																	1200	1201	1202																	1278	1279	1280																	1356	1357	1358																	1434	1435	1436																	1512	1513	1514																	1590	1591	1592																	1668	1669	1670																	1746	1747	1748																	1824	1825	1826																	1902	1903	1904																	1980	1981	1982																	2058	2059	2060																	2136	2137	2138																	2214	2215	2216																	2292	2293	2294																	2370	2371	2372																	2448	2449	2450																	2526	2527	2528																	2604	2605	2606																	2682	2683	2684																	2760	2761	2762																	2838	2839	2840																	2916	2917	2918																	2994	2995	2996																	3072	3073	3074																	3150	3151	3152																	3228	3229	3230																	3306	3307	3308																	3384	3385	3386																	3462	3463	3464																	3540	3541	3542																	3618	3619	3620																	3696	3697	3698																	3774	3775	3776																	3852	3853	3854																	3930	3931	3932																	4008	4009	4010																	4086	4087	4088																	4164	4165	4166																	4242	4243	4244																	4320	4321	4322																	4398	4399	4400																	4476	4477	4478																	4554	4555	4556																	4632	4633	4634																	4710	4711	4712																	4788	4789	4790																	4866	4867	4868																	4944	4945	4946																	5022	5023	5024																	5100	5101	5102																	5178	5179	5180																	5256	5257	5258																	5334	5335	5336																	5412	5413	5414																	5490	5491	5492																	5568	5569	5570																	5646	5647	5648																	5724	5725	5726																	5802	5803	5804																	5880	5881	5882																	5958	5959	5960																	6036	6037	6038																	6114	6115	6116																	6192	6193	6194																	6270	6271	6272																	6348	6349	6350																	6426	6427	6428																	6504	6505	6506																	6582	6583	6584																	6660	6661	6662																	6738	6739	6740																	6816	6817	6818																	6894	6895	6896																	6972	6973	6974																	7050	7051	7052																	7128	7129	7130																	7206	7207	7208																	7284	7285	7286																	7362	7363	7364																	7440	7441	7442																	7518	7519	7520																	7596	7597	7598																	7674	7675	7676																	7752	7753	7754																	7830	7831	7832																	7908	7909	7910																	7986	7987	7988																	8064	8065	8066																	8142	8143	8144																	8220	8221	8222																	8298	8299	8300																	8376	8377	8378																	8454	8455	8456																	8532	8533	8534																	8610	8611	8612																	8688	8689	8690																	8766	8767	8768																	8844	8845	8846																	8922	8923	8924																	9000	9001	9002																	9078	9079	9080																	9156	9157	9158																	9234	9235	9236																	9312	9313	9314																	9390	9391	9392																	9468	9469	9470																	9546	9547	9548																	9624	9625	9626																	9702	9703	9704																	9780	9781	9782																	9858	9859	9860																	9936	9937	9938																	10014	10015	10016																	10092	10093	10094																	10170	10171	10172																	10248	10249	10250																	10326	10327	10328																	10404	10405	10406																	10482	10483	10484																	10560	10561	10562																	10638	10639	10640																	10716	10717	10718																	10794	10795	10796																	10872	10873	10874																	10950	10951	10952																	11028	11029	11030																	11106	11107	11108																	11184	11185	11186																	11262	11263	11264																	11340	11341	11342																	11418	11419	11420																	11496	11497	11498																	11574	11575	11576																	11652	11653	11654																	11730	11731	11732																	11808	11809	11810																	11886	11887	11888																	11964	11965	11966																	12042	12043	12044																	12120	12121	12122																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Pounds Chemical Required to Remove Certain Contaminants

Contaminant to be removed	Chemical used to remove contaminant	Conversion factor mg/L (contaminant) x factor = lb/bbl chemical to add
Ca ⁺⁺	Soda ash	0.000925
Ca ⁺⁺	Sodium bicarbonate	0.000734
Mg ⁺⁺	Caustic soda	0.00115
CO ₃ ⁻²	Lime	0.00043
HCO ₃ ⁻¹	Lime	0.00043
H ₂ S	Lime	0.00076
H ₂ S	Zinc carbonate	0.00128
H ₂ S	Zinc oxide	0.000836

Due to the extreme danger associated with hydrogen sulfide (H₂S), it is recommended that a minimum of 1½ times the calculated amount of chemical be added.

6.3. Physical Properties

Bulk Volume Data

Approximate bulk volumes for three common materials.

Material	Amount	Approximate bulk volume
AQUAGEL	100 lb	1.67 ft ³
BAROID	100 lb	0.74 ft ³
Cement	94 lb	1 ft ³

Density of Common Materials

Specific gravities and densities for common materials.

Material	Specific gravity	lb/gal	lb/bbl
Barite	4.2 to 4.3	35.0 to 35.8	1470 to 1504
Calcium carbonate	2.7	22.5	945
Cement	3.1 to 3.2	25.8 to 26.7	1085 to 1120
Clays and/or drilled solids	2.4 to 2.7	20.0 to 22.5	840 to 945
Diesel oil	0.84	7.0	294
Dolomite	2.8 to 3.0	23.3 to 25.0	980 to 1050
Feldspar	2.4 to 2.7	20.0 to 22.5	840 to 945
Galena	6.5	54.1	2275
Gypsum	2.3	19.2	805
Halite (rock salt)	2.2	18.3	770
Iron	7.8	65.0	2730
Iron oxide (hematite)	5.1	42.5	1785
Lead	11.4	95.0	3990
Limestone	2.7 to 2.9	22.5 to 24.2	945 to 1015
Slate	2.7 to 2.8	22.5 to 23.3	945 to 980
Steel	7.0- to 8.0	58.3 to 66.6	2450 to 2800

6.4. Specific Materials

Saltwater Data Tables

Maximum Solubilities of Sodium Chloride

Temperature °F (°C)	% NaCl by weight (in saturated solution)
32 (0)	26.3
68 (20)	26.5
122 (50)	27.0
212 (100)	28.5

Sodium Chloride Solution Densities

Densities of aqueous sodium chloride solutions at 68°F (20°C).

sg	% NaCl by wt	Grams NaCl, 100 cm ³ solution	NaCl, lb/ft ³	NaCl, lb/gal	NaCl, lb/bbl
1.0053	1	1.01	0.628	0.84	3.52
1.0125	2	2.03	1.26	0.169	7.10
1.0268	4	4.11	2.56	0.343	14.40
1.0413	6	6.25	3.90	0.521	21.90
1.0559	8	8.45	5.27	0.705	29.61
1.0707	10	10.71	6.68	0.894	37.53
1.0857	12	13.03	8.13	1.09	45.65
1.1009	14	15.41	9.62	1.29	54.01
1.1162	16	17.86	11.15	1.49	62.58
1.1319	18	20.37	12.72	1.70	71.40
1.1478	20	22.96	14.33	1.92	80.47
1.1640	22	25.61	15.99	2.14	89.75
1.1804	24	28.33	17.69	2.36	99.29
1.1972	26	31.13	19.43	2.60	109.12

Seawater Composition Chemicals

Typical chemicals in sea water (average sg = 1.025) and gives their concentrations.

Constituent	Parts per million	Equivalent parts per million
Sodium	10440	454.0
Potassium	375	9.6
Magnesium	1270	104.6
Calcium	410	20.4
Chloride	18970	535.0
Sulfate	2720	57.8
Carbon dioxide	90	4.1
Other constituents	80	n/a

7. Metric and Standard Conversion Factors

The following table gives conversion factors used for converting one unit to another. Both metric-to- standard and standard-to-metric conversion factors are listed.

Multiply	By	To obtain
Atmospheres	14.7	pounds per square inch (psi)
	1.0132	bars
	101.32	kilopascals
Barrels US (bbl)	42	gallons US (gal)
	35	gallons (imperial)
	5.615	cubic feet (ft ³)
	159	liters (L)
	0.159	cubic meters (m ³)
	350	pounds (lb) [H ₂ O at 68°F]
Barrels/foot (bbl/ft)	42	gallons/ft (gal/ft)
	5.615	cubic ft/ft (ft ³ /ft)
	159	liters (L)
	0.159	cubic meters/foot (m ³ /ft)
	521.6	liters/meter (L/m)
	0.5216	cubic meters/meter (m ³ /m)
Barrels/minute (bbl/min)	42	gallons/minute (gal/min)
	5.615	cubic ft/minute (ft ³ /min)
	159	liters/minute (L/min)
	0.159	cubic meters/minute (m ³ /min)
Bars	0.9869	atmospheres
	14.5	pounds per square inch (psi)
	100	kilopascals
Centimeters (cm)	0.0328	feet (ft)
	0.3937	inches (in)
	0.01	meters (m)
	10	millimeters (mm)
Cubic centimeters (cm ³)	0.0610	cubic inches (in ³)
	0.0010	liters (L)
	1.0	milliliters (mL)
Cubic feet (ft ³)	0.1781	barrels (bbl)
	7.4805	gallons (gal)
	1,728	cubic inches (in ³)
	28,317	cubic centimeters (cm ³)
	28.3170	liters (L)
	0.0283	cubic meters (m ³)
Cubic inches (in ³)	16.3871	cubic centimeters (cm ³)
	0.0164	liters (L)
	0.0006	cubic feet (ft ³)
	0.0043	gallons (gal)

Multiply	By	To obtain
Cubic meters (m ³)	6.2898	barrels (bbl)
	264.17	gallons (gal)
	35.31	cubic feet (ft ³)
	61023	cubic inches (in ³)
	1,000,000	cubic centimeters (cm ³)
	1,000	liters (L)
Cubic meters/minute (m ³ /min)	6.2898	barrels/minute (bbl/min)
	264.17	gallons/minute (gal/min)
	35.31	cubic feet/minute (ft ³ /min)
	1,000	liters/minute (L/min)
Degrees, angle	60	minutes (min)
	0.0175	radians
	3,600	seconds
Degrees, temperature Celsius (°C)	(°C × 1.8) + 32	degrees Fahrenheit (°F)
Degrees, temperature Fahrenheit (°F)	(°F - 32) / 1.8	degrees Celsius (°C)
Feet (ft)	30.48	centimeters (cm)
	0.3048	meters (m)
	12	inches (in)
	0.3333	yards (yd)
Feet/minute (ft/min)	0.0167	feet/second (ft/sec)
	0.3048	meters/minute (m/min)
	0.00508	meters/second (m/sec)
Feet/second (ft/sec)	60	feet/minute (ft/min)
	18.288	meters/minute (m/min)
	0.3048	meters/second (m/sec)
Gallons, US (gal)	3785	cubic centimeters (cm ³)
	3.785	liters (L)
	0.0038	cubic meters (m ³)
	231	cubic inches (in ³)
	0.1337	cubic feet (ft ³)
	0.0238	barrels (bbl)
Gallons/minute (gal/min)	0.0238	barrels/minute (bbl/min)
	0.1337	cubic feet/minute (ft ³ /min)
	3.785	liters/minute (L/min)
	0.0038	cubic meters/minute (m ³ /min)
Grams (g)	0.0010	kilograms (kg)
	1,000	milligrams (mg)
	0.03527	ounces (oz, avoirdupois)
	0.0022	pounds (lb)
Grams/liter (g/L)	0.0624	pounds/cubic foot (lb/ft ³)
	0.0083	pounds/gallon (lb/gal)
	0.3505	pounds/barrel (lb/bbl)
	1,000	milligrams/liter (mg/L)

Multiply	By	To obtain
Inches (in)	0.0833	feet (ft)
	0.0278	yards (yd)
	25,400	microns
	25.4	millimeters (mm)
	2.54	centimeters (cm)
	0.0254	meters (m)
Kilograms (kg)	1,000	grams (g)
	0.0010	metric tons
	2.2	pounds (lb)
Kilograms/cubic meter (kg/m ³)	0.3505	pounds/barrel (lb/bbl)
	0.0083	pounds/gallon (lb/gal)
	0.0624	pounds/cubic foot (lb/ft ³)
Kilometers (km)	39,370	inches (in)
	3280.84	feet (ft)
	1,000	meters (m)
	0.6214	miles (mi)
Kilometers/hour (km/hr or kph)	54.68	feet/minute (ft/min)
	0.9113	feet/second (ft/sec)
	0.54	knots
	0.6214	miles/hour (mi/hr or mph)
	1,000	meters/hour (m/hr)
	16.6667	meters/minute (m/min)
	0.2778	meters/second (m/sec)
Kilopascals	0.1450	pounds per square inch (psi)
	0.0100	bars
	0.0099	atmospheres
Knots	1.15	miles/hour (mi/hr or mph)
	6,080	feet/hour (ft/hr)
	101.27	feet/minute (ft/min)
	1.69	feet/second (ft/sec)
	1.85	kilometers/hour (km/hr or kph)
	30.87	meters/minute (m/min)
	0.5144	meters/second (m/sec)
Liters (L)	61.03	cubic inches (in ³)
	0.0353	cubic feet (ft ³)
	0.2642	gallons (gal)
	0.0063	barrels (bbl)
	1,000	cubic centimeters (cm ³)
	0.001	cubic meters (m ³)
Liters/minute (L/min)	0.2642	gallons/minute (gal/min)
	0.0063	barrels/minute (bbl/min)
	0.0353	cubic feet/minute (ft ³ /min)
Meters (m)	1,000	millimeters (mm)
	100	centimeters (cm)
	0.001	kilometers (km)
	39.37	inches (in)
	3.28	feet (ft)
	1.0936	yards (yd)

Multiply	By	To obtain
Meters/minute (m/min)	3.28	feet/minute (ft/min)
	0.05468	feet/second (ft/sec)
	0.03728	miles/hour (mi/hr or mph)
	0.01667	meters/second (m/sec)
	1.6670	centimeters/second (cm/sec)
	0.06	kilometers/hour (km/hr or kph)
Meters/second (m/sec)	2.2369	miles/hour (mi/hr or mph)
	196.85	feet/minute (ft/min)
	3.28	feet/second (ft/sec)
	100	centimeters/second (cm/sec)
	60	meters/minute (m/min)
	0.060	kilometers/hour (km/hr or kph)
Microns	0.0010	millimeters (mm)
	0.0001	centimeters (cm)
	0.00003937	inches (in)
Miles, statute (mi)	160,934	centimeters (cm)
	1609.34	meters (m)
	1.6093	kilometers (km)
	63,360	inches (in)
	5,280	feet (ft)
	1,760	yards (yd)
Miles, nautical	6,080.27	feet (ft)
	1.1516	statute miles (mi)
	1,853.27	meters (m)
	1.8533	kilometers (km)
Milliliters (mL)	0.0010	liters (L)
Millimeters (mm)	0.0010	meters (m)
	0.10	centimeters (cm)
	0.0394	inches (in)
Ounces (oz, avoirdupois)	0.0625	pounds (lb)
	28.3495	grams (g)
	0.0283	kilograms (kg)
Pounds (lb)	16	ounces (oz, avoirdupois)
	0.0005	short tons
	453.6	grams (g)
	0.4536	kilograms (kg)
Pounds/barrel (lb/bbl)	0.047	grams/cubic inch (g/in ³)
	2.853	kilograms/cubic meter (kg/m ³)
	0.1781	pounds/cubic foot (lb/ft ³)
	0.0238	pounds/gallon (lb/gal)
Pounds/cubic foot (lb/ft ³)	0.0160	grams/cubic centimeter (g/cm ³)
	16.0185	kilograms/cubic meter (kg/m ³)
	0.1337	pounds/gallon (lb/gal)
	5.6146	pounds/barrel (lb/bbl)

Multiply	By	To obtain
Pounds/gallon (lb/gal)	0.1198	grams/cubic centimeter (g/cm ³)
	119.8260	kilograms/cubic meter (kg/m ³)
	0.0238	pounds/barrel (lb/bbl)
	7.4805	pounds/cubic foot (lb/ft ³)
Pounds/square inch (lb/in ²) (psi)	0.0680	atmospheres
	0.0689	bars
	0.0703	kilograms/square centimeter (kg/cm ²)
	6.89	kilopascals
Pounds/square inch/foot (lb/in ² /ft)	22.6203	kilopascals/meter
Square centimeters (cm ²)	0.1550	square inches (in ²)
Square feet (ft ²)	929.03	square centimeters (cm ²)
	0.0929	square meters (m ²)
	144	square inches (in ²)
	0.1111	square yards (yd ²)
Square inches (in ²)	645.16	square millimeters (mm ²)
	6.4516	square centimeters (cm ²)
Square kilometers (km ²)	0.3861	square miles (mi ²)
	100	hectares
Square meters (m ²)	10.76	square feet (ft ²)
Square miles (mi ²)	2.59	square kilometers (km ²)
	640	acres
	259	hectares
Tons, long	2,240	pounds (lb)
	1,016	kilograms (kg)
	1.016	metric tons
Tons, metric	2,204	pounds (lb)
	1,000	kilograms (kg)
	0.9842	long tons
	1.1023	short tons
Tons, short	2,000	pounds (lb)
	907.18	kilograms (kg)
	0.9072	metric tons

Date Issued	Rev. No.	Summary of Key Revisions
1-Oct-2024	D	Currently under complete review
19-Oct-2021	C	Reviewed, no changes required
04-Aug-2016	B	Document reviewed, removed all links at back of document as no longer good
10-2012	A	Initial upload into CMS
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